

Generalized multiplication tables in every fixed dimension

Tseng

Abstract

For each fixed integer $k \geq 6$, we determine, up to constants depending only on k , the number of distinct products in an arbitrary rectangular $(k+1)$ -dimensional multiplication table. We also obtain a uniform estimate for the number of integers with a product of divisors in k prescribed dyadic intervals, throughout its natural comparison range. Together with the earlier results in lower dimensions, this settles the remaining fixed-dimensional rectangular cases. A single structural profile describes the answer. An ordered saddle determines its exponential rate and equality blocks; count windows, fractional conditional persistence, and one ordinary terminal bridge determine the remaining factors. The lower bound realizes these constraints in a common family of divisor decompositions, while a causal covering argument proves that the same costs are necessary. The saddle rate is a Poisson relative entropy. For sufficiently large positive block lengths, the derivative of the persistence spectrum locates the selected environments and the adaptive targets in a family of original prime tuples contributing the full order of the harmonic volume sum. Averaging the environment before taking the fractional power incurs a strictly positive exponent loss. Arithmetic stability extends the comparison to all relative scales, including bounded and zero original block lengths.

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1 Introduction

For real numbers $3 \leq N_1 \leq \dots \leq N_{k+1}$, write

$$A_{k+1}(\mathbf{N}) = \#\{n_1 \cdots n_{k+1} : n_i \in \mathbb{N}, n_i \leq N_i\}.$$

The generalized multiplication-table problem asks for the order of this quantity, uniformly in the side lengths, with the number of factors fixed. Although there are about $N_1 \cdots N_{k+1}$ ways to select the factors, many selections have the same product. The problem is to measure this multiplicity without losing the dependence on the relative sizes of the sides. We determine the order of $A_{k+1}(\mathbf{N})$ for every fixed $k \geq 6$ and all ordered side lengths. Combined with the work of Ford and Koukoulopoulos, this completes the order-of-magnitude theory in every fixed dimension.

1.1 The arithmetic results

The divisor-interval formulation both gives a more general arithmetic result and reveals the structure of the answer. For $\mathbf{y} = (y_1, \dots, y_k)$, let

$$H^{(k+1)}(x, \mathbf{y}, 2\mathbf{y}) = \#\{n \in \mathbb{N} : n \leq x, \exists d_i \in \mathbb{N} \cap (y_i, 2y_i] \ (1 \leq i \leq k), \ d_1 \cdots d_k \mid n\}.$$

All logarithms in this paper are natural. Set $y_0 = 3$ and define

$$\ell_i(\mathbf{y}) = \log \frac{3 \log y_i}{\log y_{i-1}}, \quad L_i = \ell_i - \log 3 = \log \frac{\log y_i}{\log y_{i-1}}. \quad (1.1)$$

Thus ordered cutoffs $3 \leq y_1 \leq \dots \leq y_k$ correspond exactly to vectors $\ell \in [\log 3, \infty)^k$, or equivalently $L_i \geq 0$.

The answer is expressed by a single function Φ_k of these logarithmic lengths. Its definition, given in Section 3 and completed in (3.13), involves a finite-dimensional ordered optimization, finitely many endpoint comparisons, and a scalar conditional persistence exponent. In particular, Φ_k is defined independently of the arithmetic quantities being estimated. We first state the two arithmetic consequences; the ingredients of the profile are explained below and then specified in full.

Theorem 1.1 (Divisors in prescribed dyadic intervals). *Fix an integer $k \geq 6$. Uniformly for real parameters satisfying*

$$3 \leq y_1 \leq \dots \leq y_k, \quad x \geq 2^k \left(\prod_{i=1}^k y_i \right) y_k, \quad (1.2)$$

one has

$$\frac{H^{(k+1)}(x, \mathbf{y}, 2\mathbf{y})}{x} \asymp_k \Phi_k(\ell(\mathbf{y})).$$

Theorem 1.2 (Rectangular multiplication tables). *Fix an integer $k \geq 6$. For all real side lengths $3 \leq N_1 \leq \dots \leq N_{k+1}$, put*

$$Y_0 = 3, \quad Y_i = \max\{3, N_i/2\} \quad (1 \leq i \leq k).$$

Then

$$A_{k+1}(\mathbf{N}) \asymp_k \left(\prod_{i=1}^{k+1} N_i \right) \Phi_k(\ell(\mathbf{Y})).$$

Here $f \asymp_k g$ means that $c_k g \leq f \leq C_k g$ for positive constants depending only on k . In particular, repeated side lengths, bounded initial sides and arbitrarily large ratios of sides are all included. The dimension of the table is $k+1$, so the new range starts with seven factors. The constants are allowed to depend on the fixed dimension. The restriction on x in (1.2) is the precise range of the divisor-count comparison used in the proof; Theorem 1.2 covers the full stated range of side lengths, including those for which some $N_i/2 < 3$.

1.2 Previous work and the remaining dimensions

The classical problem goes back to Erdős [10]. Its connection with the distribution of divisors in an interval led to extensive work on $H^{(2)}(x, y, z)$, including Tenenbaum's estimates [28]. Ford

determined the order of the classical square table [12, Corollary 3]. With $Q(u) = u \log u - u + 1$, his result is

$$A_2(N, N) \asymp \frac{N^2}{(\log N)^{Q(1/\log 2)} (\log \log N)^{3/2}} \quad (N \geq 3).$$

Ford's interval estimate [12, Corollary 2], together with the multiplication-table comparison in [19, Theorem 1.1], also gives the order of all two-dimensional rectangular tables. This distinction between squares and rectangles is useful in describing the higher-dimensional problem.

Koukoulopoulos proved that for every fixed $k \geq 2$,

$$A_{k+1}(N, \dots, N) \asymp_k \frac{N^{k+1}}{(\log N)^{Q(k/\log(k+1))} (\log \log N)^{3/2}}, \quad N \geq 3, \quad (1.3)$$

see [18, Corollary 1]. His corresponding divisor theorem covers comparable logarithmic scales. Thus high-dimensional cubes were already understood. For arbitrary side lengths, his later work [19] established a uniform estimate throughout (1.2) when $2 \leq k \leq 5$, and consequently determined the order of rectangular tables with three through six factors. The same paper gives a general upper bound in every fixed dimension, a matching estimate under an additional condition on the saddle parameter, and a high-dimensional range in which the logarithmic scales are sufficiently close; see its Theorems 1.3–1.5. Its Section 2.2, Lemma 2.3 and Remark 2.4 identify finer structural constraints that can make the earlier expression too large when $k \geq 6$. The task of determining the uniform order for these remaining dimensions was also stated explicitly in [17, Chapter 6, pp. 87–88]. Theorems 1.1 and 1.2 resolve these remaining fixed-dimensional cases.

More precise questions about the square table have recently seen further progress. Sawhney announced joint work with Green giving an asymptotic formula with a periodic amplitude for the two-dimensional integer square table [27]. Their paper [15, Theorem 1.1] proves an asymptotic formula for the permutation analogue and describes the integer-table work as forthcoming. Sabuncu and Táfula [25, Theorem 1.2] study the eventual polynomial dependence on dimension for a common side length. The present results concern uniform two-sided bounds for arbitrary side lengths in each fixed dimension.

1.3 What the profile records

The exponential part of the answer is governed by an ordered saddle $s_1 \geq \dots \geq s_k$. Its maximal consecutive blocks of equal coordinates organize the constraints; we call these blocks *owners*. Within an owner, the original prime ranges retain their own lengths, and the divisor coordinates retain their own deadlines. This information is needed to distinguish several losses that can occur at the same exponential scale.

For positive lengths, the profile has the form

$$\Psi_k(L) = e^{-J(L)} B_{\text{ord}} \prod_{\substack{\text{proper} \\ \text{owners}}} (1 + d\sqrt{1+T})^{-\sigma} \prod_{\text{canonical edges } e} P_e.$$

The saddle cost J supplies the principal exponential rate. A proper owner, meaning one that does not contain the last coordinate, contributes a count-window factor: T is its total native length, σ its saddle value, and d its relative gap from the next owner. This factor keeps the transition between a fixed gap and a gap of order $T^{-1/2}$ in the same formula. The terminal owner contributes one ordinary factor B_{ord} , which includes the terminal count probability and its order-statistics bridge.

The remaining factors describe binary comparisons between a set of retained names and its complement. Their endpoint scales are obtained from a finite enumeration of legal histories on the original groups. The associated paths share a random environment: one first fixes the common gaps, averages over the private marks, and then takes a fractional moment. This order of averaging produces the exponent $\chi(\eta, \rho)$ in (3.8), defined using two independent stationary Ornstein–Uhlenbeck processes. The parameters η and ρ respectively describe the fractional power and the contribution of the common environment to the path variance. The exponent is an independent scalar object, whose existence and required uniform estimates are proved in this paper. The factors P_e retain this conditional information. Their product is established by a compatible construction and successive conditional estimates on the shared clocks.

This description builds on the order-statistics and clustering ideas of Ford and Koukoulopoulos. Fractional moments themselves already appear in [18, Section 1]. The additional issue here is to realize all the constraints simultaneously while preserving the common environment, the original coordinates and the powers attached to different owners. These requirements become essential in the high-dimensional configurations discussed in [19, Section 2.2]. The definition of Φ_k extends the positive-length expression by a specified common lift of every original coordinate and an infimum over that lift. The saddle and all its associated data are recomputed at each lift. An arithmetic stability argument proves that this definition also covers zero and bounded original lengths.

1.4 Environmental selection and contributing clocks

The profile also describes which configurations supply the arithmetic lower bound. At the physical saddle, the rate J is exactly the relative entropy of independent Poisson counts with means $L_i Z_i$ relative to those with means L_i ; see Proposition 5.17. The persistence factors describe a further selection of the common clock, related to the quenched random-wall model of Mallein and Miłoś [20]. We extend $\chi(p, \rho)$ to every positive moment order and prove that it is locally $C^{1,1}$ in p . Under the corresponding tilted law, $\partial_p \chi$ is the limiting conditional survival cost per unit time, while $\chi - p \partial_p \chi$ is the environmental entropy rate. Theorem 7.4 shows quantitatively that $\chi(\eta, \rho) > \eta/2$ for $0 < \eta, \rho < 1$. Thus averaging the clock before taking the fractional power loses a positive part of the exponent, which can change the order of the arithmetic answer.

The accepting measure used in the lower bound has mixed weights $qD^{\eta-1}$, where q is a conditional private-path probability and D is its environmental envelope. A neighboring-moment argument gives this measure the same leading concentration and entropy as the corresponding Gibbs law, with bounded Gibbs variational error. Section 17 applies the argument to the complete source, retaining its type information and regularity restrictions. For sufficiently large positive physical lengths, it constructs a family of original squarefree tuples carrying a fixed positive fraction of the harmonic volume sum. On this family, the envelope costs and the existing adaptive count and root targets lie within square-root windows of centers determined by $\partial_p \chi$. The construction restricts the same accepting measure used for the arithmetic lower bound, so its collision and prime-supply estimates continue to apply.

The entropy interpretation of the profile is completed in Section 17.6. The probability results supporting it include constant-factor joint positive-moment and deterministic-overlap laws, a full scalar large-deviation principle with an exact inverse-moment boundary, and conditional mixed-moment estimates for finitely many adapted stages. Their proofs are given in Appendices C–E.

1.5 The proof

We use Koukoulopoulos's arithmetic reduction [19, Theorem 1.7] to express the normalized divisor count as a harmonic sum of volumes of unions of divisor boxes. The main analytic statement evaluates this sum by Φ_k . Its lower bound is obtained by realizing the prescribed constraints on the original arithmetic space, retaining the same accepting density through the successive probabilistic and arithmetic comparisons. The upper bound uses causal coverings and local conditional estimates to show that the same losses are necessary. Finally, lift removal and the multiplication-table transfer give the two theorems above.

The probability arguments include the scalar persistence theorem, finite bridges with exact totals, and the simultaneous treatment of several comparisons. The ordinary order-statistics estimates enter separately. For the ordered-sum upper estimate, the method of [18, Lemma 5.3] requires an integer threshold in its band decomposition. Starting from its Lemmas 5.1 and 5.2, we give the argument with the integer rounding made explicit and recover the $A = 0$ case of [19, Lemma 5.5], including real thresholds and all positive counts. The ordinary lower estimate is the separate [19, Lemma 8.3]. The new comparison arguments and boundary transfers are proved here; the external arithmetic and order-statistics inputs are stated with their hypotheses and cited explicitly. The numbering of results cited from [19] refers throughout to arXiv:1102.3236v3; for [18] it refers to the author version linked in the bibliography.

2 Divisor volumes and the plan of proof

The two counting problems have a common geometric representation. For $3 \leq y_1 \leq \dots \leq y_k$, let $\mathcal{A}_k(\mathbf{y})$ consist of squarefree tuples $\mathbf{a} = (a_1, \dots, a_k)$ with prime factors of a_1 at most y_1 , and those of a_i , $i \geq 2$, in $(y_{i-1}, y_i]$. Thus 2 and 3 belong to the first prime range. The components are pairwise coprime, and an empty selection in any component is allowed. Define

$$\mathcal{L}^{(k+1)}(\mathbf{a}) = \bigcup_{\substack{d_1, \dots, d_k \in \mathbb{N} \\ d_1 \cdots d_j | a_1 \cdots a_j \ (1 \leq j \leq k)}} \prod_{j=1}^k [\log(d_j/2), \log d_j], \quad (2.1)$$

$$L^{(k+1)}(\mathbf{a}) = \text{vol}_k \mathcal{L}^{(k+1)}(\mathbf{a}),$$

$$K_k(\ell) = e^{-\sum_{i=1}^k h_i L_i} \sum_{\mathbf{a} \in \mathcal{A}_k(\mathbf{y})} \frac{L^{(k+1)}(\mathbf{a})}{a_1 \cdots a_k}, \quad h_i = k - i + 2. \quad (2.2)$$

Here $L_i = \ell_i - \log 3 \geq 0$ and $\log y_i = e^{L_i} \log y_{i-1}$, starting from $y_0 = 3$. The volume $L^{(k+1)}$ and the block lengths L_i are different objects. The sum in (2.2) includes all such squarefree tuples. The empty tuple alone contributes $(\log 2)^k$ before normalization.

A prime from the i th range can be assigned to one of the divisor coordinates $i, \dots, k$, or left unused. These are its h_i eligible labels. The union in (2.1) records all the resulting decompositions, counting overlapping boxes only once. This overlap, rather than the number of decompositions, is the source of the difficulty. The harmonic weight makes it possible to study the prime locations and their labels separately while retaining the original divisor union.

Koukoulopoulos's arithmetic comparison [19, Theorem 1.7] gives

$$\frac{H^{(k+1)}(x, \mathbf{y}, 2\mathbf{y})}{x} \asymp_k K_k(\ell(\mathbf{y})) \quad \text{if } x \geq 2^k y_1 \cdots y_k y_k. \quad (2.3)$$

The first prime range in that comparison begins below y_1 and includes the small primes; the choice $y_0 = 3$ above fixes only the logarithmic normalization. Bounded changes in these normalizing factors are absorbed by constants depending on k . We prove the table transfer, including bounded axes, in Section 16.

Theorem 2.1. *For every fixed $k \geq 6$, the function defined in (3.13) satisfies*

$$K_k(\ell) \asymp_k \Phi_k(\ell) \quad (\ell \in [\log 3, \infty)^k).$$

There are three parts to the proof. On positive physical lengths $Q_i \geq 1$, a covering argument gives $K_k(\log 3 + Q) \ll_k \Psi_k(Q)$. A separate construction gives $K_k(\log 3 + L) \gg_k \Psi_k(L^{L_*})$, where $L_i^{L_*} = \max(L_i, L_*)$ and $L_* = L_*(k)$ is fixed once. Finally, stability of the original harmonic sum compares the arithmetic quantities at L and L^λ . It allows the unknown threshold L_* to be removed from the definition of the answer by taking the explicit infimum in (3.13).

The same ordered saddle appears in both estimates. Its tilt extracts the exponential cost e^{-J} . Within each maximal equality block of the saddle, a shared prime clock acts as an environment for several private label constraints. This is why a fractional moment of a conditional survival probability appears: averaging the environment before taking the owner power would describe a different event. Exit-count windows contribute their own powers. The final ordinary suffix has one count variable that must simultaneously supply the root and permit path survival. The proof keeps these requirements together.

The deterministic definitions come first, followed by explicit examples and the saddle geometry. We then establish the scalar persistence and conditioned bridge estimates, construct the common accepting measure for the lower bound, and prove the upper bound directly for the full divisor union. Arithmetic stability closes the proof on the original parameter domain. We then return to the same lower source to identify the contributing clocks and interpret the profile's entropy costs in Section 17.

2.1 Size truncation of the harmonic volume sum

The arithmetic quantity in the main theorem sums all squarefree tuples in their prescribed prime blocks. The actual rectangle arguments may nevertheless start with a size-truncated tuple. The following positive weighted truncation proves this implication independently of any sharp upper bound, Poisson approximation, saddle point, or proposed profile.

Fix $H = k + 1$ and $3 \leq y_1 \leq \dots \leq y_k$. Write $V(\mathbf{a}) = L^{(k+1)}(\mathbf{a})$ and

$$S(\mathbf{y}) = \sum_{\mathbf{a} \in \mathcal{A}_k(\mathbf{y})} \frac{V(\mathbf{a})}{a_1 \cdots a_k}.$$

All prime sets here are finite; in particular all following rearrangements are identities between finite nonnegative sums. Choose one absolute constant C_W for the elementary weighted prime estimate

$$\sum_{p \leq y} \frac{\log p}{p} \leq C_W \log y \quad (y \geq 3). \quad (2.4)$$

For example the usual Chebyshev bound for $\sum_{p \leq y} \log p$ and partial summation give this estimate; see [24]. Put

$$\tau_k = 2kHC_W + 1, \quad S_{\text{tr}}(\mathbf{y}) = \sum_{\substack{\mathbf{a} \in \mathcal{A}_k(\mathbf{y}) \\ a_i \leq y_i^{\tau_k} \ (1 \leq i \leq k)}} \frac{V(\mathbf{a})}{a_1 \cdots a_k}.$$

Lemma 2.2 (Uniform positive weighted truncation). *Uniformly in the ordered cutoffs,*

$$S_{\text{tr}}(\mathbf{y}) \leq S(\mathbf{y}) \leq 2S_{\text{tr}}(\mathbf{y}). \quad (2.5)$$

The same comparison holds after multiplication by the exact normalizing factor defining K_k . Every truncated tuple has, for each i ,

$$\log(a_1 \cdots a_i) \leq \tau_k \sum_{g \leq i} \log y_g \leq k\tau_k \log y_i. \quad (2.6)$$

Proof. Let p be a prime in original block i , and remove it from that coordinate of a squarefree tuple. Inserting p back permits at most $k - i + 2 \leq H$ eligible routing labels, including the unused label. For each label the new divisor-box union is a translate of the old one. Therefore, if p does not already divide the tuple,

$$V(b_1, \dots, pb_i, \dots, b_k) \leq HV(\mathbf{b}). \quad (2.7)$$

Using the squarefree identity $\log a_i = \sum_{p|a_i} \log p$ gives the exact decomposition

$$\begin{aligned} \sum_{\mathbf{a} \in \mathcal{A}_k(\mathbf{y})} \frac{V(\mathbf{a}) \log a_i}{a_1 \cdots a_k} &= \sum_{p \text{ in block } i} \frac{\log p}{p} \sum_{\substack{\mathbf{b} \in \mathcal{A}_k(\mathbf{y}) \\ p \nmid b_i}} \frac{V(b_1, \dots, pb_i, \dots, b_k)}{b_1 \cdots b_k} \\ &\leq HS(\mathbf{y}) \sum_{p \text{ in block } i} \frac{\log p}{p} \leq HC_W \log y_i S(\mathbf{y}). \end{aligned} \quad (2.8)$$

The prime blocks are disjoint, so removal and reinsertion in this identity preserve every original prime-range restriction. In the inequality we only enlarged the inner nonnegative sum.

For the bad set $a_i > y_i^{\tau_k}$, multiply its indicator by $\log a_i / (\tau_k \log y_i) \geq 1$ and use (2.8). A union bound over i yields

$$S(\mathbf{y}) - S_{\text{tr}}(\mathbf{y}) \leq \frac{kHC_W}{\tau_k} S(\mathbf{y}) < \frac{1}{2} S(\mathbf{y}).$$

This proves (2.5). The common normalizing factor does not change its ratio. Finally (2.6) follows from the individual size bounds and the ordering of the cutoffs. $\square$

In particular, every divisor coordinate whose available primes lie in the first i original groups has logarithmic width at most $C_k \log y_i$ on the truncated tuples. Earlier contributions to a remaining coordinate obey the same bound at the last earlier cutoff. These are precisely the full-size baseline rectangles used by the actual routing covers. The constant $C_k = k\tau_k$, and its fixed powers, are absorbed into their dimension constants and fixed reserves, with the physical lengths, KKT products and owners fixed. Apply those covers first to this truncated sum, and only then perform the positive repeated- prime and cell enlargements. Lemma 2.2 supplies the upper bound for the original full sum. The size inequalities enter at the truncated squarefree stage, before the repeated-prime enlargement.

3 The structural profile

Throughout, $k \geq 6$ is fixed, $H = k + 1$, and $V_i = \sum_{g \leq i} L_g$, with $V_0 = 0$. We first define a numerical function on $L_i \geq 1$. The final definition will cover $L_i \geq 0$. The optimization dimension and every finite enumeration below depend only on k .

The four factors in the profile describe four constraints on the same family of decompositions: exponential tilting, proper exit counts, conditional binary survival, and the terminal ordinary bridge. We define them in this order. Although the final expression is a product, the underlying events share counts and environments; their joint realization and their simultaneous necessity are proved separately below.

3.1 The saddle and the original named groups

On the closed ordered simplex $1 \geq s_1 \geq \dots \geq s_k \geq 0$ put

$$G_k = s_k \log 2, \quad G_i = s_i \log(1 + \exp(G_{i+1}/s_i)), \quad (3.1)$$

$$Z_i = \exp G_i, \quad J(L; s) = \sum_i L_i + \sum_j s_j V_j - \sum_i L_i Z_i. \quad (3.2)$$

At zero products use the continuous perspective extension. Write $s = s(L)$ for the unique maximizer, whose existence and uniqueness are proved in Section 5. The geometric theorem supplies

$$\frac{1}{2} < s_k \leq s_1 < 1 - \epsilon_H, \quad \epsilon_H = \frac{2 \log 2 - 1}{2H(\log H)^2}. \quad (3.3)$$

In particular $1 \leq Z_i \leq H$. A maximal consecutive group $[a, b]$ of equal products is called an owner; write its product as σ . A proper owner has $b < k$ and $z = \exp(G_{b+1}/\sigma)$; the terminal owner has $b = k$ and $z = 1$. The exterior names of this owner are $a, \dots, b$. Its native groups are the original groups $i \in [a, b]$, of ranks $r_i = b - i + 1$. Original groups $g < a$ are its old groups, of rank $b - a + 1$. Each original group retains its own random clock.

In the divisor model, name c is coordinate d_c : primes in group i may be assigned to $c \geq i$, and V_c is its deadline. Assignments to coordinates beyond b , together with the unused primes, form the owner's compound child.

For $l < k$ define

$$\beta_l = \frac{\exp(G_{l+1}/s_l)}{1 + \exp(G_{l+1}/s_l)}.$$

The reference intensity per eligible exterior name is

$$\kappa_g = \begin{cases} Z_g \prod_{l=g}^{a-1} \beta_l / (b - a + 1 + z), & g < a, \\ (b - g + 1 + z)^{\sigma-1}, & a \leq g \leq b. \end{cases} \quad (3.4)$$

Each old group retains the intensity acquired along its original route into this owner. Averaging these coefficients would change the constraints at the individual deadlines. Set

$$E_q(z) = (q + z) \log(q + z) - z \log z, \quad g_q(q) = q - \kappa_g E_q(z).$$

3.2 An explicit enumeration of the two endpoint scores

For every native group i enumerate the edges $r_i \rightarrow q$, where $0 \leq q < r_i$ for a proper owner, and $1 \leq q < r_i$ for the terminal owner. The latter exclusion removes the degenerate ordinary edge. The retained exterior names are $B = \{b - q + 1, \dots, b\}$; B is empty when $q = 0$. The compound child is always retained.

Here is the complete named outside-context enumeration for that edge. For each name $c \in \{a, \dots, i-1\}$ choose $\tau_c \in \{V_0, \dots, V_c\}$ independently. For each $c \in B$ choose $\tau_c \in \{V_i, \dots, V_c\}$ independently. In the left completion set $\tau_c = V_{i-1}$ for all $i \leq c \leq b-q$; in the right completion set those same cuts to V_i . All other cuts are identical in the two completions. This gives a finite set of lawful pairs (v^-, v^+) . Every cut respects its original name's deadline, and the compound child remains intact. These choices specify the entire enumeration.

For either completion v and original group $1 \leq g \leq b$, let

$$A_{v,g} = \{c \in \{a, \dots, b\} : \tau_c \geq V_g\}, \quad q_{v,g} = |A_{v,g}|, \quad t_v = \max_{a \leq c \leq b} \tau_c.$$

In an old group the shallowest eligible name is a , and in native group g it is g . Call g good when $0 < q_{v,g} < r_g$ and this shallowest eligible name is absent from $A_{v,g}$. Define

$$\overline{C}_v = \sum_{g \leq b} L_g g_g(q_{v,g}), \quad S_{\text{good}}(v) = \sum_{g \text{ good}} \sqrt{L_g}. \quad (3.5)$$

The endpoint geometry gives $\overline{C}_v \geq 0$ at the physical optimizer. Both the old coefficients and the old lengths are included.

For a proper owner put $T = \sum_{a \leq i \leq b} L_i$, $d = \sigma/s_{b+1} - 1$, and

$$\widehat{R} = 1 + \frac{\sqrt{1+T}}{1+d\sqrt{1+T}}.$$

For the terminal owner set $\widehat{R} = 0$. Define

$$S_v = \begin{cases} 1 + \widehat{R} + \overline{C}_v, & t_v \leq V_b/2, \\ 1 + \overline{C}_v + S_{\text{good}}(v), & t_v > V_b/2, \end{cases} \quad X_e = \min_v S_{v-}, \quad Y_e = \min_v S_{v+}. \quad (3.6)$$

The early score includes the uncertainty of a proper exit-count window. At the late endpoint the incoming contribution cancels from the intrinsic cost; the additional square-root terms instead measure the fluctuation needed to remove an eligible shallow name while retaining deeper names. The two minima are taken separately over the same finite set of paired lawful contexts. The lower bound makes the two scores compatible by keeping the actual type increment Δ fixed and choosing the legal pair $x = \max(a_* X_e, a_* Y_e - \Delta)$, $y = x + \Delta$; see (9.5).

3.3 The independent scalar and all canonical edge factors

Let E, W be independent stationary standard Ornstein–Uhlenbeck processes, with covariance $e^{-|u-v|/2}$. For $0 < \eta, \rho < 1$ set

$$F_{\eta,\rho}(t) = \mathbb{E}_E \left[\mathbb{P}_W \{ \sqrt{\rho} E_u + \sqrt{1-\rho} W_u \geq 0 \mid 0 \leq u \leq t \mid E \}^\eta \right], \quad (3.7)$$

$$\chi(\eta, \rho) = \lim_{t \rightarrow \infty} -t^{-1} \log F_{\eta,\rho}(t). \quad (3.8)$$

The scalar theorem proves existence, local Lipschitz continuity, and $\eta/2 \leq \chi \leq 1/2$. It is independent of K_k and of all multirank history spaces. It also gives certified computation: if $b(t) = -\log F_{\eta,\rho}(t)$ and $p_g = (1 + e^{-g/2})/(1 - e^{-g/2})$, then

$$\frac{b(t)}{p_g(t+g)} \leq \chi(\eta, \rho) \leq \frac{b(t)}{t}. \quad (3.9)$$

The scalar finite Gaussian quadrature and crossing error bounds make each endpoint effective. Taking $g = 2 \log t$ gives vanishing width.

For $e = (i, r, q)$ put $a = \log(r + z)$, $b_e = \log(q + z)$, $p = (q + z)/(r + z)$, and

$$m = a - b_e p, \quad \rho_e = \frac{m^2}{m^2 + b_e^2 p(1 - p)}, \quad T_e = \frac{(1 + X_e)(1 + Y_e)}{1 + (r + z)^\sigma L_i}, \quad P_e = \min(1, T_e)^{2\chi(\sigma, \rho_e)}. \quad (3.10)$$

These laws are nondegenerate. In particular a proper zero-output edge has $z > 1$, whereas the terminal zero-output edge was excluded. Every canonical edge is used, including edges with a bounded positive factor.

3.4 An explicit ordinary suffix

Here $h_i = k - i + 2$ on the complete original chain. If the selected suffix starts at group j , then $V_{\text{pre}} = V_{j-1}$ includes all preceding owners as well as the earlier terminal groups. For $2 \leq h \leq H$ define

$$\alpha_h = \frac{\log((h-1)/\log h)}{\log h}, \quad c_w = \frac{1}{4H^2}, \quad \tau = \min \left\{ \frac{1}{4}, \frac{c_w}{2H(\log H)^2}, \frac{1}{4} \min_{2 \leq h < H} (\alpha_{h+1} - \alpha_h) \right\}.$$

The thresholds α_h are strictly increasing. If the terminal product σ lies in $[\alpha_{h_i} - \tau, \alpha_{h_i} + \tau]$ for a terminal native group, begin the suffix at that group; at most one qualifies. Otherwise begin at the first terminal native group with $e_i = h_i^\sigma \log h_i - (h_i - 1) > 0$. The KKT theorem ensures existence. Let U be the suffix length, M its largest original block, and A, B the total lengths before and after that block within the suffix. Break largest-block ties at the earliest index. Put

$$D = \sum_{i \text{ in suffix}} e_i L_i, \quad \mathcal{B}(L) = \frac{1}{\sqrt{1+U}} \min \left\{ 1, \frac{(1+B)(1+D+A)}{1+M} \right\}. \quad (3.11)$$

Here $D \geq 0$, and the suffix theorem gives $D \geq c_k V_{\text{pre}}$ and $U \geq c_k V_k$. This is the sole ordinary root factor.

The positive-length raw expression is

$$\Psi_k(L) = e^{-J(L; s(L))} \mathcal{B}(L) \prod_{j \text{ proper}} (1 + d_j \sqrt{1 + T_j})^{-\sigma_j} \prod_{e \text{ canonical}} P_e. \quad (3.12)$$

All lengths, reference costs and products in this expression refer to one and the same physical parameter vector. The rate J is a Poisson entropy, and each persistence exponent splits into an environmental selection cost and a conditional path cost. These interpretations, proved in Section 17.6, also locate, for sufficiently large positive lengths, a family of prime tuples contributing the full order of the arithmetic sum.

3.5 The profile on the full parameter domain

For $\ell_i \geq \log 3$ put $L_i = \ell_i - \log 3$, $A_k = 2(k+1)^3$, and define

$$\Phi_k(\ell) = \inf_{\lambda \geq 1} e^{A_k \lambda} \Psi_k((\max(L_i, \lambda))_{i=1}^k). \quad (3.13)$$

This completes the numerical definition with one scalar optimization; χ is the universal scalar specified in (3.8).

Lemma 16.3 gives an explicit finite interval for this last infimum and proves that $0 < \Phi_k(\ell) < \infty$. Every candidate length vector uses its own maximizer, exact equality blocks, endpoint contexts and ordinary suffix.

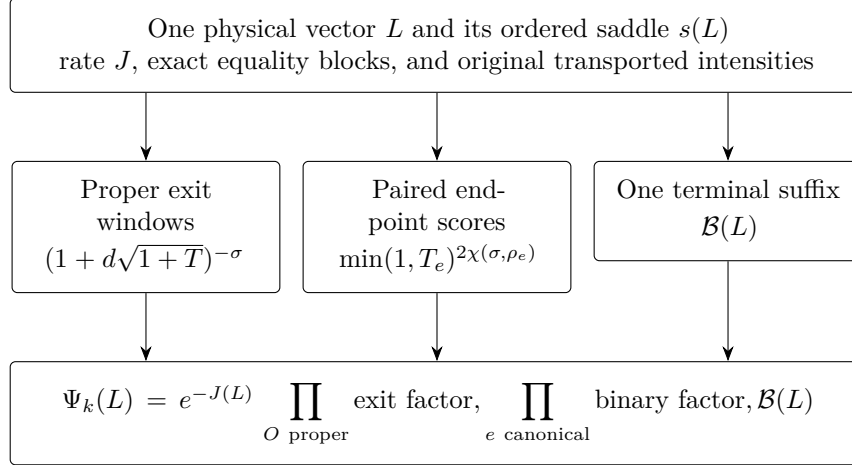


Figure 1: The deterministic construction of the positive-length profile. The arrows indicate dependence on the same saddle data. The product formula is established by the lower and upper proofs; it does not assert independence of the count and path events.

4 Reading the profile in concrete families

The exponential term alone does not distinguish all the regimes of the answer. The following families separate three further effects: the comparison of named suffixes, the count associated with leaving an owner, and the ordinary terminal constraint. They also show how several comparisons combine, first within one owner and then across different owners. Every partition into owners below is the partition of the actual maximizer of (3.2).

Throughout this section, $\log 3 + L$ means coordinatewise addition. A length vector specifies actual arithmetic cutoffs by

$$y_0 = 3, \quad \log y_i = e^{L_i} \log y_{i-1}.$$

Consequently every displayed estimate for $\Phi_k(\log 3 + L)$ is also an estimate for $H^{(k+1)}(x, \mathbf{y}, 2\mathbf{y})/x$ throughout the admissible region in Theorem 1.1. Taking $N_i = 2y_i$ for $i \leq k$ and $N_{k+1} \geq N_k$ gives the corresponding multiplication-table estimate by Theorem 1.2. The dimension is fixed in all comparisons. A constant denoted by C is chosen once, sufficiently large in terms of that dimension.

For convenient reference, write the correlation in (3.10) as

$$\rho(r, q; z) = \frac{\{\log(r + z) - p \log(q + z)\}^2}{\{\log(r + z) - p \log(q + z)\}^2 + \{\log(q + z)\}^2 p(1 - p)}, \quad p = \frac{q + z}{r + z}. \quad (4.1)$$

Here r, q, z retain their meanings in the profile, and p is the low-mark probability. The proofs of the evaluations, including the finite inequalities used to locate the constants, are given in Appendix F.

4.1 Equal cutoffs

Put $H = k + 1$ and

$$\alpha_H = \frac{\log((H - 1)/\log H)}{\log H}, \quad I_H = 1 + (H - 1)\alpha_H - H^{\alpha_H}. \quad (4.2)$$

For $L = (T, 0, \dots, 0)$, the cutoffs are all 3^{e^T} , and

$$\Phi_k(\log 3 + L) \asymp_k e^{-I_H T} T^{-3/2}. \quad (4.3)$$

To see the factors in this formula, replace the zero lengths by a fixed C . The resulting saddle has one terminal owner, with common product $\alpha_H + O_k(T^{-1})$. All binary factors have a positive constant lower bound. The ordinary suffix starts in the first group, has zero total drift, and has one group of length T , followed by bounded total length. Its root factor is $T^{-1/2}$ and its ballot factor is T^{-1} . Bounded arithmetic stability transfers the estimate back to the original equal cutoffs with all divisor coordinates retained.

4.2 Three regimes with six divisor coordinates

The next two families have a long first group, four bounded groups, and a long last group. Their different relative lengths expose different parts of the answer.

First define

$$\sigma_c = 1 - \frac{\log((7 \log 7 - 2 \log 2)/5)}{\log 7}, \quad e_h = h^{\sigma_c} \log h - (h - 1), \quad (4.4)$$

$$a = -\frac{e_7}{e_2}, \quad b = -\frac{e_6 + e_5 + e_4 + e_3}{e_2}, \quad L = (T, C, C, C, C, aT + bC). \quad (4.5)$$

The constants satisfy $\alpha_2 < \sigma_c < \alpha_3$, $54 < a < 55$, and $66 < b < 67$. The unique saddle is exactly $s_1 = \dots = s_6 = \sigma_c$. Nevertheless a binary comparison remains unsaturated. It is the first-group edge $6 \rightarrow 1$, which keeps name 6 and removes names $1, \dots, 5$. Put

$$\Delta = \sum_{h=3}^6 \left\{ h^{\sigma_c-1} (h \log h - 2 \log 2) - (h - 2) \right\} > 0.$$

The pair giving the smallest endpoint scales has cuts

	$\tau_1, \dots, \tau_5$	τ_6
left	$0, \dots, 0$	V_6
right	$V_1, \dots, V_1$	V_6

and both reference costs equal $C\Delta$. Both completions are late. On the left the first group is good, and on the right it is not; the four middle groups are good on both sides. Thus the full scores are

$$X = 1 + C\Delta + \sqrt{T} + 4\sqrt{C}, \quad Y = 1 + C\Delta + 4\sqrt{C} \quad (T \text{ sufficiently large}). \quad (4.6)$$

The distinction between reference cost and named endpoint score changes the power of T . With $\chi_c = \chi(\sigma_c, \rho(6, 1; 1))$, the answer is

$$\Phi_6(\log 3 + L) \asymp_6 e^{-J(L)} T^{-1/2-\chi_c}. \quad (4.7)$$

There is no proper-owner count. The ordinary suffix consists of the last group and has a drift proportional to its length, leaving only the power $T^{-1/2}$. All remaining binary factors are bounded below. The exponent is explicit: if $I_h^c = 1 + (h - 1)\sigma_c - h^{\sigma_c}$, then

$$J(L) = (I_7^c + aI_2^c)T + (I_6^c + I_5^c + I_4^c + I_3^c + bI_2^c)C.$$

Section 17.6 further splits $T^{-\chi_c}$ into its environmental entropy and conditional path costs. It also locates the clock and count-root values on a family of original tuples that supplies the order in (4.7).

Now move the last cutoff much farther away by taking

$$L = (T, 0, 0, 0, 0, T^3). \quad (4.8)$$

Write $\alpha = \alpha_2$. There is a unique $s_* \in (\alpha, 1)$ satisfying

$$z_* = 2^{\alpha/s_*}, \quad (5 + z_*)^{s_*-1} \{(5 + z_*) \log(5 + z_*) - z_* \log z_*\} = 5; \quad 0.54070 < s_* < 0.54072. \quad (4.9)$$

After replacing the four zero lengths by C , the actual owners are $[1, 5]$ and $[6]$. Their products tend to s_* and α , respectively. Only the proper edge $5 \rightarrow 0$ in the first group is unsaturated. Its endpoint scales are both bounded, giving $T^{-2\chi_*}$, where $\chi_* = \chi(s_*, \rho(5, 0; z_*))$. The proper count gives $T^{-s_*/2}$. The terminal length is T^3 , but its drift has order T , supplied by the earlier owner. Hence its joint ordinary and root factor is $T(T^3)^{-3/2}$. Altogether, with

$$I_2 = 1 + \alpha - 2^\alpha, \quad I_p = 1 + 5s_* + \alpha - (5 + z_*)^{s_*},$$

we obtain

$$\Phi_6(\log 3 + L) \asymp_6 e^{-I_2 T^3 - I_p T} T^{1-s_*/2-2\chi_*} (T^3)^{-3/2}. \quad (4.10)$$

The child in this proper comparison has $z_* > 1$, so $q = 0$ still leaves a nondegenerate binary mark. It is distinct from the terminal ordinary comparison.

There is a precise dimensional reason for using six coordinates in these two families. For the first construction, replacing 7 by H in (4.4) gives a product below α_2 for $3 \leq H \leq 6$, and above it for $H = 7$. For the second, the boundary sign is

$$f(H) = H^{\alpha-1}(H \log H - 2 \log 2) - (H - 2), \quad f(6) > 0 > f(7).$$

Strict concavity of f on $[2, \infty)$ makes the strict branch possible precisely for $H \geq 7$ in this two-scale family. These are thresholds for the stated constructions, not classifications of all parameter vectors.

A final contrast isolates the proper count. Fix $s = 107/200$, $\theta = 267/500$, and $z = 2^{\theta/s}$. Set

$$\begin{aligned} f_r &= (r + z)^{s-1} E_r(z) - r, & c_r &= 1 - z \log 2 (r + z)^{s-1}, \\ a &= -\frac{f_1 + f_2 + f_3 + f_4}{f_5}, & b &= \frac{ac_5 + c_1 + c_2 + c_3 + c_4}{2^\theta \log 2 - 1}, & L &= t(a, 1, 1, 1, 1, b). \end{aligned} \quad (4.11)$$

Here a and b are positive (approximately 10.958044 and 1651.936051). The exact owners remain $[1, 5]$, $[6]$, with products s, θ , but every binary factor is bounded below. Writing $J_0 = J(a, 1, 1, 1, 1, b)$, we have

$$\Phi_6(\log 3 + L) \asymp_6 e^{-t J_0 t^{-(1+s)/2}}. \quad (4.12)$$

Thus leaving an owner can impose a count penalty even when every binary comparison has sufficient endpoint room.

4.3 Two comparisons in one owner

The next family has two unsaturated edges in the same proper owner. They share its count factor, but have different endpoint scales and different correlations.

For $x > y \geq 1$, define

$$t(x, y) = 1 - \frac{\log((x \log x - y \log y)/(x - y))}{\log x}.$$

Let z be the unique solution in $(1.98455, 1.98457)$ of

$$t(14 + z, 6 + z) = t(6 + z, z), \quad \sigma = t(6 + z, z), \quad \theta = \frac{\sigma \log z}{\log 2}. \quad (4.13)$$

Then $\alpha_2 < \theta < \sigma$, with σ approximately 0.551693 and θ approximately 0.545525. For $1 \leq r \leq 14$, put

$$\kappa_r = (r + z)^{\sigma-1}, \quad f_r(q) = q - \kappa_r E_q(z), \quad g_r = f_r(r).$$

The defining equations give the exact identities

$$g_6 = 0, \quad f_{14}(6) = g_{14} > 0, \quad g_1 < 0. \quad (4.14)$$

Take $k = 15$, $U = T^3$, and specify the lengths by

$$\begin{aligned} L_1 &= T, \quad L_9 = U, \quad L_i = C \quad (2 \leq i \leq 13, i \neq 9), \\ L_{14} &= W = -\frac{g_{14}T + C \sum_{\substack{2 \leq r \leq 13 \\ r \neq 6}} g_r}{g_1}, \quad L_{15} = R = \frac{\sum_{i=1}^{14} L_i (1 - z \log 2 \kappa_{15-i})}{2^\theta \log 2 - 1}. \end{aligned} \quad (4.15)$$

All these lengths are positive; $W \asymp T$ and $R \asymp U$. The actual saddle is $(\sigma, \dots, \sigma, \theta)$, so the owners are $[1, 14]$ and $[15]$. The two equations defining W, R balance the common proper-owner direction and the terminal direction. The remaining KKT inequalities are strict.

The first rare comparison is $14 \rightarrow 6$ in group 1. Keeping the six deepest names until their deadlines gives equal bounded reference costs at its endpoints. The left completion makes the first group good, while the right completion does not. The second rare comparison is $6 \rightarrow 0$ in group 9. Its six names must survive through the earlier first group, and this creates a cost of order T at both endpoints. Precisely,

$$(X_{1,6}, Y_{1,6}) \asymp (\sqrt{T}, 1), \quad (X_{9,0}, Y_{9,0}) \asymp (T, T). \quad (4.16)$$

Every other binary factor is bounded below. Put

$$\chi_1 = \chi(\sigma, \rho(14, 6; z)), \quad \chi_2 = \chi(\sigma, \rho(6, 0; z)).$$

There is one proper count, of order $U^{-\sigma/2}$, and one ordinary root factor, of order $U^{-1/2}$. The resulting full profile is

$$\Phi_{15}(\log 3 + L) \asymp_{15} e^{-J(L)} U^{-(1+\sigma)/2} T^{-\chi_1} \left(\frac{T^2}{U} \right)^{2\chi_2}, \quad U = T^3. \quad (4.17)$$

Its exponential is explicit as well:

$$J(L) = \sum_{i=1}^{14} L_i \{1 + (15 - i)\sigma + \theta - (15 - i + z)^\sigma\} + R(1 + \theta - 2^\theta). \quad (4.18)$$

This example explains why the profile takes the product of all canonical edge factors inside an owner, but assigns a count factor only once to that owner, through the simultaneous construction on its common clock.

4.4 Two comparisons in different owners

One more family makes the different owner powers visible. Put $k = 18$, $\alpha = \alpha_2$, and, for $c > 0$, define

$$F(s; c) = (1 + e^{c/s})^{s-1} \{ (1 + e^{c/s}) \log(1 + e^{c/s}) - e^{c/s}(c/s) \} - 1.$$

Let v, u be the roots

$$F(v; \alpha \log 17) = 0, \quad G_2^* = v \log(1 + 17^{\alpha/v}), \quad F(u; G_2^*) = 0. \quad (4.19)$$

They are uniquely determined in $(0, 1)$ and satisfy $u > v > \alpha$; approximately they are $u = 0.534353736$, $v = 0.531542356$. Set

$$\begin{aligned} z_2^* &= 17^{\alpha/v}, \quad z_1^* = e^{G_2^*/u}, \quad Z_j^* = (1 + z_j^*)^{s_j^*}, \quad (s_1^*, s_2^*) = (u, v), \\ L_1 &= T, \quad L_2 = T^3, \quad L_i = 2^{i-3} T^5 \quad (3 \leq i \leq 17), \quad L_{18} = T^{11}. \end{aligned} \quad (4.20)$$

For sufficiently large T , the exact owners are $[1]$, $[2]$, $[3, 18]$. Their products satisfy

$$u_T = u + O(T^{-2}), \quad v_T = v + O(T^{-2}), \quad \tau_T = \alpha + O(T^{-6}).$$

Each of the first two owners has just the proper edge $1 \rightarrow 0$. Their endpoint scales are

$$(X_1, Y_1) \asymp (1, 1), \quad (X_2, Y_2) \asymp (T, 1).$$

The old first group supplies the left endpoint room in the second comparison. With $\chi_1 = \chi(u, \rho(1, 0; z_1^*))$ and $\chi_2 = \chi(v, \rho(1, 0; z_2^*))$, the respective binary costs are $T^{-2\chi_1}$, $T^{-4\chi_2}$, and the respective count costs are $T^{-u/2}$, $T^{-3v/2}$.

All 120 terminal binary factors have a positive constant lower bound. The ordinary suffix is the last group: its length is T^{11} and its drift has order T^5 . Its single joint factor is therefore $T^{-23/2}$. Define

$$\begin{aligned} J_0(T) &= T(1 + u + v + 16\alpha - Z_1^*) + T^3(1 + v + 16\alpha - Z_2^*) \\ &\quad + T^5 \sum_{i=3}^{17} 2^{i-3} \{ 1 + (h_i - 1)\alpha - h_i^\alpha \} + T^{11}(1 + \alpha - 2^\alpha), \quad h_i = 20 - i. \end{aligned} \quad (4.21)$$

Then $J(L) = J_0(T) + O(T^{-1})$, and

$$\Phi_{18}(\log 3 + L) \asymp_{18} e^{-J_0(T)} T^{-23/2 - u/2 - 3v/2 - 2\chi_1 - 4\chi_2}. \quad (4.22)$$

Thus the two owners contribute their own powers and counts, while the whole chain still has only one ordinary root.

For this particular construction the dimensional threshold is governed by a one-step, rather than a long-interval, secant. If

$$t_1(x) = 1 - \frac{\log\{x \log x - (x-1) \log(x-1)\}}{\log x},$$

then $t_1(17) < \alpha_2 < t_1(18)$. Together with the one-valley property in Lemma 5.18, this says that a terminal child with limiting product α_2 needs at least 17 names before the adjacent singleton can have a strictly larger product. Two such singleton owners therefore first fit at $k = 18$. This statement concerns the specified singleton construction; the example with $k = 15$ uses a different arrangement of ranks.

5 Geometry of the ordered saddle

The optimizer does more than select the exponential scale of the answer. Its equality blocks determine which count windows are narrow, and its first derivatives give the budgets available to the path constraints. We begin with a general finite rank chain, since restrictions to equality subspaces arise naturally when parameters move. We then specialize to the complete chain needed for the divisor problem.

5.1 The objective and the main conclusions

Fix integers

$$H = h_1 > h_2 > \cdots > h_m \geq 2,$$

where the one-element chain is allowed. Set

$$p_i = h_i - h_{i+1} \quad (i < m), \quad p_m = h_m - 1, \quad V_j = \sum_{i=1}^j L_i, \quad V_0 = 0,$$

with arbitrary $L_i > 0$. The symbols p_i always denote these positive integers. Child probabilities will instead be denoted by β_i .

On the ordered simplex

$$\mathcal{D} = \{s \in \mathbb{R}^m : 1 \geq s_1 \geq \cdots \geq s_m \geq 0\}$$

define, first at positive products,

$$\begin{aligned} G_m(s) &= s_m \log h_m, \\ G_i(s) &= s_i \log(p_i + \exp(G_{i+1}(s)/s_i)) \end{aligned} \quad (i < m), \quad (5.1)$$

$$\begin{aligned} Z_i(s) &= \exp G_i(s), \\ J(L; s) &= \sum_i L_i + \sum_j s_j p_j V_j - \sum_i L_i Z_i(s). \end{aligned} \quad (5.2)$$

Use the continuous extension at zero products. Indeed, induction gives $0 \leq G_i(s) \leq s_i \log h_i$: if $G_{i+1} \leq s_{i+1} \log h_{i+1}$, ordering gives $\exp(G_{i+1}/s_i) \leq h_{i+1}$. Thus when $s_i = 0$ its whole suffix is zero and $G_i = 0$, $Z_i = 1$. In particular,

$$1 \leq Z_i(s) \leq h_i \quad (s \in \mathcal{D}). \quad (5.3)$$

For $E \subseteq \{1, \dots, m-1\}$, let

$$\mathcal{D}_E = \{s \in \mathcal{D} : s_j = s_{j+1} \text{ for every } j \in E\}.$$

Every maximum below is over this *whole* feasible set. Equalities not specified by E , and all other order inequalities, remain part of the optimization. Write $s^E(L)$ for the maximizer and $J_E^*(L) = J(L; s^E(L))$. Existence and uniqueness will be proved.

Throughout, $c, C > 0$ may change from line to line and depend only on the fixed chain. In particular they do not depend on L , its ratios, the face E , or the sizes of product gaps and multipliers. The notation $A \asymp B$ means $cB \leq A \leq CB$, with this convention.

Theorem 5.1 (Global bounds). *Put*

$$c_0 = 2 \log 2 - 1, \quad \varepsilon_H = \frac{c_0}{2H(\log H)^2}. \quad (5.4)$$

For every positive length vector and every E , there is a unique maximizer on $\mathcal{D}_E$, and it satisfies

$$\frac{1}{2} < s_m^E(L) \leq s_1^E(L) < 1 - \varepsilon_H. \quad (5.5)$$

At a face maximizer define the signed cumulative multipliers

$$\lambda_j^E(L) = - \sum_{i \leq j} \partial_{s_i} J(L; s^E(L)), \quad \lambda_0^E = \lambda_m^E = 0. \quad (5.6)$$

The equality for λ_m^E follows from stationarity in the common product direction. For $j \notin E$, these multipliers are nonnegative and complementary to $s_j^E - s_{j+1}^E$. For $j \in E$ they are signed.

Theorem 5.2 (Stability across changing active faces). *Fix E , let $L' = L + \Delta$, and assume $|\Delta_i| \leq L_i/2$. Put $D = \sum_i |\Delta_i|$. Then*

$$|s_i^E(L') - s_i^E(L)| \leq C \sum_{\ell=1}^m \frac{|\Delta_\ell|}{V_{\max(i, \ell)}}, \quad (5.7)$$

$$|J_E^*(L') - J_E^*(L)| \leq CD, \quad (5.8)$$

$$|\lambda_j^E(L') - \lambda_j^E(L)| \leq CD. \quad (5.9)$$

No extra lower threshold for the lengths is required. The active inequalities outside E may appear, disappear, or remain degenerate along the perturbation.

For $x \in \mathcal{D}$, write $\text{gap}_j(x) = x_j - x_{j+1}$. Let $\bar{s} = s^E(L)$, $s^* = s^\emptyset(L)$, and set

$$a_j = \begin{cases} (-\lambda_j^E(L))_+, & j \in E, \\ 0, & j \notin E, \end{cases} \quad \mathcal{A}_E = \sum_{j < m} \frac{a_j^2}{V_j}. \quad (5.10)$$

Theorem 5.3 (Release of equalities). *Uniformly in L and E ,*

$$J_\emptyset^*(L) - J_E^*(L) \asymp \mathcal{A}_E, \quad (5.11)$$

$$\sum_i V_i (s_i^* - \bar{s}_i)^2 \asymp \mathcal{A}_E. \quad (5.12)$$

Both quantities are zero when $\mathcal{A}_E = 0$. For $E = \{j\}$, put $a = (-\lambda_j^E)_+$. If $a = 0$, then $\bar{s} = s^$. If $a > 0$, additionally*

$$\text{gap}_j(s^*) \asymp \frac{a}{V_j}, \quad J_\emptyset^*(L) - J_{\{j\}}^*(L) \asymp \frac{a^2}{V_j}. \quad (5.13)$$

The individual-gap assertion is only made for a single released edge.

We prove these estimates directly from the finite objective. The curvature controls each suffix on its own cumulative length scale; this is the reason that perturbations remain controlled when the active equality constraints change.

5.2 Concavity and the outer boundaries

The map $x \mapsto \log(p + e^x)$ is convex and increasing. Its perspective $(s, y) \mapsto s \log(p + e^{y/s})$ is convex for $s > 0$ and is increasing in y . Induction in (5.1) therefore shows that each G_i , and hence each Z_i , is convex. The continuous extension gives convexity on $\mathcal{D}$. Thus J is concave and continuous on the compact convex set $\mathcal{D}_E$, so a maximum exists.

5.2.1 The bottom plateau

Lemma 5.4. *Every maximizer on $\mathcal{D}_E$ satisfies $s_m > 1/2$.*

Proof. Suppose that $a, \dots, m$ is the maximal bottom block of equal products, with common value $\sigma \leq 1/2$. Increasing this entire block by a small positive amount is feasible. If $a > 1$, its preceding gap is strict, so no prescribed equality crosses that gap. Exact telescoping gives

$$G_i = \sigma \log h_i, \quad Z_i = h_i^\sigma \quad (i \geq a).$$

For $i < a$, set

$$\beta_i = \frac{\exp(G_{i+1}/s_i)}{p_i + \exp(G_{i+1}/s_i)}.$$

Along this bottom-block direction,

$$\frac{dG_i}{d\sigma} = \left(\prod_{i \leq b < a} \beta_b \right) \log h_a.$$

The identity

$$Z_i \beta_i = Z_{i+1} \beta_i^{1-s_i}$$

implies

$$Z_i \prod_{i \leq b < a} \beta_b = Z_a \prod_{i \leq b < a} \beta_b^{1-s_b} \leq Z_a.$$

When $\sigma = 0$, all preceding products, if present, are positive, and these formulas give the right derivative. There are no preceding terms if $a = 1$.

The derivative of the linear budget is

$$\sum_{j \geq a} p_j V_j = (h_a - 1) V_{a-1} + \sum_{i \geq a} (h_i - 1) L_i.$$

Consequently the directional derivative of J is at least

$$V_{a-1}(h_a - 1 - h_a^\sigma \log h_a) + \sum_{i \geq a} L_i(h_i - 1 - h_i^\sigma \log h_i). \quad (5.14)$$

For every real $h > 1$, one has $\sqrt{h} \log h < h - 1$. Indeed, with $t = \sqrt{h}$, the difference $t^2 - 1 - 2t \log t$ vanishes at 1 and has positive derivative $2(t - 1 - \log t)$ for $t > 1$. Every bracket in (5.14) is thus positive. Since the lengths are positive, the derivative is strictly positive. This contradicts maximality, including at $\sigma = 0$. $\square$

5.2.2 The top plateau

Lemma 5.5. *Every maximizer on $\mathcal{D}_E$ satisfies $s_1 < 1 - \varepsilon_H$.*

Proof. Let $1, \dots, b$ be the maximal top equal-product block, with common value σ . If $b < m$, put $C = h_{b+1}$ and $z = \exp(G_{b+1}/\sigma)$; otherwise put $C = z = 1$. For $i \leq b$, let $n_i = h_i - C \geq 1$. Telescoping yields

$$G_i = \sigma \log(n_i + z).$$

Here $1 \leq z \leq C$ and $n_i + z \leq H$, by the ordered recursion. Varying the top block while keeping its child products fixed gives

$$\frac{dZ_i}{d\sigma} = (n_i + z)^\sigma A(n_i, z), \quad A(n, z) = \log(n + z) - \frac{z \log z}{n + z}. \quad (5.15)$$

This formula includes the variation of $z = \exp(G_{b+1}/\sigma)$. The budget derivative is $\sum_{i \leq b} L_i n_i$.

Write

$$E_n(z) = (n + z) \log(n + z) - z \log z = (n + z) A(n, z).$$

It is increasing in z , since $\partial_z E_n = \log(1 + n/z) > 0$. Moreover

$$E_n(z) \geq E_n(1) = (n + 1) \log(n + 1) \geq 2n \log 2.$$

For the last inequality, $(n + 1) \log(n + 1)/n$ is increasing for $n > 0$, with derivative $[n - \log(n + 1)]/n^2 > 0$, and $n \geq 1$. Also $0 < A(n, z) \leq \log H$. With n, z fixed at their values at the point under consideration, for $0 \leq \tau \leq 1$,

$$\frac{d}{d\tau} ((n + z)^\tau A(n, z)) \leq H(\log H)^2.$$

This auxiliary comparison keeps z fixed. If $\sigma \geq 1 - \varepsilon_H$, it follows that

$$(n + z)^\sigma A(n, z) - n \geq n(2 \log 2 - 1) - \varepsilon_H H(\log H)^2 \geq c_0/2 > 0.$$

Thus raising the top block has strictly negative derivative for J . Lowering it a sufficiently small amount is feasible and increases J , a contradiction. This also covers $\sigma = 1$. $\square$

Let $\varepsilon = \varepsilon_H$, which lies in $(0, 1/2)$, and put

$$\mathcal{D}_\varepsilon = \{s : \varepsilon \leq s_m \leq \dots \leq s_1 \leq 1 - \varepsilon\}.$$

The preceding lemmas put every maximizer on every prescribed face in this single compact set. We will use differential estimates on the slightly larger convex set $\mathcal{D}_{\varepsilon/2}$.

5.3 Full suffix curvature and uniqueness

For $i < m$, introduce

$$a_i = G_i/s_i = \log S_i, \quad u_i = s_{i+1}/s_i, \quad \beta_i = \frac{e^{G_{i+1}/s_i}}{S_i}, \quad A_i = a_i - u_i \beta_i a_{i+1},$$

where $S_i = p_i + e^{G_{i+1}/s_i}$; set $a_m = A_m = \log h_m$. All these functions and the derivatives needed below are bounded on $\mathcal{D}_{\varepsilon/2}$. The quantities $s_i, a_i, \beta_i, 1 - \beta_i$ have positive chain-dependent lower bounds. The same holds for A_i : it is the entropy of the distribution consisting of p_i named probabilities $1/S_i$ and the one compound probability β_i . These assertions use compactness but no separation of adjacent products.

Lemma 5.6 (One perspective term controls its entire suffix). *For any $s \in \mathcal{D}_{\varepsilon/2}$, direction $z \in \mathbb{R}^m$, and i ,*

$$c \sum_{j \geq i} z_j^2 \leq D^2 Z_i(s)[z, z] \leq C \sum_{j \geq i} z_j^2. \quad (5.16)$$

Proof. Let $g_i = DG_i[z]$ and $Q_i = D^2 G_i[z, z]$. Differentiating the perspective recursion gives

$$\begin{aligned} g_m &= (\log h_m) z_m, & Q_m &= 0, \\ g_i &= A_i z_i + \beta_i g_{i+1}, \\ Q_i &= \beta_i Q_{i+1} + \frac{\beta_i(1-\beta_i)}{s_i} (g_{i+1} - u_i a_{i+1} z_i)^2. \end{aligned} \quad (5.17)$$

In particular $Q_i \geq 0$. Put $E_i = g_i^2 + Q_i$ and $w_i = g_{i+1} - u_i a_{i+1} z_i$. Then

$$g_i = a_i z_i + \beta_i w_i, \quad Q_i \geq c w_i^2 + c Q_{i+1}.$$

Thus $z_i^2 \leq C E_i$, and then $g_{i+1}^2 \leq C E_i$, so that $E_{i+1} \leq C E_i$. Iteration over the fixed number of indices, starting with $E_m = (\log h_m)^2 z_m^2$, proves

$$E_i \geq c \sum_{j \geq i} z_j^2.$$

The reverse inequality follows recursively from (5.17) and its bounded coefficients. Finally,

$$D^2 Z_i[z, z] = Z_i E_i,$$

and Z_i is bounded above and below. $\square$

Define the negative objective Hessian by $\mathcal{H}(L; s) = -D_s^2 J(L; s)$. Summing (5.16) with weights L_i gives

$$c \sum_j V_j z_j^2 \leq z^\top \mathcal{H}(L; s) z \leq C \sum_j V_j z_j^2. \quad (5.18)$$

Furthermore,

$$|\mathcal{H}_{ij}(L; s)| \leq C V_{\min(i,j)}, \quad (5.19)$$

because only terms $L_\ell Z_\ell$ with $\ell \leq \min(i, j)$ depend on both coordinates, and their second derivatives are bounded. These estimates hold for every positive length vector, without stationarity or a length hierarchy.

If two distinct maximizers existed on $\mathcal{D}_E$, concavity would make their whole connecting segment maximizing. The global boundary lemmas put that segment in $\mathcal{D}_\varepsilon$, whereas (5.18) makes the restriction to the segment strictly concave. This contradiction proves uniqueness and finishes Theorem 5.1.

5.4 An LDL inverse estimate on every equality subspace

Lemma 5.7. *Suppose that a symmetric positive definite $r \times r$ matrix A and increasing positive numbers $W_1, \dots, W_r$ satisfy*

$$c \operatorname{diag}(W) \leq A \leq C \operatorname{diag}(W), \quad |A_{ij}| \leq C W_{\min(i,j)}.$$

Then

$$|(A^{-1})_{ij}| \leq \frac{C'}{W_{\max(i,j)}}, \quad (5.20)$$

where C' depends only on r, c, C .

Proof. Perform scalar LDL elimination in the order $1, \dots, r$. Every Schur complement retains the same diagonal quadratic bounds: its quadratic form is obtained by minimizing the preceding form over the eliminated coordinates. The lower bound follows by discarding the nonnegative eliminated diagonal terms; for the upper bound choose those coordinates to be zero. Thus the pivot at index i is comparable to W_i .

The entry estimate also persists, with constants depending on the number of steps. When index i is eliminated, the update in entry $j, \ell > i$ is bounded by

$$C \frac{W_i W_i}{W_i} = CW_i \leq CW_{\min(j, \ell)}.$$

Every multiplier in the unit lower triangular elimination matrix is therefore bounded. Its inverse has bounded entries as well, by the finite nilpotent series for the strictly lower triangular part. If $A = \mathbf{L} \mathbf{D} \mathbf{L}^\top$, the i, j entry of $A^{-1} = \mathbf{L}^{-\top} \mathbf{D}^{-1} \mathbf{L}^{-1}$ is a sum over indices $t \geq \max(i, j)$ of bounded numerators divided by pivots comparable to W_t . Monotonicity of W_t proves (5.20). $\square$

Partition $\{1, \dots, m\}$ into consecutive blocks $I_a = [b_a, e_a]$. Let B map block coordinates to vectors constant on each block. The restricted Hessian is $A = B^\top \mathcal{H} B$. By (5.18), its natural quadratic weights are $\sum_{j \in I_a} V_j$, which lie between V_{e_a} and mV_{e_a} . For $a < b$, summing (5.19) over the two blocks gives $|A_{ab}| \leq CV_{e_a}$. Lemma 5.7 consequently gives

$$\left| ((B^\top \mathcal{H} B)^{-1})_{ab} \right| \leq \frac{C}{V_{\max(e_a, e_b)}}. \quad (5.21)$$

There are only finitely many such partitions, so a single constant works for all of them. The Hessian is that of the original objective restricted to its current equality subspace.

5.5 Moving optimizers and differentiation across faces

We first record the elementary KKT relations with the sign convention in (5.6). At a maximizer on $\mathcal{D}_E$, a small shift of all products in either direction is feasible by Theorem 5.1; hence $\sum_i \partial_i J = 0$. For $j \notin E$, increasing the first j coordinates is feasible even when $\text{gap}_j = 0$. Its derivative is $-\lambda_j$, so $\lambda_j \geq 0$. If $\text{gap}_j > 0$, shifting that prefix in either direction is feasible, giving $\lambda_j = 0$. Thus

$$\partial_i J = \lambda_{i-1} - \lambda_i, \quad \lambda_j \geq 0, \quad \lambda_j \text{gap}_j = 0 \quad (j \notin E). \quad (5.22)$$

The multipliers at edges in E are not sign constrained. Conversely, (5.22), the signed equalities, and concavity give the corresponding global variational inequality.

Fix E , and consider $L(t) = L + t\Delta$, $0 \leq t \leq 1$, with $|\Delta_i| \leq L_i/2$. Then

$$\frac{1}{2}V_j \leq V_j(t) \leq \frac{3}{2}V_j. \quad (5.23)$$

Let $s(t) = s^E(L(t))$. All of these true constrained maxima lie in $\mathcal{D}_\varepsilon$, by the global theorem.

Lemma 5.8. *The path $s(t)$, and all gradient components along it, are Lipschitz and hence absolutely continuous. At almost every t , its derivative is constant on each maximal block of equal current products.*

Proof. Continuity follows from compactness and uniqueness: any convergent subsequence of maximizing points as $t_n \rightarrow t$ maximizes the limiting objective, so its limit is $s(t)$.

For t, t' , the two constrained variational inequalities and strong concavity on the segment joining the maximizers give, with $d = s(t') - s(t)$,

$$cV_1\|d\|^2 \leq |(\nabla J(L(t'); s(t')) - \nabla J(L(t); s(t')) \cdot d| \leq C|t' - t|D\|d\|.$$

Here the first bound uses (5.18) and (5.23), and the second uses bounded length derivatives of the gradient. This Lipschitz bound establishes the regularity needed for the componentwise estimate below. The gradient path is also Lipschitz since the objective is smooth on the compact product set and the length path is affine.

A nonnegative absolutely continuous function has derivative zero at almost every point of its zero set. To see this, at a density point of the zero set where its derivative exists, take difference quotients along that set. The density theorem and almost-everywhere differentiability remove the remaining null sets. Apply this observation to each gap $s_i(t) - s_{i+1}(t)$. There are finitely many gaps, so at almost every common differentiability point, $\dot{s}(t)$ is constant on every current equal-product block. $\square$

At such a time, let $I_a = [b_a, e_a]$ be the current maximal blocks, and write $\dot{s} = B\dot{q}$. Both exterior gaps of a block, when present, are strictly positive. They stay positive in a neighborhood of this time, and their multipliers vanish throughout that neighborhood. No prescribed equality crosses one of these strict gaps. Consequently, keeping these block endpoints fixed, one has in the whole neighborhood

$$\sum_{i \in I_a} \partial_i J(L(t); s(t)) = 0. \quad (5.24)$$

This identity remains true if internal equalities split nearby. It may therefore be differentiated at the chosen almost-everywhere point. This is the step that avoids assuming a differentiable active-set chart.

Since J is linear in L , define

$$U_\ell(s) = \partial_{L_\ell} J = 1 + \sum_{j \geq \ell} s_j p_j - Z_\ell(s), \quad b = \sum_{\ell} \Delta_\ell \nabla U_\ell(s(t)). \quad (5.25)$$

Differentiating (5.24) gives

$$(B^\top \mathcal{H} B) \dot{q} = B^\top b. \quad (5.26)$$

The vector ∇U_ℓ has zero entries before ℓ and uniformly bounded entries thereafter. Its sum over a block can therefore be nonzero only when that block's right endpoint is at least ℓ . By (5.21), for $i \in I_a$,

$$\begin{aligned} |\dot{s}_i(t)| &\leq C \sum_{\ell} |\Delta_\ell| \sum_{b: e_b \geq \ell} \frac{1}{V_{\max(e_a, e_b)}(t)} \\ &\leq C \sum_{\ell} \frac{|\Delta_\ell|}{V_{\max(i, \ell)}(t)}. \end{aligned} \quad (5.27)$$

Indeed $e_a \geq i$, $e_b \geq \ell$, and there are at most m blocks. Integrating this almost-everywhere estimate and using (5.23) proves (5.7).

Remark 5.9. An equality with zero multiplier may persist on a set of positive measure, split, or reform. The preceding argument includes all such possibilities: tangency comes from absolute continuity of the gaps, and the differentiated stationarity equation uses only the current block's strictly separated exterior boundaries.

5.6 Values, multipliers, and own-scale parameters

The coefficients U_ℓ in (5.25) are bounded on the entire original simplex, using (5.3). Hence

$$|J(L'; s) - J(L; s)| \leq CD \quad (s \in \mathcal{D}).$$

Taking the maximum on the same $\mathcal{D}_E$ in both directions proves (5.8). In fact this value bound does not require the relative perturbation assumption.

For the signed multiplier, absolute continuity and (5.25) give almost everywhere

$$\dot{\lambda}_j = - \sum_{i \leq j} b_i + \sum_{i \leq j} \sum_a \mathcal{H}_{ia} \dot{s}_a.$$

The first term is $O(D)$. By (5.19) and (5.27), the second term is at most

$$C \sum_{i \leq j} \sum_\ell |\Delta_\ell| \sum_a \frac{V_{\min(i,a)}(t)}{V_{\max(a,\ell)}(t)}.$$

Every fraction is at most one. The finite sums are therefore $O(D)$, uniformly in all length ratios. Integration proves (5.9) and finishes Theorem 5.2. In particular the bound applies to the unrestricted signed multipliers at prescribed equality edges.

Corollary 5.10 (Own-scale exponent stability). *Let $F_i(s_i, \dots, s_m)$ be bounded and Lipschitz on $\mathcal{D}_{\varepsilon/2}$, with chain-dependent bounds. Under the hypotheses of Theorem 5.2,*

$$|F_i(s^E(L')) - F_i(s^E(L))| \leq \frac{CD}{V_i}. \quad (5.28)$$

For D bounded, each of the factors

$$(1 + L_i)^{F_i(s^E(L))}, \quad \left(\frac{1 + V_{i-1}}{1 + L_i} \right)^{F_i(s^E(L))}$$

changes by a bounded multiplicative factor when L is replaced by L' . The bound does not involve a logarithm of a more remote length.

Proof. For every suffix coordinate $j \geq i$, (5.7) is bounded above by CD/V_i , proving (5.28). Also

$$\log(1 + L_i) \leq \log(1 + V_i), \quad \left| \log \frac{1 + V_{i-1}}{1 + L_i} \right| \leq 2 \log(1 + V_i).$$

Multiply these bounds by (5.28) and use $\log(1 + V_i)/V_i \leq 1$. The bases themselves change by bounded multiplicative factors, because their numerator and denominator change by bounded additive amounts and each has a $1+$ normalization. Boundedness of F_i finishes the proof. $\square$

For example the ratio parameters $d_i = s_i/s_{i+1} - 1$ satisfy

$$|d_i(L') - d_i(L)| \leq CD/V_i,$$

since products stay away from zero. This observation and (5.9) are separate statements. They do not by themselves justify stability of a proposed persistence expression involving division by a shorter native length or a change of formula between different platforms.

5.7 Opening a single ordered-product edge

Fix $1 \leq j < m$. Let $\bar{s}$ maximize with the sole extra equality $s_j = s_{j+1}$, and let s^* be the unrestricted ordered maximizer. All other inequalities are optimized in both problems. Write $\bar{\lambda}_i = -\sum_{a \leq i} \partial_a J(L; \bar{s})$ and $a = (-\bar{\lambda}_j)_+$.

If $\bar{\lambda}_j \geq 0$, every multiplier has the sign and complementarity required for the unrestricted problem. More explicitly, for every $y \in \mathcal{D}$, summation by parts gives

$$\nabla J(L; \bar{s}) \cdot (y - \bar{s}) = - \sum_{i < m} \bar{\lambda}_i (\text{gap}_i(y) - \text{gap}_i(\bar{s})) \leq 0.$$

Concavity and uniqueness imply $\bar{s} = s^*$.

Suppose now that $a > 0$. Let $e^{(j)}$ have its first j coordinates equal to one and the rest zero. The perturbation $\bar{s} + \tau e^{(j)}$ opens only gap j ; every other gap is unchanged. It is feasible for all sufficiently small $\tau > 0$, even if some preceding or following inequalities are equalities. Its first derivative at zero is

$$\nabla J(L; \bar{s}) \cdot e^{(j)} = -\bar{\lambda}_j = a.$$

On $\mathcal{D}_{\varepsilon/2}$, its negative second derivative is at most

$$C \sum_{i \leq j} V_i \leq C m V_j.$$

We also have the useful elementary bound

$$|\bar{\lambda}_j| \leq C_0 V_j. \quad (5.29)$$

Indeed $\partial_i J = p_i V_i - \sum_{\ell \leq i} L_\ell \partial_i Z_\ell$ has absolute value at most $C V_i$, by the compact derivative bounds, and there are at most m terms in $\bar{\lambda}_j$.

Choose a sufficiently large chain-dependent C_1 and set $\tau = a/(C_1 V_j)$. By (5.29), this can be made at most $\varepsilon/2$, so the ray stays in $\mathcal{D}_{\varepsilon/2}$. Taylor's formula with integral remainder then gives

$$J(L; \bar{s} + \tau e^{(j)}) - J(L; \bar{s}) \geq a\tau - C V_j \tau^2 \geq c \frac{a^2}{V_j}. \quad (5.30)$$

For the matching bound, put $\Delta s = s^* - \bar{s}$ and $g = \text{gap}_j(s^*)$. Since the original gap is zero, $g = \Delta s_j - \Delta s_{j+1} \geq 0$. Summation by parts and complementarity at all other edges yield

$$\begin{aligned} \nabla J(L; \bar{s}) \cdot \Delta s &= - \sum_{i < m} \bar{\lambda}_i (\text{gap}_i(s^*) - \text{gap}_i(\bar{s})) \\ &= ag - \sum_{i \neq j} \bar{\lambda}_i \text{gap}_i(s^*) \leq ag. \end{aligned} \quad (5.31)$$

The neighboring nonnegative multipliers retain their negative contribution. Moreover

$$\sum_i V_i (\Delta s_i)^2 \geq V_j ((\Delta s_j)^2 + (\Delta s_{j+1})^2) \geq \frac{1}{2} V_j g^2.$$

Strong concavity integrated along the segment between the two maximizers therefore implies

$$0 \leq J(L; s^*) - J(L; \bar{s}) \leq ag - c V_j g^2 \leq C \frac{a^2}{V_j}. \quad (5.32)$$

Combining (5.30) with the bound by ag gives $g \geq ca/V_j$. Nonnegativity in (5.32) gives the reverse inequality. This proves (5.13).

The natural bounded-rate opening coordinate is thus

$$\frac{a}{\sqrt{V_j}}.$$

This is a statement about the exact optimized rate. If this coordinate diverges, using the restricted-face rate instead of the unrestricted one produces an unbounded exponential discrepancy.

5.8 Simultaneous release of a prescribed face

We now allow an arbitrary E , with the notation (5.10). All signed multipliers refer to the maximum $\bar{s} = s^E(L)$ of the original objective on that face.

5.8.1 A feasible test direction

For each j with $a_j > 0$, choose

$$\tau_j = \frac{a_j}{C_1 V_j}, \quad z = \sum_{j: a_j > 0} \tau_j e^{(j)}.$$

This opens exactly these gaps, because $z_j - z_{j+1} = \tau_j$, with $\tau_j = 0$ at all other edges. It preserves the order constraints. The argument for (5.29) gives $a_j \leq C_0 V_j$, so $\sum_j \tau_j \leq m C_0 / C_1$. Choose C_1 large enough that the entire test ray stays in $\mathcal{D}_{\varepsilon/2}$.

Its linear gain is exactly

$$\nabla J(L; \bar{s}) \cdot z = \sum_j a_j \tau_j.$$

For the quadratic cost, Cauchy–Schwarz in the finite prefix sum gives

$$\sum_i V_i z_i^2 \leq m \sum_j \tau_j^2 \sum_{i \leq j} V_i \leq m^2 \sum_j V_j \tau_j^2. \quad (5.33)$$

Taylor’s formula and (5.18), after one further chain-dependent enlargement of C_1 , yield

$$J(L; \bar{s} + z) - J(L; \bar{s}) \geq c \mathcal{A}_E. \quad (5.34)$$

The unrestricted gain is therefore at least $c \mathcal{A}_E$.

5.8.2 The upper bound and the total displacement

Put $\Delta s = s^* - \bar{s}$ and $Q = \sum_i V_i (\Delta s_i)^2$. At edges outside E the multipliers are nonnegative and complementary; at edges in E the original gap is zero. Summation by parts consequently gives

$$\nabla J(L; \bar{s}) \cdot \Delta s \leq \sum_{j: a_j > 0} a_j \text{gap}_j(s^*). \quad (5.35)$$

For $j \in E$, $\text{gap}_j(s^*) = \Delta s_j - \Delta s_{j+1}$, so

$$\sum_{j \in E} V_j \text{gap}_j(s^*)^2 \leq 2 \sum_{j < m} V_j ((\Delta s_j)^2 + (\Delta s_{j+1})^2) \leq 4Q. \quad (5.36)$$

The last bound only uses $V_j \leq V_{j+1}$. Weighted Cauchy–Schwarz in (5.35), followed by integrated strong concavity, proves

$$0 \leq J(L; s^*) - J(L; \bar{s}) \leq 2\sqrt{\mathcal{A}_E Q} - cQ. \quad (5.37)$$

Thus $Q \leq C\mathcal{A}_E$, and the rate gain is at most $C\mathcal{A}_E$. Conversely, (5.34) and (5.37) with its negative term omitted give $Q \geq c\mathcal{A}_E$ when $\mathcal{A}_E > 0$. If $\mathcal{A}_E = 0$, (5.37) forces $Q = 0$, and both optima coincide. This proves (5.11)–(5.12) and completes Theorem 5.3.

Remark 5.11. The simultaneous comparison gives a weighted total displacement, but does not assert $\text{gap}_j(s^*) \asymp a_j/V_j$ for every member of a multi-edge release. Several signed multipliers may interact through the order constraints.

5.9 The finite optimization and its product bounds

Fix integers

$$H = h_1 > h_2 > \cdots > h_m \geq 2, \quad p_j = h_j - h_{j+1} \ (j < m), \quad p_m = h_m - 1.$$

Let $L_i > 0$, $V_j = \sum_{i \leq j} L_i$, and $V_0 = 0$. The product variables belong to the ordered simplex

$$\mathcal{S} = \{1 \geq s_1 \geq s_2 \geq \cdots \geq s_m \geq 0\}.$$

For positive products define recursively

$$G_m = s_m \log h_m, \quad G_j = s_j \log(p_j + \exp(G_{j+1}/s_j)) \ (j < m), \quad (5.38)$$

$$Z_j = e^{G_j}, \quad J(\mathbf{s}) = \sum_i L_i + \sum_j s_j p_j V_j - \sum_i L_i Z_i. \quad (5.39)$$

The functions extend continuously to $\mathcal{S}$: if $s_j = 0$, the entire suffix has zero product and we set $G_j = 0$. Indeed, induction gives $0 \leq G_j \leq s_j \log h_j$, which proves continuity at those faces. A maximizer exists by compactness. The perspective of $y \mapsto \log(p + e^y)$ is convex and nondecreasing in its second argument; recursive perspective composition shows that each G_j , then each Z_j , is convex on $\mathcal{S}$. Thus J is concave. We use the unique global maximizer furnished by Theorem 5.1.

When products are positive, put

$$S_m = h_m, \quad S_j = p_j + S_{j+1}^{u_j}, \quad u_j = s_{j+1}/s_j, \quad (5.40)$$

$$\beta_j = S_{j+1}^{u_j}/S_j, \quad A_j = \log S_j - u_j \beta_j \log S_{j+1} \ (j < m), \quad A_m = \log h_m. \quad (5.41)$$

Then $G_j = s_j \log S_j$, $Z_j = S_j^{s_j}$, and $2 \leq S_j \leq h_j \leq H$. We repeatedly use the entropy increment

$$E_n(z) = (n+z) \log(n+z) - z \log z, \quad n > 0, \ z \geq 1. \quad (5.42)$$

For a nonterminal node, $A_j = E_{p_j}(S_{j+1}^{u_j})/S_j > 0$.

Lemma 5.12 (Uniform upper product bound). *Every maximizer satisfies $s_1 < 1 - \epsilon_H$, where $\epsilon_H = (2 \log 2 - 1)/(2H(\log H)^2)$.*

Proof. This is the upper bound in Theorem 5.1, applied with no prescribed equalities. $\square$

Lemma 5.13 (Canonical thresholds). *For real $h > 1$ define*

$$\alpha_h = \frac{\log((h-1)/\log h)}{\log h} = 1 - \frac{\log(h \log h/(h-1))}{\log h}.$$

Then $\alpha_h > 1/2$ and α_h is strictly increasing. In particular, $2^{\alpha_2} \log 2 = 1$.

Proof. Write $y = \log h$ and $z = y/2$. Direct calculation gives

$$\alpha_h = \frac{1}{2} + \frac{\log(\sinh z/z)}{2z}.$$

The second term is positive. Its derivative has the sign of

$$N(z) = z \coth z - 1 - \log(\sinh z/z).$$

Here $N(0+) = 0$ and $N'(z) = 1/z - z/\sinh^2 z > 0$. This proves both assertions without a discrete approximation. $\square$

Lemma 5.14 (Transported natural intensities). *For $i < j$ at positive products,*

$$Z_i \prod_{q=i}^{j-1} \beta_q = Z_j \prod_{q=i}^{j-1} \beta_q^{1-s_q} \leq Z_j. \quad (5.43)$$

If $s_1 < 1$, the inequality is strict.

Proof. The one-edge identity $Z_q \beta_q = Z_{q+1} \beta_q^{1-s_q}$ follows directly from (5.38)–(5.41). Multiplication telescopes. Since $0 < \beta_q < 1$, the conclusions follow. $\square$

Theorem 5.15 (Global product compactness). *For $H \geq 3$ and every positive length vector, every maximizer satisfies*

$$\alpha_2 < s_m \leq s_1 < 1 - \epsilon_H. \quad (5.44)$$

For the exceptional one-node chain $H = 2$, the unique maximizing product is α_2 . In particular all products, local ratios and finite positive mark probabilities used below lie in compact sets determined by H .

Proof. Lemma 5.12 already gives the upper bound. Let $a, \dots, m$ be the bottom maximal platform, with product $\sigma \leq \alpha_2$. Raising its common product is feasible. Put

$$I_{\text{in}} = \sum_{i < a} L_i Z_i \prod_{q=i}^{a-1} \beta_q.$$

The derivative in that direction is

$$(h_a - 1)V_{a-1} - I_{\text{in}} \log h_a + \sum_{i \geq a} L_i (h_i - 1 - h_i^\sigma \log h_i). \quad (5.45)$$

This also holds as a one-sided derivative at $\sigma = 0$: the older products are positive, and the bottom recursion is exactly $G_i = \sigma \log h_i$. Transport gives $I_{\text{in}} \leq h_a^\sigma V_{a-1}$. By Lemma 5.13, all brackets in (5.45) are nonnegative, and a bracket with $h_i > 2$ is strictly positive. Thus (5.45) is positive if such a rank occurs in the bottom platform. Otherwise the platform is the single rank $h_m = 2$, with $a = m > 1$ when $H \geq 3$. In that case $I_{\text{in}} < 2^\sigma V_{m-1}$ by Lemma 5.12 and the strict transport identity, so the derivative is again positive, even at $\sigma = \alpha_2$. Both alternatives contradict maximality. For the chain $H = 2$, directly differentiating $L(1 + s - 2^s)$ gives its unique maximizer α_2 . $\square$

5.10 Exact node margins and root centering

We now work at a maximizer. The preceding result removes the upper and lower faces of the ordered simplex. There are multipliers

$$\lambda_0 = \lambda_m = 0, \quad \lambda_j \geq 0, \quad \partial_j J = \lambda_{j-1} - \lambda_j, \quad \lambda_j(s_j - s_{j+1}) = 0 \quad (j < m). \quad (5.46)$$

Write $m_j = \lambda_j - \lambda_{j-1}$. A maximal platform is a maximal interval $[a, b]$ on which the products are equal. Its exterior multipliers satisfy $\lambda_{a-1} = \lambda_b = 0$.

Lemma 5.16 (Node equations and Euler root identity). *For every j ,*

$$A_j \sum_{i \leq j} L_i Z_i \prod_{q=i}^{j-1} \beta_q = p_j V_j + m_j. \quad (5.47)$$

Let $r_j = s_j/s_1$, $a_j = \log S_j$ and $\mu_j = L_j Z_j$. Then

$$T_1 := \sum_j r_j p_j V_j = \sum_j c_j \mu_j, \quad c_j = r_j a_j. \quad (5.48)$$

Moreover $0 < c_-(H) \leq c_j, Z_j \leq c_+(H) < \infty$.

Proof. Differentiation of the recursion gives $\partial G_j / \partial s_j = A_j$ and $\partial G_i / \partial G_j = \prod_{q=i}^{j-1} \beta_q$ for $i < j$. Substitution into $\partial_j J$ and (5.46) proves (5.47). Every G_i is homogeneous of degree one in the products. Consequently

$$\sum_j s_j \partial_j J = \sum_j s_j p_j V_j - \sum_i L_i Z_i G_i.$$

The left side is zero because $\sum_j s_j m_j = \sum_{j < m} \lambda_j(s_j - s_{j+1}) = 0$. Division by s_1 proves (5.48). Finally $2 \leq S_j \leq H$ and (5.44) bound r_j above and below, which gives the stated constants. $\square$

Proposition 5.17 (Poisson entropy at the saddle). *For positive original lengths, let $s = s(L)$ be the ordered optimizer in (3.2). With $\mathcal{Q}(z) = z \log z - z + 1$,*

$$J(L; s) = \sum_{i=1}^k L_i \mathcal{Q}(Z_i). \quad (5.49)$$

Equivalently, J is the relative entropy of independent Poisson counts with means $L_i Z_i$ with respect to independent counts with means L_i .

Proof. The upper and lower simplex faces are inactive, so dilation of all products is feasible in both directions, even on an equality face. Homogeneity of the G_i and stationarity give

$$0 = \sum_j s_j \partial_{s_j} J = \sum_j s_j p_j V_j - \sum_i L_i Z_i G_i.$$

Substitute $G_i = \log Z_i$ into (3.2). For a single count, $D_{\text{KL}}(\text{Pois}(Lz) \parallel \text{Pois}(L)) = L \mathcal{Q}(z)$; relative entropy adds over independent coordinates. $\square$

The identity describes the physical tilt that supplies the exponential factor of the profile. The persistence factors describe a second selection, on the shorter time scale of the conditional bridges; Section 17.6 identifies its entropy cost.

5.11 Complete unit-step chains: sign and effective time

In this section only, assume

$$H \geq 3, \quad m = H - 1, \quad h_j = H - j + 1, \quad p_j = 1 \quad (1 \leq j \leq m).$$

The *natural native node drift* is

$$e_j = Z_j A_j - 1. \quad (5.50)$$

The rank here is one. It must not be replaced by the larger rank of an entire equal-product platform with its child left unsplit.

Lemma 5.18 (A secant threshold with one valley). *For $x \geq 2$ put*

$$D(x) = x \log x - (x - 1) \log(x - 1), \quad t(x) = 1 - \frac{\log D(x)}{\log x}.$$

For $2 \leq x \leq y$,

$$t(x) \leq \max\{\alpha_2, t(y)\}, \quad (5.51)$$

strictly when $2 < x < y$. More quantitatively, if $y \leq H$, $x \geq 2 + \eta$ and $y \geq x + \eta$, then

$$\max\{\alpha_2, t(y)\} - t(x) \geq \Delta_H(\eta) := \frac{\eta^2 \log(2 \log 2)}{2H^2(1 + \log H)(\log H)^2}. \quad (5.52)$$

Proof. Set $l(x) = \log(x/(x - 1))$. Direct differentiation yields

$$t'(x) = \frac{F(x)}{xD(x)(\log x)^2}, \quad (5.53)$$

$$F(x) = D(x) \log D(x) - x \log x l(x), \quad (5.54)$$

$$F'(x) = l(x) \log D(x) + \log x (1/(x - 1) - l(x)) > 0. \quad (5.55)$$

Here $D(x) \geq 2 \log 2 > 1$, and $\log(1 + u) < u$ proves the second term positive. Thus t first decreases and then increases, allowing either monotone case, with no flat interval. Its maximum on $[2, y]$ is attained at an endpoint and $t(2) = \alpha_2$. This proves (5.51) and its strict form.

For the quantitative assertion, throughout $[2, H]$,

$$F'(v) \geq \frac{\log(2 \log 2)}{H}, \quad vD(v)(\log v)^2 \leq H(1 + \log H)(\log H)^2.$$

The latter bound follows from $D(v) = \int_{v-1}^v (1 + \log w) dw \leq 1 + \log H$. If $F(x) \geq 0$, integrate t' over $[x, x + \eta]$, using the linear lower bound for $F(v)$ based at x , and then use monotonicity up to y . If $F(x) \leq 0$, integrate backwards over $[x - \eta, x]$; F is nonpositive on $[2, x]$, so $t(2) \geq t(x - \eta)$. Both integrations give exactly (5.52). $\square$

Theorem 5.19 (No negative native node drift). *On the complete unit-step chain, every positive length vector has*

$$e_j \geq 0 \quad (1 \leq j \leq m), \quad e_m > 0. \quad (5.56)$$

If a is the first index of a maximal platform, then $e_a > 0$ when $a > 1$. In addition, for every nonterminal index that is not first in its platform,

$$e_j \geq c_H := 2^{\Delta_H(\sqrt{2}-1)} - 1 > 0. \quad (5.57)$$

There is no assumed lower bound on length ratios, gaps between platform products, or nonzero multipliers.

Proof. Let $[a, b]$ be a maximal platform with product σ . At its first node $m_a = \lambda_a \geq 0$. With $I_{\text{in}} = \sum_{i < a} L_i Z_i \prod_{q=i}^{a-1} \beta_q$, the exact node equation and transport give

$$\lambda_a = A_a(L_a Z_a + I_{\text{in}}) - V_a \leq (Z_a A_a - 1)V_a = e_a V_a. \quad (5.58)$$

This proves $e_a \geq 0$. If $a > 1$, strict transport makes the inequality strict, so $e_a > 0$.

For $b < m$ put $C = h_{b+1}$ and $z = \exp(G_{b+1}/\sigma)$; for $b = m$ put $C = z = 1$. Within the platform $S_i = h_i - C + z$ decreases by one at each step. At every node its immediate child weight is $S_i - 1$, including the last node of the platform. Therefore

$$A_i = D(S_i)/S_i, \quad e_i = S_i^{\sigma-1} D(S_i) - 1 = \exp((\sigma - t(S_i)) \log S_i) - 1. \quad (5.59)$$

The first-node sign gives $\sigma \geq t(S_a)$; global compactness gives $\sigma > \alpha_2$. Lemma 5.18 propagates $\sigma \geq t(S_i)$ through the platform. At the terminal node, $e_m = 2^{s_m} \log 2 - 1 > 0$.

For $i > a$ with $i < m$, we have $S_a \geq S_i + 1$. If $b = m$, then $S_i = h_i \geq 3$. Otherwise $u_b = s_{b+1}/\sigma > 1/2$ and $S_{b+1} \geq 2$, whence $z > \sqrt{2}$ and $S_i \geq 1 + z > 1 + \sqrt{2}$. Thus $S_i \geq 2 + (\sqrt{2} - 1)$ in both cases. Apply (5.52) with $\eta = \sqrt{2} - 1$ and then (5.59), using $S_i \geq 2$, to obtain (5.57). $\square$

Theorem 5.20 (Exact first-node drift scale). *For the first index a of any maximal platform on the complete chain,*

$$e_a \asymp_H \frac{\lambda_a + V_{a-1}}{V_a}. \quad (5.60)$$

For $a = 1$ this is the exact equality $e_1 = \lambda_1/L_1$. When $a > 1$, set

$$q_- = H^{-H}, \quad q_+ = (1 - 1/H)^{\epsilon_H} < 1.$$

Then (5.60) holds with explicit lower and upper comparison constants $1 - q_+$ and q_-^{-1} . Consequently, with the infinite-quotient convention,

$$1 + \min\{L_a, e_a^{-2}\} \asymp_H 1 + \min\left\{L_a, \left(\frac{V_a}{\lambda_a + V_{a-1}}\right)^2\right\}. \quad (5.61)$$

For a non-first, nonterminal node, the left effective time is at most $1 + c_H^{-2}$.

Proof. On the unit-step chain $1/H \leq \beta_q \leq 1 - 1/H$. Also $\epsilon_H < 1 - s_q < 1$. Formula (5.43) implies, when $a > 1$,

$$q_- V_{a-1} \leq I_{\text{in}}/Z_a \leq q_+ V_{a-1}.$$

There are fewer than H factors in each transported term; the upper bound uses at least one factor. Write $w = I_{\text{in}}/(Z_a V_{a-1}) \in [q_-, q_+]$. The exact equality in (5.58), before using transport, is

$$\lambda_a = e_a(L_a + w V_{a-1}) - (1 - w)V_{a-1}, \quad (5.62)$$

$$e_a = \frac{\lambda_a + (1 - w)V_{a-1}}{L_a + w V_{a-1}}. \quad (5.63)$$

Its numerator lies between $(1 - q_+)(\lambda_a + V_{a-1})$ and $\lambda_a + V_{a-1}$; its denominator lies between $q_- V_a$ and V_a . This proves the stated comparison. At $a = 1$ there is no incoming intensity and (5.47) gives the exact equality directly. Taking inverse squares and truncating by L_a preserves comparison by fixed constants, including the zero-drift case. The final assertion follows from (5.57). $\square$

Remark 5.21. The effective time in (5.61) identifies a possible long one-ended persistence scale. It does not evaluate reverse reserves, terminal bands, or conditional persistence with a private incoming state. In particular, it is not a claim that native packets can be multiplied independently.

5.12 A hierarchy of maximal platform lengths

We return to an arbitrary sparse chain $H = h_1 > \dots > h_m \geq 2$. No sign assertion for its individual native node drifts is needed. For a maximal equal-product platform $[a, b]$, let its common product be σ . Define

$$(C, z) = (h_{b+1}, S_{b+1}^{u_b}) \quad (b < m), \quad (5.64)$$

$$(C, z) = (1, 1) \quad (b = m), \quad (5.65)$$

$$n_i = h_i - C \quad (a \leq i \leq b), \quad n = n_a, \quad S = S_a = n + z, \quad (5.66)$$

$$A_* = E_n(z)/S, \quad \mathbf{e}_i = (n_i + z)^{\sigma-1} E_{n_i}(z) - n_i. \quad (5.67)$$

The symbol $\mathbf{e}_i$ is a *whole-platform* drift: it uses the entire exterior rank n_i , with the child intact. It is different from e_i in (5.50).

Lemma 5.22 (Exact platform balance). *With $I_{\text{in}} = \sum_{i < a} L_i Z_i \prod_{q=i}^{a-1} \beta_q$,*

$$\sum_{i=a}^b L_i \mathbf{e}_i = n V_{a-1} - A_* I_{\text{in}}. \quad (5.68)$$

If $a > 1$, then

$$I_{\text{in}} = \frac{\beta_{a-1}}{A_{a-1}} (p_{a-1} V_{a-1} - \lambda_{a-2}) \leq \frac{\beta_{a-1} p_{a-1}}{A_{a-1}} V_{a-1}. \quad (5.69)$$

Proof. The sum of budget derivatives over $[a, b]$ is $n V_{a-1} + \sum_{i=a}^b n_i L_i$. On the platform, $G_i = \sigma \log(n_i + z)$ and its derivative is $E_{n_i}(z)/(n_i + z)$, including the variation in z . The derivative for an older block is $A_* \prod_{q=i}^{a-1} \beta_q$. The sum of the KKT derivatives on the platform is $\lambda_{a-1} - \lambda_b = 0$. Substituting these derivatives proves (5.68).

At the single node $a-1$, the multiplier margin is $m_{a-1} = \lambda_{a-1} - \lambda_{a-2} = -\lambda_{a-2} \leq 0$. Equation (5.47), followed by its routing factor β_{a-1} , is exactly (5.69). This step retains the actual mixture of all older incoming intensities. $\square$

Lemma 5.23 (Whole-platform entropy surplus). *If $a > 1$, then*

$$n - A_* \frac{\beta_{a-1} p_{a-1}}{A_{a-1}} \geq \frac{n p_{a-1}}{2H^2(1 + \log H)^2} > 0. \quad (5.70)$$

This estimate is independent of the size of the strict product gap before the platform.

Proof. For real $r > 0$ and $x \geq 1$ let $D(r, x) = E_r(x)/r$. Integrating $1 + \log(x+t)$ gives

$$D(r, x) = 1 + \log x + \frac{1}{r} \int_0^r \log(1+t/x) dt, \quad (5.71)$$

$$1 + \log x + \frac{r}{2(x+r)} \leq D(r, x) \leq 1 + \log(x+r). \quad (5.72)$$

For the lower bound use $\log(1+t/x) \geq t/(x+t) \geq t/(x+r)$. Write $p = p_{a-1}$, $u = u_{a-1}$ and $w = S^u$. Then $1 \leq w \leq S$, $p+w \leq H$, and

$$\frac{\beta_{a-1} p}{A_{a-1}} = \frac{w}{D(p, w)}, \quad \frac{n}{A_*} = \frac{S}{D(n, z)} \geq \frac{S}{1 + \log S}.$$

Since $x/(1 + \log x)$ is nondecreasing for $x \geq 1$,

$$\begin{aligned} \frac{n}{A_*} - \frac{\beta_{a-1}p}{A_{a-1}} &\geq \frac{w}{1 + \log w} - \frac{w}{D(p, w)} \\ &= \frac{w[D(p, w) - (1 + \log w)]}{(1 + \log w)D(p, w)} \geq \frac{p}{2H(1 + \log H)^2}. \end{aligned} \quad (5.73)$$

Finally (5.72) implies $E_n(z) \geq n$, so $A_* \geq n/H$. Multiplication of (5.73) by A_* proves (5.70). Strict separation of the products was used only for the zero exterior multiplier, not to obtain its numerical constant. $\square$

Theorem 5.24 (Maximal-platform length hierarchy). *At every global ordered KKT maximizer on an arbitrary sparse chain, each maximal platform $[a, b]$ satisfies*

$$V_{a-1} \leq C_H \sum_{i=a}^b L_i, \quad C_H = 2H^3(1 + \log H)^3. \quad (5.74)$$

For $a > 1$, its signed incoming whole-platform drift satisfies

$$A_* I_{\text{in}} - nV_{a-1} \leq -\frac{np_{a-1}}{2H^2(1 + \log H)^2} V_{a-1}. \quad (5.75)$$

The estimates require no positive length-ratio or product-gap bound.

Proof. Combining (5.69) and (5.70) gives (5.75), and then (5.68) gives

$$\sum_{i=a}^b L_i \mathfrak{e}_i \geq \frac{np_{a-1}}{2H^2(1 + \log H)^2} V_{a-1}.$$

For every summand, $1 \leq n_i + z \leq H$, $0 < \sigma < 1$ and (5.72) give $|\mathfrak{e}_i| \leq H(1 + \log H)$. Hence

$$V_{a-1} \leq \frac{2H^3(1 + \log H)^3}{np_{a-1}} \sum_{i=a}^b L_i \leq C_H \sum_{i=a}^b L_i.$$

For $a = 1$, $V_0 = 0$, so (5.74) is immediate. $\square$

Corollary 5.25 (A long block in the final platform). *Let $[a, m]$ be the final maximal platform, and select a largest block d within it. Then*

$$L_d \geq \frac{V_m}{(H - 1)(1 + C_H)}. \quad (5.76)$$

Proof. Set $W = \sum_{i=a}^m L_i$. Theorem 5.24 gives $V_m \leq (1 + C_H)W$. The platform has at most $H - 1$ blocks, so $L_d \geq W/(H - 1)$. $\square$

Remark 5.26. The hierarchy is for whole maximal equal-product platforms. It does not imply $V_{j-1} \ll_H L_j$ at an internal node of a platform. Nor does (5.75) make several native blocks within that platform independent. Opposite whole-platform drift signs can still cancel, and incoming retained words still carry private data.

5.13 An unread uncolored root reservoir

This section is an elementary conditional local-mass statement for the natural Poisson model at the current KKT optimizer. It is not a path-survival theorem. In block i take an independent homogeneous Poisson point process of intensity Z_i over length L_i , giving $N_i \sim \text{Pois}(\mu_i)$ with $\mu_i = Z_i L_i$. Any independent private marks can be attached to these points. Let

$$V = V_m, \quad X_1 = \sum_i c_i N_i - T_1, \quad c_i = r_i \log S_i.$$

The centering $T_1 = \sum_i c_i \mu_i$ is exact by Lemma 5.16.

Choose deterministically a block d satisfying $L_d \geq \kappa_H V$ for some fixed $\kappa_H > 0$. Either a globally largest block or the block in Corollary 5.25 does so. In block d , choose a deterministic subinterval of length τL_d , where $0 < \tau_0 \leq \tau \leq \tau_1 < 1$ are fixed. Denote its uncolored count by N_R and its mean by $\mu_R = \tau \mu_d$.

Let $\mathcal{O}$ be the sigma-field of all points and private marks *outside* that interval. Then N_R is independent of $\mathcal{O}$. Write $N_i^{\text{out}} = N_i$ for $i \neq d$ and $N_d^{\text{out}} = N_d - N_R$, with corresponding means $\mu_i^{\text{out}} = \mu_i$ and $\mu_d^{\text{out}} = (1 - \tau)\mu_d$. The target x must be measurable in the uncolored part of $\mathcal{O}$. Assume, for fixed $A, B < \infty$,

$$|x| \leq A \log(2 + V), \quad \mathcal{C}_B = \bigcap_i \{|N_i^{\text{out}} - \mu_i^{\text{out}}| \leq B\sqrt{1 + L_i}\}. \quad (5.77)$$

The reservoir and its count have not been read by the target or by any previously selected outside cost.

Theorem 5.27 (Uniform conditional root atom). *Define the $\mathcal{O}$ -measurable integer*

$$n_*(\mathcal{O}) = \left\lfloor \frac{T_1 + x - \sum_i c_i N_i^{\text{out}}}{c_d} \right\rfloor. \quad (5.78)$$

For $L_d \geq L_0(H, A, B, \kappa_H, \tau_0, \tau_1)$, on $\mathcal{C}_B$ it is nonnegative and

$$|n_* - \mu_R| \leq C\sqrt{L_d}, \quad (5.79)$$

$$\mathbb{P}(N_R = n_* \mid \mathcal{O}) \asymp (1 + V)^{-1/2}. \quad (5.80)$$

The constants depend only on the displayed fixed parameters. On $N_R = n_*$,

$$x - c_d < X_1 \leq x. \quad (5.81)$$

For every nonnegative $\mathcal{O}$ -measurable random variable F , including arbitrarily rare outside private-word weights,

$$\mathbb{E}[F \mathbf{1}_{\mathcal{C}_B} \mathbf{1}_{\{N_R = n_*\}}] \asymp (1 + V)^{-1/2} \mathbb{E}[F \mathbf{1}_{\mathcal{C}_B}]. \quad (5.82)$$

Proof. Subtract $c_d \mu_R$ inside (5.78). Exact centering then gives

$$n_* - \mu_R = \frac{x - \sum_i c_i (N_i^{\text{out}} - \mu_i^{\text{out}})}{c_d} + \varepsilon, \quad -1 < \varepsilon \leq 0.$$

There are at most $H - 1$ blocks. On $\mathcal{C}_B$,

$$\sum_i \sqrt{1 + L_i} \leq C_H \sqrt{1 + V} \leq C_{H, \kappa_H} \sqrt{L_d}, \quad \log(2 + V) \leq C_{H, \kappa_H} \sqrt{L_d}$$

when $L_d \geq 1$. Uniform bounds for c_i prove (5.79). Since $\mu_R \geq c_H \tau_0 L_d$, taking L_0 sufficiently large makes n_* positive.

For completeness, if $n = \mu_R + v$ with $|v| \leq C\sqrt{\mu_R}$, then

$$n \log(n/\mu_R) - n + \mu_R = O_C(1), \quad n! \asymp \sqrt{n} (n/e)^n.$$

Uniform Stirling inequalities therefore put $e^{-\mu_R} \mu_R^n / n!$ between fixed positive multiples of $\mu_R^{-1/2}$. Because $\mu_R \asymp L_d \asymp V$, this proves (5.80) pointwise on every admitted outside fiber. The floor in (5.78) gives (5.81) exactly. Finally condition on $\mathcal{O}$ and use the pointwise conditional estimate to obtain (5.82). $\square$

Proposition 5.28 (Compatibility with exact margin-carry bands). *Assume $m \geq 2$. Suppose, in addition to the hypotheses of Theorem 5.27, that a subsequent retained word satisfies the exact recursions*

$$X_{j+1} = (X_j - Y_j)/u_j \quad (j < m) \tag{5.83}$$

and the bands are represented exactly as

$$Y_j = m_j + y_j + \xi_j - \xi_{j-1}/u_{j-1}, \quad j < m, \quad \xi_0 = 0, \tag{5.84}$$

where the initial term with u_0 is absent. Choose the uncolored root target $x = \sum_{j < m} r_j y_j$. Then

$$X_m = \frac{X_1 - x}{r_m} - \lambda_{m-1} - \frac{\xi_{m-1}}{u_{m-1}}. \tag{5.85}$$

The identity is unaffected by which block supplies the root count.

Proof. Since $r_{j+1} = r_j u_j$, recursion (5.83) gives $r_m X_m = X_1 - \sum_{j < m} r_j Y_j$. Complementarity in (5.46) gives $\sum_{j \leq m} r_j m_j = 0$ and $m_m = -\lambda_{m-1}$, hence $\sum_{j < m} r_j m_j = r_m \lambda_{m-1}$. The carry terms telescope:

$$\sum_{j < m} r_j (\xi_j - \xi_{j-1}/u_{j-1}) = r_{m-1} \xi_{m-1}.$$

Substitute these equalities and $x = \sum_{j < m} r_j y_j$ to obtain (5.85). The argument contains no assumption about the position of the selected reservoir. $\square$

Remark 5.29 (The additional conditional obligation). After conditioning on $N_R = n_*$, the reservoir points are uniform order statistics on the fixed interval and their private marks retain their original law. A later packet P that reads that reservoir needs a further pointwise conditional estimate at the prescribed count. For example, if on $\mathcal{C}_B$,

$$cG(\mathcal{O}) \leq \mathbb{E}[P \mid \mathcal{O}, N_R = n_*(\mathcal{O})] \leq CG(\mathcal{O}),$$

then one more conditioning step extends (5.82) with P on the left and G on the right. A marginal scalar estimate averaged over N_R cannot substitute for this hypothesis. Neither an ordinary bridge factor nor a fractional persistence factor is part of Theorem 5.27.

6 Fractional Gaussian persistence

This section proves the probability estimates used in the divisor-box arguments. All conditional probabilities below integrate the marks and retain the gap or Gaussian environment. In particular, a fractional power is taken *after* any permitted mark window has been summed. The conditioning and projection arguments preserve this convention.

Write $\mathbb{P}$ and $\mathbb{E}$ for probability and expectation. Comparison constants are uniform on the compact parameter sets explicitly specified below. They may depend on fixed curve amplitudes, endpoint boxes, density margins, and regularity tolerances. None depends on the length of the process, its allowed reserve, or a count in the stated central range.

6.1 The Gaussian exponent and effective finite-time bounds

Let X, Y be independent stationary standard Ornstein–Uhlenbeck processes with covariance $\mathbb{E}X_s X_t = e^{-|s-t|/2}$. For $0 < \lambda, \rho < 1$ put

$$A_t = \{\sqrt{\rho}X_s + \sqrt{1-\rho}Y_s \geq 0 : 0 \leq s \leq t\},$$

$$q_t(X) = \mathbb{P}_Y(A_t \mid X), \quad F_{\lambda, \rho}(t) = \mathbb{E}_X q_t(X)^\lambda. \quad (6.1)$$

We suppress the parameters when they are fixed. Finite-grid assertions are passed to continuous paths by decreasing events on nested dense grids and bounded convergence.

Proposition 6.1 (Existence and effective bounds). *The limit*

$$\chi(\lambda, \rho) = \lim_{t \rightarrow \infty} -t^{-1} \log F_{\lambda, \rho}(t)$$

exists. If $b(t) = -\log F(t)$ and $p_g = (1 + e^{-g/2})/(1 - e^{-g/2})$, then, for $t, g > 0$,

$$\frac{b(t)}{p_g(t+g)} \leq \chi(\lambda, \rho) \leq \frac{b(t)}{t}, \quad \frac{\lambda}{2} \leq \chi(\lambda, \rho) \leq \frac{1}{2}. \quad (6.2)$$

Thus $g = 2 \log t$, $t > 1$, gives an effective interval of width $O(\log t/t)$. Finite Gaussian integrals give arbitrarily accurate certified values of the endpoints of this interval.

Proof. The private survival indicators on two adjacent time intervals are increasing functions of a Gaussian vector with nonnegative covariances. Gaussian association [22], in the finite form proved in Lemma B.1, first in Y and then in X , gives

$$F(s+t) \geq F(s)F(t). \quad (6.3)$$

The function b is nonnegative, locally bounded and subadditive. Writing $T = nt + r$, $0 \leq r < t$, proves that its limiting slope exists and equals $\inf_{t>0} b(t)/t$.

For completeness, the Gaussian Hölder inequality used here is the following finite-dimensional result: if a centered Gaussian block vector has covariance bounded above by $\text{diag}(p_i \Sigma_{ii})$, then $\mathbb{E} \prod_i f_i(Z_i) \leq \prod_i (\mathbb{E} f_i(Z_i)^{p_i})^{1/p_i}$ for nonnegative functions. This is [6, Theorem 1(i)]. Consider N intervals of length t , separated by gaps g . The OU Markov property and conditional expectation at their endpoints imply that the covariance of linear functionals on blocks i, j is bounded in absolute value by $r^{|i-j|}$ times their standard deviations, where $r = e^{-g/2}$. Consequently, by $2ab \leq a^2 + b^2$,

$$\text{Var} \left(\sum_i L_i \right) \leq \left(1 + 2 \sum_{k \geq 1} r^k \right) \sum_i \text{Var}(L_i) = p_g \sum_i \text{Var}(L_i).$$

This is covariance domination, uniformly in the grids and N . Drop survival in the intervening gaps. Private Gaussian Hölder gives $q_{\text{joint}} \leq \prod_i q_i^{1/p_g}$. Applying the same inequality to the environmental expectation gives

$$F(Nt + (N-1)g) \leq \mathbb{E} \prod_i q_i^{\lambda/p_g} \leq F(t)^{N/p_g}.$$

Only one factor p_g remains. Taking $N \rightarrow \infty$ proves the first inequality of (6.2).

The composite process in (6.1) is itself a standard OU process. Brownian time change and reflection give its ordinary survival probability

$$a(t) = \pi^{-1} \arcsin(e^{-t/2}). \quad (6.4)$$

The inequalities $q^\lambda \geq q$ and Jensen give $a(t) \leq F(t) \leq a(t)^\lambda$, proving the range of χ and also $b(t) \leq t/2 + \log \pi$. This last bound makes the width assertion explicit.

Here are the finite-grid error bounds, to specify the claimed effective meaning. At mesh size $\Delta = t/n$, let G_h denote the fractional grid survival moment with common threshold h . Conditional on two composite endpoints $x, y \geq h$, the intervening crossing probability is $\exp[-xy/\sinh(\Delta/2)]$. Therefore

$$\begin{aligned} F(t) &\leq G_0, \\ F(t) &\geq G_h - \{n \exp[-h^2/\sinh(\Delta/2)]\}^\lambda. \end{aligned} \quad (6.5)$$

Indeed the conditional error probability is first raised to λ ; Jensen bounds its expectation by its annealed probability to that power. Translation of the composite Gaussian grid gives

$$|G_h - G_0| \leq \left\{ \frac{|h|}{\sqrt{2\pi}} \sqrt{1 + n \tanh(\Delta/4)} \right\}^\lambda. \quad (6.6)$$

The squared translation norm is $h^2 \mathbf{1}^T \Sigma^{-1} \mathbf{1} = h^2 [1 + n \tanh(\Delta/4)]$, where $\Sigma_{ij} = e^{-|i-j|\Delta/2}$. Equal-covariance Gaussian total variation is at most that norm divided by $\sqrt{2\pi}$. Taking, for example, $h = \Delta^{1/4}$ makes both errors vanish. On a bounded cube the Gaussian density is uniformly continuous and the orthant integrand is bounded and monotone; displaced-threshold upper and lower cube sums, followed by Gaussian tail bounds, give certified finite quadratures. This completes the effective construction without any assumption about a divisor-volume asymptotic. $\square$

6.2 Tilted environments and the private endpoint filter

Write $\nu_{t,\lambda}$ for the environmental probability measure with density $q_t^\lambda/F(t)$ relative to the OU law. We require endpoint estimates under this measure, including the private endpoint after it has been conditioned to survive against the actual environment.

Lemma 6.2 (Order of the tilted laws). *For $0 \leq \lambda \leq 1$,*

$$\nu_{t,0} \preceq_{\text{st}} \nu_{t,\lambda} \preceq_{\text{st}} \nu_{t,1}. \quad (6.7)$$

For each $a < 1/2$,

$$\sup_{t \geq 0, 0 \leq u \leq t} \mathbb{E}_{\nu_{t,\lambda}} e^{aX_u^2} < \infty. \quad (6.8)$$

A nonnegative decreasing environmental path functional has tilted expectation at most its ordinary expectation.

Proof. On a finite grid the OU precision matrix has nonpositive off-diagonal entries, so its density is multivariate totally positive of order two (MTP₂). Set $Z = -Y$ and $a_\rho = \sqrt{\rho/(1-\rho)}$. The set $\{z_i \leq a_\rho x_i \forall i\}$ is closed under coordinatewise minimum and maximum. Its indicator is MTP₂. Multiplication by the MTP₂ Gaussian density and marginalization over Z show that $q(x)$ is MTP₂. One can insert an independent product Gaussian density in x to ensure integrability; its factors cancel in the defining inequality. Thus $\varphi_\Sigma(x)q(x)^\lambda$ is MTP₂ and is associated. The closure and association properties used here are given in [11, Proposition 3.4 and Section 3]. Association under this tilted measure, applied to an increasing function and $q^{1-\lambda}$, proves the right comparison in (6.7). Ordinary Gaussian association proves the left one. The decreasing-functional assertion is the same left comparison.

Under $\nu_{t,1}$ rotate to $Z = \sqrt{\rho}X + \sqrt{1-\rho}Y$ and an independent standard OU process V . Only Z is conditioned to be positive. Its density at time u is

$$\frac{\varphi(z)\mathbb{P}_z(\tau_0 > u)\mathbb{P}_z(\tau_0 > t-u)}{a(t)}\mathbf{1}_{\{z>0\}}, \quad \mathbb{P}_z(\tau_0 > v) = 2\Phi\left(\frac{z}{\sqrt{e^v-1}}\right) - 1.$$

For each side of length at least one the latter probability is at most $Cze^{-v/2}$; a shorter side contributes at most one. Separating the cases with zero, one or two long sides and using (6.4) bounds the density by $C(1+z)^2\varphi(z)$, uniformly in u, t . The rotation gives uniform Gaussian exponential moments for X_u under $\nu_{t,1}$. Apply (6.7) to the increasing function $e^{a(x_+)^2}$ and the decreasing function $e^{a(x_-)^2}$ separately. This proves (6.8); dense grids give the path assertions. $\square$

Lemma 6.3 (Uniform private filter estimate). *On compact subsets of $0 < \lambda, \rho < 1$ there exist $a > 0$ and $C < \infty$ such that*

$$\mathbb{E}_X \left[e^{aX_t^2} \left(\mathbb{E}_Y[\mathbf{1}_{A_t} e^{aY_t^2} \mid X] \right)^\lambda \right] \leq CF(t). \quad (6.9)$$

The same assertion holds with simultaneous Gaussian weights at both initial and terminal endpoints, after decreasing a .

Proof. Fix the environmental path and let the private wall be $Y_s \geq -a_\rho X_s$. The filtered endpoint density has the form $\mu_v(y) \propto \varphi(y)g_v(y)$, where g_v is increasing and log-concave. Indeed the conditional OU bridge mean is increasing in its endpoint, and its constraints are convex; marginal log-concavity is Prékopa's marginalization theorem [23, Theorem 6, p. 342], first on a finite grid and then by a limit. Thus μ_v is 1-strongly log-concave and its mode m_v is nonnegative. For any such density with mode m ,

$$|Z - m| \preceq_{\text{st}} |N(0, 1)|, \quad |\mathbb{E}Z - m| \leq \sqrt{2/\pi}. \quad (6.10)$$

To verify this directly, both functions $\mu(m+r)e^{r^2/2}$ and $\mu(m-r)e^{r^2/2}$ decrease for $r \geq 0$. Their sum therefore gives likelihood-ratio domination of the radial half-normal density after normalization. The density is also dominated in monotone likelihood-ratio order by $m + |N|$: on $y \geq m$ its ratio to $e^{-(y-m)^2/2}$ decreases, while on $y < m$ the comparison density is zero. OU transition kernels are TP₂; composition with killing indicators and integration preserve this property. The likelihood-ratio convention allowing zero densities agrees with [8, Theorem 1 and Remark 16]; kernel composition here follows from the MTP₂ marginalization just used. Consequently this order propagates through a killed OU segment. Raising a wall also increases its normalized endpoint law, by association under the original truncated MTP₂ path measure.

Consider a segment of length $v \in [1, 2]$, and put $r = e^{-v/2}$, $B = \max(0, \sup_{\text{segment}}(-a_\rho X))$. Raise the incoming law to $m + |N|$ and the wall to B . If $rm \geq B + 2$, an OU-noise tube of radius one ensures survival with a fixed positive probability. The unconditional endpoint above rm has a uniform Gaussian tail, so its conditional mean is at most $rm + C$. If $rm < B + 2$, restrict the initial value to $[e(B+3), e(B+3)+1]$ and use the same noise tube. The restriction has probability at least $ce^{-C(B+1)^2}$. Dividing the unconditional Gaussian endpoint tail by this probability and integrating yields a conditional mean at most $rm + C(B+1)$. Equation (6.10) therefore gives

$$m_{\text{new}} \leq e^{-v/2}m + C(B+1). \quad (6.11)$$

An initial segment of length at most one is handled only once: raise its wall to its supremum and retain a fixed initial interval above $e(B+3)$. The same tube and tail argument bounds its mode by $C(B+1)$. There is no iteration with a contraction tending to one.

Partition backwards from the terminal time into intervals I_j of length at most two. Iterating (6.11), including the initial truncation, gives

$$m_t(X) \leq C \left(1 + \sum_{j \geq 0} e^{-j/2} \sup_{s \in I_j} (-a_\rho X_s)_+ \right) =: \mathcal{M}_t(X). \quad (6.12)$$

OU suprema on bounded intervals have uniform Gaussian tails. Weighted Cauchy–Schwarz and Jensen imply $\sup_t \mathbb{E} e^{c\mathcal{M}_t^2} < \infty$ for some $c > 0$; independence of the intervals is not needed. The majorant is decreasing in X , so Lemma 6.2 transfers this bound to $\nu_{t,\lambda}$. Finally, for a small, (6.10) gives

$$\mathbb{E}[e^{aY_t^2} \mid A_t, X] \leq (1 - 4a)^{-1/2} e^{2am_t(X)^2}.$$

Multiply by q_t^λ , use (6.8) and the preceding mode bound, and apply Cauchy–Schwarz under $\nu_{t,\lambda}$. This proves (6.9). Reversibility treats the other endpoint; Cauchy–Schwarz first inside the private law and then in the environmental law treats both endpoints simultaneously. The private law was never replaced by a fresh Gaussian after survival was imposed. $\square$

Theorem 6.4 (Constant-factor Gaussian survival). *Uniformly on compact subsets of $0 < \lambda, \rho < 1$,*

$$F(s)F(t) \leq F(s+t) \leq CF(s)F(t), \quad C^{-1}e^{-\lambda t} \leq F(t) \leq e^{-\lambda t}. \quad (6.13)$$

The second comparison is valid for $t \geq 1$. There exist compatible, sufficiently wide fixed positive endpoint boxes for both Gaussian channels that can be imposed at a constant cost. These are the boxes constructed below, or fixed boxes containing them; this assertion does not prescribe an arbitrary narrow endpoint box.

Proof. The Mehler density relative to two independent standard Gaussians is

$$k_r(x, y) = \frac{1}{\sqrt{1-r^2}} \exp \left(\frac{2rxy - r^2(x^2 + y^2)}{2(1-r^2)} \right) \leq \frac{1}{\sqrt{1-r^2}} \exp \left(\frac{r(x^2 + y^2)}{2(1+r)} \right).$$

Choose a fixed gap g so large that $r = e^{-g/2}$ makes the exponent smaller than the coefficient allowed in Lemma 6.3. Drop survival across the gap. Apply this density bound inside the private conditional integral and then to the environmental transition. The two reference blocks are independent; (6.9) at their adjacent endpoints gives $F(s+g+t) \leq CF(s)F(t)$. On the other hand, (6.3) gives $F(s+g+t) \geq F(s+t)F(g)$. This proves the first comparison. Iteration at nt and Proposition 6.1 yield the second one.

By (6.9), excluding any of the four Gaussian endpoints from $[-R, R]$ costs at most $Ce^{-cR^2}F(t)$ in fractional mass. Choose fixed R so that their union costs at most $F(t)/2$. Add to each channel $A(e^{-s/2} + e^{-(t-s)/2})$, with any fixed $A \geq 2R$. This preserves survival and moves all retained endpoints into the fixed positive box $[A - R, A(1 + e^{-1/2}) + R]$, uniformly for $t \geq 1$. The Cameron–Martin translation formula [4], in the form including an independent Gaussian initial value given in [1, Theorem 2.1], applies after the OU-to-Brownian time change. The density of this shift depends only on those endpoints and is bounded above and below on the box, uniformly for $t \geq 1$. Change variables first in the private integral and then in the environmental integral. The positive-box event therefore retains a fixed fraction of $F(t)$. A required fixed positive lower endpoint level can be accommodated by choosing A first and then retaining this sufficiently wide box (or a fixed containing box). No restriction to an arbitrary prescribed narrower box is inferred. $\square$

6.3 Parameter stability, integrable walls, and Brownian reserves

Proposition 6.5 (Local Lipschitz continuity). *On each compact subset of $(0, 1)^2$,*

$$|\chi(\lambda, \rho) - \chi(\lambda', \rho')| \leq C(|\lambda - \lambda'| + |\rho - \rho'|).$$

Proof. Whiten a finite Gaussian grid. Its survival indicator is a fixed function f , and $q_\rho = T_s f$, where T_s is the Gaussian noise semigroup and $e^{-s} = \sqrt{\rho}$. In any unit direction Gaussian integration by parts and the one-dimensional rearrangement inequality give

$$|\nabla q_\rho| \leq \sqrt{\frac{\rho}{1-\rho}} I(q_\rho), \quad I(v) = \varphi(\Phi^{-1}v).$$

For clarity, the rearrangement step fixes the value $v = \mathbb{E}f$ and maximizes $|\mathbb{E}[Z_1 f(Z)]|$ over $0 \leq f \leq 1$; the maximizer is a half-space in the first coordinate, with value $I(v)$. Gaussian tail bounds give $I(v)^2 \leq Cv^2(1 - \log v)$, including both endpoints. Differentiating under the Gaussian integral and integrating by parts,

$$\partial_s \log \mathbb{E}q_s^\lambda = \lambda(1 - \lambda) \frac{\mathbb{E}q_s^{\lambda-2} |\nabla q_s|^2}{\mathbb{E}q_s^\lambda}.$$

Convexity of $K(\lambda) = \log \mathbb{E}q^\lambda$, with $K(0) = 0$, gives $\mathbb{E}_{q^\lambda}[-\log q] \leq -K(\lambda)/\lambda$. Since $\mathbb{E}q^\lambda \geq a(t) \geq \pi^{-1}e^{-t/2}$, both the λ derivative and, on a compact ρ interval, the ρ derivative are bounded in absolute value by $C(t+1)$, independently of the grid dimension. Smooth approximation justifies the derivatives when f is an indicator. Integrating these bounds, passing to dense grids, dividing by t , and taking $t \rightarrow \infty$ proves the assertion. $\square$

For a piecewise smooth shift h on $[0, t]$, define

$$\begin{aligned} \|h\|_{\mathcal{H}}^2 &= h(0)^2 + \int_0^t (h' + h/2)^2 ds, \\ J_t(h) &= |h(0)/2 - h'(0)| + |h'(t) + h(t)/2| + \int_0^t |h/4 - h''| ds. \end{aligned}$$

Derivative jumps are included in the last total variation.

Lemma 6.6 (Bounded-action walls). *Replacing the zero composite OU wall by a deterministic wall b changes $F(t)$ by a factor bounded above and below whenever $J_t(b/\sqrt{\rho}) + \|b/\sqrt{\rho}\|_{\mathcal{H}}^2$ is bounded. An endpoint-restricted version holds with compatible sufficiently wide boxes chosen after the wall bound, or with the endpoint boxes supplied by Theorem 6.4 translated together with the path. It is not a statement for arbitrary prescribed unchanged endpoint boxes.*

Proof. The Cameron–Martin linear functional is

$$L_h(X) = (h(0)/2 - h'(0))X_0 + (h'(t) + h(t)/2)X_t + \int_0^t (h/4 - h'')X_s ds.$$

Environmental translation alone gives exactly $F_b(t)/F(t) = \mathbb{E}_{\nu_{t,\lambda}} \exp[-L_h(X) - \|h\|_{\mathcal{H}}^2/2]$, with $h = b/\sqrt{\rho}$ and the consistent translation sign. The absolute variation of the coefficients of L_h is bounded. Jensen with its normalized absolute coefficient measure and (6.8) bounds both exponential signs uniformly. Jensen also supplies a positive lower bound, or equivalently use the bound for $\mathbb{E}_\nu |L_h|$. Mixed endpoint moments, followed by Cauchy–Schwarz and truncation, give the same conclusion for compatible sufficiently wide boxes chosen after the fixed wall bound. Equivalently, translate the endpoint boxes in the same environmental change of variables. A fixed feasible interior margin and a finite initial or terminal tube permit the sufficiently wide compatible central boxes used below; their constants are chosen after the wall and drift bounds. This argument does not change the power λ . $\square$

Corollary 6.7 (Brownian one-end estimate). *Let independent Brownian channels have environmental variance σ_E^2 and private variance σ_C^2 , bounded above and below by positive constants. Put $\rho = \sigma_E^2/(\sigma_E^2 + \sigma_C^2)$. From reserve $h \geq 1$, over duration n , with $|\delta|\sqrt{n} \leq D$, the conditional private survival probability Q satisfies*

$$\mathbb{E}Q^\lambda \asymp B_n(h) := \min\{1, ((1+h)/\sqrt{n})^{2\chi(\lambda,\rho)}\}. \quad (6.14)$$

The upper bound allows a fixed soft wall $-h - gs^\gamma$, $0 < \gamma < 1/2$. For a sufficiently large fixed minimum reserve, the lower bound permits a strong wall $-h + gs^\gamma$, positive central terminal risk, and compatible sufficiently wide central terminal boxes of the separate channels. Choose these boxes after the compact-law, wall, and drift bounds; their locations and widths are then fixed uniformly in n, h .

Proof. For $h^2 < n$, scale from time h^2 and use Lamperti time $u = \log(s/h^2)$. Its length is $\log(n/h^2)$. Reserve, power wall and bounded scaled drift become bounded multiples of

$$e^{-u/2}, \quad e^{-(1/2-\gamma)u}, \quad e^{-(\log(n/h^2)-u)/2}.$$

Their J and Hilbert norms are bounded. Drop the initial time interval for the upper bound and apply Lemma 6.6 and Theorem 6.4. For the lower bound prepare, on a fixed multiple of h^2 , both channels in positive central boxes using a fixed Brownian tube. Brownian scaling makes its probability and its endpoint transition density bounds uniform. Thereafter use the endpoint version of those lemmas. The preparation has fixed cost before the fractional power as well as after it. A large fixed minimum reserve makes the normalized strong-wall amplitude as small as required. If h is comparable to or larger than $\sqrt{n}$, a fixed-scale positive tube and monotonicity give a constant lower bound; the upper bound is one. Reflection of both channels proves the same assertions for the opposite risk sign. $\square$

6.4 Independent exponential gaps: relative coupling

Let C_i be iid marks in a fixed finite alphabet, with probabilities in a compact subset of its open simplex. Their weights have two values,

$$w(C_i) = l - bB_i \in [w_*, l], \quad p = \mathbb{P}(B_i = 1), \quad \omega = l - bp,$$

where all parameters range over compact sets, $b, w_* > 0$, and p stays away from zero and one. Independent gaps E_i have exponential mean τ in a positive compact interval. Write

$$S_j = \sum_{i \leq j} (w(C_i) - E_i), \quad \delta = \omega - \tau, \quad \rho_\tau = \frac{\tau^2}{\tau^2 + b^2 p(1-p)}, \quad F_n(h) = \mathbb{E} \mathbb{P}_C \{ \min_{j \leq n} S_j \geq -h \mid E \}^\lambda. \quad (6.15)$$

The following arguments also apply to $-S$, with all monotonicities reversed together. Assume initially $|\delta|\sqrt{n} \leq D$.

Lemma 6.8 (Crude comparisons before the exponent is known). *Uniformly for $h \geq 1$, $u \geq 0$,*

$$F_n(h) \geq c_D n^{-1/2}, \quad F_n(h+u) \leq C_D(1+u)F_n(h), \quad F_{2n}(h) \geq c_D F_n(h). \quad (6.16)$$

The drift bound for a doubled interval is enlarged by a fixed factor. The assertions remain valid for a fixed soft sub-square-root power wall, with a fixed polynomial in $1+u$ allowed in the middle inequality.

Proof. We detail the elementary annealed preparation used here. For the centered walk stopped on leaving $[-h_0, B]$, upper overshoots are bounded and lower overshoots have exponential tails; for the reversed walk these roles are interchanged. Optional stopping of the walk gives an upper-exit probability comparable to $1/B$, from a fixed reserve $h_0 > 0$. Stopping its square minus the variance times time bounds the mean exit time by CB : on an upper exit its squared displacement is $O(B^2)$ and that exit has probability $O(1/B)$; on a lower exit the second moment is bounded. Truncation first justifies both stopping identities. By Markov's inequality, a sufficiently large fixed multiple of B^2 retains a fixed fraction of the upper exits. From height B , a positive scaled tube over a further comparable time has probability bounded below by the functional central limit theorem, uniformly on our compact laws. It can end above any prescribed fixed multiple of B . The lower-overshoot memoryless law and bounded upper overshoot provide uniformity at entrance. The stopped centered gap and mark martingales have second moments at most $C\mathbb{E}\tau_{\text{exit}} \leq CB$. Their probability of exceeding RB before the retained entrance is therefore at most $C/(R^2B)$. Choosing fixed R large retains a fixed fraction of the B^{-1} entrance probability with both channel sums central; the subsequent tube preserves centrality at the final time. Small drift is handled by changing the exponential mean on the retained central gap-sum tube: if $m = O(B^2)$ and $|\delta|\sqrt{m} = O_D(1)$, its log density ratio is $O_D(1)$ there. The same proof, or its sign reversal, therefore gives a preparation of length $O((1+u)^2)$ from fixed reserve to height at least u , of annealed probability at least $c_D/(1+u)$.

For $u \leq \sqrt{n}$, let p_u be its conditional private probability. Concatenating with an independent n -step survival from $h+u$, and using $\mathbb{E}p_u^\lambda \geq \mathbb{E}p_u$, gives $F_{n+m}(h) \geq c_D(1+u)^{-1}F_n(h+u)$. Time monotonicity gives the middle inequality. The same preparation and tube give ordinary survival at least $c_D/\sqrt{n}$, hence the first inequality. For $u > \sqrt{n}$ these two inequalities and $F_n(h+u) \leq 1$ suffice.

For doubling, remove environments whose centered prefix gap sum is unfavorable by more than $R\sqrt{n}$. Survival is monotone in the opposite direction to this unfavorable event, so product association bounds its F_n -weighted mass by its unweighted probability times F_n . Choose R large. On the retained environments an appropriate favorable private centered endpoint of size $(R+D+c)\sqrt{n}$ has a fixed positive probability. Its mark event is monotone in the same direction as survival, so their intersection retains that fraction of the conditional survival probability. Its ending reserve is at least $c\sqrt{n}$. An independent further n steps survive from this height with annealed probability at least a constant. If f is the prefix mark probability and Z its completed

probability as a function of the future environment, then $0 \leq Z \leq f$ and $\mathbb{E}Z \geq cf$, whence $\mathbb{E}Z^\lambda \geq cf^\lambda$. Integration proves doubling. All finite small lengths are absorbed using compactness and a fixed positive gap/mark event. For a soft wall use $(s+t)^\gamma \leq s^\gamma + t^\gamma$ after a prefix; its additional reserve is polynomial in the prefix length. The preparation and favorable continuation remain sufficient. No value of χ is used in this proof. $\square$

Lemma 6.9 (Relative cost of a bad block). *Let $\mathcal{D}$ depend on the first $b \leq n/2$ slots, including any coupling auxiliaries. Then*

$$\mathbb{E}\mathbb{P}_C(\text{survival}, \mathcal{D} \mid \text{extended environment})^\lambda \leq C_D(1+b)^2 \mathbb{P}(\mathcal{D})^{\lambda/4} F_n(h). \quad (6.17)$$

If instead $\mathcal{D}$ begins after $n/4$, its relative cost is at most $C_D \mathbb{P}(\mathcal{D})^\lambda F_n(h)$. Fixed fractions in these statements may be changed, with corresponding constants. The same bounds hold for the Gaussian comparison walk and for both risk signs.

Proof. If positive jumps are bounded, discard prefix survival and use the endpoint bound $S_b \leq lb$. Independence of the remainder, Jensen over the prefix, and Lemma 6.8 give the stronger upper bound $C(1+b)\mathbb{P}(\mathcal{D})^\lambda F_n(h)$. For unbounded positive jumps divide positive prefix endpoints into unit bins $[k, k+1)$. Their probabilities are bounded both by $\mathbb{P}(\mathcal{D})$ and by $C \exp[-c \min(k^2/b, k)]$, with the fixed $O_D(\sqrt{b})$ drift displacement absorbed in the constants and a first group of bins. Apply the reserve comparison bin by bin. Using the geometric mean of the two probability bounds and summing the Gaussian/exponential tail proves (6.17); the displayed exponent $\lambda/4$ and quadratic polynomial are deliberately weaker than necessary. Negative endpoints can only decrease the available reserve. The Gaussian case has the same tail estimate. For a late bad block retain necessary survival on the first $\lfloor n/8 \rfloor$ slots, independent of that block. Jensen and finitely many doublings prove the stated relative bound. $\square$

Lemma 6.10 (Relative control of the initial coupling block). *Fix the compact laws and the bounded scaled-drift range in Lemma 6.9, and let the fractional exponent satisfy $\lambda \in [\lambda_*, 1]$, where $\lambda_* > 0$ is fixed. Fix an integer $b \geq 1$. Couple the centered exponential and binary channels separately to their scalar Gaussian channels on the first b slots, independently of all later blocks. The environmental coupling is outside the private conditional probability, and the private coupling is inside it. For every prescribed $\varepsilon > 0$ there is a finite H_b , depending only on these fixed data and ε , such that, whenever $n \geq 2b$ and $h \geq H_b$, the following assertion holds for each marginal survival functional covered by Lemma 6.9. There is an initial good event $\mathcal{G}_b$ on which the discrepancy of the two initial risk paths is at most $H_b/8$, and*

$$\mathbb{E}[\mathbb{P}_{\text{private}}(\text{survival}, \mathcal{G}_b^c \mid \text{extended environment})^\lambda] \leq \varepsilon F_n(h).$$

The same H_b works for both risk signs. A prescribed fixed strong or soft curve on this initial finite interval may also be absorbed in $H_b/8$ by increasing H_b before lengths vary.

Proof. Let M_b be the maximum of the absolute centered partial sums of both original channels on the first b slots and of the absolute Gaussian suprema on the corresponding time interval. Enlarge its definition by the fixed compact coefficients needed to reconstruct the risk. A union bound using exponential moments of the gaps, boundedness of the binary increments, and the Gaussian maximal tail gives constants $C_b, c_b > 0$ such that

$$\mathbb{P}\{M_b > v\} \leq C_b e^{-c_b v} \quad (v \geq C_b).$$

The constants are independent of n and h . The deterministic drift on the first b slots is bounded in terms of the fixed data, because $n \geq 2b$ and the full-horizon scaled drift is bounded. Choose a fixed compact-data constant C_0 so that, on $\mathcal{G}_b = \{M_b \leq H_b/C_0\}$, the two initial risk paths differ by at most $H_b/8$; make the same choice large enough to pay the initial drift. A fixed curve on this finite interval has a finite maximum, which is paid by the final increase of H_b .

The bad event reads only the first b slots and their coupling auxiliaries. Lemma 6.9 therefore gives

$$\mathbb{E}[\mathbb{P}_{\text{private}}(\text{survival}, \mathcal{G}_b^c \mid \text{extended environment})^\lambda] \leq C(1+b)^2(C_b e^{-c_b H_b/C_0})^{\lambda/4} F_n(h).$$

After b has been fixed, choose H_b so that its coefficient is at most ε , uniformly for $\lambda \geq \lambda_*$. This choice does not depend on a persistence exponent or on the eventual reserve h . Since $h \geq H_b$, a change by $H_b/8$ lies between the reserves $7h/8$ and $9h/8$. The argument bounds the initial error only on $\mathcal{G}_b$; it has assigned no deterministic bound to any exponential gap. The finitely many experiments $n < 2b$ are outside this lemma and are handled by their explicitly supported finite cases. $\square$

Theorem 6.11 (Reference one-end theorem). *For $|\delta|\sqrt{n} \leq D$, $h \geq 1$, and either sign of S ,*

$$F_n(h) \asymp_D \min\{1, ((1+h)/\sqrt{n})^{2\chi(\lambda, \rho_\tau)}\}. \quad (6.18)$$

The upper estimate allows a fixed soft power wall of exponent below $1/2$. For $h \geq H_0$ the lower estimate may retain a fixed strong curve, a fixed positive central unreserved endpoint, and central gap, binary, and full-color endpoints. The central boxes are chosen compatible and sufficiently wide after the compact laws, wall, D , and D' are fixed, before the coupling accuracy and minimum reserve are chosen. For this endpoint version take $h \leq D'\sqrt{n}$; larger reserves are not needed for the endpoint assertion.

Proof. We use a finite strong approximation with explicit failure tails, not an unconditional limiting invariance principle. The finite scalar coupling of Sakhanenko [26, Theorem 1, p. 5], in the finite iid form of Waudby-Smith, Larsson, and Ramdas [30, Lemma 4.2, p. 12], followed by Markov's inequality (as in their equation (17)), gives

$$\mathbb{P}\left\{\max_{j \leq L} \left| \sum_{i \leq j} (Z_i - G_i) \right| \geq z\right\} \leq (1 + s_* \sqrt{Lv}) e^{-cs_* z},$$

where $v = \text{Var}(Z)$, $c > 0$ is universal, and s_* is strictly below the Sakhanenko parameter. We check this uniformly for the two centered channels $Z = E - \tau$ and $Z = w(C) - \omega$ before conditioning. Write $\tau \in [\tau_-, \tau_+]$, $b \in [b_-, b_+]$, and $p \in [p_-, 1 - p_-]$ for their fixed compact bounds. Set

$$v_- = \min\{\tau_-^2, b_-^2 p_-(1 - p_-)\} > 0, \quad v_+ = \max\{\tau_+^2, b_+^2/4\}, \quad t_0 = (2\tau_+)^{-1}.$$

For $E = \tau U$ with U exponential of mean one,

$$\mathbb{E}|E - \tau|^3 e^{t_0|E - \tau|} \leq e^{1/2} \tau_+^3 \int_0^\infty (u+1)^3 e^{-u/2} du = 158 e^{1/2} \tau_+^3.$$

For the binary channel $|Z| \leq b_+$, so the corresponding moment is at most $b_+^3 e^{t_0 b_+}$. Let M be the larger of these two bounds, $t_1 = \min\{t_0, v_-/(2M)\}$, and $s_* = t_1/2$. Then $t_1 \mathbb{E}|Z|^3 e^{t_1|Z|} \leq v_-/2 \leq v$; Definition 2.1 and (6) of [30] give a Sakhanenko parameter at least $t_1 > s_*$. These estimates also

hold for $-Z$. For any fixed $A > 0$ and $L \geq 2$, choose C_A with $cs_*C_A > A + 1/2$ and take $z = C_A \log L$. The displayed tail is at most $(1 + s_*\sqrt{v_+})L^{1/2 - cs_*C_A} \leq CL^{-A}$. Increasing C_A if needed gives error $C_A \log L$ with failure at most C_AL^{-A} , uniformly in the compact laws. The bounded initial block is treated separately below; no conditioned coupling is asserted.

Couple gaps and private binary marks separately, using independent couplings on successive dyadic blocks, starting at a fixed scale L_0 . Include Brownian interpolation oscillations in the bad events; their Gaussian bounds have the same arbitrarily high polynomial decay after increasing C_A . A block intersecting the horizon either ends by $n/2$ or begins after $n/4$, apart from a bounded initial case. Apply Lemma 6.9 on both marginal processes. The total fractional failure mass is at most

$$CF_n(h) \sum_{L \geq L_0, L \text{ dyadic}} L^{2 - A\lambda/4}, \quad (6.19)$$

and the analogous bound holds with Brownian survival in place of F_n . Choose A first, and then L_0 , so this is any prescribed small fraction of the corresponding marginal survival moment.

First fix a bounded-action envelope and the relative reserve interval $[7h/8, 9h/8]$. Fix its Gaussian and compatible endpoint lower constants. Choose the coupling tail exponent and initial dyadic scale so the later relative error sum is below the required fraction of these constants. Apply Lemma 6.10 at that fixed initial scale; choose its minimum reserve last. On the resulting good event the initial discrepancy is at most $h/8$ and the later discrepancy is at most $C \log(1 + s/L_0) + C \log^2(1 + s/L_0)$. After Lamperti transformation the latter is a bounded-action curve of Lemma 6.6. Increasing the minimum reserve reduces its normalized amplitude; it does not enlarge the chosen relative reserve interval. The initial bad event is paid at the same marginal survival scale, not by a deterministic bound on gaps. Finally choose the bounded-horizon threshold. None of these choices uses the discrete normalization being proved.

The coupling is used on an extended probability space: the coupled gap and Brownian environment is outside the private conditional probability; the independent mark/Brownian coupling is inside it. The two marginal conditional probabilities consequently remain the genuine $q_D(E)$ and $q_B(W)$. On good paths the deterministic inclusions give

$$q_D \leq q_{B,\text{soft}} + \Delta_D, \quad q_{B,\text{hard}} \leq q_D + \Delta_B.$$

Subadditivity and (6.19) absorb the first error against the discrete moment and the second against the Brownian moment. Corollary 6.7 proves (6.18), without presupposing its normalization on either side. Reserves below the fixed threshold are handled by the preparation and reserve comparisons of Lemma 6.8. The same argument retains Brownian strong curves and endpoint boxes; its errors are still bounded by the flat relative majorant.

Only the binary mark was coupled. Conditional on the entire binary path, the full colors in each category remain independent with their conditional probabilities. If the binary endpoint is central, their means differ from the unconditional color means by $O(\sqrt{n})$. Enlarging a fixed central color window once, the multinomial variance bound makes all colors central with conditional probability at least $3/4$, uniformly for every retained binary path. This is a constant loss before the fractional power. It requires no unstated multivariate KMT theorem. $\square$

6.5 Exact Gamma conditioning and relative local regularity

Let $T_n = \sum_{i=1}^n E_i$ and condition on $T_n = A > 0$, where $|A - n\tau| \leq D\sqrt{n}$ and $n \geq 1$. Write $0 < \tau_- \leq \tau \leq \tau_+ < \infty$ for the fixed gap-mean range. All such conditioning is by a regular conditional density. Write $\mathbb{E}_A$ for the resulting environmental expectation. The private marks remain independent of the gaps and of one another.

Lemma 6.12 (Selected-gap density). *For a deterministic proper set I of $m < n$ coordinates, the conditional density relative to independent mean- τ exponentials is exactly*

$$\frac{f_{n-m}(A - \sum_{i \in I} E_i)}{f_n(A)}, \quad f_k(x) = \frac{x^{k-1} e^{-x/\tau}}{\tau^k \Gamma(k)} \mathbf{1}_{\{x > 0\}}. \quad (6.20)$$

There is a fixed threshold $N_\Gamma = N_\Gamma(D, \tau_-, \tau_+) \geq \max\{1, (2D/\tau_-)^2\}$ such that the following two upper bounds hold for $n \geq N_\Gamma$: if $m \leq cn$, $c < 1$, the ratio is bounded above uniformly on its whole support; for any proper selected set it has the weaker upper bound $C_D \sqrt{n}$. If $m \leq cn$, it is bounded below by a positive constant on $|\sum_I E_i - m\tau| \leq R\sqrt{n}$ for $n \geq n_0(c, D, R) \geq N_\Gamma$, with constants depending on c, D, R and the compact gap-mean range. This threshold ensures that the displayed central set lies inside the support of the remaining Gamma density. In the exact-mean case $A = n\tau$, both upper bounds hold for every $n \geq 1$. The central lower bound still requires its stated threshold. No uniform upper bound is asserted for a near-central bounded- n window that permits $A \downarrow 0$.

Proof. Independence and convolution of Gamma densities give (6.20). Stirling's formula, uniformly for τ in its compact interval, gives

$$\sup_x f_k(x) \leq Ck^{-1/2}, \quad f_k(k\tau + z\sqrt{k}) \asymp_R k^{-1/2} \quad (|z| \leq R, k \geq N(R, \tau_-)).$$

For $k = 1$ the supremum is $1/\tau$ and the same upper bound holds. For $n \geq N_\Gamma$, enlarge the threshold for the preceding central estimate with $R = D$; the denominator in (6.20) is then comparable to $n^{-1/2}$, and $A \geq n\tau/2$. If $A = n\tau$, the explicit density and Stirling's bounds instead give $f_n(n\tau) \asymp n^{-1/2}$ for every $n \geq 1$, uniformly in τ . The asserted bounds follow; on the central selected set the remaining total has displacement $O(\sqrt{n})$ from a mean with shape comparable to n . This is one joint density for all selected coordinates, not independence of selected blocks after conditioning. $\square$

Proposition 6.13 (One-end exact-total bridge). *For fixed D, D' , the compact laws, and a fixed curve as below, there are N_{one}, H_0 such that the following holds for $n \geq N_{\text{one}}$. Suppose $|\omega - \tau|\sqrt{n} \leq D$, $|A - n\tau| \leq D\sqrt{n}$, and $H_0 \leq h \leq D'\sqrt{n}$. The flat conditional survival probability has λ moment at most $CB_n(h)$, with the exponent of (6.18). A stronger event has moment at least $cB_n(h)$ and can retain a fixed curve $g \min(j, n-j)^\gamma$, $0 < \gamma < 1/6$, positive unreserved terminal risk in $[a_0\sqrt{n}, a_1\sqrt{n}]$, and central full-color windows. The constants and windows can be fixed uniformly on the displayed parameter range. One may enlarge H_0 after fixing N_{one} so that $H_0 > D'\sqrt{N_{\text{one}}}$; the reserve range then itself enforces the length threshold. The flat upper bound alone also holds for bounded n , by increasing its constant.*

Proof. All density comparisons below are made after taking $N_{\text{one}} \geq N_\Gamma$ and the required central-support thresholds. Keeping just the first $\lfloor n/4 \rfloor$ survival constraints and using Lemma 6.12 proves the upper bound. For the lower bound first assume $h \leq \varepsilon\sqrt{n}$. On $m = \lfloor n/4 \rfloor$ slots retain the strong reference family in Theorem 6.11, with

$$a_0\sqrt{n} \leq S_m \leq a_1\sqrt{n}, \quad |T_m - m\tau| \leq R\sqrt{n}, \quad |N_c(m) - \pi_c m| \leq R\sqrt{n} \quad \text{for every color } c. \quad (6.21)$$

Its conditional mark probability f has $\mathbb{E}f^\lambda \geq cB_n(h)$. The selected-gap density has a positive lower bound on these boxes. Fix one retained prefix environment and one retained prefix label word. For $k = n - m$, the remaining exact gap total $A - T_m$ is central. Represent these gaps as

$$E'_i = (A - T_m) \frac{W_i}{W_1 + \dots + W_k}, \quad W_i \text{ iid exponential of mean one.}$$

The centered polygonal partial gap sums, divided by $\sqrt{k}$, converge to τ times a Brownian bridge plus their prescribed central linear drift. Independently, centered color counts converge to multinomial Brownian motion on the zero-sum hyperplane. Its weight projection has positive variance. Both convergences are uniform over the central boxes and compact laws: every parameter sequence has a convergent subsequence; characteristic-function expansion identifies its Gaussian finite-dimensional limits, and the uniform fourth-moment bound $\mathbb{E}|\sum_{i=1}^k Z_i|^4 \leq C(k^2 + k)$ gives tightness, including polygonal intervals shorter than $1/k$. Normalizing by $k^{-1} \sum W_i$ preserves convergence by the law of large numbers.

The starting risk is in a fixed positive compact interval after scaling. Choose a continuous color path in the zero-sum hyperplane whose weight projection carries it to a fixed positive final window while remaining strictly positive; choose the zero centered gap bridge. Each sufficiently small tube about these paths has positive limiting probability. The multinomial covariance is nondegenerate on its hyperplane, so all the required color paths lie in its support. A finite set of tubes covers the compact possible prefix endpoints and remaining central drifts. The minimum probability is positive, and uniform convergence transfers a fixed fraction to the finite bridges. Enlarging the final color windows once absorbs the prefix deviations. On the tube the risk is at least $c\sqrt{n}$ on the continuation, which dominates gn^γ for large n . On the first quarter the imposed one-sided curve is already the required global curve. Thus every retained prefix word has joint remaining-environment-and-mark continuation probability at least $c > 0$.

For each prefix environment let Z be the full successful mark probability as a function of the remaining environment. Exactly

$$0 \leq Z \leq f, \quad \mathbb{E}_{\text{remainder}} Z \geq cf, \quad \mathbb{E}_{\text{remainder}} Z^\lambda \geq cf^\lambda. \quad (6.22)$$

The last inequality follows from $z^\lambda \geq f^{\lambda-1}z$ for $0 \leq z \leq f$. Integrate against the central lower density in Lemma 6.12. This proves the lower bound. When h is comparable to $\sqrt{n}$, a fixed positive tube directly gives a constant lower bound; alternatively use a smaller fixed multiple of $\sqrt{n}$ and monotonicity. This is an annealed continuation for each prefix word, not a quenched lower bound for every remaining environment. Central full-type windows are retained; no individual exact type is asserted to have constant probability. The length threshold also includes the uniform tube-convergence and endpoint-support thresholds after the windows are fixed. Indeed $l - \omega = bp$ is bounded below by some $\beta > 0$, whereas $A \leq n\omega + 2D\sqrt{n}$. Thus $nl - A \geq \beta n - 2D\sqrt{n}$, which exceeds $a_1\sqrt{n}$ for sufficiently large n . For the opposite sign the corresponding available endpoint is $A - n(l - b) \geq b(1 - p)n - 2D\sqrt{n}$, and $b(1 - p)$ has the same kind of positive compact lower bound. Thus a positive terminal window is not demanded beyond the possible mark total in either orientation. The retained central color tubes above give its actual support. No lower bound is obtained by compactness at excluded small n . For the flat upper bound in the bounded range, use $q \leq 1$ and $B_n(h) \geq n^{-1/2}$ for $h \geq 1$. $\square$

Fix a small $\epsilon > 0$, $0 < \gamma < 1/6$, and put $d_n(i) = \min(i, n - i)$. Let REG_K consist of the following inequalities for every $0 \leq i < j \leq n$ and every color c :

$$\begin{aligned} |T_j - T_i - \tau(j - i)| &\leq \epsilon(j - i) + K[1 + d_n(i)^\gamma + d_n(j)^\gamma], \\ |N_c(i, j) - \pi_c(j - i)| &\leq \epsilon(j - i) + K[1 + d_n(i)^\gamma + d_n(j)^\gamma]. \end{aligned} \quad (6.23)$$

Lemma 6.14 (Relative regularity under the exact total). *Assume $n \geq 1$, $A > 0$, $|A - n\tau| \leq D\sqrt{n}$, $|\omega - \tau|\sqrt{n} \leq D$, and $K \geq 0$. If $Q_{\text{bad}}(E)$ is the conditional mark probability of flat survival and REG_K^c , then*

$$\mathbb{E}_A Q_{\text{bad}}^\lambda \leq Ce^{-cK} B_n(h), \quad h \geq 1. \quad (6.24)$$

The constants are uniform in the stated near-centered regime.

Proof. Choose a fixed $N_{\text{REG}} \geq \max\{32, N_{\Gamma}, (D/\epsilon)^2\}$, enlarging it for the fixed-fraction comparisons below, and first take $n \geq N_{\text{REG}}$. For a single bad interval $(i, j]$, exponential moments and Chernoff give under the independent reference law

$$\mathbb{P}(\mathcal{D}_{i,j}) \leq C \exp\{-c\epsilon^2(j-i) - c\epsilon K[1 + d_n(i)^\gamma + d_n(j)^\gamma]\}. \quad (6.25)$$

This follows by choosing a fixed tilt z with $\log \mathbb{E}e^{z(E-\tau)} \leq Cz^2$ and $C|z| \leq \epsilon/2$; the finite color variables are even bounded. We give a partition that preserves the survival scale under conditioning.

For $j \leq n/4$, retain only survival to $n/2$. The event and the bad interval use at most half the gaps. Lemma 6.12 bounds their conditional moment by the independent reference moment, to which Lemma 6.9 applies. Its upper bound is $C(1+j)^2 \mathbb{P}(\mathcal{D}_{i,j})^{\lambda/4} B_n(h)$. For $i \geq n/8$, retain necessary survival on the first $\lfloor n/16 \rfloor$ slots. It is disjoint from the bad interval; their union omits a fixed fraction of the gaps. After one joint density comparison they are independent under the reference law, giving $C \mathbb{P}(\mathcal{D}_{i,j})^\lambda B_n(h)$. The remaining intervals have $i < n/8$, $j > n/4$ and length at least $n/8$. For a proper gap interval use the $C\sqrt{n}$ density bound in Lemma 6.12; for the entire gap set its total is fixed and its linear tolerance is satisfied for large n . Mark distributions are unaffected by the gap total. Thus all these remaining errors are exponentially small, even after their λ powers.

In the early region, the polynomial in j is summable against e^{-cKj^γ} and the interval-length factor. In the late region near the terminal end sum in $n-i$; away from both ends the additional e^{-cKn^γ} absorbs the number of intervals. There are at most n^2 macroscopic intervals, and their exponential errors are smaller than $e^{-cK} B_n(h)$ since $B_n(h) \geq cn^{-1/2}$. Subadditivity of the λ power proves (6.24). For $1 \leq n < N_{\text{REG}}$, every gap sub-sum is at most $A \leq N_{\text{REG}}\tau_+ + D\sqrt{N_{\text{REG}}}$, and every color count is at most N_{REG} . Consequently all of (6.23) hold deterministically whenever

$$K \geq K_{\text{fin}} := 1 + 2N_{\text{REG}}\tau_+ + D\sqrt{N_{\text{REG}}} + N_{\text{REG}}.$$

For $0 \leq K < K_{\text{fin}}$ use $Q_{\text{bad}} \leq 1$ and $B_n(h) \geq N_{\text{REG}}^{-1/2}$; increasing C to at least $e^{cK_{\text{fin}}} \sqrt{N_{\text{REG}}}$ proves the same estimate. This handles the entire bounded- n range even when $A \downarrow 0$, without a Gamma density comparison. This proof does not subtract an unconditional error larger than the survival moment. $\square$

6.6 Complete cells, every internal ordering, and Harris's inequality

We specify the deterministic projection because its converse is needed when a relative bad event is removed. Take r independent forward positions U_i uniform on $[0, v]$, with v an integer. Fix an integer $N \geq 1$, and set

$$J_i = N + \left\lfloor \frac{v-N}{v} U_i \right\rfloor, \quad J_i \in \{N, \dots, v-1\}. \quad (6.26)$$

These are independent uniform retained cells and $U_i - 1 < J_i \leq U_i + N$. Let the slope per cell be $c > 0$. With reverse coordinates $\tau_i = v - U_i$, $\xi_i = v - J_i$, use strict before-jump deficits

$$R_{\text{full}}(t) = ct - \sum_{\tau_i < t} w(C_i), \quad R_{\text{cell}}(t) = ct - \sum_{\xi_i < t} w(C_i).$$

The displacement $\tau_i - N \leq \xi_i < \tau_i + 1$ and positivity of all weights give exactly

$$R_{\text{cell}}(t) \leq c + R_{\text{full}}(t - 1), \quad R_{\text{full}}(t) \leq cN + R_{\text{cell}}(t - N). \quad (6.27)$$

The inequalities apply respectively for $t \geq 1$ and $t \geq N$; earlier times have bounds c and cN . Integer before-jump tests control the entire cell path since its deficit increases linearly between jumps. In particular a cell flat cap implies a continuous flat cap with only cN extra reserve.

There are $n = r + 1$ spacings between the r genuine positions and both endpoints. In physical units their law is that of n exponential gaps conditioned on total cv . Introduce one independent dummy mark at the last spacing. Auxiliary after-jump and genuine before-jump risks differ by at most l at each tested time; removing the dummy changes each color total by at most one. Fixed reserve and central-window adjustments therefore transfer Proposition 6.13 to these actual positions. This transfer uses $n \geq N_{\text{one}}$ after those adjustments; it makes no endpoint-support assertion for excluded short clocks. The last empty gap is retained throughout. This assertion concerns central windows, not a prescribed exact type or a prescribed dummy color.

Lemma 6.15 (Permutation-invariant regularity projection). *There is a cell event depending only on the occupancy and the color counts in each cell, implied by (6.23), such that its occurrence implies the required color interval bounds, with a larger fixed coefficient, for every ordering within cells. It also bounds each cell occupancy by a fixed constant times one plus its distance to the nearer physical endpoint raised to γ .*

Proof. Here is a full deterministic formulation. Let ordered physical positions $x_i = T_i$, $x_0 = 0, x_n = A$, be mapped to nondecreasing cells J_i satisfying $|aJ_i - x_i| \leq \Delta_0$, with fixed a, Δ_0 ; this includes the reverse version of (6.26). Put $D_A(t) = \min(t, A - t)$. Applying the gap inequalities in (6.23) to $(0, i]$ and $(i, n]$ gives

$$(\tau - \epsilon)d_n(i) - K[1 + d_n(i)^\gamma] \leq D_A(x_i) \leq (\tau + \epsilon)d_n(i) + K[1 + d_n(i)^\gamma].$$

The inequality $Kx^\gamma \leq \eta x + C_{\eta, \gamma}K^{1/(1-\gamma)}$ therefore implies

$$C^{-1}d_n(i) - B_K \leq D_A(x_i) \leq Cd_n(i) + B_K. \quad (6.28)$$

The single-gap bound gives $x_{i+1} - x_i \leq \tau + \epsilon + K[1 + d_n(i)^\gamma + d_n(i+1)^\gamma]$. If a physical cut t is within Δ_0 of $[x_i, x_{i+1}]$, subtract this bound from the lower bound in (6.28) and absorb its sublinear term. Thus

$$d_n(i)^\gamma \leq CD_A(t)^\gamma + B'_K. \quad (6.29)$$

This includes $i = 0, n - 1$.

Let $I(s) = \#\{i : J_i \leq s\}$, $g_s = I(s) - I(s - 1)$, $D(s) = D_A(as)$, and let $N_c(s, t)$ count color c in cells $(s, t]$. A cell cut as is within Δ_0 of $[x_{I(s)}, x_{I(s)+1}]$. Equations (6.23) and (6.29) therefore imply

$$\begin{aligned} |N_c(s, t) - \pi_c[I(t) - I(s)]| &\leq \epsilon[I(t) - I(s)] + K_1[1 + D(s)^\gamma + D(t)^\gamma], \\ g_s &\leq K_1[1 + D(s)^\gamma]. \end{aligned} \quad (6.30)$$

For the second assertion, the first and last positions of an occupied cell are separated by at most $a + 2\Delta_0$; the gap lower bound then bounds $g_s - 1$, using (6.29) at the adjacent cell cuts. Furthermore, for each rank u belonging to that cell,

$$D(s) \leq Cd_r(u) + B''_K. \quad (6.31)$$

This follows from (6.28) and the bounded displacement; $|d_n(u) - d_r(u)| \leq 1$. It can be tested solely from occupancy counts, since the smallest $d_r(u)$ in a cell's consecutive rank range occurs at one of its rank endpoints.

Define the cell regularity event by (6.30) and (6.31). For an arbitrary rank interval in any internal ordering, remove its two partial endpoint cells. Apply the first bound of (6.30) to the remaining complete-cell interval. The two omitted parts have total error at most their occupancies, bounded by the second assertion. Convert their physical endpoint distances back to rank distances by (6.31). Adjacent cuts differ by a fixed amount, and $|d_r(u+1) - d_r(u)| \leq 1$. Consequently, for every order,

$$|N_c(u, v) - \pi_c(v - u)| \leq \epsilon(v - u) + K_2[1 + d_r(u)^\gamma + d_r(v)^\gamma].$$

If both endpoints are in one cell its entire contribution is controlled by that cell's occupancy. The same argument covers the first and last occupied cells and empty cuts. All constants are fixed before n varies. $\square$

Theorem 6.16 (Critical complete-cell family). *Suppose $n = r + 1 \asymp T$, $|cv - n\tau| \leq D\sqrt{n}$, $|\omega - \tau|\sqrt{n} \leq D$, and the retained-cell density is bounded strictly below a fixed $\zeta < 1$. Fix $0 < \gamma < 1/6$. For $H_0 \leq h \leq D'\sqrt{T}$ there is a family depending only on complete colored cell occupancies whose conditional mark probability f satisfies*

$$\mathbb{E}f^\lambda \geq c \min\{1, ((1+h)/\sqrt{T})^{2\chi(\lambda, \rho_\tau)}\}. \quad (6.32)$$

*The same family has the following properties: the ordinary forward occupancy bounds $G_s \leq s + 1$ and $G_s \leq \zeta s + C_0$; a reverse flat cap with reserve $h + O(1)$; a prescribed fixed strong physical curve; the complete-cell regularity of Lemma 6.15, hence color regularity for all internal orderings; a fixed positive central window for the unreserved final risk; and central full-color windows. The constants C_0, N, H_0 and all window widths are fixed. The assertion uses $n \geq N_{\text{cell}}$, with a fixed threshold chosen after these data and containing the adjusted N_{one} and central-support thresholds. If $n \geq c_*T$ is the fixed comparability bound, one may finally enlarge $H_0 > D'\sqrt{N_{\text{cell}}/c_*}$ so that its reserve range already enforces this length restriction. More explicitly, the two reverse restrictions may be written $R_{\text{cell}}(k) \leq h + C_1$ and $R_{\text{cell}}(k) \leq h + C_g - g \min(k, v - k)^\gamma$ at every integer cut. The flat extra reserve C_1 is independent of the selected amplitude g .*

If a necessary competitor event skips at most $J_d \leq C(1+d)^a$ slots and retains a deterministic free block of a fixed positive fraction of n , of length at most $c_1 n$ with $c_1 < 1$, its cell conditional probability q_d obeys

$$\mathbb{E}q_d^\lambda \leq C(1+d)^C \min\{1, ((1+h)/\sqrt{T})^{2\chi(\lambda, \rho_\tau)}\}. \quad (6.33)$$

When a skip is too large to leave that block, the same bound holds after increasing the fixed polynomial exponent.

Proof. First the ordinary occupancy conditions have fixed positive probability. For independent retained cells with density at most $\zeta_0 < \zeta$, a binomial exponential bound gives $\mathbb{P}(G_s > \zeta s + C_0) \leq Ce^{-cC_0 - cs}$. Choose C_0 large. For $G_s \leq s + 1$, the first N cells are empty, and the same negative-drift bound summed over $s \geq N$ has arbitrarily small total when fixed N is large. After deleting these fixed cells the density still has a uniform margin for all sufficiently large v . Thus their intersection has probability $c_{\text{occ}} > 0$. Both events are increasing when any labeled forward position is moved rightward.

The order of transferring the strong event is essential. Apply Proposition 6.13 first to a stronger rank curve. Subtract (6.24); on the remaining family the deterministic comparisons (6.28)–(6.29) convert this curve to the desired physical curve. The single-gap bound controls

before-jump interpolation as well. Now drop REG and all proof-only prefix boxes, retaining the physical strong and flat conditions and the position-independent total-type and final-risk windows. This larger event still has the same lower fractional moment and is monotone in each independent original forward position, for every fixed mark word. If its conditional mark probability is $Q(U)$, Harris's product-measure inequality gives

$$\mathbb{E}[\mathbf{1}_{\text{occ}} Q(U)^\lambda] \geq \mathbb{P}(\text{occ}) \mathbb{E} Q(U)^\lambda. \quad (6.34)$$

It is applied to independent labeled positions, not to the dependent Dirichlet spacings. For one coordinate the inequality is the identity $2 \text{Cov}(f, g) = \mathbb{E}[(f(Z) - f(Z'))(g(Z) - g(Z'))] \geq 0$; induction proves the product version. Total-type restrictions do not disturb monotonicity because they do not depend on the positions.

Project by (6.26). Equation (6.27) and the Hölder modulus of $\min(t, v - t)^\gamma$ give a cell strong event and a separate cell flat cap. The latter has only the fixed extra reserve c , independent of the subsequent curve amplitude. By the converse in (6.27), a bad cell family lies, for every compatible latent position vector, in continuous flat survival with $cN + O(1)$ extra reserve and failure of a stronger continuous regularity envelope. Lemma 6.14 bounds its fractional mass. Subtract this second REG loss and use Lemma 6.15. Pointwise comparisons are taken before fractional powers: if $Q_{\text{full}}(U) \leq Q_{\text{cell}}(J(U))$, then integration gives the lower direction immediately; the converse bad-event containment gives the upper direction for the error. No incorrect reversal of conditional Jensen is used.

Here is a noncircular choice of constants. Fix the retained-cell deletion, density margin, and c_{occ} . Fix central endpoint boxes and a small fraction of their flat comparison constant. Choose regularity tolerances and K so that (6.24) pays for both REG losses and the occupancy factor. Then choose the rank-curve amplitude needed in the cell projection. Choose the relative KMT accuracy and finally enlarge H_0 to dominate its fixed initial error and make the normalized curve cost small. This is possible because $\gamma < 1/2$; its normalized amplitude decreases as H_0 grows. Relative error estimates are uniform for reserves at least one, so the earlier choices do not change. Finally fix N_{cell} above all length thresholds, and, if desired, make the final reserve enlargement stated in the theorem. This proves (6.32).

For a competitor, discarding J_d positive bounded jumps increases its necessary reserve by at most lJ_d . At the same rank index, replacing latent positions by cell centers changes only the two endpoint times by a fixed amount, not by an amount proportional to the rank. Its free block is therefore majorized by reference survival with reserve $h + C + lJ_d$. Since $n \geq N_{\text{cell}} \geq N_\Gamma$, Lemma 6.12 and Theorem 6.11 give (6.33). If the skip exceeds a fixed fraction of T , use $q_d \leq 1$; a larger fixed power of $1 + d$ absorbs one since $B_T(h) \geq cT^{-1/2}$. This completes the same-family assertion. $\square$

In the critical application $h = H_0 + M/4$, $0 \leq M \leq \sqrt{T}$, $\lambda = \theta + O_D(T^{-1/2})$, and $\rho_\tau = \rho_c + O_D(T^{-1/2})$. Proposition 6.5 shows that the factors in (6.32) and (6.33) are uniformly comparable to

$$\min\{1, ((1 + M)/\sqrt{T})^{2\chi(\theta, \rho_c)}\}.$$

Indeed the error in its logarithm is $O_D(\log T/\sqrt{T})$. The assertions hold for every fixed central first count; mixing a fixed central Poisson slab of positive probability preserves the lower bound. Here the original length and each retained central count are taken above the fixed N_{cell} threshold, as required by the lower construction.

6.7 Positive drift: an effective prefix and uniform continuation

Continue with (6.15), now $0 < \delta = \omega - \tau \leq \delta_0$, and condition on $T_n = n\tau$. Put

$$B_{n,\delta}(h) = \min\{1, [(1+h)(\delta + n^{-1/2})]^{2\chi(\lambda, \rho_\tau)}\}. \quad (6.35)$$

Theorem 6.17 (Positive-drift complete-cell family). *For fixed $\gamma \in (0, 1/6)$ and fixed curve and occupancy requirements, there are $\delta_0, H_0, D_0 > 0$ such that, for $0 < \delta \leq \delta_0$, $\delta\sqrt{n} \geq D_0$, $h \geq H_0$, the flat exact-total survival moment is at most $CB_{n,\delta}(h)$, and a complete-cell family with the properties of Theorem 6.16 has moment at least $cB_{n,\delta}(h)$. Its final unreserved risk can be restricted to $[\delta n - C\sqrt{n}, \delta n + C\sqrt{n}]$, and its full color counts are central. Deterministic competitor skips satisfy the corresponding upper bound $C(1+J)^C B_{n,\delta}(h)$.*

Proof. Fix constants A_0, K_0 , and set $m = \lfloor A_0/\delta^2 \rfloor$. Choose D_0 later so $m \leq n/8$, and decrease δ_0 to absorb rounding. On this effective prefix $\delta\sqrt{m}$ is bounded. Apply Theorem 6.11, retaining

$$S_m \in [K_0/\delta, C_{A_0, K_0}/\delta], \quad |T_m - m\tau| \leq C_{A_0, K_0}/\delta, \quad |N_c(m) - \pi_c m| \leq C_{A_0, K_0}/\delta. \quad (6.36)$$

Its conditional probability f has moment at least $c \min(1, [(1+h)\delta]^{2\chi})$. For $h\delta$ bounded below, a fixed tube at the effective scale gives this constant lower bound directly; large reserves help. Lemma 6.12 bounds the prefix density above and below on (6.36). The upper survival bound already follows by retaining only this prefix. For this application the complete total is exactly $n\tau$, so the upper comparison uses the all- n exact-mean clause. The central lower comparison uses its length threshold, enforced by the eventual choices of large D_0 and small δ_0 .

Fix its environment and a retained label word. Let $k = n - m$ and $R = n\tau - T_m$. Retain remaining full-color totals within fixed $C_1\sqrt{k}$ windows around $k\pi_c$. Their joint multinomial probability is bounded below. Conditional on any such type, use independent reference parameters equal to its proportions π'_c and to $\tau' = R/k$. The binary sequence is the uniform fixed-count permutation and the gaps are a Dirichlet bridge. The new parameters satisfy

$$\pi'_c = \pi_c + O(C_1/\sqrt{n}), \quad \tau' = \tau + O(C_{A_0, K_0}/(\delta n)), \quad \delta' := \sum_c \pi'_c w(c) - \tau' \geq \delta/2,$$

after D_0 is large in terms of the constants already fixed. Also $\delta' \leq C\delta$. The complete endpoint then belongs to the stated $\delta n + O(\sqrt{n})$ window and is at least $c\delta n$.

We verify the required continuation estimates under these two exact conditions. A selected initial or terminal block of at most $k/2$ gaps has density bounded by a constant relative to the new reference law. Indeed its full total is exactly $R = k\tau'$ with τ' in a positive compact interval, so this is again the all- k exact-mean upper bound. The corresponding binary density ratio is

$$\frac{\mathbb{P}\{\text{Bin}(k-j, p') = u - U_j\}}{\mathbb{P}\{\text{Bin}(k, p') = u\}}, \quad u = kp'.$$

Its denominator is comparable to $k^{-1/2}$ and numerator at most $C(k-j)^{-1/2}$, so it too is bounded. The two ratios multiply by independence. Specifying all colors leaves this same binary bridge law.

For independent reference increments $X = w(C) - E$, uniform exponential moments and Taylor expansion give, for a fixed small $a > 0$,

$$\mathbb{E} \exp[-a\delta(X - \delta'/2)] \leq 1.$$

The exponential supermartingale therefore gives, from $x \geq K_0/\delta$,

$$\mathbb{P}\{\exists j \geq 0 : x + \sum_{i \leq j} X_i < x/2 + \delta'j/2\} \leq e^{-c\delta x}. \quad (6.37)$$

The bounded initial-half density transfers this estimate to the first half of the conditioned continuation. For its late half, the maximal Bernstein estimate for each independent centered channel is

$$\mathbb{P}\{\max_{j \leq k/2} |\sum_{i \leq j} Z_i| > z\} \leq C \exp[-c \min(z^2/k, z)].$$

Apply the same bounded density to the initial half and to the reversed terminal half. Both centered totals are fixed to zero; these halves cover the complete bridge, giving

$$\mathbb{P}\{\max_{j \leq k} |S'_j - \delta' j| > c_2 \delta k\} \leq C e^{-c \delta^2 k}. \quad (6.38)$$

There is no factor k from a union over individual times.

The global strong curve on the continuation is at most $g(m+j)^\gamma$. At its beginning $gm^\gamma = o(\delta^{-1})$, and its derivative is $O(\delta^{2-2\gamma}) = o(\delta)$ as $\delta_0 \rightarrow 0$. Thus the positive linear margin in (6.37) dominates it on the first half. On the late half $gn^\gamma = o(\delta n)$ uniformly when $\delta\sqrt{n} \geq D_0$, so (6.38) suffices. Choose K_0 so $Ce^{-cK_0} < 1/4$, then D_0 so the late failure is less than $1/4$ and all parameter comparisons hold. Every retained prefix word and remaining type now has conditional continuation probability at least $1/2$, averaged over its remaining exact gaps. The central type window has fixed positive mass. Thus the inequalities (6.22) hold again, proving the strong exact-total lower bound after integrating the central prefix density.

For completeness, regularity is not inferred from endpoint centrality. Before exact conditioning, the same effective-prefix argument and (6.37) prove flat reference survival comparable to (6.35): use the near-centered theorem when $\delta\sqrt{n}$ is bounded, and the effective prefix otherwise. Hence time doubling and polynomial reserve comparison hold uniformly throughout this positive-drift layer. Repeat the three-region proof of Lemma 6.14 with the unchanged exponential interval tails. It gives a relative bad moment at most $Ce^{-cK} B_{n,\delta}(h)$ under the exact total. The ordinary occupancy, physical projection, and two-stage REG/Harris argument are now precisely those in Theorem 6.16. The proof-only effective-prefix boxes are dropped before Harris. The final endpoint and total-type windows are position independent and remain.

Constants can be fixed in the following order: occupancy/deletion requirements; effective-prefix constants A_0, K_0 and their endpoint fraction; central type widths; REG tolerance and K ; required curve amplitude and relative coupling accuracy; minimum reserve H_0 ; then large D_0 , small δ_0 , and the length threshold. The reference relative errors are uniform in $h \geq 1$. Enlarging H_0 and shrinking δ_0 make normalized curve costs small without changing the earlier fixed success fraction. The two displayed parameter inequalities enforce $n \geq (D_0/\delta_0)^2$; take these final choices so that this exceeds every required length threshold. Competitor blocks have enlarged reserve $h + C + lJ$ and bounded selected-gap density. Use the reference upper bound; if J is a fixed fraction of the effective time $\min(n, \delta^{-2})$, the trivial bound one is absorbed by a larger polynomial. This proves all assertions. $\square$

In the ordinary boundary application, $n \asymp T$ and the count-specific drift equals $d/\kappa + O(T^{-1/2})$, with κ bounded above and below. The reference parameters differ from their critical values by $O(d + T^{-1/2})$. Proposition 6.5 permits the critical exponent in the scale

$$\min\{1, [(1+h)(d + T^{-1/2})]^{2\chi_c}\}.$$

Indeed $d \log(1/d)$ and $T^{-1/2} \log T$ are bounded. The region of bounded $d\sqrt{T}$ is already covered by Theorem 6.16; no intermediate drift range is omitted.

6.8 Physical Poisson time and weighted insertion

Let Π be a homogeneous Poisson process of rate κ on $[0, s]$, where κ ranges over a fixed positive compact interval. Its points carry independent marks of the preceding law. For fixed $c > 0$ write

$$S(t) = \sum_{u \in \Pi, u \leq t} w(C_u) - ct, \quad \tau = c/\kappa, \quad \rho_\tau = \frac{\tau^2}{\tau^2 + \text{Var}(w)}.$$

Before-jump and after-jump conventions differ by at most l and will always be allowed that fixed reserve adjustment.

Theorem 6.18 (Physical one-end upper bound). *Fix $0 < \gamma < 1/2$, $g \geq 0$, $H \geq 1$. Given Π , let q be the conditional mark probability that either $S(t)$ or $-S(t)$ stays above $-H - gt^\gamma$ for every $0 \leq t \leq s$. If $|\kappa\omega - c|\sqrt{s} \leq D$, then*

$$\mathbb{E}q^\lambda \leq C_{D,g,\gamma} \min\{1, [(1+H)/\sqrt{s}]^{2\chi(\lambda,\rho_\tau)}\}. \quad (6.39)$$

The correlation is the arrival-gap correlation ρ_τ displayed above. The bound also permits a fixed logarithmic wall in place of the power wall.

Proof. We first note why the soft-wall form of Theorem 6.11 has its stated uniformity. After a bad prefix of length b , $(b+j)^\gamma \leq b^\gamma + j^\gamma$ changes reserve by only a fixed polynomial in b . Thus the relative bad-prefix proof is unchanged up to its fixed polynomial. Under Lamperti transformation from time H^2 , the wall becomes $gH^{2\gamma-1}e^{-(1/2-\gamma)u}$, whose action and coefficient variation are bounded for $H \geq 1$. The relative KMT proof and its absorption therefore hold for both signs; no fixed unconditional REG error is subtracted to prove this upper bound.

Take $m = \lfloor \kappa s/2 \rfloor$; bounded s is absorbed in constants. The Poisson lower-tail probability $\mathbb{P}(N_s < m)$ is at most $e^{-c_0 s}$. Extend the process beyond s if necessary. Its first m scaled arrival gaps $E_i = c(t_i - t_{i-1})$ are independent exponentials of mean τ . The extension does not alter the conditional mark probability on $[0, s]$.

On $N_s \geq m$, physical survival imposes necessary inequalities at these first m arrivals. For the sign S , at a relevant test time the accumulated weight is at most lj ; hence survival implies $ct \leq lj + H + gt^\gamma$. Since $\gamma < 1$, it follows that $t \leq C(j+H+1)$ and consequently $gt^\gamma \leq C(j^\gamma + H+1)$. Thus the necessary rank event has reserve $C(H+1)$ and a fixed soft-power coefficient. For the sign $-S$, no arrival-time upper bound is needed when $-S(t) \geq 0$: its rank lower wall is automatic. When $-S(t) < 0$, one has $ct < \sum w \leq lj$, which again bounds the allowed physical soft wall by a fixed rank power wall. This separate argument avoids reversing an invalid time inequality.

At a fixed full environment the mark probability of this necessary rank event depends only on the first m gaps. Dropping the restriction $N_s \geq m$ in its expectation gives

$$\mathbb{E}q^\lambda \leq \mathbb{E}q_{\text{rank}}^\lambda + e^{-c_0 s}.$$

Here $|\omega - \tau|\sqrt{m} \leq C_D$. Apply Theorem 6.11 with its soft wall to the iid arrival gaps. Since $m \asymp s$ this gives (6.39). Finally the displayed scale is at least $cs^{-1/2}$, because $\chi \leq 1/2$ and $H \geq 1$, so the exponential tail is absorbed. A logarithm is bounded by a fixed constant plus any fixed positive power t^γ , proving the last assertion. $\square$

Corollary 6.19 (Effective physical time). *If the physical mean $d = \kappa\omega - c \geq 0$ is sufficiently small, the same full-horizon necessary event has fractional moment at most*

$$C \min\{1, [(1+H)(d + s^{-1/2})]^{2\chi(\lambda,\rho_\tau)}\}.$$

The bound is valid for either risk sign and the preceding soft walls.

Proof. Retain only the deterministic interval of length $s_0 = c_0 \min(s, d^{-2})$, with $d^{-2} = \infty$ when $d = 0$. Then $d\sqrt{s_0}$ is bounded. Apply Theorem 6.18; $s_0^{-1/2} \asymp d + s^{-1/2}$. When s_0 is bounded the conclusion follows from the trivial upper bound one. Restricting to this interval is a necessary condition for full survival, so no continuation lower bound is asserted or required. $\square$

Proposition 6.20 (Palm bounds with the survival weight retained). *Let $T \geq 1$ and let $q_H(\Pi)$ be a physical conditional survival probability of the preceding kind on $[0, T]$. Suppose its proved upper bound is $CB_T(H)$, where either*

$$B_T(H) = \min\{1, [(1+H)/\sqrt{T}]^{2\chi}\} \quad \text{or} \quad B_T(H) = \min\{1, [(1+H)(d+T^{-1/2})]^{2\chi}\}.$$

For $t \in [0, T]$ set

$$Z(t) = 1 + \sum_{u \in \Pi, u \geq t} e^{-(u-t)}, \quad \mathcal{A} = 1 + C_0 \sup_{0 \leq t \leq T} \frac{Z(t)^5}{(2+t)^4}, \quad N_T = |\Pi|.$$

For each integer $m \geq 0$ and every fixed $a > 1/4$,

$$\begin{aligned} \mathbb{E}[Z(t)^m q_H^\lambda] &\leq C_m B_T(H), \\ \mathbb{E}[N_T Z(t)^m q_H^\lambda] &\leq C_m T B_T(H), \\ \mathbb{E}[(T + N_T) \mathcal{A}^a q_H^\lambda] &\leq C_a T B_T(H). \end{aligned} \tag{6.40}$$

Moreover for $k \geq 0$,

$$\mathbb{E}[(T + N_T) q_{H+k+\lceil \log \mathcal{A} \rceil}^\lambda] \leq C(1+k)^C T B_T(H). \tag{6.41}$$

Proof. Adding j deterministic points changes every risk by at most jl , regardless of their marks. Integrating the new marks therefore gives the pointwise inequality

$$q_H(\Pi + \{u_1, \dots, u_j\}) \leq q_{H+jl}(\Pi). \tag{6.42}$$

It holds for either sign. The proved survival upper bounds imply $\mathbb{E} q_{H+jl}^\lambda \leq C(1+j)^C B_T(H)$, since $H \geq 1$ and $2\chi \leq 1$.

Expand $Z(t)^m$ and group monomials by their distinct Poisson points. The multivariate Mecke identity for j distinct points says that

$$\mathbb{E} \sum_{u_1, \dots, u_j \in \Pi}^{\neq} G(u_1, \dots, u_j, \Pi) = \kappa^j \int_{[0, T]^j} \mathbb{E} G(u_1, \dots, u_j, \Pi + \{u_1, \dots, u_j\}), d\mathbf{u}.$$

For a Poisson process this follows directly by expanding its factorial probabilities $e^{-\kappa T} (\kappa T)^n / n!$, choosing the j ordered distinct points, and reindexing $n - j$, with the weight evaluated in the resulting environment. In each monomial the insertion integrals have bounded mass, since every point carries at least one factor $e^{-(u-t)} \mathbf{1}_{\{u \geq t\}}$. Use (6.42) after insertion to obtain the first bound in (6.40). One additional Mecke point without a decaying factor gives a factor κT and proves the second; coincidences with existing monomial points only reduce the number of integrations.

For $j \leq t < j+1$ one has $Z(t) \leq eZ(j)$. Hence $\mathcal{A}^a \leq C_a [1 + \sum_{j \geq 0} (2+j)^{-4a} Z(j)^{5a}]$. Bound the noninteger moment by a larger integer moment and sum the convergent series, since $4a > 1$. This proves the last assertion of (6.40). Finally, on $\lceil \log \mathcal{A} \rceil = j$ one has $\mathcal{A}^a \geq e^{a(j-1)}$. Apply the weighted estimate with reserve $H + k + j$, separately in each bin, to get

$$\mathbb{E}[(T + N_T) q_{H+k+\lceil \log \mathcal{A} \rceil}^\lambda] \leq CT \sum_{j \geq 0} e^{-a(j-1)} B_T(H + k + j) \leq C(1+k)^C T B_T(H).$$

The reserve in the weighted estimate depends on the bin j ; a single unweighted crowding tail would not justify this step. This proof also shows why arbitrary stopped depths may be controlled by the global crowding majorant without claiming that a stop remains fixed when unseen Poisson points are deleted. $\square$

The probability results proved here are inputs about conditional mark families. Their use for divisor-box volumes requires the separate positive complete-cell kernel, geometric cover, and terminal-count arguments. In particular no pointwise identification of an actual divisor union with a fractional survival probability is used in any proof in this section.

7 Environmental selection and mixed source weights

7.1 Positive moment orders for the same scalar model

Let E, W be independent stationary standard Ornstein–Uhlenbeck processes with covariance $e^{-|u-v|/2}$. We use the same conditional survival as in (6.1), now allowing all positive moment orders. For $0 < \rho < 1$, set

$$\beta = \sqrt{\rho/(1-\rho)}, \quad q_t(E) = \mathbb{P}_W\{W_s \geq -\beta E_s \ (0 \leq s \leq t) \mid E\}, \quad S_t = -\log q_t(E). \quad (7.1)$$

For every $p > 0$ define

$$F_{p,\rho}(t) = \mathbb{E}q_t^p, \quad b_t(p, \rho) = -\log F_{p,\rho}(t), \quad \chi(p, \rho) = \lim_{t \rightarrow \infty} b_t(p, \rho)/t. \quad (7.2)$$

Theorem C.2 proves existence and, locally uniformly for positive p and $0 < \rho < 1$,

$$b_t(p, \rho) = t\chi(p, \rho) + O(1), \quad t \geq 1. \quad (7.3)$$

For $0 < p < 1$ this is the scalar in (3.8). The ordinary probability from (6.4) is

$$a(t) = \mathbb{E}q_t = \pi^{-1} \arcsin(e^{-t/2}), \quad -\log a(t) = t/2 + O(1). \quad (7.4)$$

Every continuous environmental path is bounded on a compact interval; a sufficiently high private OU tube gives $q_t(E) > 0$. The initial constraint gives $q_t(E) < 1$. Thus S_t is finite almost surely. These moments describe the selection of the common environment. Mallein and Miłoś [20, Theorem 1.6] established quenched and annealed OU persistence exponents for admissible initial data, with a strict gap between them for a nonzero wall amplitude. Here the stationary moment spectrum, its constant-factor estimates and its derivatives will locate the clocks supporting the arithmetic lower bound.

7.2 A dimension-free curvature estimate

Lemma 7.1 (Gaussian convex-probability curvature). *Let $C \subset \mathbb{R}^d$ be convex with positive Gaussian measure and let*

$$A(x) = -\log \mathbb{P}\{Z + \beta x \in C\}, \quad Z \sim N(0, I_d), \quad \beta > 0.$$

Then

$$0 \preceq D^2 A(x) \preceq \beta^2 I_d, \quad |\nabla A(x)|^2 \leq 2\beta^2 A(x). \quad (7.5)$$

Under $d\nu_p \propto e^{-pA} d\gamma_d$, $p > 0$,

$$\text{Var}_{\nu_p}(A) \leq 2\beta^2 \mathbb{E}_{\nu_p} A. \quad (7.6)$$

Proof. Let U_x have law $N(\beta x, I_d)$ conditioned to lie in C . Differentiating the Gaussian convolution gives

$$D^2 A(x) = \beta^2 \{I_d - \text{Cov}(U_x)\}.$$

The variance inequality of Brascamp and Lieb [3] for a strongly log-concave measure gives $\text{Cov}(U_x) \preceq I_d$. For the hard-boundary approximation, a convex set of positive Gaussian measure has nonempty interior, and replacing it by its closure changes no Gaussian integral. Use the potentials

$$V_{N,x}(u) = \frac{1}{2}|u - \beta x|^2 + N \text{dist}(u, C)^2.$$

Smooth convex approximations to the squared distance allow the classical inequality with $D^2 V_{N,x} \succeq I_d$. First remove the smoothing and then let $N \rightarrow \infty$; Gaussian domination passes the means and second moments to the conditioned law. This proves the covariance bound for the possibly unbounded hard convex set. Covariance is positive semidefinite, which proves the other bound. We use the standard form

$$\text{Var}_\nu f \leq \mathbb{E}_\nu \langle \nabla f, (D^2 V)^{-1} \nabla f \rangle, \quad d\nu \propto e^{-V} dx;$$

see [5, equation (1.3)]. Since $A \geq 0$ and its Hessian is bounded above by $\beta^2 I_d$,

$$0 \leq A(x - \beta^{-2} \nabla A(x)) \leq A(x) - \frac{|\nabla A(x)|^2}{2\beta^2}.$$

Finally, $|x|^2/2 + pA(x)$ has Hessian at least I_d . Apply Brascamp–Lieb once more and use the gradient estimate. $\square$

Theorem 7.2 (Regularity of the positive spectrum). *For $p > 0$ and $0 < \rho < 1$, $\chi(p, \rho)$ is nondecreasing and concave in p , and belongs to $C_{\text{loc}}^{1,1}$ in that variable. It is jointly locally Lipschitz in (p, ρ) , and $\partial_p \chi$ is jointly continuous. Locally,*

$$|\partial_p \chi(p, \rho) - \partial_p \chi(p, \rho')| \leq C|\rho - \rho'|^{1/2}. \quad (7.7)$$

Moreover, uniformly on compact positive parameter sets,

$$\partial_p b_t(p, \rho) = t \partial_p \chi(p, \rho) + O(\sqrt{t}), \quad t \geq 1. \quad (7.8)$$

Proof. On a finite grid, simultaneously whitening the two equal covariance matrices gives the cone $\Sigma^{-1/2} \mathbb{R}_+^d$ in Lemma 7.1. With $b = -\log \mathbb{E} e^{-pA}$,

$$b' = \mathbb{E}_{\nu_p} A, \quad -b'' = \text{Var}_{\nu_p}(A) \leq 2\beta^2 b'.$$

Concavity and $b(0) = 0$ imply $0 \leq b'(p) \leq b(p)/p$. For $p \leq 1$, $q^p \geq q$; for $p \geq 1$, Jensen gives $\mathbb{E} q^p \geq (\mathbb{E} q)^p$. Hence

$$0 \leq b(p) \leq \max(1, p)[- \log a(t)] \leq \max(1, p)(t/2 + \log \pi). \quad (7.9)$$

Thus b/t has uniformly locally bounded first derivatives and uniformly Lipschitz first derivatives for $t \geq 1$. For nested grids whose union is dense and includes both endpoints, the conditional grid probabilities decrease to $q_t(E)$ by path continuity. Their negative logarithms therefore increase to S_t . On each compact positive order interval, the functions $x^j e^{-px}$, $j = 0, 1, 2$, are bounded uniformly in $x \geq 0$. Dominated convergence consequently passes the unnormalized moments and their first two order derivatives to the path limit. The normalizers are positive, so the same is

true of the derivatives of b_t . Equicontinuity, pointwise convergence and integration of the first derivatives prove the $C^{1,1}$ assertion. The constant-factor estimate (7.3), proved in Appendix C, and difference quotients of length $t^{-1/2}$ give (7.8).

For the dependence on ρ , the finite-grid semigroup calculation in the proof of Proposition 6.5 gives

$$\left| \partial_u \log \mathbb{E} q_\rho^p \right| \leq C \frac{\mathbb{E}[q_\rho^p(1 - \log q_\rho)]}{\mathbb{E} q_\rho^p}, \quad u = -\frac{1}{2} \log \rho.$$

The calculation uses $|p(1-p)|$, so it applies to all compact positive order intervals. By concavity, the ratio is at most $1 + b_t(p, \rho)/p$; (7.9) bounds it by $C(1+t)$. A slightly smaller positive exponent dominates the logarithmic integrand when removing smooth approximation and passing to dense grids. Integrating the parameter bounds and taking the limiting slope proves joint local Lipschitz continuity. Finally compare difference quotients at ρ, ρ' with step $|\rho - \rho'|^{1/2}$, using the uniform Lipschitz bound for $\partial_p \chi$ in p . This proves (7.7) and joint continuity. $\square$

7.3 A neighboring-moment localization principle

Lemma 7.3. *Let $J \Subset (0, \infty)$ be a compact interval, let $\eta \in \text{int } J$, and let $R \in [0, 1]$, $\tau \geq 0$. Suppose, for the same random variable R at every $p \in J$,*

$$C^{-1} e^{-\tau \chi(p, \rho)} \leq \mathbb{E} R^p \leq C e^{-\tau \chi(p, \rho)}. \quad (7.10)$$

Put $d\nu_R = R^\eta d\mathbb{P} / \mathbb{E} R^\eta$. Then

$$\nu_R\{ |-\log R - \tau \partial_\eta \chi(\eta, \rho)| > u \} \leq C_1 \exp \left[-c_1 \min \left\{ \frac{u^2}{1 + \tau}, u \right\} \right], \quad u \geq 0. \quad (7.11)$$

The constants are locally uniform in the indicated parameters. Zeros of R have zero ν_R -mass.

Proof. For $|h|$ smaller than a fixed distance to ∂J ,

$$\mathbb{E}_{\nu_R} e^{h(-\log R - \tau \partial_\eta \chi)} = e^{-h\tau \partial_\eta \chi} \frac{\mathbb{E} R^{\eta-h}}{\mathbb{E} R^\eta} \leq C^2 e^{C_2 \tau h^2},$$

by the $C^{1,1}$ Taylor bound of Theorem 7.2. Apply Chernoff in both directions with $|h|$ comparable to $\min\{u/(1+\tau), 1\}$. $\square$

7.4 A strictly positive cost of annealing first

Theorem 7.4 (Strict extensive annealing loss). *For every $0 < \eta, \rho < 1$,*

$$\chi(\eta, \rho) - \frac{\eta}{2} \geq \frac{\rho}{16\pi} \min\{\eta, 1 - \eta\} > 0. \quad (7.12)$$

Proof. Let $\mathbf{P}_t$ be the environmental OU law and $\mathbf{Q}_t$ its law conditional on joint survival. Then

$$f_t := \frac{d\mathbf{Q}_t}{d\mathbf{P}_t} = \frac{q_t}{a(t)}.$$

For $H_t = \int_0^t E_s ds$, the unconditioned variance is

$$v_t = 4t - 8(1 - e^{-t/2}) \leq 4t.$$

Complete $Z = \sqrt{\rho}E + \sqrt{1-\rho}W$ to an independent orthogonal Gaussian rotation. Only Z is conditioned to stay positive. At time s its density on $(0, \infty)$ is proportional to

$$\varphi(z)h_s(z)h_{t-s}(z), \quad h_u(z) = \mathbb{P}_z(\tau_0 > u).$$

Both factors are increasing, so the conditional law dominates a positive half-normal. Hence

$$\mathbb{E}_{\mathbf{Q}_t} H_t \geq \sqrt{2\rho/\pi} t =: m_0 t.$$

Under $\mathbf{Q}_t$, H_t is sub-Gaussian about its mean with variance proxy v_t . To verify this, append H_t to a finite Gaussian survival grid. After any linear tilt, the survival constraint is convex, and Brascamp–Lieb bounds the variance of H_t by its original Gaussian variance. Integrating the second derivative of the log moment generating function proves the claim. The finite-grid normalizers decrease to the positive number $a(t)$; Gaussian integrability passes the means and exponential moments to the limit.

For $B_t = \{H_t \geq m_0 t/2\}$, the two sub-Gaussian estimates give

$$\mathbf{P}_t(B_t) \leq e^{-\rho t/(16\pi)}, \quad \mathbf{Q}_t(B_t^c) \leq e^{-\rho t/(16\pi)}.$$

Hölder on B_t and B_t^c yields

$$\begin{aligned} \int f_t^\eta d\mathbf{P}_t &\leq \mathbf{Q}_t(B_t)^\eta \mathbf{P}_t(B_t)^{1-\eta} + \mathbf{Q}_t(B_t^c)^\eta \mathbf{P}_t(B_t^c)^{1-\eta} \\ &\leq 2e^{-\rho t \min(\eta, 1-\eta)/(16\pi)}. \end{aligned}$$

Use $F_{\eta,\rho}(t) = a(t)^\eta \int f_t^\eta d\mathbf{P}_t$, take negative logarithms, divide by t , and let $t \rightarrow \infty$. $\square$

Remark 7.5 (Exact information identity). With the standard Rényi divergence [29],

$$-\log \mathbb{E} q_t^\eta = -\eta \log a(t) + (1-\eta) D_\eta(\mathbf{Q}_t \| \mathbf{P}_t). \quad (7.13)$$

Thus $\chi - \eta/2$ is a nonzero information rate for the change in environmental law caused by survival. Theorem 7.4 gives a positive lower bound for this rate.

7.5 Mixed weights and Gibbs variational error

We now consider the weight that arises when an accepted private path is divided by an environmental envelope. Let $J = [p_-, p_+] \Subset (0, 1)$, $\eta \in \text{int } J$, and

$$0 \leq q \leq 1, \quad 0 < D \leq 1, \quad q \leq C_0 D, \quad C_0 \geq 1.$$

For fixed $\tau \geq 0$ and $0 < \rho < 1$, the same q, D satisfy, for every $p \in J$,

$$\mathbb{E} q^p \asymp \mathbb{E} D^p \asymp e^{-\tau \chi(p, \rho)}. \quad (7.14)$$

Set

$$R = \frac{q}{D}, \quad w = R D^\eta = q D^{\eta-1}, \quad Z_\eta = \mathbb{E} w, \quad d\hat{\nu}_\eta = \frac{w}{Z_\eta} d\mathbb{P}. \quad (7.15)$$

On the support of this measure write $S = -\log q$, $T = -\log D$. On $\{q = 0\}$ all integrands with a strictly positive power of q are defined as zero.

Lemma 7.6 (Mixed-power stability). *Uniformly for $p \in J$ and a in a compact subset of $(0, \infty)$,*

$$\mathbb{E}[R^a D^p] \asymp e^{-\tau\chi(p, \rho)}. \quad (7.16)$$

Proof. Put $A_p = \mathbb{E}D^p$, $d\nu_p = D^p d\mathbb{P}/A_p$ and $r = R/C_0 \in [0, 1]$. Then

$$\mathbb{E}_{\nu_p} r^p = \frac{\mathbb{E}q^p}{C_0^p \mathbb{E}D^p} \geq \kappa > 0.$$

For $a \leq p$, $r^a \geq r^p$; for $a \geq p$, Jensen gives $\mathbb{E}r^a \geq (\mathbb{E}r^p)^{a/p}$. Hence

$$\kappa^{\max(1, a/p)} \leq \mathbb{E}_{\nu_p} r^a \leq 1.$$

Multiplication by $C_0^a A_p$ proves the result. In fact this lemma alone is valid on any compact positive p -interval. $\square$

Theorem 7.7 (Mixed spectrum and localization). *The normalization satisfies $Z_\eta \asymp e^{-\tau\chi(\eta, \rho)}$. If*

$$s < 1, \quad p := \eta - s - t \in \text{int } J, \quad (7.17)$$

then, uniformly on compact subsets of this region,

$$\log \mathbb{E}_{\hat{\nu}_\eta} e^{sS+tT} = \tau[\chi(\eta, \rho) - \chi(\eta - s - t, \rho)] + O(1). \quad (7.18)$$

Both S and T satisfy (7.11) under $\hat{\nu}_\eta$. In addition,

$$\hat{\nu}_\eta\{\log(D/q) > v\} \leq C e^{-v} \quad (v \geq 0), \quad \log(D/q) \geq -\log C_0, \quad (7.19)$$

and

$$D_{\text{KL}}(\hat{\nu}_\eta \| \mathbb{P}) = \tau[\chi(\eta, \rho) - \eta \partial_\eta \chi(\eta, \rho)] + O(\sqrt{1 + \tau}). \quad (7.20)$$

Proof. The identity

$$RD^\eta e^{sS+tT} = R^{1-s} D^{\eta-s-t}$$

and Lemma 7.6 prove (7.18). Neighboring values of s or t and the same Chernoff argument as in Lemma 7.3 give concentration about $\tau \partial_\eta \chi$.

On $\{D/q > e^v\}$ one has $w \leq e^{-v} D^\eta$. Divide expectations by Z_η and use (7.14); this proves the tail bound. In particular $\mathbb{E}_{\hat{\nu}} |\log R| = O(1)$. Integrating the concentration bound gives $\mathbb{E}_{\hat{\nu}} T = \tau \partial_\eta \chi + O(\sqrt{1 + \tau})$. Finally,

$$\log \frac{d\hat{\nu}}{d\mathbb{P}} = -\eta T + \log R - \log Z_\eta,$$

which proves (7.20). $\square$

Proposition 7.8 (Two Gibbs comparisons and near-minimality). *Let*

$$d\nu_\eta^D = \frac{D^\eta}{\mathbb{E}D^\eta} d\mathbb{P}, \quad d\nu_\eta^q = \frac{q^\eta}{\mathbb{E}q^\eta} d\mathbb{P}.$$

Then

$$\frac{d\hat{\nu}_\eta}{d\nu_\eta^D} \leq C, \quad \frac{d\hat{\nu}_\eta}{d\nu_\eta^q} \leq C, \quad D_{\text{KL}}(\hat{\nu}_\eta \| \nu_\eta^D), D_{\text{KL}}(\hat{\nu}_\eta \| \nu_\eta^q) \leq \log C. \quad (7.21)$$

Consequently the actual mixed measure has $O(1)$ excess in each of the two finite-time Gibbs variational problems:

$$0 \leq D_{\text{KL}}(\hat{\nu}_\eta \| \mathbb{P}) + \eta \mathbb{E}_{\hat{\nu}_\eta} T + \log \mathbb{E}D^\eta \leq \log C, \quad (7.22)$$

$$0 \leq D_{\text{KL}}(\hat{\nu}_\eta \| \mathbb{P}) + \eta \mathbb{E}_{\hat{\nu}_\eta} S + \log \mathbb{E}q^\eta \leq \log C. \quad (7.23)$$

Proof. The two density ratios are respectively

$$R \frac{\mathbb{E} D^\eta}{Z_\eta}, \quad R^{1-\eta} \frac{\mathbb{E} q^\eta}{Z_\eta}.$$

They are bounded because $R \leq C_0$ and $0 < \eta < 1$. The normalization estimates give the claimed uniform constants. Relative entropy is nonnegative and is at most the logarithm of an upper density bound. Expand the two relative entropies; integrability was proved in Theorem 7.7. $\square$

Remark 7.9. The bound relative to ν_η^q uses $\eta \leq 1$; the power-stability and mixed-moment formulas themselves have a larger positive-parameter range. Appendix E treats finitely many conditional stages and groups of comparisons on a shared clock.

The conditioned bridges and accepting measure constructed next provide the fixed neighboring-moment windows needed to apply these estimates to original prime tuples in Section 17.

8 Conditioned marked bridges

The scalar persistence estimates must now be used with the totals that actually occur in the arithmetic construction. Conditioning on those totals changes the law of both ends of a path. The useful comparison is one joint likelihood ratio, followed by integration of the remaining bridge before taking the fractional power. This order also permits an entire inherited measure, rather than a single entrance state, to pass through a new clock.

8.1 The exact binary bridge

Fix a compact family of marks $w_+ > w_- > 0$, their difference $b = w_+ - w_- > 0$, speed $c > 0$, gap mean $\tau > 0$, and $\eta, \pi \in (0, 1)$, with all positive quantities and complementary probabilities bounded away from zero. Let $n \geq 1$, $k = n\pi$ be an integer, and $P = (n + 1)\tau$. The clock G consists of n ordered uniform points in $[0, P]$. Independently, the low marks occupy a uniform k -subset of its slots. For either orientation $\varepsilon \in \{-1, 1\}$, write

$$Z(t) = x + \varepsilon\{S(t) - ct\}, \quad y - x = \varepsilon(nw_+ - bk - cP), \quad (8.1)$$

where $S(t)$ is the sum of the marks at or before t . Both one-sided values at a jump are included in every wall condition. Put

$$\rho = \frac{(c\tau)^2}{(c\tau)^2 + b^2\pi(1-\pi)}, \quad \chi = \chi(\eta, \rho), \quad Q_n(x, y) = \min \left\{ 1, \left(\frac{(1+x)(1+y)}{n} \right)^{2\chi} \right\}. \quad (8.2)$$

All $n + 1$ gaps, including the final unmarked gap, are present.

For a deterministic grid with bounded boundary density, including $0, P$, write $d_P(t) = \min(t, P - t)$. Given a slope $\vartheta > 0$ and $0 < \zeta < 1/2$, the condition REG_K is

$$|N_a((s, t]) - \lambda_a(t - s)| \leq \vartheta(t - s) + K\{1 + d_P(s)^\zeta + d_P(t)^\zeta\}, \quad \lambda_- = k/P, \quad \lambda_+ = (n - k)/P, \quad (8.3)$$

for both colors and every grid pair $s \leq t$.

Theorem 8.1 (Exact binary bridges with all endpoint reserves). *Fix $g \geq 0$ and $\zeta < \gamma < 1/2$. There are fixed h_0, K such that, for every n , both signs, and all legal $x, y \geq h_0$,*

$$\mathbb{E}_{G\mathbb{P}_{\text{marks}}} \{Z(t) \geq g d_P(t)^\gamma \mid 0 \leq t \leq P, \text{REG}_K \mid G\}^\eta \asymp Q_n(x, y). \quad (8.4)$$

The upper bound holds on dropping both the curve and regularity. There is no upper bound on the reserves. One may replace the rates in (8.3) by prescribed rates differing from them by at most $C/\sqrt{n}$, after changing the fixed constants.

We first establish the relative regularity needed in its proof.

Lemma 8.2 (Relative one-end regularity). *For the independent exponential/binary reference law with either risk sign and bounded scaled drift on m steps, set*

$$F_m(h) = \min\{1, ((1+h)/\sqrt{1+m})^{2\chi}\}.$$

The one-end lower family from Theorem 6.11, with its strong curve, compatible central separate totals, and positive inner endpoint, can also retain two-ended local gap and color regularity, at a fixed fraction of its η -moment, uniformly for $h_{\text{pre}} \leq h \leq C_{\text{res}}\sqrt{m}$. After a fixed reserve adjustment its curve and regularity hold in physical time.

Proof. Let D read only the first $j \leq m/2$ gaps and marks. Write the positive part of the signed prefix height in unit bins $[r, r+1)$. Discarding initial survival, the necessary remaining wall has reserve at most $h+r+1$. The one-end upper gives

$$F_{m-j}(h+r+1) \leq CF_m(h)(1+r)^{2\chi_{\max}},$$

uniformly for all $h \geq h_0$. If p_r is the private prefix mass of D and that bin, independence and fractional subadditivity give

$$\mathbb{E}q_D^\eta \leq CF_m(h) \sum_{r \geq 0} (1+r)^{2\chi_{\max}} \mathbb{E}p_r^\eta.$$

Concavity and Cauchy–Schwarz imply

$$\mathbb{E}p_r^\eta \leq \mathbb{P}(D, \text{bin } r)^\eta \leq \mathbb{P}(D)^{\eta/2} \mathbb{P}(H_j^+ \geq r)^{\eta/2},$$

with the evident adjustment for $r = 0$. The signed increments have uniform exponential moments and prefix mean $O(\sqrt{j})$. Their upper tail above a fixed multiple of $\sqrt{j}$ is $C \exp\{-c \min(r^2/(1+j), r)\}$. Summation proves

$$\mathbb{E}q_D^\eta \leq C(1+j)^C \mathbb{P}(D)^{\eta/2} F_m(h). \quad (8.5)$$

This treats the unbounded positive signed heights in the negative-jump orientation as well.

For all rank intervals $0 \leq i < j \leq m$ require

$$\left| \sum_{i < l \leq j} (E_l - \tau) \right| \leq \epsilon_0(j-i) + K_r[1 + d_m(i)^\zeta + d_m(j)^\zeta],$$

$$|N_a(i, j) - \pi_a(j-i)| \leq \epsilon_0(j-i) + K_r[1 + d_m(i)^\zeta + d_m(j)^\zeta].$$

A fixed exponential tilt bounds the probability of any failure D_{ij} by

$$C \exp\{-c(j-i) - cK_r[1 + d_m(i)^\zeta + d_m(j)^\zeta]\}.$$

For $j \leq m/2$, use (8.5); its polynomial is summable against the depth terms. For $i \geq m/4$, retain necessary survival on the first $\lfloor m/8 \rfloor$ steps, independent of the failure, and apply the one-end upper. Otherwise $j-i > m/4$; drop survival, using $F_m(h) \geq c(1+m)^{-\chi_{\max}}$ to absorb its reciprocal and the $O(m^2)$ choices into the length tail. Fractional subadditivity yields

$$\mathbb{E}q_{\text{REG bad}}^\eta \leq Ce^{-cK_r} F_m(h). \quad (8.6)$$

This is relative to the rare survival scale. If q is the retained strong one-end mass, then $(q - q_{\text{bad}})^\eta \geq q^\eta - q_{\text{bad}}^\eta$. Choose K_r after its lower constant. At least half the moment and all its endpoint restrictions remain.

Here is the conversion to physical time. Endpoint gap inequalities imply $i \leq C(T_i + B_K)$ and $m - i \leq C(T_m - T_i + B_K)$. At a physical interval's endpoints use the adjacent outer arrival indices. Rank depths are at most the corresponding physical depths times a fixed constant plus C_K . Applying gap regularity to the complete intervening gaps and the two endpoint gaps gives

$$|\Delta t - \tau \Delta \text{rank}| \leq C\epsilon_0 \Delta \text{rank} + C_K[1 + d_{T_m}(s)^\zeta + d_{T_m}(t)^\zeta].$$

Eliminate the rank difference using $\tau \geq \tau_-$ and choose ϵ_0 small. Combining with the color inequality gives any prescribed positive physical slope. One endpoint mark changes a count by at most one.

For the strong curve the right endpoint of the gap following rank j is at most $C(j + 1) + C_K[1 + (j + 1)^\zeta]$. Choose its leading rank-curve coefficient before K_r . The resulting physical-curve conversion error is bounded by $\epsilon_{\text{curve}} j^\gamma + C'_K$, the first term being paid by that coefficient. Apply the rank construction with reserve $h - C'_K$, including the fixed adverse pre-jump adjustment. For $h \geq 2C'_K + h_{\text{pre}}$ its moment is comparable to $F_m(h)$; adding C'_K back to the actual risk pays the intercept. A fixed length threshold keeps this adjustment inside the positive inner endpoint box. No individual gap has been bounded deterministically. $\square$

Proof of Theorem 8.1. Write $B(G; x)$ for the flat exact-type mass. For its upper, if both reserves exceed $\sqrt{n}$, use $B \leq 1$. If both are at most $2\sqrt{n}$, expose the first and reversed last $m = \lfloor n/8 \rfloor$ marks and their distinct gaps. The two reference drifts are opposite, with

$$\delta = \varepsilon(w_+ - b\pi - c\tau) = \frac{y - x + \varepsilon c\tau}{n}, \quad (8.7)$$

and bounded scaled magnitude. The type part of the joint density (B.1) bounds the private mass pointwise by C_{QLQR} . Its gap part transfers the η -moment to independent reference ends. Theorem 6.11 gives

$$\mathbb{E}B^\eta \leq C \min\{1, ((1+x)/\sqrt{n})^{2\chi}\} \min\{1, ((1+y)/\sqrt{n})^{2\chi}\} \leq CQ_n(x, y).$$

If $x \leq \sqrt{n}$ and $y > 2\sqrt{n}$, then $\delta \asymp y/n > 0$, apart from bounded n . Expose only a prefix of length $m = \lfloor c_0 n^2 / (1+y)^2 \rfloor$. When this exceeds a fixed scalar threshold, it is at most $n/4$ and $\delta\sqrt{m} \leq C$. Dropping all subsequent tests and using (B.1) and the one-end upper gives

$$\mathbb{E}B^\eta \leq C \min\{1, ((1+x)/\sqrt{m})^{2\chi}\} \leq CQ_n(x, y).$$

If m is bounded, $y \gg n$, and $Q_n(x, y) \gg 1$; the trivial upper suffices. Reverse the entire bridge for the opposite asymmetric case, reversing all $n + 1$ gaps and exchanging the signs and reserves. All remaining bounded n are immediate.

For the lower, first restrict $h_0 \leq x, y \leq C_{\text{res}}\sqrt{n}$, where C_{res} is any fixed constant. On the same two exposed ends use Lemma 8.2 and the strong one-end lower. In the positive-jump direction the rank reserve $x - w_+$ enforces every forward pre-jump value; in the reverse direction minima occur after the negative marks. Choose the endpoint boxes wide and compatible after C_{res} , the laws, and the curves. The two actual inner heights are in $[a\sqrt{n}, A\sqrt{n}]$; separate outer gap sums and color counts are central.

For every retained complete outer state the middle has $r = n - 2m$ marks and $r + 1$ gaps, a compact gap mean and interior type. If its duration is T' and its endpoints are x_L, y_R , its mean pre-jump height before mark $j + 1$ is exactly

$$(1 - j/r)x_L + (j/r)y_R - \varepsilon \frac{cT'(r - j)}{r(r + 1)}, \quad 0 \leq j < r. \quad (8.8)$$

The final endpoint has height y_R . The correction is $O(1)$ for both signs. Lemma B.2, separately for its $r + 1$ gaps and r colors, gives a fixed positive probability of remaining above $a\sqrt{n}/2$, which pays the global curve gP^γ for large n .

An ordinary exact-type bridge has uniformly small regularity failure:

$$\mathbb{P}(\text{REG}_K^c) \leq Ce^{-cK}. \quad (8.9)$$

Indeed its two colors are independent sets of uniform points with fixed respective counts. Each interval count is binomial of mean at most $C|I|$. Bernstein gives a bound $C \exp\{-c|I| - cK[1 + d(s)^\zeta + d(t)^\zeta]\}$ for its failure. Bounded grid density and $\sum_{l \geq 0} e^{-cl^\zeta} < \infty$ sum all pairs. Add the piece endpoints to the grid if necessary. Choose K so the failure is less than half the middle's constant success probability. Piecewise empirical centers differ from the full centers by $O(n^{-1/2})$; start with a smaller slope, absorb this difference for large n , and split a tested interval at the at most two junctions. Piece depths are no larger than global depths, so the finitely many junction terms are absorbed by the intercept.

Conditional on the outer clocks put $a = q_L q_R$, their retained reference private mass. The complete private success mass U , as a function of the middle clock, obeys

$$0 \leq U \leq Ca, \quad \mathbb{E}_{\text{middle}} U \geq ca.$$

The first inequality uses the all-support type density upper. The second uses its central lower and the constant ordinary middle success for each complete outer word. Therefore

$$\mathbb{E}_{\text{middle}} U^\eta \geq (Ca)^{\eta-1} \mathbb{E}_{\text{middle}} U \geq c' a^\eta, \quad (8.10)$$

with zero interpreted separately. Transfer the one joint outer gap density and average the independent reference ends. The result is $\gg F_n(x)F_n(y) \asymp Q_n(x, y)$ in this fixed central box.

If both reserves exceed $\sqrt{n}$, a small tube for the whole exact bridge remains above half its chord, pays the global curve, and has constant ordinary probability. Delete ordinary regularity failure using (8.9). Its fractional moment is at least its ordinary probability, proving the saturated case.

It remains, after reflection, that $x < \sqrt{n}$ and $y > K_{\text{asym}}\sqrt{n}$. The drift (8.7) is positive and comparable to y/n . For small δ take

$$m = \lfloor c_1 \delta^{-2} \rfloor, \quad x' = \min(x, C_2 \sqrt{m}), \quad y' = y - x + x'.$$

Lowering the entrance this way lowers the whole path. Choose K_{asym} large enough that $m \leq n/8$. Only the one-end threshold h_{pre} is required for x' . The regular strong one-end family retains

$$T_{\text{pre}} \asymp m, \quad a\sqrt{m} \leq H \leq A\sqrt{m}, \quad |T_{\text{pre}} - m\tau| + |K_{\text{pre}} - m\pi| \leq C\sqrt{m},$$

and has moment

$$\mathbb{E} q_{\text{pre}}^\eta \gg F_m(x') \asymp Q_n(x, y). \quad (8.11)$$

The comparison follows from $m^{-1/2} \asymp \delta \asymp y/n$; when entrance clipping occurs both sides are bounded below. For each retained complete prefix set

$$\begin{aligned} n' &= n - m, & P' &= P - T_{\text{pre}}, & k' &= k - K_{\text{pre}}, \\ M' &= (n' - k')w_+ + k'w_-, & v' &= (y' - H)/P' = \varepsilon(M'/P' - c). \end{aligned}$$

The errors are $O(\delta^{-1})$, whereas $y' \asymp n\delta$. Increasing K_{asym} makes $n\delta^2$ uniformly large and yields

$$v' \asymp \delta, \quad Hv' \geq a_0 > 0, \quad P'(v')^2 \geq R_{\text{tail}}. \quad (8.12)$$

Residual type and gap parameters stay compact. These are estimates for every retained prefix, not assertions about a typical one.

Use the positive compact speed $c_{\text{half}} = c + \varepsilon v'/2 = (c + M'/P')/2$. On the same remaining word,

$$H/2 + \varepsilon(S_{\text{rem}}(s) - c_{\text{half}}s) = Z_{\text{rem}}(s) - (H + v's)/2.$$

Lemma B.3, with entrance $H/2$, drift $v'/2$ and fixed scaled reserve $a_0/4$, gives constant ordinary probability of $Z_{\text{rem}}(s) \geq (H + v's)/2$. By (8.12),

$$(H + v's)/2 \geq c_*\sqrt{m}(1 + s/m), \quad d_P(T_{\text{pre}} + s)^\gamma \leq C_*m^\gamma(1 + s/m)^\gamma.$$

For m above a fixed threshold this pays the entire remaining curve, uniformly even when n/m is arbitrarily large. Retain ordinary tail regularity by (8.9), and join it to the prefix. Residual rates differ from full rates by $O(\sqrt{m}/n + 1/n) = O(n^{-1/2})$; the prefix's center conversion uses its fixed large effective-time threshold. Apply the single prefix density (B.1). The full mass again satisfies $0 \leq U \leq Cq_{\text{pre}}$ and $\mathbb{E}_{\text{rem}}U \geq cq_{\text{pre}}$, where the whole exact remaining clock and type are integrated. Equation (8.10) and (8.11) prove the lower.

If this effective m is bounded, $\delta \geq \delta_0 > 0$ and, for large n , the physical drift $v = (y - x)/P \geq v_0 > 0$. Apply the half-chord construction to the whole bridge. Choose the final h_0 so $h_0v_0/4 \geq 1$ and

$$h_0/2 + (v_0/2)t \geq gt^\gamma \quad (t \geq 0).$$

The finite supremum of $gt^\gamma - v_0t/2$ permits this choice. Continuation and ordinary regularity have constant joint probability; $Q_n(x, y)$ is also of constant order. This avoids using a rare-prefix estimate at one step.

For clarity, first fix the regular strong one-end family, its own threshold, and c_1, C_2 . Its endpoint boxes fix a_0 . Then choose continuation constants and K_{asym} , then the central-box constants, then a large effective-time threshold exceeding the curve and prefix thresholds and $(h_{\text{pre}}/C_2)^2$. This fixes v_0 . Only now choose the final h_0 , the ordinary middle/tail regularity intercepts, and a large horizon threshold. For the remaining bounded n , enlarge h_0, K so the bounded possible excursions, curve and counts are covered. This last enlargement changes no internal prefix construction, which required only h_{pre} . Finally, prescribed rate differences $O(n^{-1/2})$ are absorbed into an initially smaller slope at large n , and into K at bounded n . $\square$

8.2 Whole inherited measures and endpoint boxes

The next assertion concerns a measure fixed before the new clock is read. It includes old private words and the complete terminal type window in the same integral.

Theorem 8.3 (A whole inherited measure on one clock). *Fix common $n, P, w_+, w_-, c, \varepsilon$ as above. Let μ be a finite nonnegative measure on old states and exact types, independent of the new clock G . Each state has legal endpoints satisfying*

$$X \leq x \leq C_{\text{box}}X, \quad Y \leq y \leq C_{\text{box}}Y, \quad \varepsilon \leq k/n \leq 1 - \varepsilon,$$

where X, Y exceed a sufficiently large fixed threshold. Let $W = \mu(1)$ and integrate the conditional private masses over this whole measure, before taking the power. Put $T = (1 + X)(1 + Y)/n$. For a sufficiently small fixed $t_0 > 0$, define

$$\Psi_{\text{box}} = \begin{cases} 1, & T \geq t_0, \\ T^{2\chi(\eta, \rho_0)}, & T < t_0, \end{cases} \quad q_0 = \frac{w_+ - cP/n}{b}, \quad \rho_0 = \frac{(c\tau)^2}{(c\tau)^2 + b^2q_0(1 - q_0)}. \quad (8.13)$$

In the second branch with $W > 0$, q_0 lies in a fixed compact subinterval of $(0, 1)$. The whole flat mass $q(G)$ satisfies

$$\mathbb{E}q(G)^\eta \asymp W^\eta \Psi_{\text{box}}. \quad (8.14)$$

The same comparison holds when every constituent bridge retains the strong curve and regularity of Theorem 8.1. For $W = 0$ both sides are zero.

Proof. Time reversal reduces to $\varepsilon = 1$. First observe a pointwise coordinate monotonicity on the same clock. If two lawful pairs satisfy $x_2 \geq x_1$ and $y_2 \geq y_1$, couple their low subsets using the first k_1 and first k_2 indices of one uniform permutation. The endpoint relation gives $b(k_2 - k_1) = (x_2 - x_1) - (y_2 - y_1)$. If $k_2 \geq k_1$, the path difference lies between the two nonnegative endpoint differences. If $k_2 < k_1$, it is the entrance difference plus b times the number of changed low marks already seen. Thus

$$B_{k_1, x_1}(G) \leq B_{k_2, x_2}(G) \quad \text{whenever } x_1 \leq x_2, y_1 \leq y_2. \quad (8.15)$$

It holds at every physical time.

In the saturated branch $q \leq W$. With $\nu = \mu/W$, concavity gives

$$\mathbb{E}q^\eta \geq W^\eta \int \mathbb{E}B_{k,x}^\eta d\nu.$$

Each integrand is bounded below by Theorem 8.1, since its endpoint product is at least t_0 . This also proves the strong/regular lower there.

In the rare branch the endpoint identity gives

$$|k/n - q_0| \leq C(X + Y)/n \leq C'T.$$

Choose t_0 so this forces $q_0 \in [3\varepsilon/4, 1 - 3\varepsilon/4]$. Set $U = nw_+ - cP$ and choose lawful enclosing bridges

$$\begin{aligned} x_- &= X/2, & k_- &= \left\lfloor \frac{x_- + U - Y/2}{b} \right\rfloor, & y_- &= x_- + U - bk_- \in [Y/2, Y/2 + b), \\ x_+ &= 2C_{\text{box}}X, & k_+ &= \left\lfloor \frac{x_+ + U - 2C_{\text{box}}Y}{b} \right\rfloor, & y_+ &= x_+ + U - bk_+ \in [2C_{\text{box}}Y, 2C_{\text{box}}Y + b). \end{aligned}$$

Choose the reserve threshold so $X/2, Y/2$ meet the bridge theorem and $b \leq Y/2$. Then $x_- \leq x \leq x_+$ and $y_- \leq y \leq y_+$. Their type proportions differ from q_0 by $O((X + Y)/n + 1/n) = O(T)$; after

decreasing t_0 and increasing the fixed length threshold they lie in $[\epsilon/2, 1 - \epsilon/2]$. The floor choices are therefore genuine admissible types. Equation (8.15) gives, on the one actual clock,

$$WB_-(G) \leq q(G) \leq WB_+(G).$$

Their endpoint products are comparable to T . The local Lipschitz estimate for χ gives $|\chi_\pm - \chi_0| \leq CT$. Since $T|\log T|$ is bounded, both bridge moments are comparable to Ψ_{box} . Taking the power proves the flat comparison. The same exponent argument makes every constituent $Q_n(x, y)$ comparable to Ψ_{box} . Normalized-measure concavity and Theorem 8.1 give the strong/regular lower; the upper drops these extra conditions. No monotonicity of regularity under the subset coupling is needed. $\square$

Corollary 8.4 (A central terminal band). *Let an old-state measure κ be independent of G and have mass m_κ . In state ω , let the admissible private counts form an integer interval J_ω containing $\asymp r$ integers, $r \geq 1$, all with $|k - n\beta| \leq C_{\text{type}}\sqrt{n}$. Assume the common endpoint boxes of Theorem 8.3. Give type k its original mass $p_k = \mathbb{P}\{\text{Bin}(n, \beta) = k\}$ and let q be the entire old-state and terminal-band mass. Then, with*

$$\rho_\beta = \frac{(c\tau)^2}{(c\tau)^2 + b^2\beta(1 - \beta)},$$

one has

$$\mathbb{E}q^\eta \asymp m_\kappa^\eta \min(1, r/\sqrt{n})^\eta \min(1, T^{2\chi(\eta, \rho_\beta)}).$$

This also holds with the strong curve and regularity.

Proof. Uniform central Stirling gives $\sum_{k \in J_\omega} p_k \asymp \min(1, r/\sqrt{n})$ in every state. Thus $W \asymp m_\kappa \min(1, r/\sqrt{n})$. In the rare branch, $|q_0 - \beta| \leq C(T + n^{-1/2})$ and $T \geq c/n$. The exponent change times $|\log T|$ is bounded by $C(T|\log T| + n^{-1/2} \log n)$. In the other branch the factors are of constant order. Apply Theorem 8.3. $\square$

For clarity, the count-resolved version has a direct formulation. On a native block of rate ν , duration L , high mark a , low mark $a - b$, and child probability β , put $A = a - b\beta$ and $e = \nu A - c$. Given $N = n$ and entrance x , let

$$v_k = x + an - bk - cL, \quad M_n(x, I) = \sum_{k: v_k \in I} \mathbb{P}\{\text{Bin}(n, \beta) = k\}.$$

The private terminal kernel has the exact expansion

$$K_G(x, I) = \sum_{k: v_k \in I} \mathbb{P}\{\text{Bin}(n, \beta) = k\} B_k(G; x). \quad (8.16)$$

For a nonempty band $I = [y, y + \Delta]$, with $2b \leq \Delta \leq C(1 + y)$, central selected types, and $h_0 \leq x \leq C_{\text{res}}\sqrt{n}$ and $h_0 \leq y \leq y + \Delta \leq C_{\text{res}}\sqrt{n}$, monotonicity sandwiches this mass between $M_n B_{k_{\text{max}}}$ and $M_n B_{k_{\text{min}}}$. Consequently

$$\mathbb{E}_{G|n} K_G(x, I)^\eta \asymp \min(1, \Delta/\sqrt{n})^\eta \min\{1, ((1 + x)(1 + y)/n)^{2\chi_*}\}. \quad (8.17)$$

The exponent may be evaluated at either extreme selected type. Their proportions differ by $O(n^{-1/2})$, so its Lipschitz change times $\log n$ is bounded. Sufficient count-resolved conditions are

$$|n - \nu L| \leq C_N \sqrt{L}, \quad |eL| \leq C_e \sqrt{L}, \quad h_0 \leq x, y, \quad x, y + \Delta \leq C_{\text{res}} \sqrt{n}.$$

Indeed $k - \beta n = [x + A(n - \nu L) + eL - v_k]/b = O(\sqrt{L})$. Then $\tau = L/(n + 1) = \nu^{-1} + O(L^{-1/2})$ and $\pi = \beta + O(L^{-1/2})$; one may use $\rho = A^2/(A^2 + b^2\beta(1 - \beta))$ in (8.17). Every selected type is central here; an uncolored central count alone would not verify that condition.

8.3 Allocating a fixed fraction of the endpoint budget

Corollary 8.5 (Contracted chords). *On the same exact bridge write*

$$Z(t) = \ell(t) + B(t), \quad \ell(t) = (1 - t/P)x + (t/P)y, \quad B(t) = \varepsilon\{S(t) - (t/P)M\}, \quad M = nw_+ - bk.$$

For $\theta \in [\theta_0, 1]$, $\theta_0 > 0$, replacing the wall condition in (8.4) by $\theta\ell(t) + B(t) \geq gdp(t)^\gamma$ preserves its comparison to the original $Q_n(x, y)$, after increasing the fixed reserve threshold. The whole-measure version of Theorem 8.3 also holds, with θ allowed to depend on the old state and type, provided that measure and selection remain independent of G .

Proof. The exact same-word identity is

$$\theta\ell(t) + B(t) = \theta x + \varepsilon\{S(t) - c_\theta t\}, \quad c_\theta = \theta c + (1 - \theta)M/P.$$

This speed is a convex combination of two positive compact rates. Theorem 8.1 applies with endpoints $\theta x, \theta y$ and unchanged type, clock and regularity centers. If $T = (1 + x)(1 + y)/n$ is bounded below, both its factor and the original factor are bounded below. Otherwise

$$|c_\theta - c| = (1 - \theta)|y - x|/P \leq C(x + y)/n \leq CT.$$

Lipschitz continuity gives $|\chi_\theta - \chi| \leq CT$, while $\theta_0^2 T \leq (1 + \theta x)(1 + \theta y)/n \leq T$. Boundedness of $T|\log T|$ proves the factor comparison. For the whole measure, the contracted event is a subset of the original event since $\ell \geq 0$, proving the upper. Normalize the measure, apply concavity, and integrate the individual lower for the lower. This changes an auxiliary path inequality, without changing any original type probability or likelihood. $\square$

There is also a literal complete-cell version of the flat wall and regularity statement. Requiring them only at cell boundaries enlarges the continuous lower event. Conversely, if widths are bounded by $\Delta_{\max}$, a boundary wall implies continuous survival after increasing both reserves by $c\Delta_{\max}$: use the preceding boundary for positive jumps and the following boundary for negative jumps. The new Q is comparable to the old one. The boundary tests read only cell/color counts; their conditional private mass depends only on cell occupancies. Projection of uniform positions to these occupancies therefore preserves that mass and its fractional moment exactly.

8.4 Ordinary bridges with physical curves

The terminal ordinary factor has no private binary variance. We establish its path estimate separately rather than taking a degenerate limit of ρ in a fractional theorem.

Lemma 8.6 (Flat constant-mark bridge). *Let n uniform points lie in $[0, P]$, and let $Z(t) = x + \varepsilon(wN(t) - ct)$ with legal endpoint $y = x + \varepsilon(nw - cP)$. For w, c in fixed positive compact ranges,*

$$\mathbb{P}\{Z(t) \geq 0 \ (0 \leq t \leq P)\} \asymp \min\{1, xy/(1 + n)\}, \quad x, y \geq \max(w, 1).$$

For arbitrary nonnegative endpoints its upper holds with $(1 + x)(1 + y)$ in place of xy .

Proof. It suffices to take $\varepsilon = 1$. Scale height by w and time by w/c , and put $l = cP/w$, $a = x/w$, $b' = y/w = a + n - l$. Reversing the ordered uniform points gives precisely the constraints $\xi_j \geq (j - b')/l$. By [19, Lemma 8.3, pp. 53–54], for integer $n \geq 1$ and real $u, v \geq 1$ with $u + v - n \geq 1$,

$$n! \text{vol}\{0 \leq \xi_1 \leq \dots \leq \xi_n \leq 1 : \xi_j \geq (j - u)/v\} \asymp \min\{1, u(u + v - n)/n\}.$$

Use $u = b'$, $v = l$, so the other endpoint is a . When $l < 1$ and $a, b' \geq 1$, survival is automatic and the displayed factor is bounded below. The case $n = 0$ is deterministic. Raising both reserves by w proves the upper for arbitrary nonnegative endpoints, with the same elementary small-duration cases. Reflection proves the negative-jump assertion and reverses all $n + 1$ gaps. $\square$

Theorem 8.7 (Ordinary strong bridge with all reserves). *Fix positive compact w, c, τ , put $P = (n+1)\tau$, and take $n \geq 1$ uniform points in $[0, P]$. Let $Z(t) = x + \varepsilon(wN(t) - ct)$ have legal endpoint y . Fix $g > 0$, $0 < \zeta < \gamma < 1/2$, a regularity slope $\vartheta > 0$, and a grid of bounded boundary density. There are fixed h_0, K such that, for all legal $x, y \geq h_0$ and both signs, let $\mathcal{A}_{x,y}$ be the joint event*

$$\begin{aligned} Z(t) &\geq gd_P(t)^\gamma \quad (0 \leq t \leq P), \\ |N(t) - N(s) - (n/P)(t-s)| &\leq \vartheta(t-s) + K[1 + d_P(s)^\zeta + d_P(t)^\zeta] \quad \text{for all grid pairs.} \end{aligned}$$

Then

$$\mathbb{P}(\mathcal{A}_{x,y}) \asymp \beta_n(x, y) := \min\{1, (1+x)(1+y)/(1+n)\}. \quad (8.18)$$

The regularity may hold at all physical pairs. Rates within $C/\sqrt{1+n}$ of n/P are also permitted. The same result holds for a fixed contracted chord $\theta\ell + B$, with $\theta \in [\theta_0, 1]$.

Proof. The upper is Lemma 8.6. First prove the lower in a fixed central box $h_0 \leq x, y \leq C_{\text{res}}\sqrt{n}$. Rescale to mark and speed one, changing only compact constants and curve parameters. Write the duration as L and $\lambda = n/L$. Then $|\lambda - 1| \ll n^{-1/2}$ and $L \asymp n$. The uniform clock is a rate- λ Poisson process conditioned on $N(L) = n$; this single count atom has size $\asymp n^{-1/2}$.

Two elementary consequences of Lemma 8.6 are

$$K_\lambda(t; u, v) \leq \frac{C(1+u)(1+v)}{(1+t)^{3/2}}, \quad (8.19)$$

$$\mathbb{P}_u^{(1)}(\text{survival to } t) \leq C \min\{1, (1+u)/\sqrt{1+t}\}. \quad (8.20)$$

Here K is a killed Poisson endpoint atom, zero unless its count is attainable, and both orientations are allowed. For counts at least $t/2$, multiply the conditional ballot upper by the Poisson mode $C/\sqrt{1+t}$ and retain its denominator. Smaller counts have an exponential Poisson lower tail; bounded t is immediate. To prove (8.20), sum the ballot upper at rate one: the positive part of the terminal height has expectation at most $u + \sqrt{t} + 1$. For $u > \sqrt{t}$ use the trivial bound.

On a first-half dyadic annulus $[d, 2d]$, a strong-wall failure while surviving flat entails a first visit to height at most $l = g(2d)^\gamma + O(1)$. At that stopped path the likelihood relative to rate one is

$$\exp\{\varepsilon(Z(t) - x) \log \lambda - (\lambda - 1 - \log \lambda)t\}.$$

It is bounded by a constant depending on C_{res} : $|\log \lambda| \ll n^{-1/2}$, $x \leq C_{\text{res}}\sqrt{n}$, and $l \ll n^\gamma$. Equation (8.20) until the visit, followed by the remaining atom (8.19), and division by the one central count atom and the flat-ballot lower, give a relative loss at most

$$C_{C_{\text{res}}}(1+l)/\sqrt{d}. \quad (8.21)$$

Explicitly, before division by the flat probability the bridge loss is at most $C(1+x)(1+y)(1+l)/(n\sqrt{d})$, and its ratio to $\beta_n(x, y)$ is bounded in this central range.

For positive jumps no such visit occurs before $h_0/2$ once $g(h_0/2)^\gamma \leq h_0/4$. For negative jumps a visit before $h_0/2$ requires at least $3h_0/4 + (x - h_0) + O(1)$ arrivals, whereas the mean is at most $11h_0/20$ for large n . Chernoff and (8.19) give relative loss $C(1+gh_0^\gamma)e^{-ch_0}$. Reverse the argument

at the right end. The remaining annuli sum to $C_{C_{\text{res}}}(gh_0^{\gamma-1/2} + h_0^{-1/2})$. Choose h_0 so the total loss is less than one quarter.

Regularity is retained relatively to this same flat event. For a first-half interval failure $A_{s,t}$, condition on its prefix and apply (8.19) to the remaining endpoint atom. After dividing by the count atom and flat probability,

$$\frac{\mathbb{P}_{\text{bridge}}(\text{flat}, A_{s,t})}{\mathbb{P}_{\text{bridge}}(\text{flat})} \leq C_{C_{\text{res}}} \frac{\mathbb{E}_{\lambda}[(1 + Z(t))_+ \mathbf{1}_{A_{s,t}}]}{1 + x}.$$

A fixed Chernoff parameter gives

$$\mathbb{P}_{\lambda}(A_{s,t}) \leq 2e^{-c(t-s)-cK(1+s^{\zeta}+t^{\zeta})}.$$

The second moment of $(1 + Z(t))_+$ is at most $C(1 + x + t)^2$. Cauchy-Schwarz makes the last two displays summable over first-half grid pairs, with total Ce^{-cK} . Reverse for the last half. For crossing pairs use their exact binomial interval law: either the length is at least $L/4$ or both endpoint depths are at least $L/4$. Its exponential tail absorbs the $O(L^2)$ pairs and the reciprocal flat probability, which is at most $C(1 + n)$. Hence

$$\mathbb{P}_{\text{bridge}}(\text{flat}, \text{REG}_K^c) \leq Ce^{-cK} \mathbb{P}_{\text{bridge}}(\text{flat}). \quad (8.22)$$

Choose K after the preceding strong-wall constant. This proves the central case by relative deletion.

We need an independent-gap prefix with deterministic marks. For m independent exponential gaps, suppose $|\varepsilon(w - c\tau)|\sqrt{m} \leq D$ and $h_{\text{pre}} \leq h \leq C_{\text{pre}}\sqrt{m}$. There is an event of probability

$$\gg \min\{1, (1 + h)/\sqrt{1 + m}\} \quad (8.23)$$

on which the prefix has a strong one-ended physical curve, local regularity, central total duration and actual endpoint in $[a\sqrt{m}, A\sqrt{m}]$. Here is a direct construction. Conditional on $T_m = t$, the first $m - 1$ arrivals are uniform on $[0, t]$ and the genuine m th mark is at the right endpoint. Choose an actual endpoint H in a sufficiently wide fixed positive $\sqrt{m}$ box and set $t = (mw - \varepsilon(H - h))/c$. These durations occupy a central Gamma window of width $\asymp \sqrt{m}$ and thus constant probability by Stirling. Its constants are chosen after the reserve and drift bounds. The inner bridge has $m - 1$ marks, m gaps and pre-jump endpoint $H - \varepsilon w$. Subtract a linear height $a'\sqrt{m}t'/t$, where $0 < a' < a/2$. This changes the speed by $\varepsilon a'\sqrt{m}/t$ and leaves positive central endpoints. The central result just proved gives the order in (8.23). In the first half its strong two-ended curve suffices; in the second half the added linear height is at least $a'\sqrt{m}/2$ and dominates the one-ended curve. The genuine final mark and its bounded endpoint adjustment are included before fixing h_{pre} . This proves the prefix claim without private-mark persistence.

For the complete exact clock, exposing its first m gaps has density

$$\frac{f_{n+1-m,\tau}(P - T_m)}{f_{n+1,\tau}(P)}.$$

For $m \leq n/8$ this has an all-support upper and, for sufficiently large n , a positive lower on $|T_m - m\tau| \leq C\sqrt{m}$. These are the exact-reference-mean bounds of Lemma 6.12; the lower is used only above its support and length threshold.

If $x, y \geq \sqrt{n}$, use on the same clock the surrogate speed $c_0 = nw/P$ and endpoints $x_0 = y_0 = \sqrt{n}$. Its centered path is $\sqrt{n} + \varepsilon w(N(t) - nt/P)$. The central result has constant probability, and its event is a subset of the desired event since the actual chord is larger.

Otherwise, outside a sufficiently large fixed central box, reverse to $x < \sqrt{n}$, $y > K_{\text{asym}}\sqrt{n}$. The reference drift $\delta = \varepsilon(w - c\tau) = (y - x + \varepsilon c\tau)/n$ is positive. For small δ set $m = \lfloor c_1 \delta^{-2} \rfloor$ and $x' = \min(x, C_2 \sqrt{m})$, $y' = y - x + x'$. Choose K_{asym} so $m \leq n/8$. The prefix just constructed has probability $\asymp \min(1, (1 + x')/\sqrt{m}) \asymp \beta_n(x, y)$ and retains $T_m \asymp m$, $H \asymp \sqrt{m}$ and a central gap sum. For every retained prefix, the remaining duration $P' = P - T_m$ and physical drift $v' = (y' - H)/P'$ satisfy

$$v' \asymp \delta, \quad Hv' \geq a_0 > 0, \quad P'(v')^2 \geq R_{\text{tail}}.$$

The estimates follow exactly from the $O(\sqrt{m})$ prefix errors and $n\delta^2 \gg 1$. Apply Lemma B.3 with equal marks to the residual clock, speed $(c + (n - m)w/P')/2$, entrance $H/2$ and drift $v'/2$. It has constant probability of the half-chord bound

$$Z_{\text{rem}}(s) \geq (H + v's)/2 \geq c_* \sqrt{m}(1 + s/m).$$

Since $d_P(T_m + s)^\gamma \leq C_* m^\gamma (1 + s/m)^\gamma$, a fixed large effective-time threshold pays the whole global curve. Ordinary exact-count regularity has failure Ce^{-cK} , so retain it at a constant cost. Prefix and tail center discrepancies are $O(m^{-1/2})$ and $O(\sqrt{m}/n)$; reserve a small slope for them and join the inequalities at the junction. Integrate the residual ordinary success and then the single prefix density. This proves the asymmetric lower above that effective-time threshold.

Below it the drift is bounded below by $\delta_0 > 0$, and for sufficiently large n the whole physical drift is at least $v_0 > 0$. Use the same half-chord argument on the whole bridge, choosing the final h_0 so $h_0 v_0/4 \geq 1$ and $h_0/2 + v_0 t/2 \geq gt^\gamma$ for all t . The probability and β_n are then both of constant order. The order of constants is: prefix boxes and thresholds; continuation and asymmetric cutoff; effective-time threshold; then final h_0 , regularity intercepts and horizon threshold n_0 . Include the length-independent requirement

$$h_0 \geq 2 + \sup_{s \geq 0} (gs^\gamma - c_{\min}s).$$

For the remaining $n < n_0$, put all points in a fixed sufficiently short initial interval in the positive orientation. Before its end $Z(t) \geq h_0 - c_{\max}\delta_{\text{short}}$; after the last point $Z(t) = y + c(P - t)$ pays the global curve by the preceding requirement. This event has a fixed positive probability on the bounded range. Reflect it for negative jumps. Increase only the output regularity intercept to make regularity automatic there. This does not feed a reserve enlargement back into the effective-time cutoff.

To obtain all physical pairs, first use a unit mesh and sandwich an arbitrary interval between its inward and outward mesh approximations. Their lengths differ by at most two and endpoint depths by at most one, so the desired inequalities follow with a fixed larger intercept and an initially smaller slope. The permitted center change follows as in the binary case. Finally

$$\theta \ell(t) + B(t) = \theta x + \varepsilon(wN(t) - c_\theta t), \quad c_\theta = \theta c + (1 - \theta)nw/P.$$

This speed stays positive compact; apply the theorem with endpoints $\theta x, \theta y$. Its ordinary factor differs only by constants. $\square$

Corollary 8.8 (Fixed entrance and a nonnegative terminal surplus). *Fix $w > 0$, $g \geq 0$, $0 < \zeta < \gamma < 1/2$ and $\vartheta > 0$. There are fixed $h, K, H, \epsilon_0 > 0$ such that, for n uniform points on $[0, U]$ and $v = nw - U$ with $0 \leq v \leq \epsilon_0 U$, the joint event*

$$h + wN(t) - t \geq gd_U(t)^\gamma,$$

$$|N(t) - N(s) - (n/U)(t - s)| \leq \vartheta(t - s) + K[1 + d_U(s)^\zeta + d_U(t)^\zeta],$$

$$|N(t) - (n/U)t| \leq H + \vartheta d_U(t)$$

has probability $\asymp (1 + v)/(1 + U)$, for all sufficiently large U . The same holds on a fixed complete-cell mesh of bounded width.

Proof. Use Theorem 8.7 with endpoints $h, h + v$. First fix h , then choose ϵ_0 so $\epsilon_0 h$ is sufficiently small. Since $n \asymp U$, its factor is $\asymp (1 + v)/(1 + U)$. Take regularity slope initially below $\vartheta/2$. Setting $s = 0$ or $t = U$ and using the exact total bounds the endpoint discrepancy by $(\vartheta/2)d_U(t) + K'[1 + d_U(t)^\zeta]$. The inequality $K'z^\zeta \leq (\vartheta/2)z + H$ gives the third condition at no probability cost. A boundary-only wall implies a continuous flat wall after a fixed reserve increase, which proves the mesh upper; the continuous lower already implies the mesh lower. $\square$

The all-physical regularity also yields, for $T_0 = 0, T_{n+1} = P$,

$$|T_j - T_i - \frac{P}{n+1}(j - i)| \leq \vartheta_{\text{gap}}(j - i) + K_{\text{gap}}[1 + d_n(i)^\zeta + d_n(j)^\zeta], \quad d_n(i) = \min(i, n + 1 - i).$$

Indeed the rank-count difference at arrival endpoints is at most one. End-interval instances of physical regularity show that time and rank depths are comparable up to an additive constant, by absorbing Kz^ζ into a small linear term. Solve the resulting interval inequality; replacing P/n by $P/(n + 1)$ costs a fixed constant. This includes both boundary gaps.

Optional named regularity is compatible with the ordinary lower. Conditional on each retained uncolored regular clock, refine its slots by a uniform exact full type in a compact simplex. A category count in an interval is hypergeometric with mean $p_a N(I)$. Start with uncolored slope and intercept K_0 fixed. Then $N(I) \leq C|I| + CK_0[1 + d(s)^\zeta + d(t)^\zeta]$. Hypergeometric Bernstein, with a named intercept K_1 sufficiently larger than K_0 , bounds each failure by

$$C \exp\{-c|I| - cK_1[1 + d(s)^\zeta + d(t)^\zeta]\}.$$

Summation over grid pairs makes its probability as small as desired uniformly for every retained uncolored clock. Thus this refinement retains a fixed fraction before any outer mixture; it does not assert retention of an arbitrary additional private future test.

8.5 One root count and the ordinary terminal suffix

Theorem 8.9 (Joint ordinary root and path estimate). *Fix H , a compact product interval $[\sigma_-, \sigma_+] \subset (0, 1)$, $C_\zeta < \infty$, and $W \geq \log H$. Let the occupied alphabets be $H \geq a_1 > \dots > a_m \geq 2$, with positive lengths L_i , and put*

$$\nu_i = a_i^\sigma, \quad w_i = \log a_i, \quad c_i = a_i - 1, \quad e_i = \nu_i w_i - c_i, \quad V = \sum_i L_i, \quad D = \sum_i e_i L_i \geq 0.$$

Read independent rate- ν_i Poisson clocks in reverse block order $m, \dots, 1$, with jumps w_i and downward speed c_i , obtaining $R(0) = 0$ and $X = R(V) = \sum_i w_i N_i - \sum_i c_i L_i$. Let $E = D + \zeta$, where $|\zeta| \leq C_\zeta \sqrt{1 + D}$ is fixed independently of these clocks. For a largest block i_ , with a fixed tie rule, set*

$$M = L_{i_*}, \quad A = \sum_{i < i_*} L_i, \quad B = \sum_{i > i_*} L_i, \quad \beta_D(h) = \min \left\{ 1, \frac{(1 + h + B)(1 + h + D + A)}{1 + M} \right\}.$$

There are fixed V_0, h_0 such that, uniformly for $V \geq V_0$ and all $h \geq h_0$,

$$\mathbb{P}\{h + R(t) \geq 0 \ (0 \leq t \leq V), E - W \leq X \leq E\} \asymp (1 + V)^{-1/2} \beta_D(h). \quad (8.24)$$

The upper holds without the lower length threshold. The same estimate holds conditionally on any complete external state independent of the native clocks. That state may set h using private prefix labels; when the target is required to be uncolored, E is set by its uncolored data. The root count selected in the proof does not read h .

Proof. The all-count Poisson estimates are

$$\sup_n p_\mu(n) \ll (1 + \mu)^{-1/2}, \quad \sup_n \frac{p_\mu(n)}{1 + n} \ll (1 + \mu)^{-3/2}. \quad (8.25)$$

The first is Stirling. For the second, use $n \geq \mu/2$ to retain its denominator, and the exponential lower tail otherwise. At a fixed central distance from a sufficiently large mean, $p_\mu(n) \gg (1 + \mu)^{-1/2}$.

Condition on all counts except $N_* = N_{i_*}$ and write

$$B_{\text{act}} = \sum_{i > i_*} (w_i N_i - c_i L_i), \quad A_{\text{act}} = - \sum_{i < i_*} (w_i N_i - c_i L_i).$$

The largest block's actual entrance and exit reserves are $x = h + B_{\text{act}}$ and $y = h + X + A_{\text{act}}$. If either is negative there is no survival. Otherwise discard all other path tests and use Lemma 8.6. At most $1 + W/\log 2$ reservoir counts lie in the root window. The two bounds (8.25) separately give

$$C(1 + M)^{-1/2}, \quad C(1 + M)^{-3/2} [1 + h + (B_{\text{act}})_+][1 + h + D + |\zeta| + (A_{\text{act}})_+].$$

The side sums are independent and $\mathbb{E}|B_{\text{act}}| \ll 1 + B$, $\mathbb{E}|A_{\text{act}}| \ll 1 + A$ by their Poisson means and variances. Also $1 + D + |\zeta| \ll 1 + D$. Average the bounds separately, then take their minimum. Since $M \leq V \leq mM$, this is the upper in (8.24), with constants independent of h . No centrality condition has been imposed in this upper.

For the lower retain all other counts in the independent windows

$$|N_i - \nu_i L_i| \leq C_0 \sqrt{1 + L_i}, \quad i \neq i_*. \quad (8.26)$$

Choose C_0 so their joint mass is bounded below, using Chebyshev and the fixed number of blocks. For each retained vector select one count

$$N_* = \left\lfloor \frac{\sum_i c_i L_i + D + \zeta - \sum_{i \neq i_*} w_i N_i}{w_*} \right\rfloor. \quad (8.27)$$

Then $E - w_* < X \leq E$, hence the root window holds. The exact mean identity $\sum_i w_i \nu_i L_i = \sum_i c_i L_i + D$ gives

$$|N_* - \nu_* M| \leq C \left(1 + |\zeta| + \sum_{i \neq i_*} \sqrt{1 + L_i} \right) \leq C' \sqrt{1 + M},$$

since $D \ll V \ll M$. Choose V_0 so this is a nonnegative central count. Its pointwise probability is $\gg (1 + M)^{-1/2}$. On the complete selected vector,

$$|w_i(N_i - \nu_i L_i)| \leq C_1 \sqrt{1 + L_i} \quad \text{for all } i. \quad (8.28)$$

Conditioned on this vector, the original uniform positions of the blocks are independent. Neither the windows nor the selected root read h .

We next control their paths. The thresholds $\alpha_a = \log((a-1)/\log a)/\log a$ strictly increase with a , by Lemma 5.13. Choose fixed disjoint small threshold neighborhoods. At most one occupied block i_0 is near its threshold; if present, one can fix $g_0 > 0$ so

$$e_i \leq -g_0 \ (i < i_0), \quad e_i \geq g_0 \ (i > i_0), \quad |e_{i_0}| \leq g_0/8.$$

Away from all occupied thresholds, every present drift has magnitude at least some fixed $g_1 > 0$. These choices follow from finite threshold separation and $|e'_a(\sigma)| \leq H(\log H)^2$. A single occupied alphabet is included by its one near-threshold case.

For a block with outward positive drift at least g , sufficiently large L_i and (8.28) imply actual endpoint increment at least $gL_i/2$. From a fixed reserve, Lemma 8.6 gives constant survival probability, since $N_i \ll 1 + L_i$. For bounded L_i , both selected count and duration are bounded, so the possible loss is deterministically bounded. Concatenating at most H blocks gives an outward side path bounded below by a fixed $-K_{\text{side}}$ with constant conditional probability. Read positive groups forward and negative groups backward. This event is independent of the chosen $h \geq h_0$. Since $E \geq D - C_\zeta \sqrt{1 + D} \geq -C$, choosing h_0 once makes both endpoint reserves sufficient.

If i_0 is present, write L_0, A_0, B_0 for its length and the lengths on its two sides, and

$$A_{\text{mean}} = -\sum_{i < i_0} e_i L_i, \quad B_{\text{mean}} = \sum_{i > i_0} e_i L_i, \quad D = -A_{\text{mean}} + B_{\text{mean}} + e_0 L_0.$$

Here $A_{\text{mean}} \asymp A_0$, $B_{\text{mean}} \asymp B_0$. Their actual values differ by $O(\sqrt{1 + A_0})$ and $O(\sqrt{1 + B_0})$. The actual central junction heights are

$$x = h + B_{\text{act}}, \quad y = h + D + \zeta + \epsilon_{\text{root}} + A_{\text{act}}, \quad -W \leq \epsilon_{\text{root}} \leq 0.$$

For all $h \geq h_0$, choose h_0 so that

$$x \asymp 1 + h + B_0, \quad y \asymp 1 + h + D + A_0, \quad x, y \geq \log H. \quad (8.29)$$

Indeed $g_0 t - C\sqrt{1+t} \geq g_0 t/2 - C'$ and $D - C_\zeta \sqrt{1+D} \geq D/2 - C''$; the fixed remainders are absorbed while retaining at least $h/2$. This is uniform in an unbounded h .

The central count is at most $C(1 + L_0)$. Its conditional ballot probability is therefore at least

$$c\beta_0(h), \quad \beta_0(h) = \min\{1, (1 + h + B_0)(1 + h + D + A_0)/(1 + L_0)\}.$$

At count zero the positive endpoints suffice directly. The central bridge and two outward side events use disjoint positions after all counts are fixed; their junctions are the actual values above. Their joint conditional success is at least $c\beta_0(h)$.

If $i_* = i_0$, this is the desired factor. If $i_* < i_0$, then $A_{\text{mean}} \geq g_0 M$ and $L_0 \leq M$, whence $B_{\text{mean}} \geq 7g_0 M/8 + D$ and $B_0 \gg M$. If $i_* > i_0$, similarly $A_{\text{mean}} + D \geq 7g_0 M/8$ and $D + A_0 \gg M$. In either case $\beta_0(h_0) \gg 1$, so monotonicity in h supplies constant success, which dominates $\beta_D(h) \leq 1$. These cases also cover ties. If there is no near-threshold block, the negative group followed in reverse and the positive group followed forward have constant joint success. Their junction agrees because their exact increments add to X . All-negative nonempty groups are excluded by $D \geq 0$; empty one-sided groups cause no difficulty. Multiplying the conditional success by the constant outside-count mass and the one atom in (8.27) proves the lower. $\square$

With a fixed reserve $h = h_0$, Theorem 8.9 gives the intrinsic scales $(1 + B, 1 + D + A)$. Its $D = \zeta = 0$ specialization gives

$$(1 + V)^{-1/2} \min\{1, (1 + A)(1 + B)/(1 + M)\}.$$

For a single occupied block this is $(1 + V)^{-3/2}$. Thus root centering and ordinary survival use the same count even in the most restrictive case.

More generally, if $\mathcal{F}_{\text{pre}}$ contains the full old word and all its clocks, the theorem applies conditionally whenever the native clocks are still independent with the displayed laws. For every nonnegative $\mathcal{F}_{\text{pre}}$ -measurable weight A_0 ,

$$\mathbb{E}[A_0 \mathbf{1}_{\{\text{root}, \text{survival}\}}] \asymp \mathbb{E}[A_0 (1 + V)^{-1/2} \beta_D(h)].$$

This retains the rare shared prefix measure. It neither replaces h by its mean nor moves a native average through an earlier fractional power. The lower threshold $V \geq V_0$ cannot be removed for arbitrary allowed translates: at vanishing length a fixed negative translate can put the root window between all attainable counts. The fixed width $W \geq \log H$ ensures that the root selection always hits its lattice at large length.

9 Joint count constraints and the lower source

We prove the lower bound before making any use of the upper bound. There are three distinct measures in the argument: the original categorical law on fine words, the natural Poisson law on the original clocks, and the squarefree harmonic measure on prime tuples. The same bounded accepting density is retained when passing between them. Transport retains the density itself, and comparison classes retain their original conditional word law.

The construction proceeds from joint counts to paths, then transports the accepted fine-word measure and finally assigns actual primes. The finite reference and endpoint arguments are proved in Appendix G, the measure construction in Appendix H, and the full original-history estimates in Appendix I. The bridge upper estimates used to control local cost moments concern the scalar processes of the preceding sections and are independent of the arithmetic upper bound.

Theorem 9.1 (Actual lower at one fixed admissible lift). *For each fixed k there are constants $L_* = L_*(k) \geq 1$ and $c_k > 0$ such that, for every $\ell_i \geq \log 3$, putting*

$$L_i = \ell_i - \log 3, \quad L_i^+ = \max(L_i, L_*),$$

one has

$$K_k(\ell) \geq c_k \Psi_k(L^+). \tag{9.1}$$

Here Ψ_k is the same-vector raw expression (3.12); its saddle, owners, endpoint scores and scalar factors are all evaluated at L^+ . The conclusion is uniform in all original length ratios, strict product gaps and equality faces. The constant L_* is chosen once from k , and is not optimized as a function of ℓ .

9.1 Physical data and compatible endpoint pairs

For the moment all original physical lengths are large. Their unique ordered saddle is s , their maximal products are $\sigma_1 > \dots > \sigma_m$, and we set

$$u_j = \frac{\sigma_{j+1}}{\sigma_j}, \quad d_j = u_j^{-1} - 1, \quad r_j = \frac{\sigma_j}{\sigma_1}.$$

These ratios and the original fine-label probabilities lie in fixed compact ranges. An owner is a deterministic group of indices whose original physical blocks remain separate clocks. Let T_j be owner j 's native total. On a proper owner use the literal width convention

$$R_j = \max\{R_0, \min(\sqrt{T_j}, d_j^{-1})\}, \quad w_j = R_j/\sqrt{T_j}, \quad j < m. \quad (9.2)$$

The large-length threshold makes $w_j \leq 1$. Also $d_j R_j \leq C_k$ and $w_j \asymp_k (1 + d_j \sqrt{1 + T_j})^{-1}$. The fixed-width floor avoids an unsupported lattice band.

For an owner write its actual normalized history cost as $C_v = a_0 - I_v$ and its output as $Y = a_0 - b_0$. The source uses the *intrinsic* cost

$$K_v = C_v - b_0 = Y - I_v. \quad (9.3)$$

On the displaced full-name count box, native completed endpoint histories have finite scales

$$S_v^I = \begin{cases} h + R_j + \overline{C}_v, & t_v \leq T_{\text{old+native}}/2, \quad j < m, \\ h + \overline{C}_v, & t_v \leq T_{\text{old+native}}/2, \quad j = m, \\ h + \overline{C}_v + S_{\text{good}}(v), & t_v > T_{\text{old+native}}/2. \end{cases} \quad (9.4)$$

The terminal early score is constructed at zero common root target. The early comparison uses the positive reference budget to pay the incoming strict-exit subtraction; in the late row it cancels exactly in (9.3). All contexts are lawful named contexts of the original chain, including the genuine old groups.

Choose the fixed rare threshold in the finite endpoint theorem. The selected nondegenerate native edges form a set $\mathcal{I}$, with at most one selected edge per original physical block. Unselected edge factors are bounded above and below by fixed constants. For $i \in \mathcal{I}$ its original risk is

$$c_i t - a_i N_i(t) + b_i N_{B_i}(t), \quad a_i = \log S_i, \quad b_i = \log B_i, \quad c_i > 0.$$

The original marks $a_i, a_i - b_i$ are positive and nondegenerate. Let X_i^I, Y_i^I be the minima of (9.4) over its left and right completed contexts. Conditional on its actual total n_i and actual subset count k_i , put

$$\Delta_i = c_i L_i - a_i n_i + b_i k_i, \quad x_i^- = \max\{a_* X_i^I, a_* Y_i^I - \Delta_i\}, \quad y_i^- = x_i^- + \Delta_i. \quad (9.5)$$

Here $a_* > 0$ is fixed and small. The count-box theorem makes this a legal pair in a common positive box of scales X_i^I, Y_i^I . Its difference is the *actual type increment*, while the deterministic scales specify the common box.

The finite scalar theorem gives the factor

$$\Psi_i^I(\eta) = \begin{cases} \left(\frac{(1 + X_i^I)(1 + Y_i^I)}{1 + n_i} \right)^{2\chi(\eta, \rho_i^0)}, & \text{below the fixed rare threshold,} \\ 1, & \text{otherwise,} \end{cases}$$

$$\rho_i^0 = \frac{(c_i L_i / (n_i + 1))^2}{(c_i L_i / (n_i + 1))^2 + b_i^2 p_i^0 (1 - p_i^0)}, \quad (9.6)$$

where $p_i^0 = (a_i - c_i L_i / n_i) / b_i$ on the rare branch. There p_i^0 is in a fixed interior interval; bounded supported experiments are treated separately and contribute a fixed factor. All $n_i + 1$ gaps are

retained. Deterministic-mark ordinary bridges are a separate theorem, not the substitution $\rho = 1$ in (9.6).

The endpoint construction is proved in Appendix G. The bridge estimates apply with compatible endpoint boxes chosen after the fixed wall constants. Both near-central Gamma upper bounds require $n \geq N_\Gamma$, and the central lower bound also requires $n \geq n_0(c, D, R)$ and feasible remaining support; at the exact reference mean the upper bounds hold for every sample size. The fixed lift is chosen so that every original-block count meets the thresholds required here.

9.2 The whole coupled count measure and its mixed moment

The source assertion concerns one joint full-leaf count measure. Conditional on every original uncolored total in its own central window, the diagonal strict-exit construction and the fresh-name endpoint repair give a count-only measure μ on complete fine types with

$$W := \mu(\text{all types}) \asymp_k \prod_{j < m} w_j. \quad (9.7)$$

The diagonal splitting variables are genuine independent two-name splits before conditioning their bands. The subsequent acceptance may couple every block. The entire band is summed in (9.7). Terminal names can first be projected out of this count measure and then filled by their original uniform conditional law after the root count is fixed. The latter completion has constant conditional mass for each retained prefix and admitted root vector. It changes neither the uncolored root nor any proper coarse word.

For each complete type ω and selected block define its conditional private source mass $q_i(\omega, G_i)$ using the allocated strong curve, binary REG and full named REG at the pair (9.5). Named REG is supplied by uniform refinement of *both* true mark lists, conditional on that binary word. There is a nonnegative uncolored cost $D_i(G_i)$, chosen before any target is read, such that for $\eta = \sigma_{\text{owner}(i)}$,

$$q_i \leq CD_i, \quad \mathbb{E}_{G_i} q_i^\eta \geq c\Psi_i^I(\eta), \quad \mathbb{E}_{G_i} D_i^\eta \leq C\Psi_i^I(\eta), \quad (2 + n_i)^{-D_0} \leq D_i \leq 1. \quad (9.8)$$

The upper cost is the joint original-class free-bank envelope, uniform in the complete type and in every permitted logarithmic recoloring. For depths beyond its active cutoff the bank test is one, and the summable coefficient tail pays its contribution to the last moment in (9.8).

The local bounds immediately give the mixed moment which will be needed with different owner powers:

$$\mathbb{E}_{G_i} [q_i D_i^{\eta-1}] \asymp_k \Psi_i^I(\eta). \quad (9.9)$$

Indeed $q_i D_i^{\eta-1} \leq CD_i^\eta$ gives the upper bound. Writing $q_i^\eta = (q_i D_i^{\eta-1})^\eta (D_i^\eta)^{1-\eta}$ and applying Hölder gives

$$\mathbb{E}[q_i D_i^{\eta-1}] \geq \frac{(\mathbb{E} q_i^\eta)^{1/\eta}}{(\mathbb{E} D_i^\eta)^{(1-\eta)/\eta}},$$

which gives the lower bound in (9.9).

For a *fixed complete type*, different original native blocks have independent clocks and uniform multiset words. We may therefore multiply (9.9) at their respective owner powers. Only afterwards do we integrate the whole measure μ by Tonelli, retaining the dependence among type coordinates, exits, and old lists. Define

$$D_j^{\text{own}} = w_j \prod_{i: \text{owner}(i)=j} D_i \quad (j < m), \quad D_m^{\text{own}} = \prod_{i: \text{owner}(i)=m} D_i. \quad (9.10)$$

The width powers multiply (9.7) as $\prod w_j \prod w_j^{\sigma_j^{-1}} = \prod w_j^{\sigma_j}$. This is the source of the proper-owner count exponents.

Appendix H proves the joint type construction and these local bounds, including the uniform refinement of both mark lists. The count measure is integrated only after the complete type has been fixed in each path estimate.

9.3 One prefix-measurable root with a fixed small endpoint

All selected terminal binary edges lie in the negative-drift head. The remaining unread suffix is ordinary. Let its total length be U , its largest original block have length M , and let A, B be its preceding and following lengths within the suffix. Its mean is $D \geq 0$. The genuine hierarchy gives the macro reservoir and the required mean comparison with the full prefix length V_{pre} . Set

$$\mathcal{B}_I = (1 + U)^{-1/2} \min \left\{ 1, \frac{(1 + B)(1 + D + A)}{1 + M} \right\}. \quad (9.11)$$

This is the ordinary factor in (3.11).

Let $x = x(G_{\text{pre}})$ be prefix-uncolored measurable with $0 \leq x \leq A_{\text{tar}}(k) \log(2 + V_{\text{pre}})$. The constant $A_{\text{tar}}(k)$ is fixed by the literal floors and cascade range below; it is unrelated to the lift-envelope coefficient A_k in (3.13). The prefix paths and their pairs are fixed before evaluating x . Conditional on the complete prefix word, keep the other suffix counts central and select the one reservoir count

$$n_*^x = \left\lfloor \frac{\sum_{i \in \text{suffix}} c_i L_i + E_x - \sum_{i \neq *} w_i n_i}{w_*} \right\rfloor, \quad (9.12)$$

$$E_x = D + \frac{x - \Delta_{\text{pre}}}{r_m}, \quad w_i = \log(r_i + 1), \quad c_i = r_i.$$

The symbol Δ_{pre} is the actual uncolored prefix offset in the exact root identity. On the count support, $|E_x - D| \leq C\sqrt{1 + D}$. The count in (9.12) is nonnegative and central after the fixed threshold is chosen; its probability is comparable to $(1 + U)^{-1/2}$. In an unconditional Poisson experiment this is an acceptance test on the already sampled total. It does not change a fixed arithmetic tuple. It gives the true root $X_1 = x + \epsilon$, where $-C_k \leq \epsilon \leq 0$.

The terminal intrinsic endpoint theorem supplies the ordinary local scales $X_0^I \asymp 1 + P$ and $Y_0^I \asymp 1 + B$, where P is the full length preceding its weak block. The old incoming strict exit is absent from Y_0^I . If the weak block is the reservoir, the new count changes its increment by $-w_*(n_*^x - n_*^0)$: the smaller legal left representative grows by at most $O(1 + x)$ and the right representative decreases while retaining its fixed positive lower bound. The inequality $\log(2 + V_{\text{pre}}) \leq C X_0^I$ pays the left change. If it is not the reservoir, its local count is unchanged. The zero-root completion used in this argument only certifies that the prefix types have legal boxes; the actual experiment uses (9.12) anew.

At the selected counts, impose the ordinary allocated strong wall and REG at the intrinsic pair and then perform the true fine-name refinement. Strong suffix blocks use their fixed signed drift and REG; the unique nearly centered block uses the ordinary exact bridge. This gives the following *joint* statement, before any private prefix integration:

$$\mathbb{E}_{\text{suffix}} F_0^{I,x}(\omega_{\text{pre}}, G_{\text{suffix}}) \asymp_k \mathcal{B}_I F_{\text{pre}}^I(\omega_{\text{pre}}). \quad (9.13)$$

The upper bound follows from the intrinsic full-stop context's fixed reserve ordinary wall and its all-count ballot estimate. The lower constructs the type box and strong wall jointly on the same

root fiber. The one root atom occurs once in (9.13). Combining it with (9.9) gives

$$\mathbb{E} \left[f_0^{I,x}(G) \prod_{j=1}^m (D_j^{\text{own}})^{\sigma_j-1} \middle| \mathbf{n}_{\text{pre}} \right] \asymp_k \mathcal{B}_I \prod_{j < m} w_j^{\sigma_j} \prod_{i \in \mathcal{I}} \Psi_i^I(\sigma_{\text{owner}(i)}). \quad (9.14)$$

Here the original prefix totals $\mathbf{n}_{\text{pre}}$ are fixed in their central windows; the uniform comparison permits their later integration. The private mass $f_0^{I,x} = \mathbb{E}_\pi F_0^{I,x}$ retains all coupled types and the true root acceptance.

The ordinary bridge theorem and the same-count construction in Appendix H prove (9.13) uniformly in the displayed range of prefix-measurable targets.

10 Transport of the accepting measure

10.1 Adaptive targets and exact density

Put

$$y_j = -\log D_j^{\text{own}} \geq 0, \quad x = \sum_{j=1}^m r_j y_j. \quad (10.1)$$

Literal floors first bound the y_j by dimension-only multiples of $\log(2 + T_j)$. The hierarchy then bounds all transfers below by predetermined multiples of the same logarithms. Construct the all-transfer envelopes for that entire range, and enlarge the final source threshold last. The literal floors remain fixed throughout this order of choices.

In proper owner j choose its admitted native macro bank and the two actual fine names: the immediate child's first shallow name c_j and that block's eligible shallow exterior destination t_j . Write A_{j+1} for the logarithm of the child's first effective alphabet and $b_j = u_j A_{j+1}$. With $e_0 = 0$, set

$$e_j = \left\lceil \frac{y_j + A_j e_{j-1}}{b_j} \right\rceil, \quad \delta_j = b_j e_j - A_j e_{j-1} - y_j \in [0, b_j]. \quad (10.2)$$

Uniformly choose e_j occurrences of c_j in that bank and recolor them t_j . The banks are disjoint original blocks. Relative broad bank admission, performed inside the local owner-power norm, ensures enough source occurrences while retaining a fixed proportion of that norm.

For target bank counts k'_c, k'_t , the exact unrestricted output density relative to the original fine law is

$$\left(\frac{p_c}{p_t} \right)^e \frac{\binom{k'_t}{e}}{\binom{k'_c + e}{e}}, \quad \frac{p_c}{p_t} = \exp((u_j - 1)A_{j+1}). \quad (10.3)$$

On the logarithmically broad source bank its logarithm is bounded by $C(e/\log(2 + T_j) + e^2/N_{\text{bank}}) \leq C_k$. Discarding preimages preserves the upper density bound. If ν_y is the push-forward of the actual accepted source measure, define

$$F_y^I = \frac{1}{C_R} \frac{d\nu_y}{d\pi}, \quad 0 \leq F_y^I \leq 1, \quad \mathbb{E}_\pi F_y^I(G, \cdot) = f_0^{I,x}(G)/C_R. \quad (10.4)$$

The last equality is pointwise in the full uncolored environment, including its actual root indicator.

The necessary path support is reconstructed from the inverse source type. For a flipped selected block of length P , every output type has the same inverse type and hence the same pair x_0, y_0 . If $L_{\text{flip}}(t)$ counts its recolorings, exactly

$$\tilde{Z}(t) = x_0 + (c + be/P)t - aN(t) + bN_{B,\text{target}}(t) = Z_{\text{source}}(t) - b[L_{\text{flip}}(t) - et/P]. \quad (10.5)$$

Both endpoints remain x_0, y_0 . Macro interiority gives $|L_{\text{flip}}(t) - et/P| \leq CeP^{-\gamma} \min(t, P - t)^\gamma$. The preassigned spare source curve absorbs this at the final threshold. Thus the transported necessary banks retain the original source scales. Earlier recolorings may change a later old list while preserving its native points. The same-word geometry below transfers full fine-history support.

The exact grouped outputs satisfy

$$\begin{aligned} Z_j^y &= y_j + \eta_j^y - \eta_{j-1}^y / u_{j-1}, & j < m, \\ Z_m^y &= y_m + \epsilon / r_m - \eta_{m-1}^y / u_{m-1}, \\ \eta_j^y - \eta_j^0 &= \sum_{l \leq j} (\sigma_l / \sigma_j) \delta_l = O_k(1), & 0 \leq E_y := \sum_{j < m} d_j \eta_j^y \leq C_k. \end{aligned} \quad (10.6)$$

The already chosen common root supplies the terminal target, and the root suffix is left unrecolored. For two retained fine words in the same environment,

$$\frac{\pi(\omega)}{\pi(\omega')} = \exp(E_y(\omega) - E_y(\omega')). \quad (10.7)$$

This bounded likelihood ratio applies to the full chain.

The exact recoloring and density calculations, with their support conditions, are given in Appendix H.

10.2 Direct complete cells and the evaluated nominal first mass

The preceding construction is made directly on the final complete-cell grid. For total n in a block of length L , its exact uncolored occupancy law is

$$\mathbb{P}(g \mid n) = n! \prod_c \frac{(|c|/L)^{g_c}}{g_c!}. \quad (10.8)$$

This is the push-forward of the original n uniform points and all $n + 1$ gaps. Define count acceptances, walls, REG, macro banks, costs, targets and the root count on cell words from the outset.

For either one-sided jump convention the loss in extending a boundary wall into a cell is at most the linear speed times its fixed mesh. It does not depend on the number of points in the cell. For REG, round each interval endpoint inward and outward: the length changes by at most two meshes, the reference mean by a fixed amount, and $|d(t)^\zeta - d(t')^\zeta| \leq |t - t'|^\zeta$. Thus boundary REG gives continuous REG for *every* lift of the same word, with fixed extra constants. The exact-type bank proof applies to the boundary events, and the corresponding q_i, D_i are constant on each uncolored cell fiber. Consequently q_i^η and $q_i D_i^{\eta-1}$ are preserved exactly by (10.8). The uniform recoloring kernel commutes with every within-cell slot permutation. The resulting F_y^I is therefore cell-invariant and independent of the actual prime values later assigned to those slots.

Let $g = (g_{i,c})$, $h_i = k - i + 2$, and

$$W_g = \prod_{i,c} \frac{|c|^{g_{i,c}}}{g_{i,c}!}, \quad Z_F(g) = \sum_\omega F_y^I(g, \omega).$$

The normalized nominal counting mass is $\mathcal{N}(F) = e^{-\sum_i h_i L_i} \sum_g W_g Z_F(g)$. Expanding the independent natural Poisson cell law of intensity Z_i and then the original fine categorical law gives

the exact tilt identity

$$\mathcal{N}(F) = e^{-J(L;s)} \mathbb{E}_{\text{nat}} \mathbb{E}_{\pi} \left[F_y^I e^{-E_y} \prod_{j=1}^m (D_j^{\text{own}})^{\sigma_j-1} e^{(1/r_m - \sigma_1)\epsilon} \right]. \quad (10.9)$$

The expression is zero off the accepted root/count support. To check the target powers, with $q_j = \sigma_j/\sigma_m - 1$ use $q_j - q_{j+1}/u_j = d_j$ and $(1/r_m - \sigma_1)r_j - q_j = 1 - \sigma_j$. The deterministic part of the Poisson expansion is the current $J(L; s)$; no limiting rate is substituted. The identity is algebraic at fixed products and will remain valid on the frozen grid. The root error and E_y in (10.9) are bounded, whereas the logarithmic targets are not discarded as constants. Equations (9.14) and (10.4) now give

$$\mathcal{N}(F) \asymp_k e^{-J(L;s)} \mathcal{B}_I \prod_{j < m} w_j^{\sigma_j} \prod_{i \in \mathcal{I}} \bar{\Psi}_i^I(\sigma_{\text{owner}(i)}). \quad (10.10)$$

Here the natural-count factors are obtained as follows. If $\mu_i = Z_i L_i$ and $|n_i - \mu_i| \leq C\sqrt{1 + \mu_i}$, then on the compact rare branch the changes in p_i^0, ρ_i^0 are $O(\mu_i^{-1/2})$. The compact Lipschitz bound for χ gives

$$|\log \Psi_i^I - \log \bar{\Psi}_i^I| \leq C \frac{\log(2 + \mu_i)}{\sqrt{\mu_i}} \leq C.$$

A threshold crossing is harmless since both bases are then comparable to the fixed threshold. The finitely many original Poisson central windows have constant joint mass. Integrating them pays no new type atom and no extra root atom.

We finally compare the centered natural correlation with the original mark correlation in (3.10). This comparison is made only on the compact rare branch. Fix a selected edge $r \rightarrow q$, write $L = L_i$, and put

$$\lambda = (r+z)^\sigma, \quad c = r-q, \quad a = \log(r+z), \quad b_e = \log(q+z), \quad p = \frac{q+z}{r+z}, \quad m = a - b_e p.$$

Its centered natural type and finite-mean correlation are

$$\bar{p} = \frac{a - c/\lambda}{b_e}, \quad \bar{\rho} = \frac{(cL/(\lambda L + 1))^2}{(cL/(\lambda L + 1))^2 + b_e^2 \bar{p}(1 - \bar{p})}. \quad (10.11)$$

In general $\bar{p} \neq p$. For every paired context in (3.5), its right and left reference costs differ by

$$\bar{C}_{v+} - \bar{C}_{v-} = L\{c - \lambda(a - b_e p)\} = L\nu, \quad \nu = c - \lambda m.$$

Both reference costs are nonnegative. Hence $Y_e \geq L\nu$ if $\nu \geq 0$, and $X_e \geq L|\nu|$ if $\nu < 0$, even though the endpoint minima are taken separately. Since $L \geq 1$ and the natural intensity is bounded above and below, the positive score floors give

$$|\nu| \leq CT_e, \quad T_e \geq \frac{c_k}{1+L}, \quad \bar{p} - p = -\frac{\nu}{b_e \lambda}, \quad |\bar{\rho} - \rho_e| \leq C(|\nu| + L^{-1}).$$

Here the last estimate uses the compact rare parameter range; it is not a statement at a degenerate ordinary law. Proposition 6.5 therefore gives, for T_e below a fixed rare threshold,

$$|\chi(\sigma, \bar{\rho}) - \chi(\sigma, \rho_e)| |\log T_e| \leq C(T_e + L^{-1}) |\log T_e| \leq C.$$

The L^{-1} term retains the finite denominator $\lambda L + 1$. Thus the two powers of T_e are comparable by fixed constants. Replacing the fixed positive score constants in (9.4) by those in (3.6) also changes each factor by fixed constants. Branch crossings and the finitely many saturated edges have bounded factors. This proves that the product in (10.10) is precisely the binary profile product up to the required dimension-dependent constants.

Appendix H gives the complete-cell construction and verifies the count and slot invariances used in this calculation.

11 Original histories and the volume lower bound

11.1 The full original fine-history row

We state the geometric and class conclusion with its normalization explicit. Fix one cell occupancy, its original legal fine-word space Ω_g , a retained anchor ω , and any original history h . Let $Q_h(\omega)$ be its actual comparison class and $\Phi_g(h)$ its actual geometric multiplier. All original permutations, zero pivots, ties and distinct positive pivot choices remain in this set. Define

$$\begin{aligned} I_h &= \log \Phi_g(h) + \log \frac{\pi(Q_h(\omega))}{\pi(\omega)}, & R_y &= \sum_{j=1}^m Z_j^y, & C_y(h) &= R_y - I_h, \\ q_h^{\text{rel}} &= \mathbb{E}_\pi \left[F_y^I(\omega') \frac{\pi(\omega)}{\pi(\omega')} \middle| Q_h(\omega) \right]. \end{aligned} \quad (11.1)$$

There is a fixed $0 < \delta \leq 1$, chosen from the finite geometry and scalar ranges and independently of roughness, such that

$$\sum_{h \text{ original}} e^{-\delta C_y(h)} (q_h^{\text{rel}})^\delta \leq C_k \left(\prod_j D_j^{\text{own}} \right)^\delta. \quad (11.2)$$

We describe the argument, keeping the conditioning and the geometric charges explicit; the finite estimates are established in Appendix I.

First, legal parent/child projection gives the exact residual-entropy identity and a nonnegative rank-delay correction. A positive mixed entropy loss is charged only when at least two child-touching cells survive and one is mixed with an outside name. Proper-owner branching normal form pays pure-cell lifetimes. The terminal owner uses a floating last nonsingleton spine, retains the expired-first and ghost terms, and then makes a lawful same-time permanent-0 comparison. Its zero-mean named discrepancy is bounded by REG on the actual floating interval. The reference fraction and the spread/depth budgets reserved by the global STAR theorem pay this discrepancy; this comparison is applied with those reserved budgets.

Second, synchronize cuts only within an original physical block, using the ordered simplex when several owner cuts share it. Strong entropy concavity pays spread; actual empirical contrasts are supported on the moved intervals. Endpoint interpolation retains the reference, good-count and genuine mixed-time budgets and allocates the available endpoint reserve once over the finitely many native coordinates. Old weak, proper-input and wrong-name directions are paid by their reference charges. Each remaining selected native direction uses its allocated intrinsic wall. A maximum-weight lawful endpoint vertex pays the root flag without contracting its coefficient. The same-word target identity (10.5) and exact carry (10.6) then give, for the appropriate endpoint

signature of an arbitrary fine history,

$$C_y(h) \geq \sum_{j \in \mathcal{E}(h)} y_j + c_k \Lambda(h) - C_k.$$

Here $\mathcal{E}(h)$ is the set of early-completion owners and Λ retains all positive movement, delay, pure/mixed lifetime, reference/good-count and strong-wall depth penalties required below.

Third, inspect the *original* class before using a bank. On a block which is FULL away from its actual endpoint strips, condition on the complete target type and the literal revealed assignments. The remaining word has its exact original multiset law, and its two necessary bank events admit one joint Gamma/type comparison. Their conditional probability is bounded by the corresponding all-transfer bank envelope. Blocks which are fully exposed pay only their own count and floor polynomials; a comparator-FULL block which is not original-class FULL forces a pure, off-spine or genuine mixed penalty. Late non-FULL endpoint groups have the displaced square-root charge. Early completion keeps the entire y_j instead of attempting to pay a huge exposed suffix by a short prefix budget. These are precisely the signatures preserved in $\Lambda(h)$.

The full inverse-type acceptance, including path acceptance and future child weights, is summed before the δ power. Bank costs satisfy $q_{i,d} \leq D_i \exp(b_* d^\gamma)$ at active depths, and the literal floor handles exposed/far depths. Choose b_* below a fixed fraction of the geometric depth coefficient. The remaining positive charges sum all original pivot multiplicities. Erased projected times are matched to actual physical deadlines and keep their nonnegative distance penalty. Added deadline representatives may repeat in an enlarged comparison row; original full histories keep their distinct pivots. Finally (10.7) supplies the true relative likelihood in (11.1). This proves (11.2) for the same transported density.

By summing (10.6),

$$R_y = \sum_j y_j + \epsilon/r_m - E_y, \quad e^{R_y} \prod_j D_j^{\text{own}} = e^{\epsilon/r_m - E_y} \leq 1. \quad (11.3)$$

Write $M_h(F; \omega) = \sum_{\omega' \in Q_h(\omega)} F_y^I(\omega')$. There is the exact identity

$$\Phi_g(h) M_h(F; \omega) = e^{I_h} q_h^{\text{rel}}. \quad (11.4)$$

Multiplying (11.2) by $e^{\delta R_y}$ and using (11.3) proves the unnormalized arithmetic row

$$\sum_h \Phi_g(h)^\delta M_h(F; \omega)^\delta \leq C_k \quad (F_y^I(\omega) > 0). \quad (11.5)$$

Bounded complete-cell geometric discrepancies multiply its constant once per original cut. Positivity extends the row to bounded accepting densities.

Appendix I proves the projection, synchronization, and full-class estimates just used. In particular, the terminal spine's same-time comparison is paid by the global reference and depth budgets; it does not assert a conditional closure for an arbitrary child weight. The exact class identity also appears in the collision calculation of Appendix J.

11.2 Pointwise squarefree supply and one Hölder inequality

Summing the named boundary REG gives an uncolored occupancy tube

$$g_{i,t} \leq A_0 + A_1 \min(t, v_i - t + 1) \quad (11.6)$$

with constants fixed before roughness and the prime grid. Recoloring does not change g . Choose a fixed rough-prime cutoff and greedy complete prime cells of harmonic mass at most their fixed mesh $\eta_i = \log(k - i + 2)/(k - i + 1)$. For a cell D with g slots, the successive-distinct-drawing bound is

$$1 \geq \frac{g!e_g(D)}{\eta_i^g} \geq \exp \left[-\frac{2g^2}{\eta_* \min D} \right]. \quad (11.7)$$

The roughness choice makes every elementary factor positive. Cell minima grow at least exponentially in their index while (11.6) grows linearly. Thus for every retained occupancy,

$$c_k W_g \leq S_g := \sum_{\mathbf{a} \in \mathcal{A}(g)} \frac{1}{a_1 \cdots a_k} \leq W_g. \quad (11.8)$$

The reverse tube and Mertens' difference estimate also put every retained tuple inside one fixed allowed size truncation. Its auxiliary size exponent is fixed from the tube.

Let $B_\omega(\mathbf{a})$ be the original divisor box of side $b = \log 2$. On the *same* product of harmonic tuple measure and Lebesgue measure define

$$f(\mathbf{a}, t) = \sum_{\omega \in \Omega_g} F_y^I(\omega) \mathbf{1}_{B_\omega(\mathbf{a})}(t), \quad Z_F = \sum_{\omega} F_y^I(\omega).$$

Then $\int f = b^k S_g Z_F \geq c W_g Z_F$. The correlated prime-cell kernel applies to this cell-invariant bounded weight. Its finite class identity is

$$\sum_h \Phi_g(h)^\delta \sum_{\omega} F_y^I(\omega) M_h(F; \omega)^\delta = \sum_{\omega} F_y^I(\omega) \sum_h \Phi_g(h)^\delta M_h(F; \omega)^\delta.$$

By (11.5) and the kernel theorem, $\int f^{1+\delta} \leq C W_g Z_F$. Hölder on $\{f > 0\}$ gives

$$\text{meas}\{f > 0\} \geq \frac{(\int f)^{(1+\delta)/\delta}}{(\int f^{1+\delta})^{1/\delta}} \geq c_k W_g Z_F. \quad (11.9)$$

This retains the nominal first mass to the *first* power. The set $\{f > 0\}$ lies in the full original divisor union, so summing (11.9) over disjoint occupancies and using (10.10) gives the desired actual lower on consistent large grid data.

The prime-cell kernel and the pointwise supply estimates are proved in Appendix J, for the same bounded cell-invariant weight used above.

12 The physical saddle on the arithmetic grid

We complete the parameter-uniform argument by carrying the physical saddle to the complete-cell grid.

Fix the finite geometric, scalar, count, REG and occupancy constants. Choose roughness and the complete-cell grid from those constants, and let R_k bound its total length loss. Choose the lift $L_* > C_k(1 + R_k)^2$ above the remaining source thresholds. Only then lift L to L^+ , solve its KKT problem once, and call its optimizer s^+ . For final cell lengths L' one has

$$\sum_i |L'_i - L_i^+| \leq R_k.$$

Keep products, owners, categorical laws, selected native edges and the root suffix fixed from (L^+, s^+) . Map each original physical block affinely to its grid block. This preserves legal cut/deadline order and gives, uniformly over the entire original history space,

$$|\overline{C}_{L',s^+}(h') - \overline{C}_{L^+,s^+}(h)| \leq C_k R_k, \quad |J(L'; s^+) - J(L^+; s^+)| \leq C_k R_k. \quad (12.1)$$

All reference charges acquire only this fixed intercept. The same-word, class and likelihood identities remain exact on the grid.

Two exceptional endpoint templates must be handled by their exact outputs. If an owner's grid natural mean residual is $m_{\text{ref}} = O_k(R_k)$ and $\zeta_g = H_g - \mathbb{E}H_g$, then

$$\begin{aligned} C_v &= a_0 - \sum_{g \leq l} \zeta_g + \overline{C}'_v + \sum_{\text{proper}} \nu_{v,g}, \\ &= b_0 + m_{\text{ref}} + \sum_{g > l} \zeta_g + \overline{C}'_v + \sum_{\text{proper}} \nu_{v,g}. \end{aligned} \quad (12.2)$$

The all-zero template is exactly $C_v = a_0$. For the complete-deadline template $\nu = 0$ and $\overline{C}'_v = -m_{\text{ref}}$, hence $C_v = b_0$ and its intrinsic cost is exactly zero. Both small endpoints retain their original values. Every other template has reference plus good-count charge at least $c_k \sqrt{\min_i L_i^+} - C_k$, which absorbs (12.1) after the chosen lift.

The complete strict count bands retain their widths under bounded integer center translations; the final threshold makes the relevant Stirling ratios bounded. The local pair uses its actual grid increment. Frozen physical endpoint scales and all grid scalar parameters are comparable, with bounded exponent-times-log error. The newly selected grid root uses the exact grid budget. If its old prefix is nonempty, its positive incoming mean survives the bounded defect. If that prefix is empty, use the legal exact ordinary pair and its bounded root floor directly. The literal root/error/carry identities and (11.3) remain exact.

It follows that the direct cell family, full row and arithmetic transfer above apply on (L', s^+) and give

$$K_k(\log 3 + L^+) \geq c_k e^{-J(L'; s^+)} \mathcal{P}_{\text{frozen}}^I(L'; s^+) \geq c_k e^{-J(L^+; s^+)} \mathcal{P}_I(L^+). \quad (12.3)$$

Here $\mathcal{P}_I$ is (10.10)'s nonexponential factor, comparable to the factors of (3.12). The frozen likelihood comparison above applies off stationarity on the grid.

Finally the independent actual parameter-stability theorem applies to the whole harmonic volume sum at every $\ell_i \geq \log 3$. Since the lift changes each coordinate by at most the fixed L_* ,

$$K_k(\ell) \geq c_k K_k(\log 3 + L^+).$$

Combining this with (12.3) proves Theorem 9.1. The occupancy constants remain fixed after choosing roughness, and the size of L_* affects only the theorem's constants.

The complete grid comparison is proved in Appendix J. Parameter stability is proved in Section 16 directly for the original arithmetic sum.

The arithmetic comparison is ultimately applied at the original vector $\ell(\mathbf{y})$ under (2.3). The auxiliary lift of K_k does not alter the cutoffs in that hypothesis. The same observation will apply to the original table axes in Section 16.

13 Ordered exponential sums and ordinary root estimates

The proof of [18, Lemma 5.3, pp. 31–33] uses a real threshold followed by integer bands. We give the proof with an explicit integer-rounding correction in this band decomposition and recover the

$A = 0$ case of [19, Lemma 5.5] used below. The two order-statistics inputs are stated next. All constants below may depend on fixed $C > 1$ and $\mu > 1$.

Put

$$\begin{aligned}\Delta_r &= \{0 \leq \xi_1 \leq \cdots \leq \xi_r \leq 1\}, \\ S_r(u, v) &= \{\xi \in \Delta_r : \xi_j \geq (j - u)/v \ (1 \leq j \leq r)\}, \\ Q_r(u, v) &= r! \operatorname{vol}(S_r(u, v)).\end{aligned}$$

Lemma 13.1 (Flat-wall upper bound). *For an integer $r \geq 1$ and real $u, v > 0$ with $w = u + v - r > 0$,*

$$Q_r(u, v) \ll \frac{(u + 1)(w + 1)}{r}.$$

This is the upper assertion of [18, Lemma 5.1, p. 30]. Koukoulopoulos derives this statement from Ford's estimates [14, Theorem 1] and [12, Lemma 11.1]. The lower assertion of Lemma 5.1 is not used here.

Lemma 13.2 (Crowding estimate). *Fix $C > 1$. Suppose g, r, s, u, v are integers satisfying*

$$2 \leq g \leq r/2, \quad s \geq 0, \quad r \leq Cv, \quad u \geq 0, \quad u + v \geq r + 1.$$

Let R consist of the points of $S_r(u, v)$ for which there is an integer $g + 1 \leq l \leq r$ such that

$$\frac{l - u}{v} \leq \xi_l \leq \frac{l - u + 1}{v}, \quad \xi_{l-g} \geq \frac{l - u - s}{v}.$$

Then

$$\operatorname{vol}(R) \ll_C \frac{[C(s + 1)]^g (u + 1)(u + v - r)}{(g - 1)! (r + 1)!}.$$

This is [18, Lemma 5.2, p. 30]. That source obtains it from [13, Lemma 4.3], using Lemma 5.1 in place of Ford's Lemma 4.1. The stated changes are the fixed constant 10 to C , the factor $(u + v - r)^2$ to $u + v - r$, and $(g - 2)!$ to $(g - 1)!$. We use these two order-statistics estimates in precisely the forms stated above.

For any $\gamma \geq 0$, define

$$\mathcal{T}_\mu(r, v, \gamma) = \left\{ \xi \in \Delta_r : \sum_{i \leq j} \mu^{v\xi_i} \geq \mu^{j-\gamma} \ (1 \leq j \leq r) \right\}.$$

Lemma 13.3 (Corrected integer-parameter ordered-sum estimate). *Fix $C > 1$ and $\mu > 1$. For integers r, v, γ satisfying $\gamma \geq 0$ and $1 \leq r \leq Cv$, set $b = r - v$, $d = b - \gamma$, and*

$$Y = \begin{cases} b, & d \geq 1, \\ (1 - d)(\gamma + 1), & d \leq 0. \end{cases}$$

Then

$$\operatorname{vol}(\mathcal{T}_\mu(r, v, \gamma)) \ll_{C, \mu} \frac{Y}{\mu^{\mu^d} (r + 1)!}. \quad (13.1)$$

Proof. The only change of threshold from the source proof is

$$B = \left\lceil 2 + \frac{\log 32}{\log \mu} \right\rceil, \quad t = \max\{d, B\}. \quad (13.2)$$

Both d and t are integers. For $\xi \in \mathcal{T}_\mu(r, v, \gamma)$ let $D = \max_{j \leq r} (j - \gamma - v\xi_j)$. If $D < t$, all coordinates satisfy $\xi_j > (j - \gamma - t)/v$; call this part V_1 . Otherwise choose the integer $h = \lfloor D \rfloor + 1 \geq t + 1$ and an index l attaining D . Then

$$\min_{j \leq r} \left(\xi_j - \frac{j - \gamma}{v} \right) = \xi_l - \frac{l - \gamma}{v} \in [-h/v, (-h + 1)/v]. \quad (13.3)$$

Call the union of these bands V_2 . This covers also $D = t$; any overlaps at endpoints are harmless. The unrounded threshold $t_0 = \max\{d, 2 + \log 32 / \log \mu\}$ would omit points with $t_0 < D < \lceil t_0 \rceil$ when t_0 is not an integer.

If $d \geq B$, then $t = d$ and V_1 is empty: the inequality for $j = r$ would give $\xi_r > 1$. If $d < B$, then $t = B$, and Lemma 13.1 applies with $u = \gamma + B > 0$, $w = B - d \geq 1$. Thus

$$\text{vol}(V_1) \ll \frac{(\gamma + B + 1)(B - d + 1)}{(r + 1)!}. \quad (13.4)$$

Here $1/(rr!) \leq 2/(r + 1)!$. For $d \leq 0$ the numerator is at most $(B + 1)^2 Y$ and $\mu^{\mu^d} \leq \mu$. For $1 \leq d < B$ it is at most $B(B + 2)b$, while $\mu^{\mu^d} \leq \mu^{\mu^{B-1}}$. Since B depends only on μ , both cases give the required bound for V_1 , including its double exponential denominator.

Fix an integer band $h \geq t + 1$. Write

$$a = \left\lceil \frac{\log 4}{\log \mu} \right\rceil, \quad m_0 = 1 + a, \quad h_0 = h - m_0.$$

These are integers, and

$$h_0 \geq t - \left(\frac{\log 4}{\log \mu} + 1 \right) \geq 1 + \frac{\log 8}{\log \mu}. \quad (13.5)$$

We claim that there is an integer $m \geq h_0$ with $g_m = \lfloor \mu^m \rfloor < l/2$ such that

$$\xi_{l-g_m} \geq \frac{l - \gamma - 2m}{v}. \quad (13.6)$$

For completeness, order the last half of the first l summands and group their backward indices into $[g_m, g_{m+1})$. Enlarging the first group to μ^{h_0} terms and every subsequent group to $g_{m+1} - g_m$ terms gives

$$\sum_{j \leq l} \mu^{v\xi_j} \leq 2 \left(\mu^{h_0} \mu^{v\xi_l} + \sum_{\substack{m \geq h_0 \\ g_m < l/2}} (g_{m+1} - g_m) \mu^{v\xi_{l-g_m}} \right).$$

If (13.6) failed for all candidates, the band condition would therefore give

$$\sum_{j \leq l} \mu^{v\xi_j} < 2\mu^{l-\gamma} \left(\mu^{-a} + \sum_{m \geq h_0} (g_{m+1} - g_m) \mu^{-2m} \right) \leq \mu^{l-\gamma}.$$

Indeed, $\mu^{-a} \leq 1/4$, and summation by parts with $\mu^m - 1 \leq g_m \leq \mu^m$ gives

$$\sum_{m \geq h_0} (g_{m+1} - g_m) \mu^{-2m} \leq \mu^{1-h_0} + \mu^{-2h_0} \leq 2\mu^{1-h_0} \leq 1/4.$$

When there are no candidates, the first group alone bounds the sum by $\mu^{l-\gamma}/2$, so no strict-inequality argument is needed. Either case contradicts membership in $\mathcal{T}_\mu$ and proves (13.6).

Apply Lemma 13.2 with $u = \gamma + h$, $s = 2m$, $g = g_m$. The minimal-band condition places ξ in $S_r(u, v)$ and gives the required interval for ξ_l ; moreover

$$\xi_{l-g} \geq (l - \gamma - 2m)/v \geq (l - u - s)/v.$$

All parameters are integers, $g \geq 2$ by (13.5), $g < l/2 \leq r/2$, $l \geq g+1$, $s, u \geq 0$, $u+v-r = h-d \geq 1$, and $r \leq Cv$. The external lemma already takes the union over l , so there is no additional factor r . Summing only over h, m yields

$$\text{vol}(V_2) \ll_C \frac{1}{(r+1)!} \sum_{h \geq t+1} \sum_{m \geq h-m_0} (\gamma + h + 1)(h - d) \frac{[C(2m+1)]^{\lfloor \mu^m \rfloor}}{(\lfloor \mu^m \rfloor - 1)!}. \quad (13.7)$$

For fixed C, μ , Stirling's lower bound implies

$$\frac{[C(2m+1)]^{\lfloor \mu^m \rfloor}}{(\lfloor \mu^m \rfloor - 1)!} \ll_{C, \mu} \mu^{-\mu^{m+m_0}}.$$

Its logarithm is $-(\log \mu)m\mu^m + O_{C, \mu}(\mu^m \log(m+1) + m)$, which is eventually smaller than $-(\log \mu)\mu^{m+m_0}$; the finitely many remaining m are absorbed into a constant depending only on C, μ . The successive ratios of $\mu^{-\mu^q}$ for $q \geq t+1 \geq B+1$ are bounded above by a fixed number less than one. Thus the m tail is $\ll_\mu \mu^{-\mu^h}$. Writing $h = t+1+j$, each of the two linear factors grows by at most a factor $j+1$, and the same geometric domination sums $(j+1)^2$. Consequently

$$\text{vol}(V_2) \ll_{C, \mu} \frac{(\gamma + t + 2)(t + 1 - d)}{\mu^{\mu^{t+1}}(r+1)!}. \quad (13.8)$$

If $d \geq B$, its numerator is $b+2 \ll b = Y$. If $d \leq 0$, it is at most $(B+2)(B+1)Y$. If $1 \leq d < B$, it is at most $B(B+3)b$. In all cases $t+1 \geq d$, which proves the desired bound for V_2 . Together with (13.4) and the covering, this proves (13.1). $\square$

Corollary 13.4 (The $A = 0$ estimate for every count). *Fix $\mu > 1$. For all positive integers r, v and real $\gamma \geq \max\{0, r - v\}$,*

$$r! \text{vol}(\mathcal{T}_\mu(r, v, \gamma)) \leq C_\mu \min \left\{ 1, \frac{(\gamma - r + v + 1)(\gamma + 1)}{r} \right\}. \quad (13.9)$$

Proof. When $1 \leq r \leq 2v$, put $b = r - v$ and $\Gamma = \lceil \gamma \rceil$. The event increases with γ , so $\mathcal{T}_\mu(r, v, \gamma) \subseteq \mathcal{T}_\mu(r, v, \Gamma)$. Lemma 13.3 applies with $C = 2$ and $d = b - \Gamma \leq 0$. Discarding its denominator $\mu^{\mu^d} \geq 1$, and using

$$(\Gamma - b + 1)(\Gamma + 1) \leq 4(\gamma - b + 1)(\gamma + 1),$$

gives (13.9), after also using the simplex bound $\text{vol}(\mathcal{T}_\mu) \leq 1/r!$. If $r > 2v$, then $b > r/2$, $\gamma \geq b$, and $(\gamma - b + 1)(\gamma + 1)/r > 1/2$. The simplex bound alone proves the assertion with constant two in this case. Thus no restriction on r/v or centrality of the count is imposed. The singular expression $1/r$ is never used at $r = 0$. $\square$

13.1 The tail of the ordinary rectangle

The ordinary root rectangle involves a sum of exponentials of ordered positions, rather than a flat order-statistics wall. We state precisely the locally repaired input and its application to that rectangle.

Lemma 13.5 (Ordered-sum estimate in the required $A = 0$ range). *Fix $\mu > 1$. For positive integers n, v and $\gamma \geq 0$ with $\gamma \geq n - v$, let*

$$\mathcal{T}_\mu(n, v, \gamma) = \left\{ 0 \leq \xi_1 \leq \dots \leq \xi_n \leq 1 : \sum_{\ell \leq j} \mu^{v\xi_\ell} \geq \mu^{j-\gamma} \ (1 \leq j \leq n) \right\}.$$

Then

$$n! \operatorname{vol}(\mathcal{T}_\mu(n, v, \gamma)) \leq C_\mu \min \left\{ 1, \frac{(\gamma - n + v + 1)(\gamma + 1)}{n} \right\}. \quad (13.10)$$

Proof. This is Corollary 13.4, with $r = n$. $\square$

This is exactly the $A = 0$ portion of [19, Lemma 5.5], with the theorem numbering of arXiv:1102.3236v3. The proof in Section 13 includes integer rounding of the threshold and the passage to real γ . Its hypotheses include every n/v ; there is no central-count restriction. The factor $n!$ converts ordered-simplex volume to the probability for the order statistics of n independent uniform points. No assertion for $A > 0$ is needed here.

Lemma 13.6 (All-count tail of the ordinary rectangle). *Fix an integer $2 \leq h \leq H$ and $C_0 \geq 1$. Set $c = h - 1$, $w = \log h$ and $\mu = h^{1/c}$. For $v \geq 1$ an integer, let $\xi_1 \leq \dots \leq \xi_n$ be the ordered independent uniform points on $[0, 1]$, and put*

$$Z = (n - v) \log h, \quad F_n = \min_{0 \leq j \leq n} h^{-j} \left(1 + \sum_{\ell \leq j} \mu^{v\xi_\ell} \right)^c, \quad W_n = \min\{1, C_0 e^Z F_n\}. \quad (13.11)$$

There exist constants C, C_1 depending only on H, C_0 such that, for every $n \geq 0$ and $\ell \geq 0$,

$$\mathbb{P}\{W_n > e^{-\ell} \mid n\} \leq \min \left\{ 1, \frac{C(1 + \ell)(C_1 + Z + \ell)_+}{1 + n} \right\}. \quad (13.12)$$

For $\ell < 0$ the event is empty.

Proof. Suppose first $n \geq 1$ and the event is nonempty. The term $j = 0$ gives $F_n \leq 1$, hence $Z + \ell + \log C_0 > 0$. Define

$$\gamma_0 = \frac{Z + \ell + \log C_0}{\log h}, \quad \gamma = \gamma_0 + \frac{\log 2}{\log \mu}.$$

For each $1 \leq j \leq n$, the event in question implies

$$1 + \sum_{a \leq j} \mu^{v\xi_a} > \mu^{j-\gamma_0}.$$

The sum without the initial 1 is at least 1. Consequently

$$\sum_{a \leq j} \mu^{v\xi_a} > \frac{1}{2} \mu^{j-\gamma_0} = \mu^{j-\gamma}.$$

Here $\gamma \geq 0$, and the exact identity $Z = (n - v) \log h$ gives

$$\gamma - n + v = \frac{\ell + \log C_0}{\log h} + \frac{\log 2}{\log \mu} \geq 0.$$

Lemma 13.5 therefore applies in its proved $A = 0$ range. Furthermore

$$\gamma - n + v + 1 \leq C(1 + \ell), \quad \gamma + 1 \leq C(C_1 + Z + \ell)_+$$

on the nonempty event, after choosing C_1 large in terms of H, C_0 . There are only finitely many $\mu = h^{1/(h-1)}$, $2 \leq h \leq H$; the constants from Corollary 13.4 may therefore be replaced by their maximum over this fixed finite set. Its probability is at most the right side of (13.12), since $1 + n \leq 2n$.

If $n = 0$, then $F_0 = 1$ and $W_0 = \min(1, C_0 e^Z)$ is deterministic. On its nonempty event $Z + \ell > -\log C_0$. Increasing C_1 once makes the displayed right side equal to one. This proves the assertion also at count zero. Finally $W_n \leq 1$ makes the event empty for $\ell < 0$. $\square$

In the actual root application v is the number of completed harmonic cells, of width $\eta = \log h / (h - 1)$, and their budget is $U = \eta v$. Uniform latent points in the cells give exactly the variables used in (13.11); rounding changes the geometric constant C_0 by a fixed factor. Thus $Z = n \log h - (h - 1)U$ exactly. A bounded change of physical budget is kept in the external root translate, as in Section 15. If there are no cells, the completed Poisson count is identically zero and is covered directly. The subsequent Poisson maximum-atom and layer integrals use the same count.

14 Causal rising covers and the full native upper bound

This section proves the upper bound on the original divisor union. The main step treats a proper full-to-zero native transition with two small endpoint reserves. A conditional rising cover supplies both exponential half-line kernels needed by the original routing filtration. Their powers are the products of their respective owners, rather than the final product.

14.1 Notation and previously established inputs

Fix the dimension bound H . Perform the fixed bounded lift once and take its genuine ordered optimizer. Its maximal equal-product owners have products

$$0 < \sigma_* \leq \sigma_m < \cdots < \sigma_1 \leq \sigma^* < 1.$$

All constants below depend on H and the fixed conventions in the previously established estimates. Write

$$s = \sigma_1, \quad r_j = \frac{\sigma_j}{s}, \quad u_j = \frac{\sigma_{j+1}}{\sigma_j}, \quad d_j = \frac{1}{u_j} - 1 \quad (j < m). \quad (14.1)$$

The compound child weight of a proper owner is denoted by z_j . It satisfies $1 < z_* \leq z_j \leq H$, and a native slot with r eligible exterior names has original coarse law

$$\pi_j(c) = \frac{1}{r + z_j} \quad (c \text{ exterior}), \quad \pi_j(\text{child}) = \frac{z_j}{r + z_j}. \quad (14.2)$$

The following are the precise earlier inputs used in the proof. Their complete proofs are given in Appendix K; we state them together to make the parameters of the conditional cover argument explicit.

Proposition 14.1 (Geometric and scalar estimates). *The following assertions hold with constants depending only on H .*

1. *Every retained native transition is canonical $r \rightarrow q$, with full input and deepest-suffix output. Within an owner, sufficiently rare transitions have disjoint rank intervals: if one precedes another, then $q_1 \geq r_2$. Thus a zero-output transition, if present, is last. Its actual intrinsic endpoint scores are positive, and its persistence factor is*

$$\Psi_i = \min \left\{ 1, \left(\frac{(1 + X_i)(1 + Y_i)}{1 + \mu_i} \right)^{2\chi_i} \right\}.$$

The exponents and rates are in fixed compact ranges. Factors above any fixed positive rarity threshold are bounded below by a fixed positive constant.

2. *A retained nonzero-output transition has a necessary test on an original short native bank. The test reads that bank's coarse labels and the uncolored counts in its intermediate native groups. The corresponding true failure cover has volume at most $S_i e^{-h}$. Its private probability $p_i(h)$ satisfies*

$$\mathbb{E} p_i(h)^{\sigma_{o(i)}} \leq C(1 + h)^K \Psi_i.$$

The same assertion holds for a proper zero-output transition when $1 + X_i \geq c_X \sqrt{1 + L_i}$. In that case its intermediate groups are all the later native groups of its owner. These current-plus- intermediate groups are pairwise disjoint within each owner and across owners. All these estimates allow fixed point insertions at the same power. Original count middles are disjoint from the short label banks.

3. *For any proper full-to-zero transition which remains genuinely rare, let M be its native block length, A, B the preceding and following native lengths of that owner, and V_{old} its complete old physical length. Then*

$$V_{\text{old}} + A + B \leq C\tau M, \quad T_j \asymp M, \quad (14.3)$$

$$L = 1 + B, \quad R = 1 + V_{\text{old}} + A + \varrho,$$

$$\varrho = \max \left\{ \varrho_0, \frac{c_0 \sqrt{1 + M}}{1 + d_j \sqrt{1 + M}} \right\},$$

$$X_i \asymp R, \quad Y_i \asymp L, \quad |e|M \leq C(V_{\text{old}} + A + B), \quad (14.4)$$

where $e = (r + z_j)^{\sigma_j} \mathbb{E} W - r$ and $W = \log(r + z_j) - \log z_j \mathbf{1}_{\text{child}}$. The V_{old} here consists of the genuine original groups and their transported coefficients. It is not a newly sampled old list.

4. *On disjoint original end banks and a middle bank in the M block, the signed logarithmically lowered one-end estimates and the whole-band binomial maximum atom give*

$$\mathbb{E} B_w(h, n)^{\sigma_j} \leq C(1 + h)^K (1 + |n|)^K F_j^0, \quad F_j^0 = \left(\frac{\varrho}{\sqrt{1 + M}} \right)^{\sigma_j} \min \left\{ 1, \left(\frac{LR}{1 + M} \right)^{2\chi_i} \right\}. \quad (14.5)$$

Here $w \geq M/2$; a completed-word band has width ϱ , while a first-hit band may have fixed width. The left reserve is at most $h + C(1 + N_B)$ and the right reserve is at most $C(1 + h + V_{\text{old}} + A + (|n| + 1)\varrho)$. The estimates sum all original counts and keep all $n + 1$ conditional gaps. If

the larger reserve is superdiffusive, only the smaller-reserve bank of length $cM^2/\max(L, R)^2$ is retained. In the small-gap case $d_j\sqrt{M} < 1$, a left bank alone gives the same factor, since $\varrho/\sqrt{M} \asymp 1$.

5. *Fixed-dimensional original shot sums have all fixed moments. Their insertion estimates are obtained by a finite Poisson–Mecke partition before summing bank locations. A count middle occurring in an intermediate reserve group is handled by the conditional inequality*

$$\mathbb{E}[f(N)g(N)] \leq \mathbb{E}f(N) \mathbb{E}g(N) \quad (f \text{ nonincreasing, } g \text{ nondecreasing}), \quad (14.6)$$

with all other groups fixed. This includes its count weight $(1+d_j\sqrt{1+N})^{-1}$ and an increasing necessary-bank reserve.

For clarity, the bounded-scaled-drift hypothesis in (14.5) follows from (14.4): use two banks of length $M/16$ if $\max(L, R) \leq C\sqrt{M}$, and otherwise a bank of length $cM^2/\max(L, R)^2$. The original correlation parameter and the centered scalar parameter differ by $O(LR/M)$, so the exponent comparison costs only $\exp\{O((LR/M)|\log(LR/M)|)\}$. Thus (14.5) has the same current intrinsic factor as the lower bound.

Let F_j be the proper owner's count factor, at power σ_j , times all its retained native persistence factors. For the terminal owner it is just the product of its retained nondegenerate factors. Consequently

$$\prod_{j=1}^m F_j = \mathcal{C}_{\text{count}} \mathcal{P}_{\text{ret}}. \quad (14.7)$$

Each F_j has a dimension-dependent polynomial lower bound in the total length of the original native groups it reads. The terminal ordinary factor is denoted by $\mathcal{B}_I$ and is kept separate.

14.2 Actual routing capacities and two left covers

By Lemma 2.2, the full squarefree harmonic volume sum is at most twice its restriction to $a_i \leq y_i^{T_k}$ for all i . The prefix bounds (2.6) supply the dimension-constant full-size rectangles below. Fix such an original squarefree tuple and its complete uncolored data G . Its original product fine-label measure is π_G . Let $\mathcal{Q}_j$ reveal the complete first j owner routes and leave the genuine descendant alphabet compounded. Write Z_j for the actual grouped output and put

$$P_j = \sum_{\ell \leq j} Z_\ell, \quad \eta_j = \sum_{\ell \leq j} \frac{\sigma_\ell}{\sigma_j} Z_\ell, \quad X = \sum_{j=1}^m r_j Z_j. \quad (14.8)$$

The common root X is uncolored. For $j < m$ the exact identities are

$$P_j = \eta_j - \sum_{\ell < j} d_\ell \eta_\ell, \quad P_m = \frac{X}{r_m} - \sum_{\ell < m} d_\ell \eta_\ell. \quad (14.9)$$

Call a proper owner *hard* if its retained zero-output edge does not satisfy the one-end admissibility hypothesis of Proposition 14.1(2). All other owners are simple. A hard owner has one selected zero block and all its other retained edges are earlier nonzero edges. Constants may be chosen so that a hard owner has $d_j\sqrt{M} \geq 1$; the small-gap variant below also covers the bounded transition range.

Lemma 14.2 (Original projected rectangles and dual left walls). *A complete $\mathcal{Q}_j$ atom q has actual capacity*

$$\text{Vol}(q) \leq C\pi_G(q)e^{P_j(q)}. \quad (14.10)$$

At a hard owner, fix a genuine $\mathcal{Q}_{j-1}$ atom α . Read its selected zero block backwards and set

$$C_j(w) = H_B + \sum_{x < w} [\log(r + z_j) - \log z_j \mathbf{1}_{\text{child}}] - rw, \quad Z_j = C_j(M) + Y_A. \quad (14.11)$$

Then $H_B \leq CN_B$ and $Y_A \geq -C(V_{\text{old}} + A)$ wordwise. There are two genuine partial capacities,

$$\text{Vol}(Q_w^{\text{glob}}) \leq C\pi_G(Q_w^{\text{glob}})e^{C_j(w)} \text{Shot}_w, \quad (14.12)$$

$$\text{Vol}(\alpha, Q_w) \leq C\pi_G(\alpha, Q_w)e^{P_{j-1}(\alpha) + C_j(w)} \text{Shot}_w. \quad (14.13)$$

Imposing both left walls

$$C_j(w) \geq -h - \mathcal{L}(w), \quad P_{j-1} + C_j(w) \geq -h - \mathcal{L}(w), \quad \mathcal{L}(w) = A_0 \log(1 + \min(w, M - w)), \quad (14.14)$$

costs at most CSe^{-h} in the union of their failures. Their simultaneous survivors have effective reserve

$$h_{\text{eff}} = h + \min(P_{j-1}, 0) \leq h. \quad (14.15)$$

Here $S \geq 1$ is an uncolored original shot with all fixed moments; it may be chosen jointly for the finitely many owners and other native failure covers, without reading the final unread root suffix.

Proof. Fixing the observed exterior divisors gives translated original rectangles in their coordinates. All unresolved coordinates keep their original size bounds. The negative logarithm of the observed class probability is the sum of its original information marks. Subtracting the geometric budgets gives precisely P_j , proving (14.10).

For (14.12), leave all ancestor and earlier coordinates in their full rectangles. Reveal the later native word and the crossed points of the current block. The r unresolved exterior widths have common threshold $b - w$, where b is the block's original right endpoint. Their budget deficit is $rw + \sum_{\text{later}} r_g L_g$. The negative observed log mass is the corresponding mark sum, so their difference is $C_j(w)$. Every other original coordinate budget cancels. In particular no earlier private output or selected incoming count occurs in this capacity.

Instead first fix α . Its actual resolved translations give $\pi_G(\alpha)e^{P_{j-1}}$. The same rectangle on its literal child fiber gives the conditional factor $\pi_G(Q_w \mid \alpha)e^{C_j(w)} \text{Shot}_w$. Their product is (14.13). The fiber is not renormalized by a survival event. A shot using all original points dominates its selected-list shot by positivity. The two capacities are different covers, not two factorizations of one artificially independent route.

At the first failure of either wall, including $w = 0$, substitute the failed inequality into its corresponding capacity. The result is the original class probability times $e^{-h - \mathcal{L}(w)} \text{Shot}_w$. The first-failure classes form an antichain in their own reveal order. For the second cover, first sum conditional Kraft masses and then their original ancestor probabilities. Their total is at most one. Summing the fixed logarithmic shot weights gives CSe^{-h} for each cover. The two reveal orders need not be one filtration: the union bound retains all their failures. Finally their intersection is exactly (14.15). The wordwise bounds in the statement follow from positive information marks and the original physical rank budgets; no mean of an old list is used. $\square$

14.3 Conditional half-line kernels on true incoming fibers

We first record the simple-owner calculation. A proper native slot is after all ancestor exterior deadlines, so its ancestor category is deterministically child. Thus conditioning on any $\mathcal{Q}_{j-1}$ atom leaves its native labels with exactly the product law (14.2).

Lemma 14.3 (Simple-owner kernels). *Choose an original native count middle of n slots, disjoint from the owner's necessary label banks. Let $E_j(h)$ be their joint event and $p_j(h, G)$ its private probability. For every uncolored threshold ξ_j , set $\zeta_j = \eta_j - \xi_j$. Uniformly in every genuine $\mathcal{Q}_{j-1}$ atom,*

$$\begin{aligned} \mathbb{E}_{\pi_G}[e^{\zeta_j} 1_{\{\zeta_j \leq 0\}} 1_{E_j(h)} \mid \mathcal{Q}_{j-1}] &\leq C b_j(h), \\ \mathbb{E}_{\pi_G}[e^{-d_j \zeta_j} 1_{\{\zeta_j > 0\}} 1_{E_j(h)} \mid \mathcal{Q}_{j-1}] &\leq C b_j(h), \\ b_j(h) &= (1 + d_j \sqrt{1+n})^{-1} p_j(h, G). \end{aligned} \quad (14.16)$$

For a terminal owner the corresponding conditional mask mass is bounded by $b_m(h) = p_m(h, G)$.

Proof. Fix the actual ancestor atom and then the entire current coarse route outside the middle. Its masks are now fixed, and

$$\zeta_j = H_0 - \log z_j K, \quad K \sim \text{Bin}\left(n, \frac{z_j}{r + z_j}\right),$$

where H_0 may depend arbitrarily on the old list and outside word. The binomial maximum atom is at most $C(1+n)^{-1/2}$. Summing the negative exponential half-lattice gives $C(1+n)^{-1/2}$; summing the positive half-lattice gives $C/[d_j \sqrt{1+n}]$, capped by one. At $d_j = 0$ use only the probability bound. The resulting common bound is $C(1 + d_j \sqrt{1+n})^{-1}$, uniformly in H_0 . Release the outside labels only after taking this uniform bound. Their mask mass is $p_j(h, G)$ on every ancestor fiber. The terminal assertion is the same native-only observation without a count weight. Future child words were not fixed while varying K . $\square$

For a hard owner write $P = V_{\text{old}} + A$ and let $E_{<}(h)$ denote its earlier own native masks. Their probability $p_h(G_A)$ reads only original native groups strictly before the selected block. Choose a fixed D_A and put

$$\alpha_h = \min\{1, (2 + A)^{-D_A} + p_h(G_A)\}. \quad (14.17)$$

By its polynomial factor floor, D_A can be chosen so that

$$p_h \leq \alpha_h \leq 1, \quad -\log \alpha_h \leq D_A \log(2 + A) \leq C(1 + P), \quad \mathbb{E} \alpha_h^{\sigma_j} \leq C(1 + h)^K \mathcal{P}_{<}. \quad (14.18)$$

An empty earlier collection has $\alpha_h = 1$.

Lemma 14.4 (Hard-owner kernels and right stops). *On a fixed genuine ancestor atom let*

$$x_j = \xi_j - \frac{\sigma_{j-1}}{\sigma_j} \eta_{j-1} \quad (j > 1), \quad x_1 = \xi_1, \quad \zeta_j = Z_j - x_j. \quad (14.19)$$

In addition to the dual left walls, use the right wall

$$C_j(w) \geq x_j + \log \alpha_h - \log S - A_1 \log(1 + M - w). \quad (14.20)$$

Assume its deterministic gate places right-governed first hits after $M/2$. There is one uncolored coefficient $b_j(h)$, independent of the ancestor private word and of x_j , with all three properties:

1. the total right-stop capacity on the ancestor fiber is at most $C\pi_G(\alpha)e^{P_{j-1}+x_j}b_j(h)$;
2. complete surviving owner words with $\zeta_j \leq 0$ have conditional weighted mass at most $Cb_j(h)$ under weight e^{ζ_j} ;
3. complete surviving owner words with $\zeta_j > 0$ have conditional weighted mass at most $Cb_j(h)$ under weight $e^{-d_j\zeta_j}$.

The complete-survivor conclusions themselves do not require the gate. The small-gap variant omits the right wall and internal right stops.

Proof. First consider complete survivors. The exact numerical identity

$$e^\zeta 1_{\{\zeta \leq 0\}} + e^{-d_j\zeta} 1_{\{\zeta > 0\}} = e^{-x_j} \min\{e^{Z_j}, e^{x_j/u_j - d_j Z_j}\} \quad (14.21)$$

is only a test-function identity. No child volume bound is used. Set $\bar{x} = x_j - \log S$ and divide the WHOLE output $Z_j - \bar{x}$ into bands $[n\varrho, (n+1)\varrho)$, $n \in \mathbb{Z}$. The right side of (14.21) is at most $S^{d_j}w_n$, where

$$w_n = e^{-d_j n \varrho} \quad (n \geq 0), \quad w_n = e^{-(|n|-1)\varrho} \quad (n \leq -1). \quad (14.22)$$

For positive bands use the second exponential branch and for negative bands the first. This remains valid if a shifted band crosses $\zeta = 0$, since the minimum is at most either branch everywhere. If $d_j\sqrt{M} \geq 1$, both series of $w_n^{\sigma_j}$ times every fixed polynomial in $|n|$ converge uniformly.

The reflected necessary reserve follows by subtracting (14.20) from $Z_j = C_j(M) + Y_A$:

$$\begin{aligned} C + Z_j - \bar{x} - Y_A - \log \alpha_h &\leq C_H(1 + V_{\text{old}} + A) + (|n| + 1)\varrho + D_A \log(2 + A) \\ &\leq C_D(1 + V_{\text{old}} + A) + (|n| + 1)\varrho. \end{aligned} \quad (14.23)$$

Both the supplied root level and $\log S$ cancel. The small left reserve is still $h + C(1 + N_B)$: only for this necessary test we enlarge h_{eff} to h .

For a fixed band, fix the complete current old and earlier native word, the later native word, and all selected-block labels outside its middle bank. The whole-band center is now fixed. The middle child-count maximum atom is uniform in that center. The two end tests use disjoint original slots and (14.23) no longer reads the earlier private word. Release the earlier own word with its event $E_<(h)$, giving its exact mass $p_h \leq \alpha_h$. Release the true old word on the ancestor fiber with total mass one. The later word is released with its original mass, using $H_B \leq CN_B$ only in the left necessary reserve. Thus the conditional mass of the band is at most $C\alpha_h B_M(h, n)$, uniformly over the actual incoming list. With

$$T_{\text{full}}(h) = \sum_{n \in \mathbb{Z}} w_n B_M(h, n),$$

each of the two complete half-lines is at most $CS^{d_j}\alpha_h T_{\text{full}}(h)$.

At a right first hit, positivity of the jumps and bounded downward speed put the risk in a bounded strip below its wall. This does not require a bound on a tied positive prime batch. Substitution into (14.13), divided by $\pi_G(\alpha)e^{P_{j-1}+x_j}$, gives

$$C \frac{\alpha_h}{S} (1 + M - w)^{-A_1} \text{Shot}_w \quad \text{times the true conditional prefix mass.} \quad (14.24)$$

The factor α_h comes from the actual capacity, not from multiplying an unobserved earlier probability into a node. Prior survival and the first-hit strip imply the fixed- w end-bank tests and one

fixed-width middle atom. Their centers can read the fixed ancestor data, but their upper bounds are uniform. Let $T_{\text{right}}(h)$ sum these bounds with the weights and shots in (14.24). One may therefore take

$$b_j(h) = C\alpha_h[S^{d_j}T_{\text{full}}(h) + S^{-1}T_{\text{right}}(h)]. \quad (14.25)$$

This proves the three separate statements.

If $d_j\sqrt{M} < 1$, then $\varrho \asymp \sqrt{M}$ and $R \geq c\sqrt{M}$. Keep only the necessary left bank and omit the right wall. The function in (14.21) is at most one. Whole-word conditioning gives p_h times this bank's probability; its moment is F_j^0 because the count-width factor is comparable to one. This avoids any series with a decay rate tending to zero. $\square$

Lemma 14.5 (Joint moments at all owner powers). *The simple and hard costs can be chosen so that, for every subset $J \subseteq \{1, \dots, m\}$, every nonnegative integer vector (h_j) , and every fixed p ,*

$$\mathbb{E} \left[S^p \prod_{j \in J} b_j(h_j)^{\sigma_j} \right] \leq C_p \prod_{j \in J} (1 + h_j)^K F_j, \quad (14.26)$$

$$\mathbb{E} \left[(1 + N_{\text{pre}}) S^p \prod_{j \in J} b_j(h_j)^{\sigma_j} \right] \leq C_p (1 + V_{\text{pre}}) \prod_{j \in J} (1 + h_j)^K F_j. \quad (14.27)$$

Moreover $b_j(h) \leq CS^Q$ pointwise, uniformly in h , for a fixed Q . These are expectations over the ORIGINAL uncolored Poisson restrictions, not over clocks conditioned on a chosen stopping route.

Proof. For a hard owner, before attaching shots the input of α_h is strictly before its selected block. The fixed- w , fixed-band bank expression reads only that block and later native counts. Original Poisson restrictions on these groups are independent. Apply (14.18) and (14.5) at the same power σ_j , then sum the signed bands and weighted positions. This gives $\mathcal{P}_{<} F_j^0 = F_j$ with a fixed polynomial in h .

For a simple owner, a middle count in an intermediate group is not independent of the corresponding mask reserve. Condition on other groups and use (14.6) once, then integrate the disjoint groups. The same argument includes an admissible large-left zero edge. No input group leaves its own native owner. Different owners' resulting functions therefore use disjoint original Poisson restrictions. Old physical lengths in a hard reserve are deterministic and introduce no old random count into its cost.

Attach a global shot by a finite Mecke partition to each fixed bank expression before summing positions or forming any reserve envelope. Inserted native points are deleted at the cost of a bounded reserve shift; a later count raises N_B by a bounded amount; an inserted middle point decreases its inverse-count bound. The earlier mask has $\alpha_h(G + \text{points}) \leq C_k \alpha_{h+C_k}(G)$, with unchanged floor. Apply the preceding disjoint-group estimates after these fixed insertions. This proves (14.26); a raw-count insertion costs its own mean and proves (14.27). Omitted owners have factor one, so the proof holds for every subset. In particular it applies to mixed terms created by deterministic floors. There is no redistribution of powers by Holder's inequality.

Finally every private bank mass and truncated maximum-atom upper is at most one, the signed band weights have uniformly bounded sum, and the weighted right-position sum is bounded by S . The exponents d_j are bounded. This proves the pointwise bound, which will pay positive logarithms; it is not inferred from a fractional moment. $\square$

14.4 Floors, the causal gate, and the original-route tower

For a proper owner let T_j be its total native length. For the terminal owner use only its native head before the unread suffix. Choose fixed D_j so that $f_j = (2 + T_j)^{-D_j}$ satisfies $f_j^{\sigma_j} \leq CF_j$. Set

$$a_j = f_j + \sum_{h=0}^{\infty} (1+h)^{-\beta} b_j(h), \quad B_* = \sum_{j=1}^m r_j \log a_j, \quad \theta_j = \frac{1}{r_j} \sum_{\ell > j} r_\ell \log a_\ell, \quad \xi_j = \frac{X}{r_j} + \theta_j. \quad (14.28)$$

The a_j are positive uncolored numbers and need not be at most one. They are fixed before the single cover reserve h is chosen. In particular

$$b_j(h) \leq a_j(1+h)^\beta, \quad -\log a_j \leq D_j \log(2 + T_j). \quad (14.29)$$

The pointwise bound in Lemma 14.5 makes every sum finite when $\beta > 1$.

Lemma 14.6 (Mixed pooling and the no-future-scale gate). *Choose $\beta\sigma_* > K + 2$ and $\beta_* \geq m\beta$, and put*

$$L_S = 1 + \log(2S), \quad D = L_S^{\beta_*} e^{B_*}. \quad (14.30)$$

Then

$$\begin{aligned} \mathbb{E}[S^p D^s] &\leq C_p \prod_j F_j, \\ \mathbb{E}[(1 + N_{\text{pre}} + V_{\text{pre}} + |\log D|) D^s] &\leq C(1 + V_{\text{pre}}) \prod_j F_j. \end{aligned} \quad (14.31)$$

If $w_0 = X + \log D < 0$, choose

$$h = \lceil \log(2S) - w_0 \rceil. \quad (14.32)$$

At every reached hard owner, whose previous proper outputs continued, the right-hit gate holds with a fixed dimension constant.

Proof. Expand the separate sums and floors using subadditivity at each $\sigma_j < 1$. Apply (14.26) to every resulting owner subset, and sum $(1+h)^{K-\beta\sigma_j}$. Since $sr_j = \sigma_j$, this gives the first bound in (14.31). Use (14.27) for the raw count. The floors give $(-\log D)_+ \leq C \log(2 + V_{\text{pre}})$. The pointwise polynomial shot bound gives $\log_+ D \leq C(1 + \log S)$, which is absorbed by a fixed additional shot moment. This proves the second bound.

Set $\zeta_\ell = \eta_\ell - X/r_\ell - \theta_\ell$ and

$$c_j = \theta_j - \sum_{\ell < j} d_\ell \theta_\ell \quad (j < m), \quad c_m = - \sum_{\ell < m} d_\ell \theta_\ell. \quad (14.33)$$

Direct telescoping gives, including $j = m$,

$$c_j + \sum_{\ell \leq j} \log a_\ell = B_*. \quad (14.34)$$

The local level at a hard owner satisfies

$$\begin{aligned} h_{\text{eff}} + x_j &\leq h + P_{j-1} + x_j \\ &= h + X + c_j - \sum_{\ell < j} d_\ell \zeta_\ell \leq h + X + c_j. \end{aligned} \quad (14.35)$$

The first inequality holds for both signs of P_{j-1} , since $\min(P_{j-1}, 0) \leq P_{j-1}$. The last uses the true previous continuations $\zeta_\ell > 0$. The choice (14.32) yields $h + X + B_* \leq \log(2S) + C$. Therefore

$$h_{\text{eff}} + x_j + \log \alpha_h - \log S \leq C - \sum_{\ell \leq j} \log a_\ell \leq C + \sum_{\ell \leq j} D_\ell \log(2 + T_\ell). \quad (14.36)$$

All future-owner scales have canceled. At a hard owner, (14.3) implies that every preceding owner's native length is at most $V_{\text{old}} \leq CM$ and the current $T_j \leq CM$. Thus the last expression is at most $D_{\text{gate}} \log(2 + M) + C$. Choose A_1 large relative to D_{gate} and A_0 ; direct comparison of the two logarithmic walls then prevents a right-governed first hit before $M/2$. This is the required gate. No sign assumption on an incoming output and no bound on a future length were used. $\square$

Remark 14.7 (Order of constants). First fix dimension compactness, rarity and deterministic bank scales. Choose the prefix floors D_A , owner floors D_j , and then D_{gate} from those floors. Choose A_1 next, including the condition $A_1 \sigma_* > 2$ for position sums. Determine the scalar reserve exponent K after this choice, and finally choose β and β_* . If a scalar effective-bank threshold depends on A_1 , the bounded effective-length case is treated by the probability bound one and the same middle atom: its persistence ratio is bounded below by a fixed positive constant. One does not reduce the original rarity cutoff in a way which feeds back into D_{gate} . Increasing β_* only improves the gate.

Theorem 14.8 (Actual causal rising profile). *For the original complete divisor union there is a prefix-uncolored $D > 0$, unread by the final ordinary suffix, such that at its TRUE root*

$$I_{\text{actual}} \leq C J_{\beta_*}(X + \log D), \quad J_b(w) = \min\{1, C_b e^w (1 + (-w)_+)^b\}, \quad (14.37)$$

and D satisfies (14.31). All original failure words are included in this upper bound.

Proof. Use all genuine left-failure covers, keeping their union cost CSe^{-h} . On the survivors route owners in order. At a hard owner allow its internal right stop; otherwise complete its route and stop at the first $\eta_j \leq \xi_j$. If there is no proper stop, use the complete terminal word. These are actual projected branches. A later private filter is never charged at an earlier projection.

For a complete stop at owner $j < m$, the exact capacity exponent is

$$P_j = X + c_j + \zeta_j - \sum_{\ell < j} d_\ell \zeta_\ell. \quad (14.38)$$

Condition on its true $\mathcal{Q}_{j-1}$ fiber and apply the negative kernel, then integrate earlier routes backwards with their positive kernels. Each bound is uniform on every genuine preceding fiber. The result is at most $Ce^{X+c_j} \prod_{\ell \leq j} b_\ell(h)$.

For an internal hard right stop, its reference exponent is exactly

$$P_{j-1} + x_j = X + c_j - \sum_{\ell < j} d_\ell \zeta_\ell.$$

Use the separate right-stop assertion of Lemma 14.4, followed by the same backward positive kernels. It has the same upper bound. For the terminal branch, (14.9) gives $P_m = X + c_m - \sum_{\ell < m} d_\ell \zeta_\ell$. Integrate its true native mask first and then use the earlier positive kernels.

The conditional tower integrates each future child word by a uniform bound before varying the current native count. The current middle therefore retains its conditional binomial law, and ancestor atoms retain their original probabilities.

By (14.29) and (14.34), each branch is at most $Ce^{X+B_*}(1+h)^{j\beta}$. There are dimension-many branch types, so

$$I_{\text{actual}} \leq C \min\{1, Se^{-h} + e^{X+B_*}(1+h)^{m\beta}\}. \quad (14.39)$$

For $w_0 < 0$, Lemma 14.6 permits the reserve in (14.32). It satisfies $1+h \leq CL_S(1+(-w_0)_+)$, and (14.30) now gives (14.37). For $w_0 \geq 0$, use the global original rectangle and no gated cover. The case $m = 1$ has no proper frontier and follows directly from its terminal native masks. $\square$

14.5 One unread ordinary root and the arithmetic conclusion

We state explicitly the earlier one-root input, including its range. After the proved whole-original-block trimming needed by terminal nonzero banks, the unread ordinary suffix has length U , largest block M_* , side lengths A_*, B_* , and deterministic mean surplus Δ_* . It retains the same intrinsic ordinary factor

$$\mathcal{B}_I \asymp (1+U)^{-1/2} \min\left\{1, \frac{(1+B_*)(1+\Delta_*+A_*)}{1+M_*}\right\}, \quad \Delta_* \geq cV_{\text{pre}}. \quad (14.40)$$

The trimming leaves the largest root block unchanged and reads no clock needed by the new proper-owner banks. On the same original tuple its genuine endpoint and largest-block ordinary rectangles may be combined with (14.37).

Proposition 14.9 (The joint ordinary root estimate). *Let $D > 0$ and κ be arbitrary original-prefix-measurable numbers, and write $X = \kappa + (\sigma_m/s)Z_{\text{suffix}}$. If the same tuple has the profile (14.37), its combination with the ordinary suffix rectangles satisfies*

$$\mathbb{E}_{\text{suffix}}[e^{-sX} I_{\text{actual}} \mid G_{\text{pre}}] \leq CD^s e^{-\sigma_m H_-} (1+U)^{-1/2} \min\left\{1, \frac{(1+B_*)(1+H_++A_*)}{1+M_*}\right\}, \quad (14.41)$$

Here $H = (-\log D - \kappa)/(\sigma_m/s)$. This is valid for every translated root, all original counts, and the fixed polynomial profile exponent. It contains the one largest-block atom jointly with its ballot denominator, with all $n+1$ gaps. The rectangle's ordered-sum tail is Lemma 13.6, including its separate count-zero case. No additional Poisson atom is to be appended.

Proof. The same-count layer calculation, including every translate and the polynomial outgoing profile, is given in Section K.9. $\square$

Theorem 14.10 (Upper bound with the same intrinsic factor). *At the fixed lifted physical optimizer,*

$$K_k(\ell^+) \leq C_H e^{-J} \mathcal{C}_{\text{count}} \mathcal{B}_I \mathcal{P}_{\text{native}}. \quad (14.42)$$

Together with the previously established complete intrinsic arithmetic lower, this is a two-sided estimate with that same finite factor. Actual bounded-parameter stability transfers it to the original vector; the original multiplication-table and H comparisons are then applied only on their original admissible domains.

Proof. The profile coefficient of Theorem 14.8 reads no uncolored point of the unread suffix. Apply (14.41) ONCE at the actual original root. The exact prefix normalization gives $|\kappa| \leq C(1+V_{\text{pre}}+N_{\text{pre}})$. For its first alternative use $e^{-\sigma_m H_-} \leq 1$ and the first bound of (14.31). For its second alternative use

$$1+H_++A_* \leq C(1+A_*+V_{\text{pre}}+N_{\text{pre}}+|\log D|)$$

and the second bound. Average these two upper bounds separately and then take their minimum. The inequality in (14.40) compares the result with $C\mathcal{B}_I \prod_j F_j$.

All actual geometric covers preceded the positive prime-series and cell enlargement. The original exponential change of measure bounds K_k above by $C_H e^{-J} \mathbb{E}[e^{-sX} I_{\text{actual}}]$; consistently rounded harmonic budgets have only a fixed normalization error. Thus (14.7) proves (14.42) with $\mathcal{P}_{\text{ret}}$. Changing between the finitely many fixed rarity thresholds only changes a dimension constant, so the full $\mathcal{P}_{\text{native}}$ is equivalent. The current correlations were retained by Proposition 14.1 and the bank comparisons. Finally invoke the actual lower and the arithmetic bounded-lift stability. $\square$

15 Actual prime clocks and the positive core

The lower construction requires one sufficiently large fixed length lift. The upper bound has no such restriction. We prove here the precise extension needed in Proposition 16.4:

Theorem 15.1 (Positive-core upper). *For each fixed k , the numerical profile of (3.12) satisfies*

$$K_k(\log 3 + L_1, \dots, \log 3 + L_k) \leq C_k \Psi_k(L), \quad L_i \geq 1. \quad (15.1)$$

The optimizer, its owners, and all endpoint scores in this formula are taken at the displayed physical vector L . No optimizer is taken at a rounded prime-cell vector.

We give the arithmetic and probability details of this extension. The geometric covers, scalar all-count estimates, and conditional owner tower are the established inputs and theorems of Section 14. Their length thresholds concern only effective scalar banks; bounded banks are treated below. Lemma 2.2 first restricts the full arithmetic sum to the size-truncated original tuples used by those covers, at a factor of two. Its prime blocks include the original first primes 2 and 3.

15.1 Actual prime banks and their exact harmonic clock

Put $H = k + 1$. All physical saddle products, Gibbs probabilities and rates are frozen. By (3.3), all parameters needed by the nondegenerate scalar laws belong to dimension-dependent compact sets. For an original block, with either forward or reverse log-log coordinate $t(p)$, the uniform prime harmonic estimate gives

$$\left| \sum_{p: t(p) \in [a, a+v]} \frac{1}{p} - v \right| \leq C_H. \quad (15.2)$$

The constant covers open endpoints and fixed omitted initial primes. An empty terminal piece has bounded length and causes the same error. This estimate is uniform in a, v and the original block.

For clarity about the initial primes, set

$$\eta_* = \min_{2 \leq h \leq H} \frac{\log h}{h-1}, \quad Q_H \geq \max(3, 4/\eta_*).$$

Deleting one prime from an actual divisor union costs at most its H possible labels. Summing over the presence of the finitely many primes $p \leq Q_H$ costs at most

$$\prod_{p \leq Q_H} (1 + H/p),$$

a fixed dimension constant. Deletion preserves each original upper size constraint. The remaining calculation still uses the original physical optimizer and budgets: the missing harmonic mass is part of the bounded error in (15.2). This is an upper operation, not an extra roughness assumption on the input.

On a deterministic native bank I , write $H_I = \sum_{p \in I} 1/p$. The generalized inverse T of its harmonic cumulative measure obeys

$$|T(s) - s| \leq C_H \quad (0 \leq s \leq H_I), \quad H_I = |I| + O_H(1). \quad (15.3)$$

Push a homogeneous rate- Z_i Poisson process through T , retaining its iid Gibbs marks. The image is exactly the required prime process with intensity $Z_i \sum_{p \in I} p^{-1} \delta_{t(p)}$. In particular this identity permits arbitrarily many occurrences at a single prime. Disjoint deterministic actual banks are independent.

The signed wall comparisons are pointwise. For a positive-jump lower wall, the homogeneous partial sum inside a collapsed atom is at least the actual pre-jump sum. For a negative-jump wall use the actual post-jump minimum: an incomplete sum of the negative jumps is larger. The linear clock error in (15.3) and a fixed logarithmic soft allowance enlarge the reserve by $O_H(1)$. Thus an actual necessary wall implies the corresponding homogeneous necessary wall with that enlarged reserve. No multiplicity or central-count bound is involved.

The middle-binomial atom is likewise an all-count assertion: conditional on the actual middle count N_I , it is at most $C(1 + N_I)^{-1/2}$, uniformly in the band center and in every outside private word. For fixed $a > 0$,

$$\mathbb{E}(1 + N_I)^{-a} \leq C_{H,a}(1 + |I|)^{-a}. \quad (15.4)$$

For large $|I|$ this follows from the Poisson lower-tail estimate and (15.3); for bounded $|I|$ the right side has a fixed positive lower bound. The translated half-line binomial bound proves the same extension for a proper count window of width $1/d$.

All prefix shot and insertion arguments remain on these actual atomic processes. For a fixed number of inserted occurrences, condition their marks and enlarge each necessary reserve by at most the number inserted times the largest mark. The middle bound can still use the original free middle occurrences. A fixed-degree shot polynomial has a finite Mecke partition, so only a fixed finite number of such insertions occurs. Its exponentially or summably polynomially weighted intensity moments remain bounded under (15.3). The joint subset moment estimate therefore retains the original power σ_j of every owner.

15.2 Bounded effective banks

If an effective bank has length exceeding a fixed scalar threshold plus $2C_H$, its harmonic length is comparable to its physical length. Apply the sharp signed one-end bound of Theorem 6.11 after (15.3). The reserve change is harmless because the factor contains $1 + \text{reserve}$.

If an effective bank is bounded, its numerical persistence factor has a fixed positive lower bound. Indeed the selected one-end regime has $\bar{T} \asymp_H (1 + Y)/\sqrt{\ell}$, and the selected two-end regime has the product of the two normalized reserves. Since each reserve is at least one, bounded ℓ cannot give a vanishing factor. Discard that test in the upper and keep the other banks and the count middle, with the rarity cutoff fixed in the order of Remark 14.7. A bounded total native owner length also has a count factor bounded below; both weighted half-lines are at most one. These arguments are uniform for all $L_i \geq 1$.

15.3 One domination for the entire completed Poisson law

The unread ordinary root suffix is represented by equal mass cells. The following whole-law comparison handles the completion with all private powers kept inside the integrand.

Lemma 15.2 (Whole-law mass completion). *Let $0 \leq \mu_c \leq \mu_c^*$ on a finite collection of cells, and let $\Delta = \sum_c (\mu_c^* - \mu_c)$. Attach the same conditional Gibbs label law to the counts under both product Poisson measures. For every nonnegative measurable functional F of the entire environment and private word,*

$$\mathbb{E}_\mu F \leq e^\Delta \mathbb{E}_{\mu^*} F. \quad (15.5)$$

This remains true conditionally on any unchanged prefix.

Proof. On the support of the first measure its likelihood ratio is

$$\frac{d\mathbb{P}_\mu}{d\mathbb{P}_{\mu^*}}(n) = e^\Delta \prod_c (\mu_c / \mu_c^*)^{n_c} \leq e^\Delta.$$

Use the usual support convention for zero means. The identical Gibbs conditional law cancels from the ratio. Integrating F proves the claim. $\square$

For example F may contain the minimum of all actual rectangles, the true root tilt, shots, and several conditional masses raised to different owner powers. The coupling $N_c^* = N_c + \text{Pois}(\mu_c^* - \mu_c)$ represents the two laws. Inequality (15.5) compares complete integrands by their likelihood ratio; it requires no pointwise monotonicity under added points.

In our use the actual same-tuple minimum has already been formed. All prefix costs and gates are fixed before the ordinary suffix is sampled. We apply the lemma once to its whole nonnegative suffix integrand.

15.4 A bounded total cell deficit and the physical rectangle

For an ordinary block with h colors put $\eta = \log h / (h - 1)$. Greedily group successive primes above Q_H until adding the next prime would exceed mass η . If p_j begins a complete cell and p_{j+1} begins the next one, its harmonic mass H_j satisfies

$$\eta - 1/p_{j+1} \leq H_j \leq \eta, \quad H_j \geq 3\eta/4. \quad (15.6)$$

Since the prime sum is at most the corresponding integer harmonic sum,

$$H_j \leq 1/p_j + \log(p_{j+1}/p_j).$$

Our choice of Q_H gives

$$p_{j+1} \geq e^{\eta/2} p_j, \quad \sum_j (\eta - H_j) \leq \frac{1}{Q_H(e^{\eta/2} - 1)}. \quad (15.7)$$

A final partial cell and an initial clipped piece each add at most η to the deficit. Complete them too. There are at most k blocks and their rates are at most H , so the total tilted Poisson mass added is bounded solely by H . Lemma 15.2 therefore costs a fixed multiplicative constant.

If U'_i is the number of completed cells times η_i , then

$$U'_i = L_i + O_H(1). \quad (15.8)$$

Every partial sum of the cell deficits is bounded by the same constant in (15.7). Combining it with (15.2), a prime in cell j has log-log coordinate within $O_H(1)$ of $j\eta_i$. Consequently replacing actual positions by cell positions multiplies a whole expression $1 + \sum_\ell e^{t_\ell}$ by at most $e^{O_H(1)}$. Its exponent $h - 1$ is bounded, so the comparison constant is uniform in the number of occurrences. The actual original rectangle is proved before this replacement and before repeated-prime enlargement.

Under the completed Poisson law, conditional on the total $N = n$, cell indices are iid uniform indices. They can be realized by n independent uniform continuous positions on $[0, U'_i]$ and rounding to cells. For either signed minimum, use the appropriate boundary of the cell; the loss is at most its fixed linear-clock width, irrespective of the number of points in the cell. This realizes the same all- n ordinary ballot bound with all $n + 1$ spacings retained. For the actual rectangle the precise ordered-sum estimate is Lemma 13.6; its proof does not impose centrality or exclude count zero.

15.5 The unchanged root and the all-count suffix integral

Write $w_i = \log h_i$, $c_i = h_i - 1$ on the unread suffix. With its same counts and the original root coefficient r ,

$$\begin{aligned} Z_{\text{phys}} &= \sum_i (w_i N_i - c_i L_i), & Z_{\text{cell}} &= \sum_i (w_i N_i - c_i U'_i), \\ X &= \kappa + r Z_{\text{phys}} = \kappa' + r Z_{\text{cell}}, \\ \kappa' &= \kappa + r \sum_i c_i (U'_i - L_i). \end{aligned} \tag{15.9}$$

This exact identity has $\kappa' - \kappa = O_H(1)$ and preserves the root draw and the suffix's conditional Gibbs law.

Let $D > 0$ be the complete prefix cost after the causal cover. The all-count ordinary rectangle theorem of Proposition 14.9 is valid for every real translate

$$H_* = \frac{-\log D - \kappa'}{r}.$$

On the completed suffix it yields

$$CD^s e^{-t(H_*)} (1 + U')^{-1/2} \min \left\{ 1, \frac{(1 + B')[1 + (H_*)_+ + A']}{1 + M'} \right\}. \tag{15.10}$$

It also permits the established fixed polynomial outgoing envelope. The single largest-block Poisson atom and the ballot denominator were extracted jointly in this bound, and Lemma 15.2 preserves that bound.

All quantities with 1+length in (15.10) are comparable to their physical versions. For a large suffix one may use its physical largest block, whose completed length remains comparable; for a bounded suffix the numerical factors have a fixed positive lower bound. The arbitrary translate in (15.10) absorbs a grid mean negative by $O_H(1)$. The physical KKT inequality $D_{\text{nat}} \geq c_H V_{\text{pre}}$ is used only when averaging the prefix. The bounded change of κ is absorbed by $1 + D_{\text{nat}} + A$. The joint subset moment and the one-count Mecke bound then give the same physical ordinary factor as in Theorem 14.10.

15.6 Completion of the positive-core proof

Proof of Theorem 15.1. Apply every geometric cover to the original size-truncated tuple and only then enlarge its nonnegative analytic envelope. All finite KKT comparisons, original eligibility rules and causal gates are valid at positive physical lengths. The actual harmonic comparison above supplies their all-count prefix estimates. Bounded effective banks have fixed factor floors and can be omitted. The unread suffix is completed by Lemma 15.2, with its true root preserved by (15.9). Thus the complete causal proof, its different owner powers and its single root integration apply for every $L_i \geq 1$. All constants are independent of L .

It remains to identify the numerical profile. This comparison takes place at the same physical optimizer. Fixed proof reserves and window constants can be replaced by the finite scores in (3.6), because their reference and good-group terms are nonnegative. For paired contexts the reference increment is νL_i , independent of the outside context. Nonnegativity of their minima gives $|\nu| L_i \leq X_i + Y_i$. Therefore a strongly drifting or omitted saturated canonical edge contributes a factor bounded below. For a genuinely rare edge, the speed-centered and natural correlations differ by $O_H(|\nu| + 1/L_i) = O_H(T_e)$. The local Lipschitz estimate for χ and boundedness of $T_e |\log T_e|$ allow the exponent replacement at constant cost. Including all the finitely many canonical edges is therefore legitimate. The count widths similarly become $(1 + d\sqrt{1+T})^{-\sigma}$, and the explicit ordinary suffix has the same factor. This proves (15.1). $\square$

Corollary 15.3 (No unspecified lift in the definition). *The finite-infimum formula of Section 3, together with the established large-lift lower, satisfies $K_k(\ell) \asymp_k \Phi_k(\ell)$ on $[\log 3, \infty)^k$.*

Proof. Use Theorem 15.1 in Proposition 16.4. The unknown lower-source lift occurs only in that proof, not in the definition or its finite search range. The only comparison between different physical vectors is the independent arithmetic stability theorem for K_k . $\square$

16 Stability and the uniform arithmetic estimates

16.1 Stability of the original harmonic sum

The following comparison comes directly from the arithmetic sum. Write $c = \log 3$ and $m_i = k - i + 2$ as before, and fix an absolute constant C_M for which

$$\left| \sum_{p \leq t} \frac{1}{p} - \log \log t - B_1 \right| \leq C_M \quad (t \geq 3).$$

This form follows, for example, from the explicit estimates of Rosser and Schoenfeld [24]; enlarging the absolute constant covers the bounded range. Only this uniform coarse bound is used.

Theorem 16.1 (Explicit log-parameter stability). *For $\ell, \ell' \in [c, \infty)^k$ put*

$$\delta_i = \ell'_i - \ell_i, \quad D = \sum_{i=1}^k \left| \sum_{h=1}^i \delta_h \right|, \quad A = \sum_{i=1}^k m_i |\delta_i| + (k+1)(D + 2kC_M).$$

Then

$$e^{-A} K_k(\ell) \leq K_k(\ell') \leq e^A K_k(\ell).$$

In particular, for each fixed $L < \infty$, $K_k(\ell') \asymp_{k,L} K_k(\ell)$ uniformly whenever $\|\ell' - \ell\|_1 \leq L$.

Proof. Let y_i, y'_i be the associated cutoffs. Their definitions give

$$\log \log y'_i - \log \log y_i = \sum_{h \leq i} \delta_h.$$

Every prime that belongs to different blocks in the two configurations, or belongs to only one of their prime ranges, lies in

$$\mathcal{B} = \bigcup_{i=1}^k \{p : \min(y_i, y'_i) < p \leq \max(y_i, y'_i)\}.$$

Indeed, outside this union the truth values of all inequalities $p \leq y_i$ and $p \leq y'_i$ agree. Hence their least valid index, if any, agrees. Mertens's estimate and positivity imply

$$\sum_{p \in \mathcal{B}} \frac{1}{p} \leq D + 2kC_M. \quad (16.1)$$

Retain only the common primes outside $\mathcal{B}$, with their common block labels, and let S_* be their unnormalized harmonic volume sum. This number is positive, since the empty selection contributes $(\log 2)^k$. Write S, S' for the two full unnormalized sums. Inserting a selected block- i prime gives the union of at most m_i translates of the old union, one of them unchanged. Thus it increases the volume by a factor between 1 and $m_i \leq k+1$. Sum first over the common primes, and then over all selections of the remaining primes, retaining their exact reciprocal weights. This gives, for both $S_1 = S$ and $S_1 = S'$,

$$S_* \leq S_1 \leq S_* \prod_{p \in \mathcal{B}} \left(1 + \frac{k+1}{p}\right) \leq S_* \exp \left((k+1) \sum_{p \in \mathcal{B}} \frac{1}{p} \right).$$

It follows that S'/S lies between the reciprocal and the value of the last exponential; the exponent is not doubled. Finally the exact ratio of the normalizing products is $\exp(-\sum_i m_i \delta_i)$. Combine this identity with (16.1) to obtain the assertion. $\square$

Corollary 16.2 (Contraction of any bounded collection of blocks). *Fix $L \geq c$ and any set $I \subseteq \{1, \dots, k\}$. If $c \leq \ell_i \leq L$ for $i \in I$, replace those coordinates by c and leave the others unchanged, obtaining $\tilde{\ell}$. Then $K_k(\ell) \asymp_{k,L} K_k(\tilde{\ell})$ uniformly over all remaining coordinates. If L is specified in terms of k only, the constants depend on k only.*

Proof. The changed vector remains admissible and its ℓ^1 distance from the original is at most $k(L-c)$. Apply Theorem 16.1. $\square$

16.2 The lifted infimum and completion of the core estimate

Lemma 16.3 (Finite search range). *Set*

$$N = \frac{k(k+1)}{2}, \quad E = N + \frac{k-1}{2} + \frac{3}{2}, \quad C = \frac{k(k+1)}{2}, \quad c = kH, \quad a = A_k - C,$$

and

$$R = 1 + \max \left\{ 1, \max_i L_i, \frac{4E^2 c}{a^2}, \frac{2[A_k + (k-1) \log 2 + 1]}{a} \right\}.$$

The infimum in (3.13) is unchanged if restricted to $1 \leq \lambda \leq R$. It is strictly positive and finite.

Proof. There are at most N canonical edges. Since scores are at least one, $Z_i \leq H$, and $2\chi \leq 1$, each edge factor is at least $(1 + HL_i)^{-1}$. The bounds (3.3) give $0 < d_j < 1$. Each proper count factor is therefore at least $[2\sqrt{1+T_j}]^{-1}$. Finally $\mathcal{B} \geq (1 + V_k)^{-3/2}$. If $\lambda \geq \max_i L_i$, every lifted length equals λ . The rate is at most $\sum_j V_j = C\lambda$: in (3.2), $Z_i \geq 1$ and $s_j \leq 1$. Hence the objective is at least

$$2^{-(k-1)} e^{a\lambda} (1 + c\lambda)^{-E}.$$

The elementary inequality $\log(1+x) \leq \sqrt{x}$ gives $a\lambda - E \log(1+c\lambda) \geq a\lambda/2$ when $\lambda \geq 4E^2 c/a^2$. For $\lambda \geq R$ the last lower bound exceeds e^{A_k} . At $\lambda = 1$ the objective is at most e^{A_k} , because $J \geq 0$ and every factor in (3.12) is at most one. This proves the range claim. The same explicit factor floors on the bounded interval, and the elementary bound $J \leq kV_k$, bound the infimum strictly away from zero. No continuity across an owner merger is required for this argument. $\square$

Proposition 16.4. *The positive-length upper bound of Theorem 15.1, the fixed-lift lower bound of Theorem 9.1, and Theorem 16.1 imply Theorem 2.1.*

Proof. Put $L_i^\lambda = \max(L_i, \lambda)$ and $\delta_i = L_i^\lambda - L_i$, so that $0 \leq \delta_i \leq \lambda$. The exponent in Theorem 16.1 satisfies

$$\sum_i h_i \delta_i \leq \frac{k(k+3)}{2} \lambda, \quad \sum_{i=1}^k \left| \sum_{h \leq i} \delta_h \right| \leq \frac{k(k+1)}{2} \lambda.$$

Hence it is at most $B_k \lambda + C_0(k)$, where

$$B_k = \frac{k(k^2 + 3k + 4)}{2}, \quad C_0(k) = 2k(k+1)C_M, \quad A_k - B_k = \frac{3k^3 + 9k^2 + 8k + 4}{2} > 0.$$

For every $\lambda \geq 1$, stability and the positive-length upper give

$$K_k(\ell) \leq e^{C_0(k)} C_U(k) e^{B_k \lambda} \Psi_k(L^\lambda) \leq e^{C_0(k)} C_U(k) e^{A_k \lambda} \Psi_k(L^\lambda).$$

Taking the infimum proves $K_k(\ell) \ll_k \Phi_k(\ell)$. For the reverse bound, choose just the value $\lambda = L_*(k)$ from Theorem 9.1:

$$\Phi_k(\ell) \leq e^{A_k L_*} \Psi_k(L^{L_*}) \leq e^{A_k L_*} c_L(k)^{-1} K_k(\ell).$$

All constants depend only on k ; the comparison between parameter vectors is supplied by arithmetic stability. $\square$

16.3 Transfer to the multiplication table

For ordered real parameters $3 \leq N_1 \leq \dots \leq N_{k+1}$, put

$$A_{k+1}(\mathbf{N}) = \#\{n_1 \cdots n_{k+1} : n_i \in \mathbb{N}, n_i \leq N_i\}, \quad P = \prod_{i=1}^{k+1} N_i.$$

For ordered cutoffs $3 \leq t_1 \leq \dots \leq t_r$, write $\mathcal{K}_r(\mathbf{t}) = K_r(\ell(\mathbf{t}))$, where

$$\ell_i(\mathbf{t}) = \log \frac{3 \log t_i}{\log t_{i-1}}, \quad t_0 = 3.$$

The arithmetic normalization of this quantity is

$$\mathcal{K}_r(\mathbf{t}) = \prod_{i=1}^r \left(\frac{\log t_i}{\log t_{i-1}} \right)^{-(r-i+2)} \sum_{\mathbf{a} \in \mathcal{A}_r(\mathbf{t})} \frac{L^{(r+1)}(\mathbf{a})}{a_1 \cdots a_r}. \quad (16.2)$$

Here a_1 is squarefree with prime factors at most t_1 , and for $i \geq 2$ the prime factors of a_i lie in $(t_{i-1}, t_i]$. In particular, the first prime block begins at 1, not at 3. Write $\mathcal{L}^{(r+1)}(\mathbf{a})$ for the union of boxes

$$\prod_{i=1}^r [\log(d_i/2), \log d_i], \quad d_1 \cdots d_i \mid a_1 \cdots a_i \quad (1 \leq i \leq r).$$

Its volume is $L^{(r+1)}(\mathbf{a})$.

We record the consequence of Koukoulopoulos's multiplication-table and divisor-volume reductions in the normalization used here, including bounded initial axes.

Theorem 16.5 (Full table transfer). *For every fixed integer $k \geq 1$, define*

$$Y_0 = 3, \quad Y_i = \max\{3, N_i/2\} \quad (1 \leq i \leq k), \quad \lambda_i = \log \frac{3 \log Y_i}{\log Y_{i-1}}.$$

Then uniformly over $3 \leq N_1 \leq \cdots \leq N_{k+1}$,

$$A_{k+1}(\mathbf{N}) \asymp_k PK_k(\boldsymbol{\lambda}). \quad (16.3)$$

Consequently, any independently proved comparison $K_k(\ell) \asymp_k \Phi_k(\ell)$ on $[\log 3, \infty)^k$ gives

$$A_{k+1}(\mathbf{N}) \asymp_k P\Phi_k(\boldsymbol{\lambda})$$

on this entire table region. This implication requires no stability assumption on Φ_k .

Proof. We use two published reductions, whose hypotheses we record explicitly. Koukoulopoulos's Theorem 1.1 [19] gives, for every $r \geq 1$ and every $3 \leq M_1 \leq \cdots \leq M_{r+1}$,

$$A_{r+1}(\mathbf{M}) \asymp_r H^{(r+1)} \left(\prod_{i=1}^{r+1} M_i, (M_1/2, \dots, M_r/2), (M_1, \dots, M_r) \right).$$

His Theorem 1.7 [19] gives $H^{(r+1)}(x, \mathbf{t}, 2\mathbf{t})/x \asymp_r \mathcal{K}_r(\mathbf{t})$ provided $3 \leq t_1 \leq \cdots \leq t_r$ and $x \geq 2^r t_1 \cdots t_r t_r$.

If $M_1 \geq 6$, then $t_i = M_i/2 \geq 3$ and

$$2^r t_1 \cdots t_r t_r = \left(\prod_{i=1}^r M_i \right) \frac{M_r}{2} \leq \frac{1}{2} \prod_{i=1}^{r+1} M_i.$$

Thus the two reductions apply directly. Choose constants $a_r, b_r > 0$ depending only on r such that

$$a_r \mathcal{K}_r(M_1/2, \dots, M_r/2) \leq \frac{A_{r+1}(\mathbf{M})}{\prod_i M_i} \leq b_r \mathcal{K}_r(M_1/2, \dots, M_r/2) \quad (M_1 \geq 6). \quad (16.4)$$

Let $b = \#\{1 \leq i \leq k : N_i < 6\}$ and $m = k - b$. By ordering these are the first b indices. Set

$$P_{\text{tail}} = \prod_{i=b+1}^{k+1} N_i, \quad A_{\text{tail}} = A_{m+1}(N_{b+1}, \dots, N_{k+1}),$$

with $A_1(N) = \lfloor N \rfloor$. The full table contains the tail table, by taking its first b factors equal to 1. Conversely it is a union of at most $\prod_{i=1}^b \lfloor N_i \rfloor$ integer dilates of the tail table. Since $\prod_{i=1}^b N_i < 6^b$ when $b > 0$, we have in all cases

$$6^{-b} \frac{A_{\text{tail}}}{P_{\text{tail}}} \leq \frac{A_{k+1}(\mathbf{N})}{P} \leq \frac{A_{\text{tail}}}{P_{\text{tail}}}. \quad (16.5)$$

Suppose $m \geq 1$ and put $z_j = N_{b+j}/2$ for $1 \leq j \leq m$. Then $z_1 \geq 3$, and (16.4) applies in dimension m . We claim, on writing $h = \log 2$ and $D = \log 12$, that

$$h^b \mathcal{K}_m(\mathbf{z}) \leq \mathcal{K}_k(\underbrace{3, \dots, 3}_b, \mathbf{z}) \leq D^b \mathcal{K}_m(\mathbf{z}). \quad (16.6)$$

For $b = 0$ this is equality. For $b \geq 1$, every tuple on the middle side satisfies $a_1 \mid 6$ and $a_2 = \dots = a_b = 1$. The map

$$(a_1, 1, \dots, 1, a_{b+1}, \dots, a_k) \mapsto (a_1 a_{b+1}, a_{b+2}, \dots, a_k)$$

is a bijection onto $\mathcal{A}_m(\mathbf{z})$, and preserves the weight $1/(a_1 \dots a_k)$: the factors at primes 2, 3 uniquely recover a_1 . Let $\mathbf{a}'$ denote its image. Setting the first b divisor coordinates equal to 1 gives the inclusion

$$[-h, 0]^b \times \mathcal{L}^{(m+1)}(\mathbf{a}') \subseteq \mathcal{L}^{(k+1)}(\mathbf{a}).$$

Conversely each of the first b divisor coordinates is at most 6, and the projection onto the last m coordinates belongs to $\mathcal{L}^{(m+1)}(\mathbf{a}')$. Hence

$$\mathcal{L}^{(k+1)}(\mathbf{a}) \subseteq [-h, \log 6]^b \times \mathcal{L}^{(m+1)}(\mathbf{a}').$$

Taking volumes gives the factors h^b, D^b . The first b normalizing log ratios are 1; the remaining ratios and their exponents agree exactly with those in dimension m . Summing proves (16.6). Together with (16.4) and (16.5), this yields

$$3^{-b} a_m D^{-b} \mathcal{K}_k(\mathbf{Y}) \leq \frac{A_{k+1}(\mathbf{N})}{P} \leq b_m h^{-b} \mathcal{K}_k(\mathbf{Y}).$$

It remains to handle $m = 0$. The ratio $A_{\text{tail}}/P_{\text{tail}} = \lfloor N_{k+1} \rfloor / N_{k+1}$ lies in $[2/3, 1]$. At $\mathbf{Y} = (3, \dots, 3)$ the only tuples are $(a_1, 1, \dots, 1)$ with $a_1 \mid 6$, and their reciprocal weights sum to 2. Every such union has volume between h^k and D^k . Consequently

$$2h^k \leq \mathcal{K}_k(3, \dots, 3) \leq 2D^k,$$

and (16.5) gives the explicit bounds

$$\frac{\mathcal{K}_k(\mathbf{Y})}{3(6D)^k} \leq \frac{A_{k+1}(\mathbf{N})}{P} \leq \frac{\mathcal{K}_k(\mathbf{Y})}{2h^k}.$$

Taking the minimum and maximum over the finitely many possible $b \in \{0, \dots, k\}$ proves (16.3) with constants depending only on k . $\square$

Proof of Theorems 1.1 and 1.2. Theorem 2.1 and the arithmetic comparison (2.3) give the asserted estimate for $H^{(k+1)}$ throughout the stated region. For the table, apply Theorem 16.5 at the original axes and then Theorem 2.1 at $\ell(\mathbf{Y})$. Both comparisons have constants depending only on the fixed dimension. Adjacent equal cutoffs, bounded axes, and arbitrarily small strict saddle gaps are included in these same arguments. $\square$

17 Contributing clocks and the entropy of the profile

We apply the neighboring-moment estimates to one fixed accepting source from Sections 9–12. All its original physical groups, fine types, count windows, allocated walls, costs and adaptive targets are retained. The source measure below includes the regularity and bank acceptance on unselected prefix blocks as well as the selected binary comparisons.

17.1 One source for a compact interval of moment orders

Fix a dimension-dependent compact interval $J = [p_-, p_+] \Subset (0, 1)$ which contains the whole range of owner products in (3.3) in its interior. The margin from these products to the endpoints of J is fixed and positive. The physical saddle and every original mark law are held fixed when a test order $p \in J$ is varied.

For a physical vector L in the sufficiently large positive-length region, let $\mathcal{I}$ be the selected native binary comparisons of Section 9.1. There is at most one such comparison in an original physical block. If $i \in \mathcal{I}$, write e_i for its physical canonical edge and put

$$\eta_i = \sigma_{\text{owner}(i)}, \quad \rho_i = \rho_{e_i}, \quad \tau_i = -2 \log \min\{1, T_{e_i}\}, \quad (17.1)$$

where T_e, ρ_e are exactly those of (3.10). In particular $P_{e_i} = \exp\{-\tau_i \chi(\eta_i, \rho_i)\}$. The durations and correlations in this section are deterministic functions of the physical vector. Every sum involving them is over selected comparisons unless another index set is displayed. Unselected blocks have $D_i = 1$ and contribute no duration or Gaussian parameter; they still retain their actual local acceptance.

Lemma 17.1 (Uniform neighboring moments of a fixed cell source). *The complete-cell source and its costs can be chosen once, uniformly for $p \in J$, so that at every supported complete prefix type ω and admitted original count vector,*

$$\begin{aligned} 0 \leq q_i(\omega, g_i) \leq 1, \quad 0 < D_i(g_i) \leq 1, \quad q_i \leq C_k D_i, \\ \mathbb{E}_{g_i} q_i(\omega, g_i)^p \asymp_k \mathbb{E}_{g_i} D_i(g_i)^p \asymp_k \exp\{-\tau_i \chi(p, \rho_i)\}, \quad p \in J. \end{aligned} \quad (17.2)$$

Here g_i has the original exact-total cell law (H.58). The same q_i, D_i occur for every p . Each D_i reads only the uncolored cell occupancies, including their total, and is independent of the fine type after that total is fixed.

Proof. Choose the compact scalar ranges with the test order ranging over all of J . Theorem 8.1 and Corollary 8.5 then supply common wall, reserve and binary regularity constants. The refinement in Lemma H.2 reads both true mark lists and has a uniform conditional success probability for each retained binary word and full type. Choose its constants once for these same walls.

In (H.13), choose a single floor exponent satisfying $D_0 p_- > 2 \sup_{p \in J, \rho} \chi(p, \rho) + 1$ over the compact correlation range. Fix the literal exit widths and use this floor to bound the whole logarithmic target and recoloring range. Then choose the coefficient tail and the all-transfer envelopes (H.43) uniformly for $p \in J$, with their geometric depth coefficient fixed below the available history budget. Finally choose the spare strong-wall and named regularity margins, macroscopic-bank support, complete-cell allowances, and final count and root thresholds, in the order of (H.34) and the end of Section H.11. None of these later choices changes D_0 or requires evaluating a target before its envelope has been fixed.

The lower bridge moment, the upper envelope moment (H.14), and the relative macroscopic-bank retention (H.21) are uniform in this compact order range. The direct cell construction (H.57) preserves them directly for the same functions of the cell word, with the private acceptance evaluated before taking its power. The pointwise inequality $q_i \leq C_k D_i$ supplies the other sides of the two moment comparisons.

Initially these comparisons use the exact count, compatible local pair and centered correlation of (9.6). The central-count estimate (H.54) and the physical-correlation comparison following (10.11) are uniform in $p \in J$ by the compact Lipschitz bound for χ . More explicitly, on a rare

branch the additional correlation error is $O_k(T_{e_i} + L_i^{-1})$ and $T_{e_i} \geq c_k/(1 + L_i)$; its product with $|\log T_{e_i}|$ is bounded. Replacing endpoint scales by fixed comparable scales changes the effective duration by $O_k(1)$. Threshold crossings and saturated comparisons also have bounded effective duration. These observations yield (17.2) with the physical parameters (17.1).

On the actual arithmetic grid, keep the physical saddle frozen as in Section 12. Its bounded grid defects and exact local increments give the same endpoint and exponent-times-log comparisons uniformly over J . The preceding argument therefore applies directly to the grid functions. The resulting neighboring-moment inequalities determine the physical centers below. Bounded supported experiments retain their prescribed finite support and fixed positive mass throughout. $\square$

17.2 The whole prefix source and the unselected blocks

Let $\mathcal{P}$ be the set of original blocks before the unread ordinary suffix. For each admitted prefix total vector $\mathbf{n}$, let $\mu_{\mathbf{n}}$ be the one count-only measure on complete prefix types in (H.22). Its mass is uniformly comparable to $\prod_{j < m} w_j$. Conditional on a complete type, all original cell clocks are independent. At a selected block retain q_i from Lemma 17.1. At $i \in \mathcal{P} \setminus \mathcal{I}$, denote its actual private local regularity and bank acceptance by $q_i^{\text{reg}}(\omega, g_i)$. The unselected-block case of (H.21), together with the named refinement, gives

$$0 \leq q_i^{\text{reg}} \leq 1, \quad c_k \leq \mathbb{E}_{g_i} q_i^{\text{reg}}(\omega, g_i) \leq 1 \quad (17.3)$$

uniformly in the whole supported type. Equivalently, the lower fractional moment already established with $D_i = \Psi_i = 1$ implies this lower first moment by Jensen.

To include the remaining central uncolored totals explicitly, let $\nu(d\mathbf{n})$ be their original independent natural Poisson law restricted to their prescribed central windows. Put

$$\mathcal{W} = \int \mu_{\mathbf{n}}(1) \nu(d\mathbf{n}), \quad d\mathbb{P}_{\text{ref}} = \mathcal{W}^{-1} \nu(d\mathbf{n}) \mu_{\mathbf{n}}(d\omega) \prod_{i \in \mathcal{P}} d\mathbb{P}_{i, n_i}(g_i), \quad (17.4)$$

where $\mathbb{P}_{i, n_i}$ is (H.58). The same definitions can be used conditional on any one admitted total vector, by taking ν concentrated there. This reference law retains the coupled type coordinates under the count-only acceptance.

Set

$$\begin{aligned} U(\omega, g) &= \prod_{i \in \mathcal{P} \setminus \mathcal{I}} q_i^{\text{reg}}(\omega, g_i), \\ R_i &= q_i / D_i, \quad W_{\text{pre}}(\omega, g) = U(\omega, g) \prod_{i \in \mathcal{I}} R_i D_i^{\eta_i}, \\ Z_{\text{pre}} &= \mathbb{E}_{\text{ref}} W_{\text{pre}}, \quad d\mathfrak{M} = Z_{\text{pre}}^{-1} W_{\text{pre}} d\mathbb{P}_{\text{ref}}. \end{aligned} \quad (17.5)$$

The deterministic factor $\prod_{j < m} w_j^{\sigma_j - 1}$ from the owner costs cancels on normalization. Thus $\mathfrak{M}$ is the complete prefix mixed source, including its unselected blocks.

Lemma 17.2 (Integrating the unselected acceptance). *At a fixed $(\mathbf{n}, \omega)$, integrating the unselected clocks in (17.5) replaces their factor by*

$$H(\mathbf{n}, \omega) = \prod_{i \in \mathcal{P} \setminus \mathcal{I}} \mathbb{E}_{g_i} q_i^{\text{reg}}(\omega, g_i), \quad c_k \leq H \leq 1. \quad (17.6)$$

This is a bounded reweighting of the same coupled count-only type measure. In particular, uniformly on compact subsets of $s_i < 1$, $p_i = \eta_i - s_i - t_i \in \text{int } J$,

$$Z_{\text{pre}} \asymp_k \exp\left\{-\sum_{i \in \mathcal{I}} \tau_i \chi(\eta_i, \rho_i)\right\}, \quad (17.7)$$

$$\log \mathbb{E}_{\mathfrak{M}} e^{\sum_{i \in \mathcal{I}} (s_i S_i + t_i T_i)} = \sum_{i \in \mathcal{I}} \tau_i [\chi(\eta_i, \rho_i) - \chi(p_i, \rho_i)] + O_k(1), \quad (17.8)$$

where $S_i = -\log q_i$, $T_i = -\log D_i$ on the mixed support.

Proof. At a fixed complete type, the unselected factors use only their own original clocks. Their exact integral is the product in (17.6); its bounds follow from (17.3) and the fixed number of blocks. The selected clocks remain independent conditional on that type. For arbitrary fixed $a_i > 0$ and $p_i \in J$, Lemma 7.6 and (17.2) give

$$\mathbb{E}_{g_i} R_i^{a_i} D_i^{p_i} \asymp_k e^{-\tau_i \chi(p_i, \rho_i)}.$$

Multiply these integrals at the fixed type, retain H inside the whole type sum, and use its uniform upper and lower bounds. Tonelli then integrates that one coupled measure and the central totals. Taking $a_i = 1, p_i = \eta_i$ gives the normalization. For the moment numerator take $a_i = 1 - s_i$ and use the exact identity $R_i D_i^{\eta_i} e^{s_i S_i + t_i T_i} = R_i^{1-s_i} D_i^{p_i}$. Division by the normalization proves the spectrum with $O_k(1)$ error. $\square$

17.3 Clock shells, entropy, and projection

Define physical centers and an uncolored prefix event by

$$m_i = \tau_i \partial_p \chi(\eta_i, \rho_i), \quad \mathcal{G}_A = \bigcap_{i \in \mathcal{I}} \{|T_i - m_i| \leq A\sqrt{1 + \tau_i}\}. \quad (17.9)$$

Proposition 17.3 (Contributing clock shells). *Uniformly over the physical large-length region and the admitted central-count support,*

$$\mathfrak{M}(\mathcal{G}_A^c) \leq C_k e^{-c_k A}, \quad A \geq 1. \quad (17.10)$$

Both S_i and T_i satisfy

$$\mathfrak{M}\{|S_i - m_i| > u\}, \quad \mathfrak{M}\{|T_i - m_i| > u\} \leq C_k \exp\{-c_k \min(u^2/(1 + \tau_i), u)\}. \quad (17.11)$$

Moreover

$$\mathfrak{M}\{\log(D_i/q_i) > v\} \leq C_k e^{-v}, \quad \log(D_i/q_i) \geq -\log C_k, \quad (17.12)$$

and

$$D_{\text{KL}}(\mathfrak{M} \parallel \mathbb{P}_{\text{ref}}) = \sum_{i \in \mathcal{I}} \tau_i [\chi(\eta_i, \rho_i) - \eta_i \partial_p \chi(\eta_i, \rho_i)] + O_k\left(\sum_{i \in \mathcal{I}} \sqrt{1 + \tau_i} + 1\right). \quad (17.13)$$

The same entropy formula holds for projection to the complete uncolored prefix occupancies, with the corresponding reference marginal. The expected conditional entropy of the discarded type information is $O_k(1)$. The event $\mathcal{G}_A$ reads neither a fine name nor an unread ordinary-suffix clock.

Proof. The local $C^{1,1}$ estimate of Theorem 7.2, applied to (17.8), gives the two-sided Bernstein bound by Chernoff exactly as in Proposition E.3. Here the order interval has a fixed margin around every η_i . For $A \geq 1$, $\min(A^2, A\sqrt{1+\tau_i}) \geq A$; a union bound gives the shell estimate. At a fixed type, the proof of the ratio tail uses $R_i \mathbf{1}_{\{R_i < e^{-v}\}} \leq e^{-v}$. The other selected integrals and the factor H have uniform comparisons, so the same argument proves (17.12) for the complete source. In particular each $\mathbb{E}_{\mathfrak{M}}|\log R_i|$ is bounded.

Conditional on the type and selected clocks, the unselected clocks have density U/H relative to their original product law. Since $0 \leq U \leq 1$ and $H \geq c_k$,

$$\mathbb{E}_{\mathfrak{M}}[-\log U \mid \mathbf{n}, \omega, g_{\mathcal{I}}] = H^{-1} \mathbb{E}[U(-\log U) \mid \mathbf{n}, \omega] \leq (ec_k)^{-1}.$$

Thus the unselected source restrictions have bounded entropy cost. Integrating (17.11) gives $\mathbb{E}_{\mathfrak{M}} T_i = m_i + O_k(\sqrt{1+\tau_i})$. Expansion of the exact density

$$\log \frac{d\mathfrak{M}}{d\mathbb{P}_{\text{ref}}} = \log U + \sum_{i \in \mathcal{I}} \log R_i - \sum_{i \in \mathcal{I}} \eta_i T_i - \log Z_{\text{pre}}$$

proves (17.13), including the case $\mathcal{I} = \emptyset$.

For clarity, define the two Gibbs laws relative to this same full reference measure by densities proportional to $\prod_{i \in \mathcal{I}} D_i^{\eta_i}$ and $\prod_{i \in \mathcal{I}} q_i^{\eta_i}$, respectively. Their normalizers and Z_{pre} are comparable. The respective density ratios of $\mathfrak{M}$ are bounded by constants times $U \prod_i R_i$ and $U \prod_i R_i^{1-\eta_i}$, hence are uniformly bounded. Both forward Gibbs relative entropies, and the corresponding finite-time Gibbs variational excesses, are consequently $O_k(1)$. For the D -Gibbs law its density is measurable with respect to the complete uncolored occupancies (including the totals). Its conditional type law given those occupancies agrees exactly with that of $\mathbb{P}_{\text{ref}}$. The relative-entropy chain rule, as in Corollary E.4, bounds the entropy of the discarded information by $O_k(1)$ and proves the marginal formula. All physical centers remain deterministic when the original central totals are integrated in (17.4). $\square$

17.4 Restriction and transfer to original prime tuples

Theorem 17.4 (Full-order arithmetic lower family). *For each fixed k , the source constants and complete-cell conventions can be chosen as above. There are fixed $L_*(k)$ and $A(k)$ such that for every physical vector $L_i \geq L_*(k)$ there is a family $\mathcal{A}_A(L)$ of the original squarefree tuples in (2.2) satisfying*

$$e^{-\sum_i h_i L_i} \sum_{\mathbf{a} \in \mathcal{A}_A(L)} \frac{L^{(k+1)}(\mathbf{a})}{a_1 \cdots a_k} \asymp_k \Psi_k(L), \quad h_i = k - i + 2. \quad (17.14)$$

Every tuple in this family has complete-cell occupancy in $\mathcal{G}_A$ and retains the original uncolored count/root support and size restrictions. Thus the family contributes a fixed positive dimension-dependent fraction of the entire harmonic volume sum.

Proof. The joint ordinary-suffix kernel (H.59) holds pointwise on each retained complete prefix. Multiply it by the original owner cost factors and $\mathbf{1}_{\mathcal{G}_A}$. The complete prefix measure in (17.5), including U , is exactly the one which results when these factors are integrated against the private prefix mass. Equations (17.10) and (H.59) therefore bound the removed mixed source mass by $C_k e^{-c_k A}$ times the original mixed source mass.

Keep the costs, floors, targets, strong curves, count bands and root rule unchanged, and restrict the accepted source by $\mathbf{1}_{\mathcal{G}_A}$. The recoloring kernel preserves every uncolored cell occupancy, so

its transported density is exactly

$$F_A(g, \omega) = \mathbf{1}_{\mathcal{G}_A}(g) F_y^{I, \text{cell}}(g, \omega). \quad (17.15)$$

The pointwise mass identity (H.60) includes its original root indicator. In the exact nominal identity (H.52), the remaining factors involving E_y and the root error are bounded above and below. Consequently the removed nominal first mass is at most $C_k e^{-c_k A}$ of the whole first mass. Choose the fixed $A(k)$ large enough to retain a fixed positive fraction. The nominal mass remains comparable to $\Psi_k(L)$ by (10.10) and the frozen-grid comparison of Section 12.

For any retained anchor and any original history, $M_h(F_A; \omega) \leq M_h(F_y^{I, \text{cell}}; \omega)$. The original unnormalized collision row (11.5) thus retains its constant under the restriction. For every retained occupancy the pointwise squarefree supply (11.8) also remains valid. Apply the original prime-cell kernel to F_A on that occupancy. With $Z_{F_A}(g) = \sum_{\omega} F_A(g, \omega)$, it gives

$$\int f_A \geq c_k W_g Z_{F_A}(g), \quad \int f_A^{1+\delta} \leq C_k W_g Z_{F_A}(g).$$

The same Hölder inequality as (11.9) yields $\text{meas}\{f_A > 0\} \geq c_k W_g Z_{F_A}(g)$.

Here is an explicit tuple definition. For every occupancy g let $\mathcal{A}(g)$ be its original squarefree tuple set from (11.8), with the source size restrictions, and set

$$\mathcal{A}_A(L) = \bigcup_{g: Z_{F_A}(g) > 0} \mathcal{A}(g). \quad (17.16)$$

These sets have disjoint occupancies. The positive set of f_A is contained in the original divisor union for each such tuple. Summing the preceding bound over g and retaining the nominal normalization proves the lower bound in (17.14). Its upper bound follows by inclusion in the entire sum (2.2) and Theorem 15.1. Bounded grid-length losses change the normalizing exponent only by a dimension-dependent constant.

All these objects are constructed on the actual complete prime-cell grid with the physical products and owner structure frozen. The grid comparison (12.1) uses the exact budget of Section 12. Since (17.15) only removes accepted words, the original count, root, occupancy and size support is preserved. $\square$

Each $\mathbf{a} \in \mathcal{A}_A(L)$ determines its occupancies $g_{i,c}(\mathbf{a})$ by counting its prime factors in the fixed complete cells. We define $D_i(\mathbf{a}) = D_i(g_i(\mathbf{a}))$. The shell and targets are therefore actual functions of the original tuple, evaluated from its uncolored occupancies. The condition $Z_{F_A}(g) > 0$ in (17.16) expresses the existence of an accepted fine word; $q_i(\omega, g_i)$ still depends on that word's fine type.

17.5 The original adaptive targets and their physical centers

Retain (9.10), (10.1) and (10.2), with $w_m = 1$ for the terminal owner:

$$y_j = -\log w_j + \sum_{\substack{i \in \mathcal{I} \\ \text{owner}(i)=j}} T_i, \quad x = \sum_j \frac{\sigma_j}{\sigma_1} y_j.$$

Define deterministic physical centers by

$$\bar{y}_j = -\log w_j + \sum_{\substack{i \in \mathcal{I} \\ \text{owner}(i)=j}} \tau_i \partial_p \chi(\sigma_j, \rho_i), \quad (17.17)$$

$$\bar{x} = \sum_j \frac{\sigma_j}{\sigma_1} \bar{y}_j. \quad (17.18)$$

Corollary 17.5 (Target and root localization on original tuples). *Every tuple in (17.16) satisfies*

$$|y_j - \bar{y}_j| \leq C_k A \sqrt{1 + \sum_{\substack{i \in \mathcal{I} \\ \text{owner}(i)=j}} \tau_i}, \quad (17.19)$$

$$|x - \bar{x}| \leq C_k A \sqrt{1 + \sum_{i \in \mathcal{I}} \tau_i}. \quad (17.20)$$

Let L'_i be the actual complete-cell lengths, $V'_j = \sum_{i \leq j} L'_i$, and retain the physical s_i, G_i . The exact uncolored grid root on a tuple is

$$X_1^{\text{grid}}(\mathbf{a}) = \frac{\sum_i G_i \omega(a_i) - \sum_j s_j V'_j}{\sigma_1}, \quad X_1^{\text{grid}} = x + \epsilon, \quad -C_k \leq \epsilon \leq 0. \quad (17.21)$$

Here $\omega(a_i)$ denotes the number of prime factors of the squarefree integer a_i . If e_j is the original integer recoloring recursion, then it differs by $O_k(A\sqrt{1 + \sum_i \tau_i} + 1)$ from the deterministic real recursion

$$\bar{e}_0 = 0, \quad \bar{e}_j = (\bar{y}_j + A_j \bar{e}_{j-1})/b_j.$$

Proof. Sum the shell inequalities within each owner and use Cauchy–Schwarz and the dimension-bounded number of selected blocks. The ratios σ_j/σ_1 , the reciprocals of b_j , and the remaining recursion coefficients are dimension-bounded. Iterating the recursion difference, with the error in $[0, 1)$ at each ceiling, proves the stated recoloring estimate.

The first expression in (17.21) is the uncolored count/root identity (G.4) on the complete original chain, evaluated algebraically at the frozen physical coefficients and the actual grid budget. The actual reservoir count is the floor in (H.29), using this grid budget as specified in Section 12. Thus its floor error is in $[-r_m w_*, 0]$, where $r_m = \sigma_m/\sigma_1$ and $w_* \leq \log(k+1)$ is the reservoir mark. This proves the second assertion in (17.21). Recoloring fixes all uncolored counts, and the shell restriction fixes the root rule, so the identity holds on every tuple of the family. Thus the root is the complete-cell count functional with frozen marks G_i and grid budgets V'_j . $\square$

The derivative in (17.17) is with respect to the test moment order while the physical vector, mark law and correlation are fixed. The exact support and recoloring rules continue to use the adaptive targets y_j, x ; the centers describe their localization.

For the complete normalized prefix source, every fixed real linear path target $X_{\text{path}} = \sum_{i \in \mathcal{I}} a_i T_i$ also obeys

$$\log \mathbb{E}_{\mathfrak{M}} e^{h X_{\text{path}}} = \sum_{i \in \mathcal{I}} \tau_i [\chi(\eta_i, \rho_i) - \chi(\eta_i - h a_i, \rho_i)] + O_k(1) \quad (17.22)$$

for small fixed h . A deterministic count-window contribution can be added exactly; the conditional statement likewise holds at fixed admitted totals. The selected local deviation conclusion of Theorem E.5 applies when the effective durations have a common diverging scale.

Remark 17.6 (Scope of the arithmetic localization). The localization describes a full-order subfamily in the sufficiently large positive-length region, measured by harmonic divisor volume. The local deviation principle concerns the normalized prefix source. For bounded or zero physical lengths, arithmetic stability controls the total sum; localization of the contributing clocks in that region would require a separate argument.

17.6 Environmental entropy and the structural profile

The scalar exponent measures a balance between selecting a favorable clock and asking its private labels to survive. To make this balance precise, let $\mathbf{P}_t$ be the OU environmental law and write $S_t = -\log q_t$. The classical Gibbs variational formula [9, Section 1.4], applied to ηS_t , gives

$$b_t(\eta, \rho) = \inf_{\mathbf{Q} \ll \mathbf{P}_t} \{D_{\text{KL}}(\mathbf{Q} \parallel \mathbf{P}_t) + \eta \mathbb{E}_{\mathbf{Q}} S_t\}. \quad (17.23)$$

Indeed, for $d\nu_{\eta,t} = q_t^\eta d\mathbf{P}_t / F_{\eta,\rho}(t)$, subtracting b_t from the expression in braces gives $D_{\text{KL}}(\mathbf{Q} \parallel \nu_{\eta,t})$. Define

$$v(\eta, \rho) = \partial_\eta \chi(\eta, \rho), \quad d(\eta, \rho) = \chi(\eta, \rho) - \eta v(\eta, \rho). \quad (17.24)$$

Theorem 7.2 gives

$$\mathbb{E}_{\nu_{\eta,t}} S_t = tv(\eta, \rho) + O(\sqrt{t}), \quad D_{\text{KL}}(\nu_{\eta,t} \parallel \mathbf{P}_t) = td(\eta, \rho) + O(\sqrt{t}).$$

In particular $d(\eta, \rho) \geq 0$. The complete mixed source has the corresponding sums of these leading quantities by Proposition 17.3. Its selected local factors are $qD^{\eta-1}$, and its unselected acceptance has bounded entropy cost.

The full scalar large-deviation principle in Theorem D.1 makes the environmental cost explicit. Its good rate function I_ρ satisfies

$$\begin{aligned} \chi(\eta, \rho) &= \min_{v \geq 0} \{I_\rho(v) + \eta v\}, \\ \arg \min_{v \geq 0} \{I_\rho(v) + \eta v\} &= \{\partial_\eta \chi(\eta, \rho)\}. \end{aligned} \quad (17.25)$$

These identities follow from convex duality for $\Lambda_\rho(\theta) = -\chi(-\theta, \rho)$ at $\theta = -\eta$; differentiability makes its subgradient a singleton. Thus $d(\eta, \rho) = I_\rho(v(\eta, \rho))$.

For a canonical edge e in (3.10), write $\sigma_e = \sigma_{\text{owner}(e)}$ for its owner product and set

$$\tau_e = 2[-\log T_e]_+, \quad P_e = e^{-\tau_e \chi(\sigma_e, \rho_e)}. \quad (17.26)$$

Proposition 5.17 and (17.25) now express the positive-length profile as two entropy costs:

$$\begin{aligned} -\log \Psi_k(L) &= \sum_i L_i \mathcal{Q}(Z_i) - \log \mathcal{B}(L) + \sum_{j \text{ proper}} \sigma_j \log(1 + d_j \sqrt{1 + T_j}) \\ &\quad + \inf_{(v_e \geq 0)} \sum_e \tau_e \{I_{\rho_e}(v_e) + \sigma_e v_e\}. \end{aligned} \quad (17.27)$$

Here T_j denotes owner j 's native length as in (3.12). For every active edge the minimizing coordinate is $v_e = \partial_\eta \chi(\sigma_e, \rho_e)$; a coordinate with $\tau_e = 0$ has no effect. The sum separates by the proved product formula (3.12): each summand uses the scalar OU rate function for its edge. The Poisson cost is measured in the physical lengths L_i , whereas the persistence cost is measured in the effective times τ_e , which are often logarithmic in those lengths. The constant-factor bridge estimates remain necessary to recover the arithmetic order.

17.7 The loss from averaging the clock first

Let Ψ_k^{ann} be the expression (3.12) with each $\chi(\sigma_e, \rho_e)$ replaced by $\sigma_e/2$. This substitution keeps the saddle, endpoint scores, count windows and ordinary suffix fixed.

Corollary 17.7. *Uniformly on the positive-length domain,*

$$\log \frac{\Psi_k^{\text{ann}}(L)}{\Psi_k(L)} = \sum_e \tau_e \left[\chi(\sigma_e, \rho_e) - \frac{\sigma_e}{2} \right] \asymp_k \sum_e \tau_e. \quad (17.28)$$

Both sides are zero if all τ_e vanish. Along a parameter family, the ratio tends to infinity exactly when $\sum_e \tau_e \rightarrow \infty$.

Proof. The owner products are bounded away from zero and one. On the canonical nondegenerate parameter range, the correlations have a positive lower bound depending only on k . Apply Theorem 7.4 for the lower estimate and $\chi(\sigma_e, \rho_e) \leq 1/2$ for the upper estimate. $\square$

17.8 The critical six-coordinate family

In the family (4.5), take the fixed middle length C above the threshold of Theorem 17.4. The rare edge is $6 \rightarrow 1$, with owner product σ_c and correlation

$$\rho_c = \frac{(7 \log 7 - 2 \log 2)^2}{(7 \log 7 - 2 \log 2)^2 + 10(\log 2)^2}.$$

Its low-mark probability is $2/7$, so this is exactly $\rho(6, 1; 1)$ in (4.1). Put

$$\chi_c = \chi(\sigma_c, \rho_c), \quad v_c = \partial_\eta \chi(\sigma_c, \rho_c), \quad d_c = \chi_c - \sigma_c v_c = I_{\rho_c}(v_c).$$

The endpoints (4.6) give $\tau_e = \log T + O_k(1)$. Thus (4.7) becomes

$$\Phi_6(\log 3 + L) \asymp_6 e^{-J(L)} T^{-1/2} T^{-d_c} T^{-\sigma_c v_c}. \quad (17.29)$$

The last two factors respectively describe the environmental selection cost and the conditional path cost at the selected clock. Under the mixed source, the environmental entropy is $d_c \log T + O_k(\sqrt{\log T})$. Moreover, the arithmetic family in Theorem 17.4 can be chosen so that

$$-\log D = v_c \log T + O_k(\sqrt{\log T}), \quad X_1^{\text{grid}} = v_c \log T + O_k(\sqrt{\log T}). \quad (17.30)$$

Here X_1^{grid} is the uncolored count root of the complete-cell construction, including its bounded rounding error, as in Corollary 17.5. The bounded clocks contribute only bounded target terms. This locates a full-order lower family of original prime tuples. Finally the comparison that averages first is too large by a factor $\asymp T^{\Delta_c}$, where

$$\Delta_c = \chi_c - \sigma_c/2 \geq \frac{\rho_c}{16\pi} \min\{\sigma_c, 1 - \sigma_c\} > 0.$$

Since $T = \log(\log y_1 / \log 3)$, this is a power of $\log \log y_1$.

17.9 How overlap changes the environmental cost

The original comparison constraints can share a clock. A separate calculation for deterministic OU windows shows explicitly why the cost of a shared environment differs from that of independent ones. Use the conditional probabilities $q_{j,I}$ and joint pressures Ξ_A of (C.1). Theorem C.4 gives

$$\mathbb{E} \prod_j q_{j,I_j}^{p_j} \asymp \exp \left[- \int_{\mathbb{R}} \Xi_{\{j:s \in I_j^c\}}(\mathbf{p}; \boldsymbol{\beta}) ds \right]$$

for any fixed number of deterministic intervals, uniformly in their lengths and positions. If all slopes equal $\sqrt{\rho/(1-\rho)}$, then the conditional probabilities on the same window coincide, so

$$\Xi_A = \chi \left(\sum_{j \in A} p_j, \rho \right), \quad \mathbb{E} \prod_j q_{\rho, I_j}^{p_j} \asymp \exp \left[- \int_{\mathbb{R}} \chi(w_{\mathbf{p}}(s), \rho) ds \right], \quad (17.31)$$

where $w_{\mathbf{p}} = \sum_j p_j \mathbf{1}_{I_j}$ and the value $\chi(0, \rho) = 0$ denotes the region with no active interval. For $I_j = t[a_j, b_j]$, write $\Xi_{\Gamma}(\mathbf{p}) = \int \chi(\sum_j p_j \mathbf{1}_{[a_j, b_j]}(s), \rho) ds$. The finitely many active sets and Theorem 7.2 give

$$\partial_{p_j} \Xi_{\Gamma} = \int_{a_j}^{b_j} \partial_p \chi(w_{\mathbf{p}}(s), \rho) ds, \quad (17.32)$$

where in this formula $w_{\mathbf{p}} = \sum_j p_j \mathbf{1}_{[a_j, b_j]}$. The entropy rate of the joint Gibbs environment is consequently

$$\Xi_{\Gamma} - \mathbf{p} \cdot \nabla \Xi_{\Gamma} = \int \{ \chi(w_{\mathbf{p}}(s), \rho) - w_{\mathbf{p}}(s) \partial_p \chi(w_{\mathbf{p}}(s), \rho) \} ds, \quad (17.33)$$

with zero integrand outside the active union. The same leading rate holds for mixed envelopes satisfying the joint window (E.17).

For two intervals of length t with overlap h , the formula reads

$$\mathbb{E}[q_{I_1}^{p_1} q_{I_2}^{p_2}] \asymp e^{-(t-h)[\chi(p_1, \rho) + \chi(p_2, \rho)] - h\chi(p_1 + p_2, \rho)}. \quad (17.34)$$

In particular, at $p_1 = p_2 = 1/2$ it equals, up to constants, $e^{-2(t-h)\chi(1/2, \rho) - h/2}$. These deterministic-window identities explain the shared-clock effect; the arithmetic profile uses the conditioned, exact-type construction proved above.

18 Open problems

For an integer $d \geq 2$ and real side lengths $1 \leq N_1 \leq \dots \leq N_d$, write

$$\mathcal{M}_d(\mathbf{N}) = \{n_1 \cdots n_d : n_i \in \mathbb{N}, n_i \leq N_i\}, \quad A_d(\mathbf{N}) = \#\mathcal{M}_d(\mathbf{N}).$$

Variable-dimensional rectangular tables. Seek matching two-sided estimates for $A_d(\mathbf{N})$ uniformly in the dimension and the entire ordered side-length vector. For asymptotic formulae, consider joint families $(d(t), \mathbf{N}(t))$ satisfying

$$d(t) \longrightarrow \infty \quad \text{or} \quad N_{d(t)}(t) \longrightarrow \infty \quad (t \rightarrow \infty).$$

Such a theory should recover Theorem 1.2 at fixed dimension and the equal-side large-dimension results of [25].

Perfect powers. For an integer $r \geq 2$, define

$$A_d^{[r]}(\mathbf{N}) = \#(\mathcal{M}_d(\mathbf{N}) \cap \{m^r : m \in \mathbb{N}\}).$$

This counts distinct perfect r th powers. The equal-side case $d = r$ is elementary. What matching estimates hold uniformly in $(d, \mathbf{N}, r)$, and what asymptotic formulae hold along the growing families above with an integer-valued $r = r(t) \geq 2$, including fixed r ?

A Two reference-cost identities

The following identities keep the original physical groups separate. They are useful when a cost is charged to an old group, whose intensity need not satisfy an individual stationarity equation.

Lemma A.1 (Clipped margin telescope). *Use the complete chain $p_i = 1$. Let $[a, b]$ be a maximal equal-product owner, with product σ , child weight z , and original coefficients κ_g as in (3.4). Write $r_g = b - a + 1$ for $g < a$, and $r_g = b - g + 1$ for $a \leq g \leq b$. For integers $0 \leq R \leq b - a + 1$ set*

$$\Lambda_R = \sum_{g \leq b} L_g \{ \min(r_g, R) - \kappa_g E_{\min(r_g, R)}(z) \}.$$

Then

$$\Lambda_R = \lambda_{b-R} \geq 0, \quad \Lambda_0 = \lambda_{b-a+1} = 0, \quad \lambda_j = - \sum_{i \leq j} \partial_{s_i} J, \quad \lambda_0 = \lambda_k = 0.$$

Proof. For $1 \leq R \leq b - a + 1$ put $j = b - R + 1$ and differentiate in the formal direction which raises $s_j, \dots, s_b$ together. The derivatives exist in the positive product region even when this direction is not a two-sided feasible order direction. The linear budget derivative is $\sum_{l=j}^b V_l = \sum_{g \leq b} \min(r_g, R) L_g$.

For $g \geq j$, differentiating $G_g = \sigma \log(r_g + z)$, with the permanent child exponent below b fixed, gives $dz/d\sigma = -z \log z/\sigma$ and

$$\frac{dG_g}{d\sigma} = \frac{E_{r_g}(z)}{r_g + z}, \quad \frac{dZ_g}{d\sigma} = \kappa_g E_{r_g}(z).$$

For an earlier native group the routing probability into the surviving compound alphabet is $(R + z)/(r_g + z)$. For an old group it is

$$\left(\prod_{l=g}^{a-1} \beta_l \right) \frac{R + z}{b - a + 1 + z}.$$

The derivative at j is $E_R(z)/(R + z)$; propagation and multiplication by Z_g therefore give $\kappa_g E_R(z)$ in both cases. Subtraction from the linear derivative yields

$$\Lambda_R = \sum_{l=j}^b \partial_{s_l} J = \lambda_{j-1} - \lambda_b = \lambda_{b-R}.$$

Maximality of the owner gives $\lambda_b = \lambda_{a-1} = 0$, with the stated endpoint conventions. The remaining multipliers are nonnegative. The case $R = 0$ follows from the definition. $\square$

This identity concerns clipped benchmark paths. General legal rank paths also require the shifted compression argument: the capacities change between groups, so one cannot infer their nonnegativity from a global concavity assertion about the displayed sequence.

Lemma A.2 (Entropy denominator for an old mixture). *Suppose $H \geq 2$, $p > 0$, $S \geq 1$, $0 \leq u \leq 1$, and $p + S^u \leq H$. With*

$$E_p(x) = (p + x) \log(p + x) - x \log x, \quad D(p, x) = E_p(x)/p,$$

one has

$$S^{u-1}(1 + \log S) + \frac{p}{2H} \leq D(p, S^u) \leq 1 + \log H. \quad (\text{A.1})$$

Consequently, if $\kappa_{\text{av}} \geq 0$ and $\kappa_{\text{av}} \leq S^{u-1}/D(p, S^u)$, then

$$\kappa_{\text{av}}(1 + \log S) \leq 1 - \frac{p}{2H(1 + \log H)}.$$

Proof. For $x = S^u$, integration of the derivative of $t \log t$ gives

$$D(p, x) = 1 + \log x + \frac{1}{p} \int_0^p \log(1 + t/x) dt.$$

Since $\log(1 + t/x) \geq t/(x + t) \geq t/(x + p)$,

$$D(p, x) \geq 1 + u \log S + \frac{p}{2(x + p)} \geq 1 + u \log S + \frac{p}{2H}.$$

The same expression is the average of $1 + \log(x + t)$, giving the upper bound in (A.1). Put $L = \log S$ and $f(u) = 1 + uL - e^{(u-1)L}(1 + L)$. Then $f'' \leq 0$, $f(1) = 0$, and $f(0) = 1 - (1 + L)e^{-L} \geq 0$. Concavity implies $f(u) \geq 0$, which proves the lower bound. Finally divide it by $D(p, S^u)$ and use its upper bound:

$$\frac{S^{u-1}(1 + \log S)}{D(p, S^u)} \leq 1 - \frac{p}{2HD(p, S^u)} \leq 1 - \frac{p}{2H(1 + \log H)}.$$

Multiplication by the assumed bound for κ_{av} completes the proof. $\square$

In applications κ_{av} is obtained from the whole incoming count balance. The conclusion bounds that average, and does not assign an individual stationarity equation to each old physical group.

B Auxiliary probability estimates

Lemma B.1 (Finite Gaussian association). *Let Z be a centered nondegenerate Gaussian vector whose covariance matrix Σ has nonnegative entries. For bounded coordinatewise nondecreasing Borel functions f, g ,*

$$\mathbb{E}[f(Z)g(Z)] \geq \mathbb{E}f(Z)\mathbb{E}g(Z).$$

The assertion also holds whenever truncation makes both sides converge.

Proof. For smooth bounded functions with bounded derivatives, let (X_t, Y_t) have diagonal covariance blocks Σ and cross block $t\Sigma$, $0 \leq t \leq 1$. This is a positive semidefinite covariance: one may take $X_t = Z$ and $Y_t = tZ + \sqrt{1 - t^2}Z'$ with Z' an independent copy. Gaussian integration by parts gives

$$\frac{d}{dt} \mathbb{E}[f(X_t)g(Y_t)] = \sum_{i,j} \Sigma_{ij} \mathbb{E}[\partial_i f(X_t) \partial_j g(Y_t)] \geq 0.$$

The two off-diagonal covariance blocks account for the absence of a factor $1/2$. The covariance derivative identity follows by differentiating $\exp(-\langle \xi, C(t)\xi \rangle/2)$ and integrating twice by parts, first for smooth compactly supported test functions; cutoff approximation and covariance regularization give the displayed form. At $t = 0$ the two vectors are independent; at $t = 1$ they coincide. Integration proves the claim for smooth functions.

Convolution with a positive smooth product kernel preserves monotonicity. Bounded coordinatewise monotone functions have a Lebesgue-null discontinuity set, so the convolutions converge almost everywhere for the nonsingular Gaussian law. Dominated convergence proves the Borel assertion. At distinct finite Ornstein–Uhlenbeck times the covariance is positive definite; coincident times can simply be deleted. Applying this finite result on nested dense time grids and using path continuity proves its survival applications. This argument does not assert association after arbitrary conditioning. $\square$

B.1 Joint exact-total comparisons and small tubes

Let $f_{r,\tau}$ denote the density of a sum of r independent exponentials of mean τ . If n marks have exact low count k , $\pi = k/n$, and their $n+1$ gaps have total $P = (n+1)\tau$, exposing m specified marks and m distinct specified gaps has joint density

$$\frac{f_{n+1-m,\tau}(P-T)}{f_{n+1,\tau}(P)} \frac{\mathbb{P}\{\text{Bin}(n-m, \pi) = k-K\}}{\mathbb{P}\{\text{Bin}(n, \pi) = k\}} \quad (\text{B.1})$$

relative to independent reference gaps and marks. Here T, K are the exposed totals, and the ratio is zero outside its support. Independence of the two unobserved sums proves the identity. The same formula with m replaced by $2m$ applies jointly to the first m and last m marks and to the distinct gap sets

$$E_1, \dots, E_m; \quad E_{n-m+2}, \dots, E_{n+1}.$$

There are $n-2m$ residual marks and $n+1-2m$ residual gaps.

For $m \leq n/4$, or for $m = \lfloor n/8 \rfloor$ at each end, this ratio is bounded above on its entire support, uniformly on compact positive τ and interior π . On each fixed central box

$$|T - m\tau| + |K - m\pi| \leq C\sqrt{n}$$

it is bounded below positively for sufficiently large n ; in the two-end formula replace m by $2m$. Indeed the Gamma maximum density is $O((1+r)^{-1/2})$, its density at the reference mean is $\asymp (1+r)^{-1/2}$, and the corresponding binomial bounds have the same order. Central Stirling bounds give the lowers after the remaining totals are inside their support. These denominators are at their exact reference means. Near-central variants require the large-total hypotheses of Lemma 6.12.

Lemma B.2 (Exact multicolor small tube). *Fix $r, \epsilon > 0$, $0 < \tau_- \leq \tau_+ < \infty$, and $a > 0$. Let a word of length n be uniform of exact type $v = (v_1, \dots, v_r)$, $\pi_c = v_c/n \geq \epsilon$. Independently let n exponential gaps be conditioned to have total $n\tau$, $\tau \in [\tau_-, \tau_+]$. For all sufficiently large n ,*

$$\mathbb{P} \left\{ \max_{j \leq n} |T_j - j\tau| \leq a\sqrt{n}, \quad \max_{c,j} |N_c(j) - j\pi_c| \leq a\sqrt{n} \right\} \geq c_a > 0.$$

The threshold depends on the fixed data and a .

Proof. First, for both exact bridges,

$$\mathbb{P}\{\max_j |T_j - j\tau| > u\} + \sum_c \mathbb{P}\{\max_j |N_c(j) - j\pi_c| > u\} \leq C \exp\{-c \min(u^2/n, u)\}. \quad (\text{B.2})$$

To see this, expose at most $3n/4$ coordinates. Their likelihood relative to independent reference coordinates is the product of

$$\frac{f_{n-m,\tau}(n\tau - \sum_I E_j)}{f_{n,\tau}(n\tau)}, \quad \frac{\mathbb{P}\{\text{Mult}(n-m, \pi) = v - N_I\}}{\mathbb{P}\{\text{Mult}(n, \pi) = v\}}.$$

Gamma and multinomial maximum-density and central-density Stirling bounds make both ratios bounded above. This is a comparison for the whole selected set. It does not assert independence of separated selected coordinates. Independent centered gaps and centered color indicators have moment generating functions bounded by $\exp(C\theta^2)$ for $|\theta| \leq \theta_0$. The exponential supermartingale

maximal inequality gives the right side of (B.2) on the first half. Reverse the second half, using exchangeability and the zero centered complete total. This proves (B.2); $n = 1$ is immediate.

Choose a fixed integer $J \geq 2$, and split the coordinates into J blocks of lengths between $n/(2J)$ and $2n/J$. In the first $J - 1$ blocks, let the first $r - 1$ color counts vary in integer intervals of width $\delta\sqrt{n}$ about their means. Determine the remaining color count by the block length and the last block by the full type. All deviations are $O_r(J\delta\sqrt{n})$ and all counts are nonnegative for sufficiently large n . The exact conditional atom is

$$\frac{\prod_{b=1}^J \mathbb{P}\{\text{Mult}(n_b, \pi) = v_b\}}{\mathbb{P}\{\text{Mult}(n, \pi) = v\}}.$$

Central Stirling gives a lower bound $c_{J,\delta} n^{-(J-1)(r-1)/2}$ for each choice. There are at least $c_{J,\delta} n^{(J-1)(r-1)/2}$ choices. Thus this event has positive constant probability, including $r = 1$.

Likewise prescribe the first $J - 1$ block gap totals in real intervals of width $\delta\sqrt{n}$ about $n_b\tau$, with the last determined by the full total. Their density is $\prod_b f_{n_b,\tau}(A_b)/f_{n,\tau}(n\tau)$, bounded below by $c_{J,\delta} n^{-(J-1)/2}$ on a box of volume $\asymp n^{(J-1)/2}$. Its probability is positive, and the gap and color channels remain independent.

After J is fixed, choose δ small enough that all cumulative boundary deviations are at most $a\sqrt{n}/4$. Conditional on these block totals, the blocks are independent exact bridges with compact new parameters. By (B.2), their combined failure probability for tubes of radius $a\sqrt{n}/2$ about their own linear interpolants is at most $CJ \exp(-ca^2J)$ for sufficiently large n . Choose J to make this at most $1/2$, and then choose δ and the length threshold. Boundary and internal deviations sum to less than $a\sqrt{n}$. All choices precede n . $\square$

The gap and color conclusions can have different lengths: apply their separate estimates at $n + 1$ and n , respectively. In particular the lemma applies to a bridge with n marks and all $n + 1$ gaps. Its large length threshold is essential for an arbitrarily narrow tube.

B.2 Positive drift under exact conditioning

Lemma B.3 (Ordinary positive-drift continuation). *Fix compact positive ranges for two marks w_0, w_1 (which may be equal), a speed c , and $\tau = P/(n+1)$, with k/n in a fixed compact subinterval of $(0, 1)$. On $[0, P]$ take n uniform points and, independently, a uniform k -subset bearing mark w_1 ; the others bear w_0 . For $\varepsilon \in \{-1, 1\}$ put*

$$Z(t) = x + \varepsilon\{w_0 N_0(t) + w_1 N_1(t) - ct\}, \quad v = \varepsilon \left(\frac{(n-k)w_0 + kw_1}{P} - c \right).$$

Suppose $0 < v \leq v_{\max}$. For every fixed $a > 0$ there are $R(a) < \infty$ and $b(a) > 0$ such that

$$x \geq a/v, \quad Pv^2 \geq R(a) \implies \mathbb{P}\{Z(t) \geq 0 \ (0 \leq t \leq P)\} \geq b(a).$$

This probability averages both the exact clock and exact type.

Proof. Write $n_0 = n - k$, $n_1 = k$. The law is equivalently that of independent uniform positions of each color, with respective counts n_i . Use independent Poisson processes of rates $\lambda_i = n_i/P$ as a reference law Q , and call their law conditioned on the two totals B . On the first half, the exact density is

$$\frac{dB}{dQ} = \prod_{i=0}^1 \frac{p_{n_i/2}(n_i - N_i(P/2))}{p_{n_i}(n_i)}, \quad p_\mu(r) = e^{-\mu} \mu^r / r!. \quad (\text{B.3})$$

It is at most a fixed C_{den} everywhere. On $\mathcal{C}_K = \{|N_i(P/2) - n_i/2| \leq K\sqrt{n_i}, i = 0, 1\}$ it is at least $c_{\text{den}}(K) > 0$ once $n_i \geq N_0(K)$. The upper follows from Poisson maximum atoms, and the lower from central Stirling, for example first taking $n_i \geq 16K^2$. The same statements apply to the reversed half.

If w_+ bounds the marks above and τ_- bounds the gap mean below, put

$$C_{\text{exp}} = \frac{w_+^2 e^{w_+}}{2\tau_-}, \quad \kappa = \min\{v_{\text{max}}^{-1}, (2C_{\text{exp}})^{-1}\}.$$

For $0 \leq \theta \leq 1$, the Taylor remainder gives in both orientations

$$\varepsilon c\theta + \sum_i \lambda_i (e^{-\varepsilon\theta w_i} - 1) \leq -v\theta + C_{\text{exp}}\theta^2.$$

With $\theta = \kappa v$, the process $e^{-\theta Z(t)}$ is a nonnegative supermartingale. At the first exit from $(0, \infty)$ its value is at least one; the possible negative overshoot is bounded by w_+ . Stopping at the exit or a fixed time and passing to the limit proves that the exit probability under Q is at most $e^{-\theta x} \leq e^{-\kappa a}$. Thus first-half survival has Q -probability at least $p_a = 1 - e^{-\kappa a} > 0$. Choose $K \geq \max(1, \sqrt{2/p_a})$. Chebyshev gives $Q(\mathcal{C}_K^c) \leq K^{-2} \leq p_a/2$, so

$$B(\text{first-half survival}) \geq c_{\text{den}}(K)p_a/2 =: b_{\text{left}}(a) > 0.$$

For $0 \leq u \leq P/2$, let $N_i^{\text{rev}}(u)$ count points in the last interval of length u , with the strict-open convention at the moving left endpoint. The exact identity is

$$Z(P - u) = x + v(P - u) - \varepsilon M^{\text{rev}}(u), \quad M^{\text{rev}}(u) = \sum_i w_i (N_i^{\text{rev}}(u) - \lambda_i u).$$

A second-half failure therefore forces $\sup_{u \leq P/2} \varepsilon M^{\text{rev}}(u) \geq vP/2$. Including one-sided limits makes this the usual right-continuous Poisson maximal event. Its reference exponential compensator is

$$\Lambda_\varepsilon(\theta) = \sum_i \lambda_i (e^{\varepsilon\theta w_i} - 1 - \varepsilon\theta w_i) \leq C_{\text{exp}}\theta^2.$$

The exponential martingale maximal inequality and the upper density in (B.3) give

$$B(\text{second-half failure}) \leq C_{\text{den}} e^{-\kappa P v^2/4}.$$

Choose $R(a)$ to make this at most $b_{\text{left}}(a)/2$, and large enough that $Pv^2 \geq R(a)$ forces $n_i \geq N_0(K)$. It suffices for the latter that $R(a)/v_{\text{max}}^2 \geq \tau_+[N_0(K)/\epsilon + 1]$. Subtracting gives the result. All constants are fixed before the bridge or any retained prefix is chosen. $\square$

The reference rank drift is $\varepsilon(\mathbb{E}W - c\tau) = (y - x + \varepsilon c\tau)/n$. It differs from the physical drift $(y - x)/P$ by the contribution of the terminal unmarked gap; neither gap nor contribution is omitted.

C Joint positive OU pressure and deterministic overlaps

For independent stationary standard OU processes $E, W_1, \dots, W_m$, fixed slopes $\beta_j > 0$ and positive orders p_j , put

$$q_{j,I}(E) = \mathbb{P}_{W_j}\{W_{j,s} \geq -\beta_j E_s \ (s \in I) \mid E\}.$$

Write $q_{j,t} = q_{j,[0,t]}$ and let μ denote the stationary environmental OU path law. For a nonempty active set A , define

$$\Xi_A(\mathbf{p}; \beta) = - \lim_{t \rightarrow \infty} t^{-1} \log \mathbb{E} \prod_{j \in A} q_{j,[0,t]}^{p_j}, \quad \Xi_\emptyset = 0. \quad (\text{C.1})$$

The existence of this limit is part of the theorem below. This appendix proves the positive-moment inputs independently of a full LDP and independently of arithmetic estimates. All comparison constants can be chosen uniformly on compact positive order and wall-slope sets, for a fixed number of comparisons.

C.1 Association and the order of tilted environments

On a finite ordered time grid, the OU precision matrix is tridiagonal with nonpositive off-diagonal entries. Its density is MTP_2 . Here a nonnegative density f is MTP_2 when

$$f(x \wedge y) f(x \vee y) \geq f(x) f(y).$$

The product, marginalization and association properties used below are standard; see [11, Proposition 3.4 and Section 3, especially equation (3.5)]. Gaussian association also follows directly from nonnegative covariances [22].

For $q_j(E) = \mathbb{P}(W_j \geq -\beta_j E \mid E)$, set $Z = -W_j$. The indicator of $\{Z_i \leq \beta_j E_i \ \forall i\}$ is MTP_2 : its support is closed under coordinatewise minimum and maximum. Multiply by the MTP_2 density of Z and an independent product Gaussian density in E , integrate out Z , and cancel the auxiliary product density. This proves that $q_j(E)$ is MTP_2 . It is also increasing in E .

Therefore

$$d\mu_{\mathbf{p},t} \propto \prod_j q_j^{p_j} d\mu$$

is associated for every $p_j \geq 0$. If $0 \leq p_j \leq n_j$ coordinatewise,

$$\mu \preceq_{\text{st}} \mu_{\mathbf{p},t} \preceq_{\text{st}} \mu_{\mathbf{n},t}. \quad (\text{C.2})$$

The first comparison applies original Gaussian association to an increasing test and $\prod_j q_j^{p_j}$. The second applies association under $\mu_{\mathbf{p},t}$ to the same test and $\prod_j q_j^{n_j - p_j}$. For nested dense grids including the endpoints, the private grid survival probabilities decrease to the path survival probabilities. Dominated convergence passes each bounded cylindrical association and order inequality to the path law. Approximation by finite-grid maxima, followed by truncation and monotone convergence, gives the same inequalities for the endpoint and short-interval supremum functionals used below.

C.2 An exact cone time-change identity

Choose integers $n_j \geq p_j$ and write

$$U = (E, W_{j,\ell} : 1 \leq \ell \leq n_j), \quad d = 1 + \sum_j n_j.$$

Its coordinates are independent stationary standard OU processes. Set

$$C = \{(e, w) : w_{j,\ell} + \beta_j e \geq 0 \ \forall j, \ell\}, \quad p_C(t) = \mathbb{P}\{U_s \in C \ (0 \leq s \leq t)\}.$$

The environmental marginal conditioned on this event is $\mu_{\mathbf{n},t}$. All constraint signals have nonnegative covariances at all pairs of times. Consequently

$$p_C(s+t) \geq p_C(s)p_C(t). \quad (\text{C.3})$$

Lemma C.1 (Quadratic endpoint weights shorten cone time). *For $0 < a < 1/2$, set*

$$v = (1 - 2a)^{-1}, \quad f_v(h) = \log \left(1 + \frac{e^h - 1}{v} \right).$$

For $0 \leq s \leq t$,

$$\mathbb{E} \left[e^{a|U_s|^2}; U_u \in C \ (0 \leq u \leq t) \right] = v^{d/2} p_C(f_v(s) + f_v(t-s)). \quad (\text{C.4})$$

In particular,

$$\sup_{t \geq 0, 0 \leq s \leq t} \mathbb{E}[e^{a|U_s|^2} \mid U_{[0,t]} \subset C] \leq \frac{v^{d/2}}{p_C(2 \log v)} < \infty. \quad (\text{C.5})$$

Proof. Multiplication by $e^{a|U_s|^2}$ changes the $N(0, I_d)$ density of U_s into $v^{d/2}$ times the density of $N(0, vI_d)$. Given U_s , the past and future are independent reversed and forward OU branches. Starting from $\sqrt{v}G$, a branch has representation

$$e^{-h/2}(\sqrt{v}G + B_{e^h-1}).$$

Divide by $\sqrt{v}$ and scale Brownian time. Membership in the cone becomes the usual OU cone condition for a branch of duration $f_v(h)$; the remaining multiplier is positive and does not change cone membership. The two branches share a standard Gaussian initial point and recombine into a stationary OU segment of duration $f_v(s) + f_v(t-s)$. This proves (C.4).

Since $0 \leq h - f_v(h) \leq \log v$, the new duration t' differs from t by at most $2 \log v$. By (C.3) and monotonicity,

$$p_C(t) \geq p_C(t')p_C(t-t') \geq p_C(t')p_C(2 \log v).$$

This gives (C.5). The positive coordinate orthant is contained in C , so $p_C(2 \log v) \geq a(2 \log v)^d > 0$, uniformly in the positive slopes. $\square$

Apply (C.2) separately to $e^{a(E_s^+)^2}$ and $e^{a(E_s^-)^2}$. Lemma C.1 and the original Gaussian law give

$$\sup_{t \geq 0, 0 \leq s \leq t} \mathbb{E}_{\mu_{\mathbf{p},t}} e^{aE_s^2} < \infty, \quad 0 < a < 1/2. \quad (\text{C.6})$$

On a compact positive order set the dominating integers n_j can be fixed once.

C.3 The deterministic filter under positive tilts

The deterministic part of Lemma 6.3 is independent of the moment order and applies pathwise before any environmental tilt. In the notation of its proof, (6.10) and (6.11) apply to the actual private endpoint law conditioned on the whole environmental path and survival. The proof includes the possibly truncated initial law, a single initial segment of length at most one, and all subsequent segments of lengths in $[1, 2]$, retaining the incoming conditional private law at each step.

For each slope β_j and each endpoint $v \in \{0, t\}$, time reversal and (6.12) supply a nonnegative decreasing functional $M_{j,v}$ dominating the private endpoint mode, of the form

$$M_{j,v}(E) = C \left(1 + \sum_{\ell \geq 0} e^{-\ell/2} \sup_{s \in I_{v,\ell}} (-\beta_j E_s)_+ \right). \quad (\text{C.7})$$

The intervals have length at most two and are ordered inward from v ; absent intervals contribute zero. Constants are uniform when the slopes vary in a compact subset of $(0, \infty)$. The proof of Lemma 6.3 gives, for some $c > 0$,

$$\sup_{t \geq 0} \mathbb{E}_\mu e^{cM_{j,0}^2} + \sup_{t \geq 0} \mathbb{E}_\mu e^{cM_{j,t}^2} < \infty. \quad (\text{C.8})$$

These interval suprema are controlled together: if $a_\ell = e^{-\ell/2}$ and $A = \sum a_\ell$, weighted Cauchy–Schwarz bounds $(\sum a_\ell B_\ell)^2$ by $A \sum a_\ell B_\ell^2$, and Jensen bounds its exponential expectation by the supremum of the uniformly bounded short-interval OU supremum exponential moments. Since the functionals in (C.8) are decreasing, the first stochastic-order comparison in (C.2) transfers these bounds to $\mu_{\mathbf{p},t}$ for every positive order vector.

The radial bound (6.10), followed by Cauchy–Schwarz within the conditional private law, gives, for small $a > 0$,

$$\mathbb{E}_{W_j}[e^{a(W_{j,0}^2 + W_{j,t}^2)} \mid \text{survival } j, E] \leq C \exp\{Ca(M_{j,0}^2 + M_{j,t}^2)\}. \quad (\text{C.9})$$

Thus only the environmental endpoint estimate (C.6) needed extension beyond total order one; that extension was supplied by the integer replica cone and its exact time change above.

C.4 Joint endpoint weights and quasi-multiplicativity

For each private channel set

$$Q_{j,t}^{(a)}(E) = \mathbb{E}_{W_j}[\mathbf{1}_{\{\text{survival } j\}} e^{a(W_{j,0}^2 + W_{j,t}^2)} \mid E], \quad Z(t) = \mathbb{E} \prod_j q_{j,t}^{p_j}.$$

By (C.6), (C.8), (C.9), and finitely many Hölder inequalities, there are $a > 0, C < \infty$ such that

$$\mathbb{E} \left[e^{a(E_0^2 + E_t^2)} \prod_j (Q_{j,t}^{(a)}(E))^{p_j} \right] \leq CZ(t), \quad t \geq 0. \quad (\text{C.10})$$

To see the normalization explicitly, divide by $Z(t)$. Each private ratio $Q_j^{(a)}/q_j$ is bounded by (C.9); the resulting decreasing exponential majorants and the environmental endpoint exponential have uniform moments. Choose a small enough to absorb the finite positive exponents and Hölder powers.

Theorem C.2 (Joint positive pressure with constant-factor error). *For every fixed nonempty finite collection of positive p_j, β_j , the pressure (C.1) exists, is finite and positive, and*

$$Z(s)Z(t) \leq Z(s+t) \leq CZ(s)Z(t). \quad (\text{C.11})$$

Consequently

$$C^{-1}e^{-t\Xi} \leq Z(t) \leq e^{-t\Xi}, \quad t \geq 1. \quad (\text{C.12})$$

All constants are locally uniform. In particular, (7.3) holds for every positive scalar order.

Proof. For fixed E , Gaussian association in each private process gives $q_{j,s+t} \geq q_{j,s} q_{j,t} \circ \vartheta_s$. Environmental association then proves the first inequality in (C.11). Hence $b = -\log Z$ is subadditive.

If $K = \sum_j \max(1, p_j)$, association between the increasing functions $q_j^{p_j}$ and Jensen give

$$Z(t) \geq a(t)^K, \quad 0 \leq b(t) \leq K(t/2 + \log \pi).$$

Thus $\Xi = \lim b(t)/t = \inf_{t>0} b(t)/t$ exists and is finite: the usual subadditive argument applies because the displayed bound controls b on every bounded time interval, including a remainder tending to zero. Also $Z \leq \mathbb{E} q_j^{p_j}$. For $p_j \leq 1$, Jensen gives $\mathbb{E} q_j^{p_j} \leq a(t)^{p_j}$; for $p_j \geq 1$, $q_j^{p_j} \leq q_j$. Hence $\Xi \geq \min(1, p_j)/2 > 0$.

For a fixed gap g , write $r = e^{-g/2}$. The density of two separated stationary OU blocks relative to independent blocks is the endpoint Mehler ratio

$$k_r(x, y) = \frac{1}{\sqrt{1-r^2}} \exp \left[\frac{2rxy - r^2(x^2 + y^2)}{2(1-r^2)} \right] \leq \frac{1}{\sqrt{1-r^2}} e^{r(x^2+y^2)/(2(1+r))}.$$

Choose a fixed g large enough that this quadratic coefficient fits into (C.10). Drop survival across the gap, apply this bound first inside every private conditional integral and then to the environmental transition. The two reference blocks are independent, and (C.10) gives

$$Z(s+g+t) \leq C_0 Z(s) Z(t).$$

The lower product inequality also gives $Z(s+g+t) \geq Z(s+t) Z(g)$. This proves the upper bound in (C.11). Therefore

$$b(s) + b(t) - \log C \leq b(s+t) \leq b(s) + b(t).$$

Iterate at integer multiples of t and pass to the limiting slope to get

$$t\Xi \leq b(t) \leq t\Xi + \log C.$$

The same estimate can be used down to $t = 0$ after a fixed enlargement of C : for $0 \leq t \leq 1$, the preceding bound gives $Z(t) \geq a(1)^K$, while $Z(t) \leq 1$ and Ξ is locally bounded above. This explicitly supplies the uniform short-time constant needed for arbitrarily short overlap pieces. This proves the theorem. $\square$

C.5 Differentiation, concentration, and selected local deviations

Proposition C.3. *The joint pressure is nondecreasing and concave in $\mathbf{p}$ and belongs to $C_{\text{loc}}^{1,1}((0, \infty)^m)$. Under the joint tilt, let $\mathbf{Y}_t = (-\log q_j/t)_j$ and $\mathbf{y} = \nabla \Xi(\mathbf{p})$. Then*

$$\mu_{\mathbf{p},t}(\|\mathbf{Y}_t - \mathbf{y}\|_\infty > \epsilon) \leq C \exp[-ct \min\{\epsilon^2, \epsilon\}]. \quad (\text{C.13})$$

Furthermore,

$$\begin{aligned} \lim_{\delta \downarrow 0} \liminf_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{P}(\|\mathbf{Y}_t - \mathbf{y}\|_\infty < \delta) &= \lim_{\delta \downarrow 0} \limsup_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{P}(\|\mathbf{Y}_t - \mathbf{y}\|_\infty < \delta) \\ &= -[\Xi(\mathbf{p}) - \mathbf{p} \cdot \nabla \Xi(\mathbf{p})]. \end{aligned} \quad (\text{C.14})$$

The same bracket is $\lim_t t^{-1} D_{\text{KL}}(\mu_{\mathbf{p},t} \| \mu)$.

Proof. On a whitened finite grid put $A_j = -\log q_j$. Lemma 7.1 applies to each A_j with constant β_j^2 . The joint tilted potential $|x|^2/2 + \sum_j p_j A_j$ has Hessian at least I . For any $v \in \mathbb{R}^m$,

$$\begin{aligned} \text{Var}_{\mathbf{p}} \left(\sum_j v_j A_j \right) &\leq \mathbb{E}_{\mathbf{p}} \left| \sum_j v_j \nabla A_j \right|^2 \\ &\leq 2 \left(\sum_j v_j^2 \beta_j^2 \right) \sum_j \mathbb{E}_{\mathbf{p}} A_j. \end{aligned}$$

Concavity of b_t in each coordinate and nonnegativity on the coordinate face give

$$\mathbb{E}_{\mathbf{p}} A_j \leq b_t(\mathbf{p})/p_j = O(t+1).$$

Thus the Hessians of b_t/t are uniformly bounded on positive compact sets. Grid limits of the derivatives follow from slightly smaller positive exponents, which dominate polynomial factors in the A_j . Equicontinuity and integration of the gradients give $C^{1,1}$ regularity. The $O(1)$ error in Theorem C.2 also gives $\nabla(b_t/t) - \nabla \Xi = O(t^{-1/2})$ locally.

For small $|h|$ with $\mathbf{p} - h\mathbf{e}_j$ positive,

$$\mathbb{E}_{\mathbf{p},t} e^{h(S_j - ty_j)} = e^{-hty_j} \frac{Z_{\mathbf{p}-h\mathbf{e}_j}(t)}{Z_{\mathbf{p}}(t)} \leq C e^{Lth^2/2}.$$

Two-sided Chernoff and a union bound prove (C.13). For the local deviations, write exactly

$$\mathbb{P}(B_\delta) = Z_{\mathbf{p}}(t) \mathbb{E}_{\mathbf{p},t} [e^{t\mathbf{p} \cdot \mathbf{Y}_t} \mathbf{1}_{B_\delta}].$$

On B_δ , replace the exponent by $t\mathbf{p} \cdot \mathbf{y}$ with error at most $t\delta \|\mathbf{p}\|_1$. Use (C.13) and then let $\delta \downarrow 0$. Finally

$$D_{\text{KL}}(\mu_{\mathbf{p},t} \|\mu) = b_t(\mathbf{p}) - \mathbf{p} \cdot \nabla b_t(\mathbf{p}),$$

which gives the entropy rate at the selected points. $\square$

C.6 Proof of the deterministic-overlap formula

Theorem C.4. *For finitely many deterministic positive-length intervals I_j , put $A(s) = \{j : s \in I_j^\circ\}$. Then*

$$\mathbb{E} \prod_j q_{j,I_j}^{p_j} \asymp \exp \left[- \int_{\mathbb{R}} \Xi_{A(s)}(\mathbf{p}; \beta) ds \right]. \quad (\text{C.15})$$

The constants are independent of all interval lengths, locations and overlaps, and locally uniform in the positive orders and slopes for a fixed number of intervals.

Proof. Sort all interval endpoints and partition the active union into positive-length intervals J_ℓ of lengths L_ℓ , with constant active sets A_ℓ . There are at most $2m - 1$ such elementary intervals. Conditional association in each private process gives

$$q_{j,I_j} \geq \prod_{\ell: J_\ell \subset I_j} q_{j,J_\ell}.$$

After raising to positive powers, environmental association and Theorem C.2 give the lower bound

$$\mathbb{E} \prod_j q_{j,I_j}^{p_j} \geq \prod_\ell Z_{A_\ell}(L_\ell) \geq c \exp \left[- \sum_\ell L_\ell \Xi_{A_\ell} \right].$$

For the upper bound choose the fixed gap g in the proof of Theorem C.2 uniformly over all active subsets. Keep the elementary intervals with length at least $2g$ and remove $g/2$ from each end. Their cores have length $L_\ell - g \geq g$ and are pairwise separated by at least g . Drop all other survival constraints. For each private channel its active cores form an ordered subsequence. Its joint law relative to independent stationary cores is a product of endpoint Mehler ratios. Bound these as above, then do the same for the environmental core law. Each core endpoint is used at most once on each side. Apply (C.10) inside each reference core to obtain

$$\mathbb{E} \prod_j q_{j,I_j}^{p_j} \leq C \prod_{\ell: L_\ell \geq 2g} Z_{A_\ell}(L_\ell - g).$$

The total active duration deleted is at most $C_m g$. All active-set pressures have locally uniform finite upper bounds. Hence

$$\sum_{L_\ell \geq 2g} (L_\ell - g) \Xi_{A_\ell} = \sum_\ell L_\ell \Xi_{A_\ell} + O(1).$$

Theorem C.2 proves the upper bound. This also covers the case in which all elementary intervals are short. Endpoints shared by adjacent intervals are handled by continuity; the stated integrals are unaffected by sets of zero length. $\square$

D The full scalar LDP for stationary equal-rate OU processes

Let E, W be independent stationary standard OU processes with covariance $e^{-|s-t|/2}$, and retain $q_t(E)$ and $S_t = -\log q_t(E)$ from (7.1). We prove the exact negative-moment domain, regularity of its pressure, and a full large-deviation principle. The stationary initial law and equal covariance functions of the two channels are part of this model.

Theorem D.1 (Full scalar large deviations). *Fix $0 < \rho < 1$, let $\beta^2 = \rho/(1 - \rho)$ and set*

$$\theta_c = \beta^{-2} = \frac{1 - \rho}{\rho}.$$

The limit

$$\Lambda_\rho(\theta) = \lim_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{E} e^{\theta S_t} = \lim_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{E} q_t^{-\theta} \quad (\text{D.1})$$

exists and is finite for every $\theta < \theta_c$, and

$$\Lambda_\rho \in C_{\text{loc}}^{1,1}((-\infty, \theta_c)). \quad (\text{D.2})$$

For every $t > 0$, $\mathbb{E} q_t^{-\theta} = \infty$ when $\theta \geq \theta_c$. Furthermore,

$$\Lambda_\rho(\theta) \rightarrow +\infty, \quad \Lambda'_\rho(\theta) \rightarrow +\infty \quad (\theta \uparrow \theta_c), \quad \Lambda'_\rho(\theta) \rightarrow 0 \quad (\theta \rightarrow -\infty). \quad (\text{D.3})$$

The variables $Y_t = S_t/t$ satisfy a full LDP at speed t with good convex rate

$$I_\rho(y) = \sup_{\theta < \theta_c} \{\theta y - \Lambda_\rho(\theta)\}. \quad (\text{D.4})$$

Explicitly, for every Borel set $A \subset \mathbb{R}$,

$$\begin{aligned} -\inf_{A^\circ} I_\rho &\leq \liminf_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{P}(Y_t \in A) \\ &\leq \limsup_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{P}(Y_t \in A) \leq -\inf_A I_\rho. \end{aligned} \quad (\text{D.5})$$

D.1 The exact negative-moment domain

On a finite grid with covariance Σ , simultaneous whitening gives

$$q(x) = \mathbb{P}\{Z + \beta x \in C\}, \quad C = \Sigma^{-1/2} \mathbb{R}_+^n,$$

where x, Z are standard normal. Since C is a cone,

$$q = T_{\sqrt{\rho}} \mathbf{1}_C, \quad T_r f(x) = \mathbb{E} f(rx + \sqrt{1-r^2} Z).$$

We use the Gaussian reverse-hypercontractive inequality [6, Section 3.3, equation (3.14)]:

$$\|T_r f\|_s \geq \|f\|_p, \quad s < p < 1, \quad r^2 \leq \frac{1-p}{1-s}. \quad (\text{D.6})$$

Negative exponent norms mean $\|f\|_s = (\mathbb{E} f^s)^{1/s}$ for positive f . Reverse hypercontractivity originates with Borell [2]. We give the logarithmic Sobolev calculation, following the semigroup method in [21], to fix the Gaussian normalization. For the OU semigroup $P_u = T_{e^{-u}}$, set $r(u) = 1 - (1-p)e^{2u}$ and $v_u = P_u f > 0$. Then

$$\begin{aligned} \frac{d}{du} \log \|v_u\|_{r(u)} &= \frac{r'(u)}{r(u)^2} \frac{\text{Ent}(v_u^{r(u)})}{\mathbb{E} v_u^{r(u)}} \\ &\quad - (r(u) - 1) \frac{\mathbb{E}[v_u^{r(u)-2} |\nabla v_u|^2]}{\mathbb{E} v_u^{r(u)}}. \end{aligned}$$

Gross's Gaussian logarithmic Sobolev inequality [16] gives

$$\text{Ent}(v^r) \leq \frac{r^2}{2} \mathbb{E}[v^{r-2} |\nabla v|^2].$$

Because $r' = 2(r-1) < 0$, substitution reverses the direction in the first term and makes the derivative nonnegative. Start with bounded strictly positive f , pass through exponent zero by the geometric-mean limit, and approximate an indicator from above. This proves the form needed in (D.6).

For $0 < \theta < \theta_c$, take

$$s = -\theta, \quad p = 1 - \rho(1 + \theta) > 0.$$

Since a negative power reverses the norm inequality, one obtains

$$\mathbb{E} q_{\text{grid}}^{-\theta} \leq a_{\text{grid}}^{-\theta/[1-\rho(1+\theta)]}.$$

Along nested dense grids, $q_{\text{grid}} \downarrow q_t$ and $a_{\text{grid}} \downarrow a(t) > 0$. Monotone convergence therefore gives

$$\mathbb{E} q_t^{-\theta} \leq a(t)^{-\theta/[1-\rho(1+\theta)]} < \infty. \quad (\text{D.7})$$

In particular,

$$\limsup_t \frac{1}{t} \log \mathbb{E} e^{\theta S_t} \leq \frac{\theta}{2[1-\rho(1+\theta)]}. \quad (\text{D.8})$$

For the converse, the initial constraint gives $q_t(E) \leq \Phi(\beta E_0)$. The Gaussian lower-tail estimate implies that $\mathbb{E} \Phi(\beta G)^{-\theta}$ dominates a positive constant times

$$\int_{-\infty}^{\infty} x^\theta \exp\left[-\frac{\theta\beta^2 - 1}{2} x^2\right] dx.$$

It is infinite when $\theta\beta^2 \geq 1$, including equality. Thus the finite-time moment domain is exact. Section D.4 proves the corresponding divergence of the limiting pressure.

D.2 Covariance interpolation and the existence of all required limits

Consider two consecutive parts of a finite ordered OU grid. Multiply the one adjacent correlation joining them by $v \in [0, 1]$. The new covariance Σ_v has unchanged within-block entries and cross-block entries multiplied by v . Hence

$$\dot{\Sigma}_{ii} = 0, \quad \dot{\Sigma}_{ij} \geq 0 \quad (i \neq j).$$

Indeed an ordered OU grid has the representation $X_{k+1} = r_k X_k + \sqrt{1 - r_k^2} G_{k+1}$, with independent standard G_{k+1} and $0 < r_k < 1$. Replacing only the joining r_k by vr_k has exactly the stated effect on the cross-block covariance. The precision remains tridiagonal with nonpositive off-diagonal entries, including at $v = 0$. Thus its Gaussian density is MTP₂; see [11, Section 4.1]. Write

$$u_v(z) = \mathbb{P}\{N(0, \Sigma_v) \leq z\}.$$

MTP₂ and marginalization give

$$u_i \geq 0, \quad \partial_{ij} \log u \geq 0 \quad (i \neq j). \quad (\text{D.9})$$

Indeed the set $\{(w, z) : w_i \leq z_i \ \forall i\}$ is closed under coordinatewise minimum and maximum. Its indicator times the Gaussian density is MTP₂ in (w, z) , and marginalization preserves MTP₂ by [11, Proposition 3.4]. An independent product density in z can be inserted and cancelled to obtain an integrable joint density.

For $\eta > -\beta^{-2}$ set

$$H_\eta(v) = \mathbb{E}_{X_v \sim N(0, \Sigma_v)} u_v(\beta X_v)^\eta.$$

Differentiate the environmental covariance using Gaussian integration by parts, and the private covariance using its heat equation. The two contributions are

$$H'_\eta(v) = \frac{\eta}{2} \sum_{i,j} \dot{\Sigma}_{ij} \mathbb{E} \left[u^{\eta-2} \{ (1 + \beta^2) u u_{ij} + \beta^2 (\eta - 1) u_i u_j \} \right], \quad (\text{D.10})$$

where derivatives of u are evaluated at βX_v . The bracket rewrites exactly as

$$(1 + \beta^2) u^2 \partial_{ij} \log u + (1 + \eta \beta^2) u_i u_j. \quad (\text{D.11})$$

Only $i \neq j$ terms occur. By (D.9) and $1 + \eta \beta^2 > 0$, the bracket is nonnegative. Thus

$$H'_\eta(v) \geq 0 \quad (\eta > 0), \quad H'_\eta(v) \leq 0 \quad (-\beta^{-2} < \eta < 0). \quad (\text{D.12})$$

To justify differentiation for negative powers, first fix the grid. The covariance path is uniformly positive definite. After whitening, Lemma 7.1 bounds first and second spatial derivatives of $\log u$ by fixed polynomials, uniformly in v . The reverse-hypercontractive proof above applies to every Σ_v . Its ordinary orthant probability is bounded below uniformly in v : Gaussian association gives at least 2^{-n} for a grid of n points. Choose θ' with $-\eta < \theta' < \beta^{-2}$. The resulting uniform θ' inverse-moment bound and Hölder's inequality control every polynomial-weighted differentiated integrand. One may first differentiate with smooth cutoffs and then remove them using this uniform integrability. The positive-power case follows by the same polynomial bounds and $u^\eta \leq 1$.

At $v = 0$ the two blocks are independent, and at $v = 1$ their original joint law is restored. For adjacent continuous intervals choose nested dense grids with no duplicated boundary point. In each block, points approach the common endpoint from its own side; continuity then imposes that

endpoint constraint in both marginal limits. For negative powers, monotone convergence and the finite bound (D.7) justify this limit. If $M_\theta(t) = \mathbb{E}e^{\theta S_t}$, we obtain

$$M_\theta(s+t) \leq M_\theta(s)M_\theta(t), \quad 0 < \theta < \theta_c, \quad (\text{D.13})$$

whereas

$$M_\theta(s+t) \geq M_\theta(s)M_\theta(t), \quad \theta < 0. \quad (\text{D.14})$$

For positive θ , (D.7) bounds $\log M_\theta(t)$ above by $C(t+1)$ and it is nonnegative. For negative θ , (7.9) bounds it below by $-C(t+1)$ and it is nonpositive. Continuous-parameter subadditivity or superadditivity therefore proves existence and finiteness in (D.1); more precisely,

$$\Lambda_\rho(\theta) = \begin{cases} \inf_{t>0} t^{-1} \log M_\theta(t), & 0 < \theta < \theta_c, \\ 0, & \theta = 0, \\ \sup_{t>0} t^{-1} \log M_\theta(t), & \theta < 0. \end{cases} \quad (\text{D.15})$$

D.3 Differentiability throughout the finite domain

On a whitened grid, write $A = -\log q$ and

$$d\nu_\theta \propto e^{\theta A} d\gamma, \quad V_\theta(x) = |x|^2/2 - \theta A(x).$$

Lemma 7.1 gives

$$D^2 V_\theta \succeq \kappa_\theta I, \quad \kappa_\theta = \begin{cases} 1, & \theta \leq 0, \\ 1 - \theta\beta^2, & 0 < \theta < \theta_c. \end{cases} \quad (\text{D.16})$$

For $\theta \leq 0$, convexity of A gives the lower bound I_d . Brascamp–Lieb and (7.5) imply

$$\text{Var}_{\nu_\theta}(A) \leq \frac{2\beta^2}{\kappa_\theta} \mathbb{E}_{\nu_\theta} A.$$

Consequently, for $\Lambda_{t,\text{grid}} = t^{-1} \log \mathbb{E}e^{\theta A}$,

$$0 \leq \Lambda''_{t,\text{grid}}(\theta) \leq \frac{2\beta^2}{\kappa_\theta} \Lambda'_{t,\text{grid}}(\theta). \quad (\text{D.17})$$

The moment bounds on a slightly larger compact subinterval of $(-\infty, \theta_c)$ and convex secants give a uniform bound for $\Lambda'_{t,\text{grid}}$ for $t \geq 1$. Thus the second derivatives are uniformly bounded on compacts. A neighboring exponent dominates $A^j e^{\theta A}$, $j \leq 2$, when passing to dense grids. The same bounds hold for the continuous-time Λ_t .

Apply Arzelà–Ascoli to the first derivatives and use

$$\Lambda_t(b) - \Lambda_t(a) = \int_a^b \Lambda'_t(\theta) d\theta$$

together with pointwise convergence. Every subsequential derivative limit is the derivative of Λ_ρ . Hence (D.2) holds and

$$\Lambda'_t \longrightarrow \Lambda'_\rho \quad \text{locally uniformly on } (-\infty, \theta_c). \quad (\text{D.18})$$

D.4 Extensive divergence at the finite boundary

Take a unit-spaced grid of n times in $[0, t]$, with $n/t \rightarrow 1$. Its covariance is $\Sigma_{ij} = r^{|i-j|}$, $r = e^{-1/2}$. Let $Q = \Sigma^{-1}$. Both matrices have eigenvalues in $[K^{-1}, K]$, with $K = (1+r)/(1-r)$; this also follows from the explicit tridiagonal precision and row-sum bounds.

For an environmental vector satisfying $Qx \leq 0$, choose the nonnegative Chernoff vector $h = -\beta Qx$. Then

$$q_{\text{grid}}(x) = \mathbb{P}\{Z \geq -\beta x\} \leq e^{\beta h^T x + \frac{1}{2} h^T \Sigma h} = e^{-\beta^2 x^T Q x / 2}. \quad (\text{D.19})$$

Since $q_t \leq q_{\text{grid}}$, for $0 < \theta < \theta_c$ and $\delta = 1 - \theta\beta^2$,

$$\begin{aligned} M_\theta(t) &\geq \int_{\{Qx \leq 0\}} e^{\theta\beta^2 x^T Q x / 2} dN(0, \Sigma)(x) \\ &= \delta^{-n/2} \mathbb{P}\{QX \leq 0\}, \quad X \sim N(0, \Sigma). \end{aligned} \quad (\text{D.20})$$

The equality uses homogeneity of the cone. Now $QX \sim N(0, Q)$; on $[-1, 0]^n$ its density is at least

$$(2\pi K)^{-n/2} e^{-Kn/2}.$$

Thus $\mathbb{P}(QX \leq 0) \geq e^{-Cn}$, and

$$\Lambda_\rho(\theta) \geq \frac{1}{2} \log \frac{1}{1 - \theta\beta^2} - C. \quad (\text{D.21})$$

This proves divergence of Λ_ρ at θ_c ; convexity proves divergence of its derivative. The linear number of grid constraints makes the divergence extensive in time.

For the other boundary, fix $a > 0$ and set

$$p_a = \mathbb{P}(E_s \geq a \ (0 \leq s \leq 1)), \quad r_a = \mathbb{P}(W_s \geq -\beta a \ (0 \leq s \leq 1)).$$

Both are positive. Association on consecutive unit intervals gives

$$\mathbb{E} q_t^\eta \geq p_a^{[t]} r_a^{\eta[t]}, \quad \Lambda_\rho(-\eta) \geq \log p_a + \eta \log r_a.$$

First let $\eta \rightarrow \infty$, then $a \rightarrow \infty$, using $r_a \rightarrow 1$. Since $\Lambda_\rho(-\eta) \leq 0$, this yields $\Lambda_\rho(-\eta)/(-\eta) \rightarrow 0$. Convexity proves $\Lambda'_\rho(-\infty) = 0$. This completes (D.3).

D.5 The full upper and lower LDP bounds

We now apply the Gärtner–Ellis argument [7] to the pressure just constructed, giving the upper and lower bounds explicitly.

For any fixed $0 < \theta_0 < \theta_c$, Chernoff gives

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{P}(Y_t > R) \leq \Lambda_\rho(\theta_0) - \theta_0 R \rightarrow -\infty.$$

Since $Y_t \geq 0$, the laws are exponentially tight.

By continuity of Λ'_ρ and its two boundary limits, each $y > 0$ is $\Lambda'_\rho(\theta)$ for some interior parameter. Convexity then gives

$$I_\rho(y) = \theta y - \Lambda_\rho(\theta).$$

Under

$$\frac{d\mathbb{P}_{\theta,t}}{d\mathbb{P}} = e^{\theta S_t - t\Lambda_t(\theta)},$$

the mean of Y_t tends to y by (D.18), and its variance is $\Lambda_t''(\theta)/t = O(t^{-1})$. Thus $\mathbb{P}_{\theta,t}(|Y_t - y| < \epsilon) \rightarrow 1$. Changing measure back yields

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \log \mathbb{P}(|Y_t - y| < \epsilon) \geq -I_\rho(y) - |\theta|\epsilon.$$

Let $\epsilon \downarrow 0$ to obtain the local lower bound. The endpoint $y = 0$ is obtained by positive approximation if $I_\rho(0) < \infty$: convexity and lower semicontinuity then imply right continuity there. If $I_\rho(0) = \infty$ it adds no lower-bound obligation. For $y < 0$, (D.4) is infinite, by the sublinear left asymptotic of Λ_ρ .

For the upper bound, each fixed θ gives a Chernoff bound on a sufficiently small neighborhood of a point. Choose an almost-optimal supporting parameter in (D.4), cover a compact set by finitely many such neighborhoods, and use exponential tightness outside that compact set. This proves the closed-set upper bound. The rate is convex and lower semicontinuous; it dominates $\theta_0 y - \Lambda_\rho(\theta_0)$ on $y \geq 0$, so its sublevel sets are compact. Theorem D.1 is proved.

D.6 Consequences for the spectrum and tilted environments

Define the extended spectrum

$$\tilde{\chi}(\eta, \rho) = -\Lambda_\rho(-\eta), \quad \eta > -(1 - \rho)/\rho.$$

It agrees with (7.2) on positive orders. Convex duality gives

$$\tilde{\chi}(\eta, \rho) = \inf_{y \geq 0} \{I_\rho(y) + \eta y\}, \quad I_\rho(y) = \sup_{\eta > -(1-\rho)/\rho} \{\tilde{\chi}(\eta, \rho) - \eta y\}. \quad (\text{D.22})$$

The minimizer in the first formula is uniquely $\Lambda'_\rho(-\eta)$: equality in convex duality forces y into the singleton subgradient of the differentiable function Λ_ρ at $-\eta$. This proves (17.25).

The unique zero of I_ρ is $\gamma_\rho = \Lambda'_\rho(0)$, and

$$\partial_\eta \tilde{\chi}(0, \rho) = \gamma_\rho = \lim_{t \rightarrow \infty} \frac{1}{t} \mathbb{E}[-\log q_t]. \quad (\text{D.23})$$

Small positive and negative Chernoff parameters give exponential concentration about γ_ρ . Borel–Cantelli along integers and monotonicity of S_t imply $S_t/t \rightarrow \gamma_\rho$ almost surely. Jensen and (D.8) give

$$\frac{1}{2} \leq \gamma_\rho \leq \frac{1}{2(1 - \rho)}.$$

Theorem 7.4 improves the lower bound to

$$\gamma_\rho \geq \frac{1}{2} + \frac{\rho}{16\pi}, \quad (\text{D.24})$$

since for $0 < \eta \leq 1/2$ it bounds $\chi(\eta, \rho)/\eta$ from below, and this quotient tends to $\partial_\eta \tilde{\chi}(0, \rho)$.

For the pure q_t^η -tilted OU environment, Y_t has the full good rate

$$I_{\eta, \rho}(y) = I_\rho(y) + \eta y - \chi(\eta, \rho), \quad \eta > 0. \quad (\text{D.25})$$

One proves this by the same change of measure on compact sets; the nonnegative linear tilt and exponential tightness control the upper tail. Its unique zero is $\partial_\eta \chi(\eta, \rho)$. The mixed arithmetic source has the separate local conclusion in Theorem E.5.

Finally,

$$\lim_{y \rightarrow \infty} \frac{I_\rho(y)}{y} = \theta_c = \frac{1 - \rho}{\rho}. \quad (\text{D.26})$$

Every $\theta < \theta_c$ gives the lower limiting slope. For $y \geq \gamma_\rho$, negative θ cannot improve the dual supremum over its value at zero, because $\Lambda_\rho(\theta) \geq \gamma_\rho \theta$. For $\theta \geq 0$, $\Lambda_\rho(\theta) \geq 0$, so $I_\rho(y) \leq \theta_c y$. This proves the opposite inequality.

The exact Lamperti representation $E_u = e^{-u/2} B_{e^u}^E$, $W_u = e^{-u/2} B_{e^u}^W$ transfers the scalar statement to Brownian random-wall survival on $[1, R]$ with speed $\log R$. The initial law is the distribution induced by Brownian time 1.

E Adapted histories, signed moments, and finite mixtures

The mixed-power argument extends to finitely many adapted stages when the same neighboring-moment bounds hold uniformly on every true entering fiber. This section gives the conditional theorem, its entropy and localization consequences, and the complementary finite-mixture law.

E.1 The conditional window hypothesis

Fix m . For each index n , let $\mathbb{P}_n$ be obtained from an initial history and probability kernels

$$K_{i,n}(dx_i \mid h_{i-1}), \quad 1 \leq i \leq m,$$

on standard Borel spaces. The history retains all previously revealed data. Let q_i, D_i be measurable at stage i , with

$$0 \leq q_i \leq 1, \quad 0 < D_i \leq 1, \quad q_i \leq C_0 D_i.$$

For compact intervals $J_i \Subset (0, 1)$ assume, on every legal entering history,

$$\mathbb{E}[q_i^p \mid h_{i-1}] \asymp \mathbb{E}[D_i^p \mid h_{i-1}] \asymp e^{-\tau_{i,n} \chi_i(p)}, \quad p \in J_i. \quad (\text{E.1})$$

The comparison constants are uniform in n, p and the full entering history. The numbers $\tau_{i,n} \geq 0$ and the functions χ_i are deterministic once the physical parameters are fixed. Each χ_i is concave and locally $C^{1,1}$ on $\text{int } J_i$. The principal application is $\chi_i(p) = \chi(p, \rho_i)$.

Fix $\eta_i \in \text{int } J_i$ and put

$$R_i = q_i / D_i, \quad W_n = \prod_i R_i D_i^{\eta_i}, \quad Z_n = \mathbb{E} W_n, \quad d\hat{\mathbb{P}}_n = \frac{W_n}{Z_n} d\mathbb{P}_n. \quad (\text{E.2})$$

On its support define $S_i = -\log q_i$ and $T_i = -\log D_i$.

Theorem E.1 (History-stable mixed spectrum). *Under (E.1),*

$$Z_n \asymp \exp \left[- \sum_i \tau_{i,n} \chi_i(\eta_i) \right]. \quad (\text{E.3})$$

For $s_i < 1$ and $p_i = \eta_i - s_i - t_i \in \text{int } J_i$,

$$\log \mathbb{E}_{\hat{\mathbb{P}}_n} e^{\sum_i (s_i S_i + t_i T_i)} = \sum_i \tau_{i,n} [\chi_i(\eta_i) - \chi_i(p_i)] + O(1). \quad (\text{E.4})$$

The error is uniform on compact subsets of the displayed parameter region. Successive histories need not be independent; the number of stages is fixed.

Proof. For fixed $a_i > 0, p_i \in J_i$, Lemma 7.6 applies to each conditional kernel:

$$\mathbb{E}[R_i^{a_i} D_i^{p_i} \mid h_{i-1}] \asymp e^{-\tau_{i,n} \chi_i(p_i)}.$$

Integrate the last kernel first. Its uniform upper and lower constants multiply the nonnegative preceding integrand. Continue backwards and then integrate the initial history. This yields

$$\mathbb{E} \prod_i R_i^{a_i} D_i^{p_i} \asymp e^{-\sum_i \tau_{i,n} \chi_i(p_i)}. \quad (\text{E.5})$$

Take $a_i = 1, p_i = \eta_i$ for (E.3). For the numerator in (E.4), use the exact identity

$$R_i D_i^{\eta_i} q_i^{-s_i} D_i^{-t_i} = R_i^{1-s_i} D_i^{\eta_i - s_i - t_i}.$$

The exponent $1 - s_i$ is positive, so (E.5) applies. On $q_i = 0$ this numerator is zero, consistently with the support of the mixed measure. Divide by Z_n . $\square$

E.2 Gibbs comparison, entropy, and forgetting fine types

Let ν_n^D, ν_n^q have densities proportional to $\prod_i D_i^{\eta_i}$ and $\prod_i q_i^{\eta_i}$ relative to $\mathbb{P}_n$, with normalizers Z_n^D, Z_n^q .

Corollary E.2. *There is a constant independent of n such that*

$$\frac{d\hat{\mathbb{P}}_n}{d\nu_n^D} \leq C, \quad \frac{d\hat{\mathbb{P}}_n}{d\nu_n^q} \leq C.$$

In particular,

$$0 \leq D_{\text{KL}}(\hat{\mathbb{P}}_n \parallel \mathbb{P}_n) + \sum_i \eta_i \mathbb{E}_{\hat{\mathbb{P}}_n} T_i + \log Z_n^D \leq \log C, \quad (\text{E.6})$$

$$0 \leq D_{\text{KL}}(\hat{\mathbb{P}}_n \parallel \mathbb{P}_n) + \sum_i \eta_i \mathbb{E}_{\hat{\mathbb{P}}_n} S_i + \log Z_n^q \leq \log C. \quad (\text{E.7})$$

Proof. Backward integration shows Z_n, Z_n^D, Z_n^q are comparable to the same expression in (E.3). The unnormalized density ratios are

$$\prod_i R_i \leq C_0^m, \quad \prod_i R_i^{1-\eta_i} \leq C_0^{\sum_i (1-\eta_i)}.$$

The latter step uses $\eta_i < 1$. Expand the resulting bounded forward relative entropies. Neighboring exponents in Theorem E.1 give all required logarithmic integrability. $\square$

Proposition E.3 (Target concentration and entropy). *For fixed real coefficients a_i , put*

$$X_n = \sum_i a_i T_i, \quad \bar{X}_n = \sum_i a_i \tau_{i,n} \chi'_i(\eta_i).$$

Then, for h in a fixed neighborhood of zero,

$$\log \mathbb{E}_{\hat{\mathbb{P}}_n} e^{hX_n} = \sum_i \tau_{i,n} [\chi_i(\eta_i) - \chi_i(\eta_i - ha_i)] + O(1), \quad (\text{E.8})$$

and

$$\widehat{\mathbb{P}}_n\{|X_n - \bar{X}_n| > u\} \leq C \exp \left[-c \min \left\{ \frac{u^2}{1 + \sum_i a_i^2 \tau_{i,n}}, u \right\} \right]. \quad (\text{E.9})$$

Each S_i satisfies the corresponding one-coordinate bound. Moreover,

$$\widehat{\mathbb{P}}_n\{\log(D_i/q_i) > v\} \leq C e^{-v}, \quad \log(D_i/q_i) \geq -\log C_0, \quad (\text{E.10})$$

and

$$\text{D}_{\text{KL}}(\widehat{\mathbb{P}}_n \| \mathbb{P}_n) = \sum_i \tau_{i,n} [\chi_i(\eta_i) - \eta_i \chi'_i(\eta_i)] + O \left(\sum_i \sqrt{1 + \tau_{i,n}} \right). \quad (\text{E.11})$$

Proof. Take $s_i = 0, t_i = ha_i$ in (E.4); Taylor's estimate for χ_i and two-sided Chernoff give (E.9). Integrating the one-coordinate tail gives

$$\mathbb{E}_{\widehat{\mathbb{P}}_n} T_i = \tau_{i,n} \chi'_i(\eta_i) + O(\sqrt{1 + \tau_{i,n}}),$$

and similarly for S_i . To prove (E.10), integrate all future factors using their uniform bounds, use $R_i \mathbf{1}_{\{R_i < e^{-v}\}} \leq e^{-v}$ at stage i , and then integrate the current $D_i^{\eta_i}$ and all preceding factors. Divide by the lower bound for Z_n . Finally expand

$$\log \frac{W_n}{Z_n} = - \sum_i \eta_i T_i + \sum_i \log R_i - \log Z_n.$$

The ratio tails make the sum of the expected $\log R_i$ uniformly bounded, giving (E.11). $\square$

Corollary E.4 (Entropy after projection to the uncolored clocks). *Let $\mathcal{C}_n$ be a sub- σ -field on which all D_i are measurable. It may forget the fine type while retaining the complete uncolored clocks. The projected mixed and reference laws satisfy*

$$\begin{aligned} \text{D}_{\text{KL}}(\widehat{\mathbb{P}}_n |_{\mathcal{C}_n} \| \mathbb{P}_n |_{\mathcal{C}_n}) &= \sum_i \tau_{i,n} [\chi_i(\eta_i) - \eta_i \chi'_i(\eta_i)] \\ &\quad + O \left(\sum_i \sqrt{1 + \tau_{i,n}} \right). \end{aligned} \quad (\text{E.12})$$

The expected conditional entropy of the discarded information, relative to its reference conditional law given $\mathcal{C}_n$, is $O(1)$.

Proof. Since the density of ν_n^D relative to $\mathbb{P}_n$ is $\mathcal{C}_n$ -measurable, its conditional law of the discarded information is unchanged. The chain rule gives

$$\begin{aligned} \text{D}_{\text{KL}}(\widehat{\mathbb{P}}_n \| \nu_n^D) &= \text{D}_{\text{KL}}(\widehat{\mathbb{P}}_n |_{\mathcal{C}_n} \| \nu_n^D |_{\mathcal{C}_n}) \\ &\quad + \mathbb{E}_{\widehat{\mathbb{P}}_n} \text{D}_{\text{KL}}(\widehat{\mathbb{P}}_n(\cdot | \mathcal{C}_n) \| \mathbb{P}_n(\cdot | \mathcal{C}_n)). \end{aligned}$$

Both nonnegative terms are bounded by Corollary E.2. Expanding the first term and using the expectations in Proposition E.3 proves (E.12). $\square$

E.3 A local LDP under the actual mixed measure

Theorem E.5. Assume $a_n \rightarrow \infty$ and $\tau_{i,n}/a_n \rightarrow c_i > 0$. Fix $p_i \in \text{int } J_i$ and define

$$v_i = c_i \chi'_i(p_i), \quad (\text{E.13})$$

$$\mathcal{J}_\eta(\mathbf{p}) = \sum_i c_i [\chi_i(p_i) + (\eta_i - p_i) \chi'_i(p_i) - \chi_i(\eta_i)]. \quad (\text{E.14})$$

If B_δ is the event that both S_i/a_n and T_i/a_n lie within δ of v_i for every i , then

$$\lim_{\delta \downarrow 0} \liminf_n \frac{1}{a_n} \log \widehat{\mathbb{P}}_n(B_\delta) = \lim_{\delta \downarrow 0} \limsup_n \frac{1}{a_n} \log \widehat{\mathbb{P}}_n(B_\delta) = -\mathcal{J}_\eta(\mathbf{p}). \quad (\text{E.15})$$

Proof. Let $\widehat{\mathbb{P}}_{\mathbf{p},n}$ be the actual mixed measure with weight $\prod_i R_i D_i^{p_i}$. Concentration under this measure gives $\widehat{\mathbb{P}}_{\mathbf{p},n}(B_\delta) \rightarrow 1$ for fixed $\delta > 0$. The exact likelihood ratio is

$$\frac{d\widehat{\mathbb{P}}_{\eta,n}}{d\widehat{\mathbb{P}}_{\mathbf{p},n}} = \frac{Z_{\mathbf{p},n}}{Z_{\eta,n}} e^{-\sum_i (\eta_i - p_i) T_i}.$$

On B_δ , the exponent differs from $-a_n \sum_i (\eta_i - p_i) v_i$ by at most $a_n \delta \sum_i |\eta_i - p_i|$. Use (E.3), then take the two limits. Nonnegativity of (E.14) also follows from concavity. This proves the local LDP at the selected vectors. $\square$

E.4 The exact backward correction

Let $H_{m+1} = 1$ and

$$H_i(h) = \int w_i(h, x) H_{i+1}(h, x) K_i(dx | h), \quad w_i = R_i D_i^{\eta_i}.$$

The conditional kernel under the global mixed measure is exactly

$$K_i^*(dx | h) = \frac{w_i(h, x) H_{i+1}(h, x)}{H_i(h)} K_i(dx | h). \quad (\text{E.16})$$

If the entering history itself has reference law μ_0 , its marginal under the mixed measure is $H_1(h) \mu_0(dh)/Z_n$. Backward integration proves

$$H_i(h) \asymp e^{-\sum_{j \geq i} \tau_{j,n} \chi_j(\eta_j)}$$

uniformly in the entering history. Thus the future correction relative to the locally normalized mixed kernel is bounded above and below. This is the reason Theorem E.1 tolerates adaptation. If the remaining spectral cost changes by order a_n between histories, the uniform hypothesis fails and the selection cost must be retained.

E.5 Several comparisons on a shared clock

For several comparisons on a shared clock, assume a joint window on a compact box $J \in (0, \infty)^r$,

$$\mathbb{E} \prod_{j=1}^r q_j^{p_j} \asymp \mathbb{E} \prod_{j=1}^r D_j^{p_j} \asymp e^{-\tau \Xi(\mathbf{p})}, \quad q_j \leq C_0 D_j. \quad (\text{E.17})$$

Then for positive a_j in compact sets,

$$\mathbb{E} \prod_j (q_j / D_j)^{a_j} D_j^{p_j} \asymp e^{-\tau \Xi(p)}. \quad (\text{E.18})$$

Indeed, under the $\prod_j D_j^{p_j}$ tilt put $r_j = q_j / (C_0 D_j)$ and $B = \prod_j r_j^{p_j}$. Its mean has a uniform positive lower bound. With $c = \max_j (a_j / p_j)$, $\prod_j r_j^{a_j} \geq B^c$; Jensen, or $B^c \geq B$ for $c \leq 1$, proves the assertion.

Consequently the history partition and mixed-spectrum theorems extend to vector stages:

$$\log \mathbb{E}_{\widehat{\mathbb{P}}_n} e^{\sum_{i,j} (s_{ij} S_{ij} + t_{ij} T_{ij})} = \sum_i \tau_{i,n} [\Xi_i(\boldsymbol{\eta}_i) - \Xi_i(\boldsymbol{\eta}_i - \mathbf{s}_i - \mathbf{t}_i)] + O(1), \quad (\text{E.19})$$

provided $s_{ij} < 1$ and the shifted vectors remain in their joint windows. Local $C^{1,1}$ regularity gives the corresponding concentration, entropy and selected local LDP with gradients and dot products. The comparison to the D -Gibbs law holds for all positive owner coordinates. The second comparison, to the q -Gibbs law, additionally requires $\eta_{ij} \leq 1$ coordinatewise, as in the arithmetic application. The hypothesis is the joint envelope window for D_j .

E.6 An exact joint negative-moment boundary

Proposition E.6. *For a common stationary OU environment and slopes $\beta_j > 0$, fix $t > 0$. If $\theta_j \geq 0$, then*

$$\mathbb{E} \prod_j q_{j,t}^{-\theta_j} < \infty \iff \sum_j \theta_j \beta_j^2 < 1. \quad (\text{E.20})$$

For arbitrary real θ_j , the condition $\sum_j \theta_j^+ \beta_j^2 < 1$ is sufficient for finiteness.

Proof. Put $a_j = \theta_j \beta_j^2$ and $A = \sum_j a_j < 1$. Omit zero terms; if none remain the assertion is immediate. Hölder with exponents $r_j = A/a_j$ has $\sum_j r_j^{-1} = 1$ and $r_j \theta_j = A/\beta_j^2 < \beta_j^{-2}$. The scalar estimate (D.7) gives finiteness of every required factor. Conversely, $q_{j,t}(E) \leq \Phi(\beta_j E_0)$. At $E_0 = -x \rightarrow -\infty$ the joint negative-moment integral dominates a constant times

$$\int^\infty x^{\sum_j \theta_j} e^{(\sum_j \theta_j \beta_j^2 - 1)x^2/2} dx,$$

which diverges at and beyond the boundary. For arbitrary signs discard factors $q_j^{-\theta_j} \leq 1$ with $\theta_j < 0$ and apply the result to the positive coordinates. $\square$

On a whitened grid, the mixed-sign tilted potential also obeys

$$D^2 \left(\frac{|x|^2}{2} - \sum_j \theta_j A_j(x) \right) \succeq \left(1 - \sum_j \theta_j^+ \beta_j^2 \right) I.$$

This gives finite-time curvature estimates in the sufficient region. For coincident slopes and windows, the vector is diagonal, so its full LDP is exactly the scalar LDP of Theorem D.1. For distinct slopes, existence of a mixed-sign long-time pressure requires a further argument.

E.7 A complete LDP for finitely many history branches

Theorem E.7 (Finite history mixtures). *Let H_n take values in a fixed finite set $\mathcal{H}$, let $a_n \rightarrow \infty$, and assume*

$$-a_n^{-1} \log \mathbb{P}(H_n = h) \rightarrow d_h < \infty, \quad \min_h d_h = 0.$$

If, given $H_n = h$, Y_n satisfies a full LDP at speed a_n with good rate I_h , then (H_n, Y_n) has good rate

$$I(h, y) = d_h + I_h(y),$$

and Y_n has good rate

$$I_{\text{mix}}(y) = \min_{h \in \mathcal{H}} \{d_h + I_h(y)\}. \quad (\text{E.21})$$

The resulting rate can be nonconvex.

Proof. For a closed set, sum the finitely many conditional upper bounds and retain the largest exponential term. For an open set, retain one branch, use its conditional lower bound, and optimize over branches. The joint statement uses the finite discrete topology on $\mathcal{H}$. Each sublevel set is a finite union of compact sets, which proves goodness. $\square$

For example, let $H \in \{1, 2\}$ be an independent fair variable revealed at time zero and put $Y_t = S_{Ht}/t$. This is a nonanticipating random horizon. Theorem E.7 gives

$$I_{\text{mix}}(y) = \min\{I_\rho(y), 2I_\rho(y/2)\}.$$

It vanishes at γ_ρ and $2\gamma_\rho$, but is positive between them, and is therefore nonconvex. Its limiting log moment generating function is

$$\max\{\Lambda_\rho(\theta), 2\Lambda_\rho(\theta)\}.$$

Its Legendre transform cannot recover the nonconvex rate. Thus even a nonanticipating history selection need not retain the scalar convex-pressure description.

E.8 Beyond the conditional moment window

Positive neighboring moments cannot determine every signed tail of an envelope ratio. To see this, take any D_n with the required positive spectrum and, independently, set

$$R_n = \begin{cases} 1, & \text{with probability } 1 - e^{-An}, \\ e^{-Bn}, & \text{with probability } e^{-An}, \end{cases} \quad q_n = D_n R_n, \quad A, B > 0.$$

For every $p > 0$,

$$\mathbb{E} q_n^p = \mathbb{E} D_n^p [1 - e^{-An} + e^{-(A+pB)n}],$$

so all positive moment exponents agree. Under the actual mixed law proportional to $R_n D_n^\eta$, however,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}_{\text{mixed}} (D_n/q_n)^s = \max\{0, (s-1)B - A\}. \quad (\text{E.22})$$

This example exhibits the additional ratio rate that can occur beyond $s = 1$, outside the window in (7.17).

For a full source LDP, an informative additional coordinate is

$$X_n = -a_n^{-1} \log D_n, \quad U_n = a_n^{-1} \log(C_0 D_n/q_n) \geq 0.$$

If this pair is finite-valued and its reference law has a good full LDP on $[0, \infty)^2$ with rate $\mathcal{I}$, then the law weighted by $q_n D_n^{\eta-1}$ has good rate

$$\widehat{\mathcal{I}}_\eta(x, u) = \mathcal{I}(x, u) + \eta x + u - \inf_{z, v \geq 0} \{\mathcal{I}(z, v) + \eta z + v\}. \quad (\text{E.23})$$

Indeed $q_n D_n^{\eta-1} = C_0 e^{-a_n(\eta X_n + U_n)}$. The nonnegative continuous tilting cost gives its normalization by the Laplace principle, and the usual compact upper and local lower estimates give (E.23); exponential tightness follows from that of the reference law. The joint reference LDP is a new input. If $q_n = 0$ can occur, the value $U_n = \infty$ and its support must be handled separately.

F Proofs of the worked examples

This appendix derives the evaluations in Section 4 from the saddle and the named endpoint definitions. We first record the two observations that permit exact face calculations and the passage from positive vectors to arithmetic families with repeated cutoffs.

F.1 Stationarity and bounded coordinates

Suppose a candidate saddle has a proper owner $[1, n]$ with product s , followed by a singleton terminal owner with product $\theta < s$. Write $z = 2^{\theta/s}$, $r_i = n - i + 1$, and $\kappa_r = (r + z)^{s-1}$. On this face the two common-direction stationarity equations are

$$\sum_{i=1}^n L_i \{\kappa_{r_i} E_{r_i}(z) - r_i\} = 0, \quad (\text{F.1})$$

$$L_{n+1}(2^\theta \log 2 - 1) = \sum_{i=1}^n L_i \{1 - z \log 2 \kappa_{r_i}\}. \quad (\text{F.2})$$

For $1 \leq j < n$, the internal multiplier is

$$\lambda_j = \sum_{i=1}^j L_i \{\kappa_{r_i} [E_{r_i}(z) - E_{n-j}(z)] - (r_i - n + j)\}. \quad (\text{F.3})$$

These formulas follow by differentiating $G_i = s \log(r_i + z)$. When differentiating in s , the child product θ is held fixed, so the derivative of z must be included. For example,

$$\frac{\partial}{\partial s} (r + 2^{\theta/s})^s = (r + z)^{s-1} E_r(z).$$

For the prefix in (F.3), the child has weight $n - j + z$; taking the difference of the corresponding entropy increments gives the displayed formula. Its sign convention agrees with (5.46). Thus non-negative internal multipliers, zero exterior multipliers, and strict separation of the two products verify all KKT conditions. Theorem 5.1 then identifies the candidate as the unique global saddle. The same calculation for a single terminal owner uses $z = 1$, with common-direction balance

$$\sum_i L_i \{h_i^s \log h_i - (h_i - 1)\} = 0. \quad (\text{F.4})$$

We use the word *saturated* below in the asymptotic sense that a binary factor has a positive lower bound independent of the parameter tending to infinity. It need not equal one. If one

endpoint score is $\gg L_i$, the other is at least one, and the native intensity remains in a fixed compact interval, the factor for group i is saturated. Every edge on a bounded positive clock is saturated for the same reason.

Choose $C \geq L_*(k)$, where $L_*(k)$ is the fixed threshold in Theorem 9.1. On vectors all of whose coordinates exceed this threshold, that theorem and Theorem 15.1 give $K_k \asymp_k \Psi_k$; Theorem 2.1 then gives $\Phi_k \asymp_k \Psi_k$. Replacing a fixed number of zero coordinates by C changes the total log-parameter distance by a bounded amount. Theorem 16.1 therefore transfers the arithmetic estimate to the original vector. This proves the required statements about Φ_k without any assertion about where its defining infimum is attained. In particular, a raw saddle is never assigned to a vector with zero coordinates.

F.2 A finite rational calculation used below

All decimal endpoints in this appendix denote exact rational numbers. We give the finite inequalities explicitly and describe a single rational procedure for verifying them. For $1 \leq x \leq 2$, let $w = (x - 1)/(x + 1)$. Then

$$2 \sum_{j=0}^{N-1} \frac{w^{2j+1}}{2j+1} \leq \log x \leq 2 \sum_{j=0}^{N-1} \frac{w^{2j+1}}{2j+1} + \frac{2w^{2N+1}}{(2N+1)(1-w^2)}. \quad (\text{F.5})$$

This follows by integrating the geometric series for $2/(1-w^2)$, and bounding all remaining denominators below by $2N+1$. General positive rational arguments are reduced to $[1, 2]$ by extracting powers of two and, if necessary, taking reciprocals. For $0 \leq x < N+2$, Taylor's series gives

$$\sum_{j=0}^N \frac{x^j}{j!} \leq e^x \leq \sum_{j=0}^N \frac{x^j}{j!} + \frac{x^{N+1}}{(N+1)!} \frac{1}{1 - x/(N+2)}. \quad (\text{F.6})$$

Indeed, successive omitted terms have ratio at most $x/(N+2)$. Reduce a nonnegative argument to $[0, 1/4]$, then square back; negative arguments are handled by reciprocation.

The bounds below result from 48 terms in (F.5), degree 40 in (F.6), and outward rounding to the rational grid $10^{-40}\mathbb{Z}$ after each operation. Interval addition, multiplication, and division use their elementary rational endpoint formulas; divisions are used only when zero is excluded from the denominator interval. Monotonicity supplies interval logarithms and exponentials. Thus each table is a finite rational inequality, with explicit remainders. Exact zeros used in the constructions are instead proved algebraically from their defining equations.

The finite sums and interval operations specified here suffice to reproduce every bound below. Neither numerical optimization nor floating-point equality tests are needed.

F.3 Equal cutoffs

Evaluate the positive vector $Q = (T, C, \dots, C)$. At a common product s , equation (F.4) is strictly increasing in s , and gives $s = \alpha_H + O_k(T^{-1})$. For $1 \leq j < k$, its multiplier is

$$\lambda_j = T \left[H^{\alpha_H-1} \{ H \log H - (H-j) \log(H-j) \} - j \right] + O_k(1). \quad (\text{F.7})$$

The term in brackets is positive. To prove this, apply strict convexity of $x \log x$: the secant over $[H-j, H]$ has larger slope than the secant over $[1, H]$. The latter equals $H^{1-\alpha_H}$, since $H^{\alpha_H} \log H = H - 1$. Multiplication by jH^{α_H-1} gives the claim. All internal multipliers are therefore positive for large T , and common-direction stationarity proves the asserted single owner.

For each first-group binary edge $k \rightarrow q$, $1 \leq q < k$, the left completion retains exactly q names through the first group. Its reference cost is

$$T\{q - H^{\alpha_H-1}(q+1)\log(q+1)\} + O_k(1) \gg_k T.$$

The inequality follows because $(q+1)\log(q+1)/q$ is strictly increasing and the corresponding quantity at $q = k$ is $H^{1-\alpha_H}$. The other clocks are fixed, so all binary factors are saturated. The ordinary suffix starts in group 1, because $|s - \alpha_H| < \tau$ for large T . Equation (F.4) gives $D = 0$, while $U = T + (k-1)C$, $M = T$, $A = 0$, and $B = (k-1)C$. Formula (3.11) gives $T^{-3/2}$. Finally $J(Q) = I_H T + O_k(1)$. The bounded-coordinate argument above proves (4.3).

F.4 The critical family with six divisor coordinates

Use σ_c, e_h, a, b from (4.4)–(4.5). Put

$$f_h(q) = q - h^{\sigma_c-1}(q+1)\log(q+1), \quad \Delta_h(Q) = h^{\sigma_c-1}(h\log h - Q\log Q) - (h - Q).$$

Thus $f_h(0) = 0$, $f_h(h-1) = -e_h$, and $\Delta_h(Q) = f_h(Q-1) - f_h(h-1)$. The finite inequalities needed here are

$$\begin{aligned} e_2 &> 0, & e_h &< 0 \quad (3 \leq h \leq 7), \\ \Delta_7(2) &= 0, & \Delta_7(Q) &> 0 \quad (3 \leq Q \leq 6), & \Delta_h(Q) &> 0 \quad (3 \leq h \leq 6, 2 \leq Q < h). \end{aligned} \quad (\text{F.8})$$

The zero is exactly the definition of σ_c . The rational bounds of §F.2 give

quantity	enclosure or lower bound
σ_c	(0.540128277594, 0.540128277595)
e_2	(0.00790656, 0.00790657)
a	(54.82474304, 54.82474305)
b	(66.42545992, 66.42545993)
$\min\{\Delta_h(Q) : (h, Q) \neq (7, 2) \text{ in (F.8)}\}$	> 0.1079

They also give $\sigma_c - \alpha_2 > 1/(8 \cdot 7^3(\log 7)^2)$ and $\alpha_3 - \sigma_c > 1/(8 \cdot 7^3(\log 7)^2)$. Since the right side is an upper bound for the profile's τ , the ordinary suffix is exactly the last group. The signs of e_h also follow from these two inequalities and the strict increase of α_h .

The common-direction balance holds exactly by the definitions of a, b . For $1 \leq j \leq 5$, the prefix derivative gives

$$\lambda_j = \sum_{i=1}^j L_i \Delta_{8-i}(7-j).$$

For $j \leq 4$, each summand is positive by (F.8). At $j = 5$, the first summand is zero and $\lambda_5 = C \sum_{h=3}^6 \Delta_h(2) > 0$. Together with $\lambda_6 = 0$, these verify the full KKT system. In particular the fixed positive buffers keep the last internal multiplier positive; the actual family is not an owner-merging boundary.

We next verify both endpoint minima, rather than just exhibit one favorable pair. For the edge $6 \rightarrow 1$, the only freely chosen cut is $\tau_6 \in \{V_1, \dots, V_6\}$. If $\tau_6 < V_6$, there is no negative contribution from the final group. The function f_h is strictly concave, vanishes at zero, and is positive at $h-1$ for $h \geq 3$. It is therefore positive at every intermediate integer. The first-group contribution is $\gg T$ on both sides. If $\tau_6 = V_6$, use total balance to subtract the full surviving configuration.

Since $\Delta_7(2) = 0$, both reference costs are exactly $C \sum_{h=3}^6 \Delta_h(2) = C\Delta$. The completion is late, and its good groups are exactly those listed before (4.6). Hence the two scores in that equation are the separate minima for all sufficiently large T .

For the other edges, a completion with $\tau_6 < V_6$ again has cost $\gg T$: a retained name survives the first group, and no negative last-group contribution occurs. If $\tau_6 = V_6$, at least one name survives every group, and total balance gives

$$\overline{C}_v = \sum_{i=1}^5 L_i \Delta_{8-i}(q_{v,i} + 1). \quad (\text{F.9})$$

Every summand is nonnegative. In group 1, an edge with $q \geq 2$ has left cost $\gg T$. In a middle group, a proper subset of the names survives on the left, giving left cost $\gg C$. These bounds saturate every remaining edge. The ordinary factor has $U = M = L_6$, $A = B = 0$, and $D = e_2 L_6$, and is therefore $\asymp T^{-1/2}$. Formula (3.10) applied to (4.6) supplies $T^{-\chi_c}$. This proves (4.7) and the stated explicit expression for $J(L)$.

For completeness, let $\sigma_c(H) = 1 - \log((H \log H - 2 \log 2)/(H - 2))/\log H$. The values of $\sigma_c(H) - \alpha_2$, for $H = 3, 4, 5, 6$, lie respectively in

$$\begin{aligned} &(-0.117568, -0.117566), \quad (-0.056865, -0.056864), \\ &(-0.024370, -0.024368), \quad (-0.003499, -0.003498). \end{aligned}$$

Thus $H = 7$ is the first admissible integer in this critical first/last-group construction.

F.5 The separated family with six divisor coordinates

It is useful to check the dimensional boundary for the same family with $n = k - 1$ and $H = n + 2$. Let $\alpha = \alpha_2$ and define

$$f(H) = H^{\alpha-1}(H \log H - 2 \log 2) - (H - 2).$$

The rational calculation gives

$$0.025 < f(6) < 0.026, \quad -0.110 < f(7) < -0.109, \quad 2\alpha - 1 < \alpha(1 - \alpha) \log 2. \quad (\text{F.10})$$

For real $x \geq 2$, direct differentiation gives

$$f''(x) = x^{\alpha-2} \left[\alpha(\alpha - 1) \log x + 2\alpha - 1 - \frac{2 \log 2 (1 - \alpha)(2 - \alpha)}{x} \right] < 0.$$

Indeed, the first two terms already have negative sum by (F.10). Since $f(2) = 0$, strict concavity and the two signs imply $f(H) > 0$ for integer $3 \leq H \leq 6$ and $f(H) < 0$ for integer $H \geq 7$.

At fixed $\theta = \alpha$, the derivative of $(n + 2^{\theta/s})^s$ is $(n + z)^{s-1} E_n(z)$, and its second derivative is strictly positive. For instance, with $c = \theta \log 2$, $p = z/(n + z)$, and $A = \log(n + z) - p \log z$, it equals

$$(n + z)^s \{A^2 + c^2 p(1 - p)/s^3\} > 0.$$

Thus the equation $(n + z)^{s-1} E_n(z) = n$ has at most one root. Its left side minus n equals $f(H)$ at $s = \alpha$ and is positive at $s = 1$, since $E_n(z) = \int_0^n (1 + \log(z + t)) dt > n$. For $H \geq 7$ this proves existence of a unique root above α . For $k = 6$ the interval $(0.54070, 0.54072)$ isolates it by opposite signs at its endpoints. For $H \leq 6$, the last multiplier on the terminal diagonal has

positive limiting coefficient $f(H)$; the other prefix coefficients are positive by concavity as well. This proves the dimensional assertion for this family.

Now take $Q = (T, C, C, C, C, B)$, $B = T^3$, with $k = 6$. Solve (F.1)–(F.2) with $n = 5$. After division by T and B , respectively, the limiting equations are (4.9) and $2^\theta \log 2 = 1$. Their Jacobian is triangular with positive diagonal entries. The implicit function theorem gives

$$s = s_* + O(T^{-1} + T/B), \quad \theta = \alpha + O(T/B). \quad (\text{F.11})$$

The exterior multiplier is zero by the first equation. For $1 \leq j < 5$, the leading internal multiplier is

$$T \left[(5 + z_*)^{s_*-1} \{ (5 + z_*) \log(5 + z_*) - (5 + z_* - j) \log(5 + z_* - j) \} - j \right].$$

Strict convexity of $x \log x$, compared with its secant on $[z_*, 5 + z_*]$, makes this positive. The remaining terms are $O(1)$. The products have a fixed limiting gap, so these are the full KKT conditions for precisely the two asserted owners.

The first edge $5 \rightarrow 0$ has no outside cuts. On the left all five names end at zero; the completion is early and has bounded score, since the proper exit gap has a positive limit. On the right they end at $V_1 = T$, a late cut for the proper owner. Its reference cost is $T\{5 - (5 + z)^{s-1} E_5(z)\} = O(1)$, by (F.1), and it has no good group. Thus both endpoint scores are $\asymp 1$. For a first-group edge with $q \geq 1$, the left cost is

$$T\{q - (5 + z_*)^{s_*-1} E_q(z_*)\} + O(1) \gg T.$$

This follows from strict convexity of $E_q(z_*)$ in q , or equivalently from strict increase of $E_q(z_*)/q$, with equality only at rank 5. The middle clocks are bounded. The terminal singleton has no binary edge. Hence the sole unsaturated binary factor has order $T^{-2\chi_*}$, and the sole proper count has order $T^{-s_*/2}$. The use of the limiting products in these powers is valid by (F.11) and the local Lipschitz continuity of χ .

Finally, (F.2) gives

$$D = c_* T + O(1 + T^2/B), \quad c_* = 1 - z_* \log 2 (5 + z_*)^{s_*-1} > 0.$$

The rational bounds give $0.4403 < c_* < 0.4404$; positivity alone is needed. Thus $\mathcal{B}_{\text{ord}} \asymp TB^{-3/2}$. At the limiting products the objective is $I_2 B + I_p T + O(1)$. Taylor expansion at those products has linear error $O(1)$ and quadratic error $O(T|s - s_*|^2 + B|\theta - \alpha|^2) = O(1)$. The mixed derivatives are $O(T)$, so their contribution is also bounded. This proves $J(Q) = I_2 B + I_p T + O(1)$ and (4.10), including the transfer from Q to the original repeated cutoffs.

F.6 The positive ray

For (4.11), equations (F.1)–(F.2) hold by the definitions of a, b . Direct use of (F.3) gives the following enclosures:

j	1	2	3	4
λ_j/t	(1.77059, 1.77060)	(2.97343, 2.97344)	(3.37910, 3.37911)	(2.61456, 2.61457)

The products are strictly separated, so this proves the exact saddle.

Here the complete endpoint verification is small enough to specify directly. Represent the cut V_d of name c by its integer index d_c . For an edge (i, q) , choose

$$d_c \in \{0, \dots, c\} \quad (c < i), \quad d_c \in \{i, \dots, c\} \quad (6 - q \leq c \leq 5),$$

and set $d_c = i - 1$ on the left and $d_c = i$ on the right for $i \leq c \leq 5 - q$. This is exactly the paired-template definition of the profile, including zero cuts and ties. With $w = (a, 1, 1, 1, 1)$, each cost is the explicit expression

$$\frac{\overline{C}_v}{t} = \sum_{g=1}^5 w_g \left[q_g - (6 - g + z)^{s-1} E_{q_g}(z) \right], \quad q_g = \#\{c : d_c \geq g\}. \quad (\text{F.12})$$

The following table gives rational lower bounds for its minimum over all templates on the indicated side. The template count makes the finite set in each row explicit.

i	q	side	templates	lower bound
1	0	right	1	0.5490
1	1	left	5	2.4949
1	2	left	20	3.3791
1	3	left	60	2.9734
1	4	left	120	1.7705
2	0	left	2	0.5490
2	1	left	8	0.6686
2	2	left	24	0.4464
2	3	left	48	0.1751
3	0	left	6	0.4791
3	1	left	18	0.4284
3	2	left	36	0.1837
4	0	left	24	0.3280
4	1	left	48	0.1805
5	0	left	120	0.1455

For example, the first row has all cuts at V_1 , so its cost is $a[5 - (5 + z)^{s-1} E_5(z)]$. Every other row is obtained by the same explicit sum (F.12) over the displayed independent integer choices, with $\log, \exp$ bounded by (F.5)–(F.6). Thus each edge has one endpoint score $\gg t$ and the other at least one, proving saturation of all fifteen factors. The proper count is $\asymp t^{-s/2}$. The terminal group has $U = M = bt$ and $D = (2^\theta \log 2 - 1)bt \asymp t$, so its ordinary factor is $\asymp t^{-1/2}$. Homogeneity gives $J(L) = tJ_0$. All lengths tend to infinity, proving (4.12) directly.

F.7 Two comparisons within one owner

We first justify all fixed constants in (4.13). Let $D_{r,q}(z) = (E_r(z) - E_q(z))/(r - q)$ and $t_{r,q}(z) = 1 - \log D_{r,q}(z)/\log(r + z)$. Differentiation gives

$$t'_{r,q}(z) = \frac{\log D_{r,q}(z)}{(r + z)\{\log(r + z)\}^2} - \frac{\log(r + z) - \log(q + z)}{(r - q)D_{r,q}(z)\log(r + z)}. \quad (\text{F.13})$$

On $I = [1.98455, 1.98457]$, the rational bounds give

$$\begin{aligned} (t_{14,6} - t_{6,0})(1.98455) &\in (-0.000000199838, -0.000000199837), \\ (t_{14,6} - t_{6,0})(1.98457) &\in (0.000000161388, 0.000000161389), \\ 0.01806066 &< (t_{14,6} - t_{6,0})'(z) < 0.01806187 \quad (z \in I). \end{aligned}$$

The intermediate value theorem and strict monotonicity therefore give exactly one root in I . The same calculation on I gives

$$\alpha_2 < \theta < \sigma, \quad -0.166 < g_1 < -0.165, \quad 1.604 < g_{14} < 1.605, \quad \min_{1 \leq r \leq 14} (1 - z \log 2 \kappa_r) > 0.1574. \quad (\text{F.14})$$

The identities $g_6 = 0$ and $f_{14}(6) = g_{14}$ follow exactly from the two secant equations, since $(r+z)^{\sigma-1}(E_r - E_q) = r - q$ is equivalent to $\sigma = t_{r,q}(z)$. With $\Delta_r = \kappa_r(E_r - E_6) - (r - 6)$, further positive bounds are

r	7	8	9	10	11	12	13
Δ_r	> 0.1725	> 0.2765	> 0.3250	> 0.3277	> 0.2916	> 0.2222	> 0.1238

The two coefficients in W and the leading coefficient in R are enclosed by

$$\begin{aligned} 9.6779 &< -g_{14}/g_1 < 9.6822, \\ 25.0555 &< -\frac{\sum_{2 \leq r \leq 13, r \neq 6} g_r}{g_1} < 25.0707, \\ 39.1859 &< \frac{1 - z \log 2 \kappa_6}{2^\theta \log 2 - 1} < 39.214. \end{aligned}$$

Thus all lengths in (4.15) are positive. Each $f_r(q)$ is strictly concave in q . Its zero at $q = 0$ and the equalities just proved imply

$$f_6(q) > 0 \quad (0 < q < 6), \quad f_{14}(q) > g_{14} \quad (6 < q < 14), \quad f_{14}(q) > 0 \quad (0 < q \leq 14). \quad (\text{F.15})$$

The lengths W, R give exactly (F.1)–(F.2) for $n = 14$. For an internal index j , put $q = 14 - j$ in (F.3). If $j < 8$, the first term is $T[f_{14}(q) - g_{14}] > 0$, while the others are $O(C)$. If $j = 8$, the first term vanishes exactly and the other seven terms are the strictly positive middle margins displayed above, multiplied by C . If $9 \leq j \leq 13$, the ninth group gives $U f_6(q) > 0$, with all other terms $O(T + C)$. Since $U = T^3$, all internal multipliers are positive for large T . The exterior multipliers vanish and $\sigma > \theta$, proving the asserted exact owner partition.

Only groups 1, 9, 14 have growing clocks in the proper owner; all other binary factors are automatically saturated. For the edge $(i, q) = (1, 6)$, keep names 9, $\dots$, 14 until their own deadlines. Comparing the two completions with the full configuration and using $f_{14}(6) = g_{14}$, both reference costs equal

$$C \sum_{r=7}^{13} \{\kappa_r(E_r - E_6) - (r - 6)\}.$$

The cuts are late. The left side makes the first group good, and both sides make precisely the seven intermediate groups of ranks 7, $\dots$, 13 good. In particular the displayed pair gives $X_{1,6} \ll \sqrt{T}$, $Y_{1,6} \ll 1$.

To prove the lower bound for $X_{1,6}$, consider any legal completion. If between one and five names survive through the rank-six group, its cost is $U f_6(q') \gg U$, and all other possibly negative contributions have total magnitude $O(T + C)$. If none survives, the rank-one compensating length W contributes nothing, while the first-group contribution is $g_{14}T$. If all six survive but name 14 ends before its deadline V_{14} , the rank-one negative contribution is again absent, leaving $g_{14}T + O(C)$. In the remaining case name 14 reaches V_{14} ; the completion is late and the first

group is good on the left. Its score is then at least $\sqrt{T}$. These cases exhaust the templates and prove $(X_{1,6}, Y_{1,6}) \asymp (\sqrt{T}, 1)$.

For $(i, q) = (9, 0)$, all six deep names end at V_8 or V_9 . At least these six names survive the first group. Since the number there lies between 6 and 14, strict concavity and equal endpoint values give a contribution at least $g_{14}T$. In group 9, the left and right contributions are respectively $f_6(0)U = f_6(6)U = 0$. No name survives into the group of length W . The remaining costs are $O(C)$. These bounds apply to every completion. Conversely, ending every earlier name at zero gives costs $O(T + C)$, and the good-group correction is at most $\sqrt{T} + 11\sqrt{C}$: the long rank-six group has either no survivor or all its names. The exit reserve is bounded. Thus both endpoint scores have order T .

It remains to exclude additional unsaturated edges on the three growing clocks. In group 1, the edge $q = 0$ has a bounded left score and right reference cost $g_{14}T$, so it is saturated. For $1 \leq q \leq 5$, the left completion has at most five survivors in group 9. A nonempty surviving set there costs $\gg U$; an empty set leaves $Tf_{14}(q) + O(C) \gg T$, with no rank-one compensation. For $7 \leq q \leq 13$, the entire possible negative rank-one compensation is $g_1W = -g_{14}T + O(C)$. The first-group cost still leaves $T[f_{14}(q) - g_{14}] + O(C) \gg T$, and the rank-six contribution is nonnegative. In group 9, each edge with $q \geq 1$ has left cost $Uf_6(q) + O(T + C) \gg U$. Finally, group 14 has only $q = 0$. Its left completion keeps name 14 through group 9, but supplies no negative rank-one contribution. If fewer than six names survive group 9, its cost is $\gg U$; otherwise the first group contributes at least $g_{14}T$, up to bounded errors. Since $W \asymp T$, this also saturates the edge. These cases cover every canonical edge.

The proper native length is $\asymp U$, and its exit gap is a fixed positive constant, so its count is $U^{-\sigma/2}$. The terminal singleton has length $R \asymp U$ and fixed positive drift coefficient $2^\theta \log 2 - 1$. Its factor is $U^{-1/2}$. Inserting the two endpoint pairs in (3.10) proves (4.17). Direct substitution into (3.2) gives (4.18).

F.8 Two comparisons in different owners

Write $z = e^{c/s}$, $Z = (1 + z)^s$, and $A = \log(1 + z) - z \log z / (1 + z)$. The rank-one equation is $F(s; c) = ZA - 1$. At fixed $c > 0$, differentiation gives

$$\partial_s F(s; c) = Z \left[A^2 + \frac{c^2}{s^3} \frac{z}{(1 + z)^2} \right] > 0. \quad (\text{F.16})$$

Moreover $F(1; c) > 0$, because $(1 + z) \log(1 + z) - z \log z > 1$ when $z > 1$. The rational inequalities

$$t_1(17) - \alpha \in (-0.00026097, -0.00026095), \quad t_1(18) - \alpha \in (0.00374983, 0.00374984)$$

give $F(\alpha; \alpha \log 17) < 0$. The roots of (4.19) are isolated by

$$0.531542356069 < v < 0.531542356071, \quad 0.5343537358 < u < 0.5343537362. \quad (\text{F.17})$$

For the first interval substitute its endpoints in $F(s; \alpha \log 17)$; for the second use the entire first interval in G_2^* , and substitute the second pair of endpoints in $F(s; G_2^*)$. The signs are respectively negative and positive under the rational bounds of §F.2. Equation (F.16) proves uniqueness, and the enclosures prove the two strict gaps.

On the proposed face, set $s_1 = u_T$, $s_2 = v_T$, $s_3 = \cdots = s_{18} = \tau_T$. Solve

$$\partial_{s_1} J = 0, \quad \partial_{s_2} J = 0, \quad \sum_{j=3}^{18} \partial_{s_j} J = 0. \quad (\text{F.18})$$

After division by T, T^3, T^{11} , respectively, the limiting system is (4.19) together with $2^\alpha \log 2 = 1$. Its Jacobian is triangular when the terminal variable is solved first. Its diagonal entries are nonzero by (F.16) and $\frac{d}{d\tau}(2^\tau \log 2)|_{\tau=\alpha} = \log 2$. The normalized perturbation in the second equation is $O(T^{-2})$, and in the last it is $O(T^{-6})$; the first is an exact native equation with perturbed child. Applying the implicit function theorem successively gives

$$u_T - u = O(T^{-2}), \quad v_T - v = O(T^{-2}), \quad \tau_T - \alpha = O(T^{-6}). \quad (\text{F.19})$$

We verify all internal terminal KKT inequalities. At the limiting products their leading coefficients, after division by T^5 , are

$$\Lambda_j = \sum_{i=3}^j 2^{i-3} \left[h_i^{\alpha-1} \{h_i \log h_i - (19-j) \log(19-j)\} - (j-i+1) \right], \quad 3 \leq j \leq 17. \quad (\text{F.20})$$

Rational lower bounds for these fifteen expressions are in the middle column below. The old groups contribute $O(T^3)$, and parameter perturbations give smaller errors; hence $\lambda_j/T^5 = \Lambda_j + O(T^{-2}) > 0$. The first two equations of (F.18) give $\lambda_1 = \lambda_2 = 0$, and its last equation gives $\lambda_{18} = 0$. This verifies the entire KKT system and the exact owner partition.

The same table records the finite signs needed to saturate all terminal binary edges. For $3 \leq h \leq 17$, put

$$\delta_{h,q} = h - 1 - q - h^{\alpha-1} \{h \log h - (q+1) \log(q+1)\}, \quad 1 \leq q \leq h-2. \quad (\text{F.21})$$

The last column bounds $\min_{1 \leq q \leq h-2} |\delta_{h,q}|$ from below. Each entry is obtained by the finite sum (F.20) or the $h-2$ values in (F.21), with the rational series bounds already specified.

j	lower bound for Λ_j	h	lower bound for $\min_q \delta_{h,q} $
3	0.000739	3	0.13787
4	0.010639	4	0.16404
5	0.068703	5	0.11999
6	0.299317	6	0.02515
7	1.046964	7	0.10933
8	3.195478	8	0.00702
9	8.902046	9	0.05276
10	23.236999	10	0.06103
11	57.730006	11	0.04796
12	137.702459	12	0.02228
13	316.297189	13	0.01094
14	696.898091	14	0.01397
15	1448.473445	15	0.01498
16	2693.099320	16	0.00755
17	3533.706046	17	0.000739

In particular all these lower bounds exceed $7/10000$.

For the singleton owner [1], there is just one paired template, and no good group. Its left reference cost is zero. Its right reference cost is also zero by $\partial_{s_1} J = 0$. The left cut is early and the right cut is late. Since its exit gap has a positive limit, $X_1 = 1 + \hat{R}_1 \asymp 1$, $Y_1 = 1$.

The singleton owner [2] also has just one pair, with cuts V_1 and V_2 . Let A_2 be its native entropy increment, so that $A_2 Z_2 \rightarrow 1$. Its left reference cost is

$$\bar{C}_{2,-} = T(1 - A_2 Z_1 \beta_1), \quad 1 - A_2 Z_1 \beta_1 \longrightarrow 1 - (\beta_1^*)^{1-u} > 0. \quad (\text{F.22})$$

Here the identity $Z_1\beta_1 = Z_2\beta_1^{1-s_1}$ follows directly from the recursion, and accounts for the transported old intensity. The right reference cost is zero by $\partial_{s_2}J = 0$. The left cut is early since $T \ll T^3$, and the right cut is late. No good group is possible in rank one. Thus $X_2 \asymp T$ and $Y_2 = 1$. The two binary ratios are $\asymp T^{-1}$ and $\asymp T^{-2}$, respectively. Their powers tend to $2\chi_1, 2\chi_2$. Local Lipschitz continuity and (F.19) allow replacement by those limiting powers at a bounded multiplicative cost. The proper count factors are $T^{-u/2}$ and $T^{-3v/2}$.

For any terminal edge in group i , changing from its left to its right completion changes cuts only across that group. Therefore the reference-cost difference is exactly $L_i\delta_{h_i,q}(\tau_T)$, where the notation in (F.21) now uses τ_T instead of α . The finite lower bounds and (F.19) keep its absolute value $\gg L_i$. If the difference is positive, every right reference cost is $\gg L_i$, because every left reference cost is nonnegative. If it is negative, every left cost is $\gg L_i$. Thus one entire side has this lower bound before taking the two separate minima. All $\sum_{h=3}^{17}(h-2) = 120$ binary factors are saturated; no enumeration of their many individual contexts is required.

The ordinary suffix starts in group 18, since $\tau_T \rightarrow \alpha_2$. The common terminal balance gives

$$D = (2^{\tau_T} \log 2 - 1)T^{11} = d_0T^5 + O(T^3) + O(T^{-1}), \quad d_0 = \sum_{i=3}^{17} 2^{i-3} \{h_i - 1 - h_i^\alpha \log h_i\} > 0. \quad (\text{F.23})$$

Each summand of d_0 is positive because $\alpha_h > \alpha_2$ for $h > 2$. To obtain the error bound, the old groups have total length $O(T^3)$, and changing the product in the nonfinal terminal groups costs $O(T^5|\tau_T - \alpha|) = O(T^{-1})$. Now $U = M = T^{11}$, $A = B = 0$, so $\mathcal{B}_{\text{ord}} \asymp T^{-11/2}T^5/T^{11} = T^{-23/2}$.

Finally, $J_0(T)$ in (4.21) is precisely the objective at the limiting products (u, v, α) . Its residual first derivative is zero in the first direction, $O(T)$ in the second, and $O(T^5)$ in the common terminal direction. Together with (F.19), the linear change is $O(TT^{-2} + T^5T^{-6}) = O(T^{-1})$. For the quadratic change, differentiating the recursion on a compact neighborhood gives diagonal bounds $O(T)$, $O(T^3)$, $O(T^{11})$. Mixed derivatives involving the first direction are $O(T)$, and the remaining mixed derivative is $O(T^3)$: only groups preceding both differentiated nodes can contribute. Their products with the respective increments are no larger than $O(T^{-1})$. Hence $J(L) - J_0(T) = O(T^{-1})$. All coordinates tend to infinity, and the positive-vector comparison proves (4.22).

To finish the dimensional statement, suppose a terminal owner has C names including its permanent child and limiting product α_2 . A singleton immediately above it solves $F(s; \alpha_2 \log C) = 0$. By (F.16), its product exceeds α_2 exactly when $t_1(C+1) > \alpha_2$. The one-valley property of Lemma 5.18 and $t_1(17) < \alpha_2 < t_1(18)$ show that the first integer value is $C = 17$. Two preceding singleton owners require $H = 19$, or $k = 18$. The root enclosures above verify that both strict gaps are realized in this first admissible case.

G Reference geometry and endpoint estimates

This appendix proves the identities and endpoint estimates used in the construction of the lower bound. Throughout, the original physical groups remain distinct. An owner is a maximal interval of equal saddle products; when discussing an individual owner we also call it a platform. Expectations of reference costs always mean the unconditioned natural reference flow. Actual incoming lists, once selected, retain their original conditional law.

G.1 Exact carries and the full word density

We first allow a decreasing alphabet chain $H = h_1 > \dots > h_m \geq 2$, $m \geq 2$, with positive lengths L_i . Write $p_j = h_j - h_{j+1}$ for $j < m$, $p_m = h_m - 1$, and $V_j = \sum_{i \leq j} L_i$. At its ordered saddle, set

$$S_m = h_m, \quad S_j = p_j + S_{j+1}^{s_{j+1}/s_j}, \quad Z_i = S_i^{s_i}, \quad J = \sum_i L_i + \sum_j s_j p_j V_j - \sum_i L_i Z_i.$$

The compactness theorem gives $1/2 < s_m \leq \dots \leq s_1 < 1 - \epsilon_H$. Define

$$\begin{aligned} u_j &= s_{j+1}/s_j, & d_j &= u_j^{-1} - 1, & r_j &= s_j/s_1, & a_j &= \log S_j, & \beta_j &= S_{j+1}^{u_j}/S_j, \\ A_j &= a_j - u_j \beta_j a_{j+1} \quad (j < m), & A_m &= a_m, \\ \lambda_j &= -\sum_{i \leq j} \partial_i J, & \lambda_0 &= \lambda_m = 0, & m_j &= \lambda_j - \lambda_{j-1}. \end{aligned}$$

The signed margin m_j is distinct from the number m of levels. Complementarity and summation by parts give

$$\lambda_j \geq 0, \quad \lambda_j(s_j - s_{j+1}) = 0, \quad \sum_j s_j m_j = 0, \quad \sum_j m_j = 0, \quad \sum_{j < m} r_j m_j = r_m \lambda_{m-1}. \quad (\text{G.1})$$

Indeed the first sum is $\sum_{j < m} (s_j - s_{j+1}) \lambda_j$. With $\mu_i = L_i Z_i$, differentiation of the perspective recursion yields

$$A_j \sum_{i \leq j} \mu_i \prod_{i \leq q < j} \beta_q = p_j V_j + m_j. \quad (\text{G.2})$$

Euler homogeneity of $G_i = s_i \log S_i$ then gives the uncolored root identity

$$T_1 := \sum_j r_j p_j V_j = \sum_i r_i a_i \mu_i. \quad (\text{G.3})$$

For the difference, multiplied by s_1 , is $\sum_j s_j \partial_j J = 0$ by (G.1).

Use the original recursive color law on a fixed uncolored tuple. At level $j < m$, let M_j count the incoming and native slots and let C_j count those routed to the child. Put

$$Y_j = a_j M_j - u_j a_{j+1} C_j - p_j V_j, \quad X_m = a_m M_m - p_m V_m, \quad X_1 = \sum_i r_i a_i N_i - T_1.$$

Here X_1 denotes the uncolored root residual, not Y_1 . Using $M_{j+1} = C_j + N_{j+1}$ and $r_{j+1} = r_j u_j$, every child count cancels and gives

$$X_1 = \sum_{j < m} r_j Y_j + r_m X_m. \quad (\text{G.4})$$

Under the natural marked Poisson law, $\mathbb{E}Y_j = m_j$ and $\mathbb{E}X_m = m_m = -\lambda_{m-1}$. The full word probability $\pi(\omega)$ includes the terminal free names. The negative log probability of an exterior name is a_j , and that of a child route is $a_j - u_j a_{j+1}$; subsequent refinement contributes at its own level. Therefore, term by term,

$$-\log \pi(\omega) = B_{\text{geom}} + \sum_{j < m} Y_j + X_m, \quad B_{\text{geom}} = \sum_j p_j V_j. \quad (\text{G.5})$$

If $q_j = s_j/s_m - 1$, $q_m = 0$, elimination of X_m gives

$$1 = A_{\text{eff}} \pi(\omega) \exp \left(- \sum_{j < m} q_j Y_j \right), \quad A_{\text{eff}} = \exp(B_{\text{geom}} + X_1/r_m). \quad (\text{G.6})$$

Let y_j be fixed by the uncolored environment before its private refinements, and define carries by

$$Y_j = m_j + y_j + \xi_j - \xi_{j-1}/u_{j-1}, \quad \xi_0 = 0. \quad (\text{G.7})$$

The index-zero term is zero. Put $x = \sum_{j < m} r_j y_j$. If $x - C_{\text{root}} < X_1 \leq x$, then

$$X_m = (X_1 - x)/r_m - \lambda_{m-1} - \xi_{m-1}/u_{m-1}. \quad (\text{G.8})$$

Thus the terminal margin retains the last cumulative multiplier. For example, solving for the last native count by

$$N_m^* = \left\lfloor \frac{T_1 + x - \sum_{i < m} r_i a_i N_i}{r_m a_m} \right\rfloor$$

would give $C_{\text{root}} = r_m a_m$ when this count is admissible. Centrality and nonnegativity of this count are separate probability requirements; the identity alone supplies neither when L_m is small.

Since $\sum_j q_j m_j = 0$ and $q_j - q_{j+1}/u_j = d_j$, the exact cancellation is

$$\sum_{j < m} q_j Y_j = \sum_{j < m} q_j y_j + \sum_{j < m} d_j \xi_j. \quad (\text{G.9})$$

Combining this with the factor $e^{-s_1 X_1}$ gives

$$(r_m^{-1} - s_1)X_1 - \sum_{j < m} q_j Y_j = \sum_{j < m} (1 - s_j) y_j - \sum_{j < m} d_j \xi_j + (r_m^{-1} - s_1)(X_1 - x). \quad (\text{G.10})$$

Here $(r_m^{-1} - s_1)r_j - q_j = 1 - s_j$. Thus $y_j = \log(C_j^*/D_j)$ produces the powers $D_j^{s_j-1}$, with no additional multiplier-dependent rate.

G.2 Comparison of the original posteriors

Fix one complete uncolored environment G , one accepted full-chain weight $0 \leq F_G \leq 1$, and the original prior π_G on all its fine words. The carries in (G.7) can be defined for any word; bounds on them are additional support conditions. Write $E(\omega) = \sum_{j < m} d_j \xi_j(\omega)$. For two retained words,

$$\frac{\pi_G(\omega)}{\pi_G(\omega')} = e^{E(\omega) - E(\omega')}.$$

Let $Q_h(\omega)$ be any genuine original comparison class, including classes that mix parent and child cells, and put

$$q_h = \mathbb{E}_\pi[F_G(\omega') \mid Q_h(\omega)], \quad q_h^{\text{rel}} = \mathbb{E}_\pi[F_G(\omega') e^{E(\omega) - E(\omega')} \mid Q_h(\omega)].$$

If the oscillation of E on $\{F_G > 0\}$ is at most B_H , positivity on this same class gives

$$e^{-B_H} q_h \leq q_h^{\text{rel}} \leq e^{B_H} q_h. \quad (\text{G.11})$$

For the actual geometric factor Φ_h , define $I_h = \log \Phi_h + \log[\pi_G(Q_h)/\pi_G(\omega)]$. Direct substitution gives

$$e^{I_h} q_h^{\text{rel}} = \Phi_h \sum_{\omega' \in Q_h} F_G(\omega').$$

Consequently (G.11) compares the full-history sums for every fixed positive collision power. The same argument applies to an enlarged whole cell with its actual class and original geometric factor. A sufficient oscillation condition is

$$|\xi_j - c_j| \leq w_j, \quad \sum_{j < m} d_j w_j \leq B_H/2,$$

where the centers c_j are common to the environment. If $E_c = \sum d_j c_j$, the nominal mass is

$$\mathbb{E}_\pi[F_G e^{-E}] \asymp e^{-E_c} \mathbb{E}_\pi F_G.$$

The center factor is retained even when the oscillation is bounded.

Now group the levels into maximal owners $[a_\nu, b_\nu]$, with products $\sigma_1 > \dots > \sigma_M$. Put

$$u_\nu = \sigma_{\nu+1}/\sigma_\nu, \quad d_\nu = u_\nu^{-1} - 1, \quad \eta_\nu = \xi_{b_\nu}, \quad y^{(\nu)} = \sum_{j=a_\nu}^{b_\nu} y_j, \quad Z_\nu = \sum_{j=a_\nu}^{b_\nu} Y_j \quad (\nu < M).$$

Exterior multipliers vanish, and internal ratios equal one. Hence

$$Z_\nu = y^{(\nu)} + \eta_\nu - \eta_{\nu-1}/u_{\nu-1}, \quad E = \sum_{\nu < M} d_\nu \eta_\nu, \quad (\text{G.12})$$

$$Z_M := \sum_{j=a_M}^{m-1} Y_j + X_m = \frac{X_1 - x}{r_m} + \sum_{j=a_M}^{m-1} y_j - \eta_{M-1}/u_{M-1}. \quad (\text{G.13})$$

Common owner sums of the targets suffice in these formulas; individual internal targets may be redistributed while those sums and the grouped bands are maintained. If the strict exits have widths R_ν satisfying

$$0 \leq \eta_\nu \leq C_H R_\nu, \quad R_\nu \leq C_H \sqrt{1 + T_\nu}, \quad d_\nu R_\nu \leq C_H,$$

then $E = O_H(1)$. The lattice-sized floor of $\sqrt{1 + T_\nu}/(1 + d_\nu \sqrt{1 + T_\nu})$ has these properties. Internal equal-product carries have coefficient zero. In the one-owner case the full word probability is independent of the routing at fixed G , although a proper atomic projection need not be uniform.

Formula (G.10) expresses the nominal integrand as

$$\prod_{j < m} D_j^{s_j - 1} \mathbb{E}_\pi[F_G e^{-E}],$$

up to the fixed target constants and bounded root factor. Grouping retains the corresponding owner powers $\sigma_\nu - 1$. Evaluating this expression requires the conditional first-mass estimate at each stage with its entire type mixture inside the power, retaining the coupled count acceptance in F_G .

For completeness, the distinction between a full word and its atomic projection can be seen from one parent and child. If the parent output is $Z = y + \eta$, then

$$\pi_{\text{full}}(w, v) = \pi_p(w) \pi_c(v \mid w), \quad -\log \pi_p(w) = B_p + Z(w).$$

For two accepted full words,

$$\begin{aligned}\log \frac{\pi_p(w_0)}{\pi_p(w)} &= \eta - \eta_0, \\ \log \frac{\pi_{\text{full}}(w_0, v_0)}{\pi_{\text{full}}(w, v)} &= d(\eta_0 - \eta), \\ \log \frac{\pi_c(v_0 \mid w_0)}{\pi_c(v \mid w)} &= (\eta_0 - \eta)/u.\end{aligned}$$

The child fiber supplies the cancellation even when $d = 0$. Thus full-word posterior comparison is applied after the fine-history estimate, retaining the actual child conditional mass.

G.3 A finite shift of the strict-exit targets

Let $A_\nu = \log S_{a_\nu}$ denote the mark of the shallowest exterior leaf of owner ν , and put $b_\nu = u_\nu A_{\nu+1} = \log z_\nu$ for $\nu < M$. In a native bank of owner ν , change e_ν leaves that are the immediate child's shallowest exterior into the owner's eligible shallow name. The banks lie in disjoint original native blocks. With $e_0 = 0$,

$$\Delta Z_\nu = b_\nu e_\nu - A_\nu e_{\nu-1} \quad (\nu < M), \quad \Delta Z_M = -A_M e_{M-1}.$$

For desired nonnegative shifts y_ν , choose successively

$$e_\nu = \left\lceil \frac{y_\nu + A_\nu e_{\nu-1}}{b_\nu} \right\rceil, \quad \delta_\nu = b_\nu e_\nu - A_\nu e_{\nu-1} - y_\nu \in [0, b_\nu).$$

Comparison of (G.12) before and after the change gives

$$\Delta \eta_\nu = \delta_\nu + \Delta \eta_{\nu-1}/u_{\nu-1} = \sum_{l \leq \nu} \frac{\sigma_l}{\sigma_\nu} \delta_l = O_H(1).$$

Thus an existing exit window is preserved with a fixed enlargement. If $y_\nu = O_H(\log(2 + T_\nu))$, the owner length hierarchy gives $T_l \leq C_H T_\nu$ for $l < \nu$, and the finite recursion yields $e_\nu = O_H(\log(2 + T_\nu))$. These changes fit a macroscopic native bank with a central required-leaf count once its own total is large. The weighted sum of the output changes is zero because $r_\nu b_\nu = r_{\nu+1} A_{\nu+1}$, so the private changes leave the uncolored root unchanged. Its desired target rises by $\sum r_\nu y_\nu$; the unread reservoir must therefore be selected under that target before the private construction.

G.4 Secants, compression, and the reference cost

Throughout this appendix the ambient alphabet has at most H letters; all constants may depend on H . For $z \geq 1$ put

$$E_q(z) = (q + z) \log(q + z) - z \log z, \quad E_0(z) = 0.$$

The following elementary estimate supplies both the qualitative compression and the quantitative separation needed below.

Lemma G.1 (Normalized secants). *For $1 \leq b < y \leq H$, let*

$$L(b, y) = \frac{y \log y - b \log b}{y - b}, \quad \beta_b(y) = \frac{\log L(b, y)}{\log y}.$$

Then $\beta'_b(y) \leq -1/(24H^2)$.

Proof. Set $F(t) = \log \int_0^1 e^{tu} du$, $0 \leq t \leq \log H$. Its second derivative is the variance under the density $p_t(u) = e^{tu} / \int_0^1 e^{tv} dv \geq H^{-1}$, so

$$F''(t) \geq H^{-1} \int_0^1 (u - \mathbb{E}_t u)^2 du \geq (12H)^{-1}.$$

For $t = \log y$, $\beta_1(y) = 1 - F(t)/t$, whence

$$-\beta'_1(y) = \frac{\int_0^t v F''(v) dv}{yt^2} \geq \frac{1}{24Hy} \geq \frac{1}{24H^2}.$$

For general b , writing $s = b/y$ gives

$$y\partial_y L(b, y) = h(s) = \frac{1 - s + s \log s}{(1 - s)^2}, \quad h'(s) = \frac{(1 + s) \log s + 2(1 - s)}{(1 - s)^3} < 0.$$

Indeed $\log s - 2(s - 1)/(s + 1)$ increases to zero on $(0, 1)$, its derivative being $(s - 1)^2/[s(s + 1)^2]$. Increasing b at fixed y increases L and decreases $y\partial_y L$. It therefore decreases

$$\beta'_b(y) = \frac{(y\partial_y L/L) \log y - \log L}{y(\log y)^2}.$$

Comparison with $b = 1$ proves the assertion. $\square$

Lemma G.2 (Shifted compression). *Let $0 \leq e \leq 1$ and $F_r(q) = q - (r + z)^{-e} E_q(z)$. At finitely many increasing integer capacities r , let integers d_r be nondecreasing and satisfy $0 \leq d_r \leq r$. There is an integer $R \geq 0$ such that*

$$F_r(\min(r, R)) \leq F_r(d_r) \quad \text{at every capacity.}$$

The assertion remains true if only the coefficient at the largest capacity is decreased to an arbitrary positive value. No restriction on the increments $d_{r'} - d_r$ is required.

Proof. Read capacities in increasing order, starting with benchmark $R = 0$. At capacity r , replace $R = A$ by the current count $C = d_r$ only if $F_r(A) > F_r(C)$. Monotonicity of the counts ensures $A < C \leq r$. The failing comparison means

$$L(z + A, z + C) > (z + r)^e.$$

For an earlier capacity $A < h < C$, Lemma G.1 gives $L(z + A, z + h) > (z + h)^e$; for $C \leq h \leq r$ the displayed inequality suffices, and for $h \leq A$ the clipped arguments coincide. Thus every earlier comparison is preserved. If the last coefficient is $a \leq (r + z)^{-e}$, its failing comparison instead gives $L(z + A, z + C) > 1/a \geq (z + r)^e$, which proves the same preservation. For the singleton convention $\phi_m(n) = n - 1 - m^{-e} n \log n$, $1 \leq n \leq m$, the identical argument uses $L(A, C) > m^e$; zero counts may first be replaced by one, since the entropy margin is zero at both zero and one. $\square$

Fix a maximal equal-product platform $[a, b]$ of product σ in the complete natural chain. Write

$$r_i = b - i + 1, \quad S_i = r_i + z, \quad c_i = S_i^{\sigma-1}, \quad V_i = \sum_{l \leq i} L_l, \quad T = V_b.$$

The permanent child has weight $z \geq 1$. Retain every original old physical block, with its natural coefficient

$$\kappa_i = \frac{Z_i}{S_a} \prod_{i \leq l < a} \beta_l \quad (i < a), \quad \kappa_i = c_i \quad (a \leq i \leq b). \quad (\text{G.14})$$

Its mean letter intensities are $\kappa_i(1, \dots, 1, z)$. All old blocks have capacity r_a ; native block i has capacity r_i . Put $K_{\text{in}} = \sum_{i < a} L_i \kappa_i$. The platform stationarity identities give

$$K_{\text{in}} E_{r_a}(z) - r_a V_{a-1} + \sum_{i=a}^b L_i [c_i E_{r_i}(z) - r_i] = 0, \quad (\text{G.15})$$

$$\Lambda_R := R V_{a-1} - K_{\text{in}} E_R(z) + \sum_{i=a}^b L_i \{\min(r_i, R) - c_i E_{\min(r_i, R)}(z)\} = \lambda_{b-R} \geq 0. \quad (\text{G.16})$$

Here $0 \leq R \leq r_a$ and $\lambda_{a-1} = \lambda_b = 0$. For example, at $R = r_j$ the left side is precisely the incoming buffer minus the remaining native drift, hence the multiplier λ_{j-1} . This also checks the two zero endpoint values. These coefficients describe the unconditioned natural reference; conditional estimates retain each old word's actual list.

Proposition G.3 (Reference nonnegativity). *Every lawful original coarse history on this platform alphabet has nonnegative mean total cost. More precisely, its cost has the form*

$$\overline{C} = D_{\text{partition}} + D_{\text{clock}} + \sum_v \omega_v \left(\lambda_{b-R_v} + \sum_i L'_i \Delta_{i,v} \right), \quad (\text{G.17})$$

where all terms on the right are nonnegative, $\sum_v \omega_v = 1$, and the finite list of vertices has cardinality bounded in terms of H alone.

Proof. First retain the chain of cells containing the permanent child, $A_0 \supsetneq \dots \supsetneq A_q$, with split times $\tau_0 \leq \dots \leq \tau_q$. Resolve every pure sibling immediately when it leaves this chain. This is lawful, since every name in a live cell is eligible until its split. If $d(t)$ is the number of unresolved boundaries, then d is nonincreasing, $d(t) \leq r_i$ in native block i , and the resulting star-spine cost is exactly

$$\overline{C}_{\text{sp}} = \overline{F}_{\text{full}}(\tau_0) + \sum_{l=1}^q \{\overline{F}_{A_l}(\tau_l) - \overline{F}_{A_l}(\tau_{l-1})\} \quad (\text{G.18})$$

$$= \int_0^T [d(t) - \kappa(t) E_{d(t)}(z)] dt. \quad (\text{G.19})$$

The initial value is zero by (G.15); no pure-cell lifetime or original rank delay remains in this comparison history.

The transport identity is $\kappa_i = \kappa_{i+1} \beta_i^{1-s_i}$, including the old/native transition, so κ increases on $[0, V_a]$. Define

$$\bar{c}_a = \frac{K_{\text{in}} + L_a c_a}{V_a} \leq c_a, \quad A(x) = \int_0^x (\bar{c}_a - \kappa(v)) dv.$$

Then $A \geq 0$, $A(0) = A(V_a) = 0$. Since $E_{d(t)}$ is nonincreasing, Stieltjes summation gives the exact correction

$$D_{\text{clock}} = \int_0^{V_a} (\bar{c}_a - \kappa(t)) E_{d(t)} dt = \sum_{0 < x < V_a} \Delta E_d(x) A(x) \geq 0.$$

Replace this first region by $L'_a = V_a$, $c'_a = \bar{c}_a$; for $i > a$ keep $L'_i = L_i$, $c'_i = c_i$. The suffix sums (G.16) are unchanged.

Subdivide the ordered split-time domain at the reduced physical endpoints $0, V_a, \dots, V_b$. In each closed chamber the averaged cost is affine. At a vertex every split time is an endpoint: otherwise a maximal tied group of nonendpoint coordinates can be moved in both directions without violating its order, block or deadline constraints. Thus the cost is a convex combination, with weights ω_v , of

$$\sum_i L'_i F_i(d_i(v)), \quad F_i(q) = q - c'_i E_q(z).$$

At every vertex the ranks $d_i(v)$ decrease in physical order. Lemma G.2, including its reduced largest coefficient, supplies R_v for which $\Delta_{i,v} = F_i(d_i(v)) - F_i(\min(r_i, R_v)) \geq 0$. This proves (G.17) for a star spine. In particular the total negative proper-cell increments are at most the root mean plus the positive proper-cell increments. The root mean is nonnegative: it starts and ends at zero and its slopes have one positive-to-negative sign change, as established below.

For a general coarse history let $d(t)$ count all unresolved original boundaries. Its eligible unresolved rank is at most $d(t)$, the remaining rank being original delay. A partition of masses $(1, \dots, 1, z)$ with unresolved rank at most d has entropy at most $E_d(z)$. Indeed merge two nontrivial cells and split off their smallest atom. Convexity of $x \log x$ preserves rank and does not decrease entropy. Iteration leaves one nontrivial cell, and increasing any chosen mass increases its entropy; it therefore uses the largest $d + 1$ atoms, namely the child and d exterior names. Consequently

$$D_{\text{partition}} = \int_0^T \left\{ \kappa_i E_{d(t)}(z) - \sum_{A \in \Pi(t)} \text{Ent}_{v_i}(A) \right\} dt \geq 0.$$

Apply the preceding chamber argument to the history's actual activation order to obtain (G.17). $\square$

For a proper platform, whose z is bounded away from one, the finite partition inequality has a strict gap unless its unresolved partition is one child-containing cell of full unresolved rank and singletons. It has no gap between different child-containing sets of the same cardinality. At $z = 1$ further equal-weight ties remain. Moreover, Proposition G.3 concerns this platform alphabet: old cells crossing to a name outside it belong to their larger original context.

G.5 Discrete reserves and single crossings

For groups of capacity c_j , length $L_j \geq 0$ and coefficient $0 < \kappa_j \leq 1$, put

$$g_j(q) = q - \kappa_j E_q(z), \quad \Lambda_q = \sum_j L_j g_j(\min(c_j, q)).$$

Assume $c_j + z \leq H$ and all $\Lambda_q \geq 0$.

Lemma G.4 (Discrete curvature reserve). *Let $T_a = \sum_{c_j=a} L_j$ and $W_a = \sum_{c_j>a} \kappa_j L_j$. Then*

$$\Lambda_a + \frac{1}{2} \log H T_a \geq \frac{W_a}{2H}. \quad (\text{G.20})$$

If $A < R < B$, set $G_R(a) = (\min(R, a) - A)(B - \max(R, a))/(B - A)$ for $A < a < B$. Then

$$\Lambda_R + \log H \sum_{a=A+1}^{B-1} G_R(a) T_a \geq \frac{(R - A)(B - R)}{2H} \sum_{c_j \geq B} \kappa_j L_j. \quad (\text{G.21})$$

In particular a native group of capacity greater than a contributes at least $L_i/(2H^2)$ to the right side of (G.20).

Proof. With $Q_a = E_{a+1} - 2E_a + E_{a-1}$,

$$Q_a = \int_{-1}^1 \frac{1-|t|}{z+a+t} dt \geq H^{-1},$$

and direct subtraction gives

$$D_a := 2\Lambda_a - \Lambda_{a-1} - \Lambda_{a+1} = Q_a W_a + \sum_{c_j=a} L_j [1 - \kappa_j(E_a - E_{a-1})].$$

The last bracket is at least $-\log H$. Since $2\Lambda_a \geq D_a$, this proves (G.20). The discrete Green identity expresses Λ_R as the linear interpolation of Λ_A, Λ_B plus $\sum_{a=A+1}^{B-1} G_R(a) D_a$. These endpoint values are nonnegative, $W_a \geq \sum_{c_j \geq B} \kappa_j L_j$, and $\sum_a G_R(a) = (R-A)(B-R)/2$, giving (G.21). A native coefficient is at least H^{-1} . $\square$

Lemma G.5 (Weak edges and their order). *On one platform let $g_r(q) = q - (r+z)^{-e} E_q(z)$, $0 \leq e \leq 1$, and put $\epsilon_* = (\log 2)/(192H^2)$. If $0 \leq q < r$ and $|g_r(r) - g_r(q)| \leq \epsilon_*$, then for $q < h < r$*

$$g_h(q) - g_h(h) \geq \frac{\log 2}{32H^2}. \quad (\text{G.22})$$

Two such full-input edges $r_1 \rightarrow q_1, r_2 \rightarrow q_2$ with $r_1 > r_2$ necessarily satisfy $q_1 > q_2$.

Proof. Weakness gives $|(r+z)^{-e} L(q+z, r+z) - 1| \leq \epsilon_*/(r-q)$. Its logarithm is at least $-2\epsilon_*$. Integrating Lemma G.1 between $h+z$ and $r+z$, and using $\log(h+z) \geq \log 2$, gives

$$\log[(h+z)^{-e} L(q+z, h+z)] \geq \frac{\log 2}{24H^2} - 2\epsilon_* \geq \frac{\log 2}{32H^2}.$$

Since $g_h(q) - g_h(h) = (h-q)[(h+z)^{-e} L(q+z, h+z) - 1]$, this proves the first assertion. For the second write $\beta(q, r) = \beta_{q+z}(r+z)$. Weakness implies $|\beta(q_j, r_j) - e| \leq 2\epsilon_*/\log 2$. If $q_1 \leq q_2$, increasing the lower secant endpoint and then using Lemma G.1 gives

$$\beta(q_2, r_2) - \beta(q_1, r_1) \geq \beta(q_1, r_2) - \beta(q_1, r_1) \geq \frac{1}{24H^2},$$

whereas the two weak bounds give at most $1/(48H^2)$. $\square$

Thus surviving deepest output sets are strictly nested in physical order within one platform.

We record explicitly the uniform old coefficients used in the estimates. If every natural product lies in $[\varepsilon_H, 1 - \varepsilon_H]$, then (G.14) and its transport identity give

$$H^{-H-1} =: \kappa_* \leq \kappa_l \leq 1, \quad \kappa_{l+1} - \kappa_l \geq d_* > 0 \quad (\text{G.23})$$

for consecutive old groups, where one may take

$$d_* = \min \left\{ \frac{\varepsilon_H}{H^2}, H^{-H-1} [1 - (1 - H^{-1})^{\varepsilon_H}] \right\}.$$

The first term also covers the last old/native transition. This follows by inserting the natural transport ratio $\kappa_l/\kappa_{l+1} = \beta_l^{1-s_l}$ and the bounds on its finite alphabet. In particular all old full-rank slopes decrease. For later quantitative comparisons one may fix

$$\delta_* = \frac{\kappa_*}{4H}, \quad \epsilon_0 = \min\{\delta_*/2, d_*\delta_*/(4H)\}, \quad C_{\text{len}} = H/\delta_*, \quad c_{\text{prefix}} = \frac{d_*}{2(1 + C_{\text{len}})}.$$

Lemma G.6 (Single crossing and length allocation). *Take any equal-product interval $[a, b]$, not necessarily maximal, with common product σ , child size $C = h_{b+1}$ and $z = \exp(G_{b+1}/\sigma)$; at the terminal boundary $C = z = 1$. Put $n_i = h_i - C$, $Z_i = (n_i + z)^\sigma$, and*

$$e_i = (n_i + z)^{\sigma-1} E_{n_i}(z) - n_i, \quad I_{\text{in}} = \sum_{l < a} L_l Z_l \prod_{l \leq v < a} \beta_v, \quad M_H = H(1 + \log H).$$

Then

$$\sum_{i=a}^b L_i e_i = n_a V_{a-1} - I_{\text{in}} \frac{E_{n_a}(z)}{n_a + z} - \lambda_{a-1} + \lambda_b. \quad (\text{G.24})$$

Writing $B_0 = M_H V_{a-1} + |\lambda_{a-1}| + |\lambda_b|$, the absolute value of the left side is at most B_0 . The signs of e_i have at most one negative-to-positive change. Except for at most one index, $|e_i| \geq \gamma_H$, where

$$d_H = (24H^2)^{-1}, \quad \gamma_H = 1 - 2^{-d_H/3} > 0.$$

Consequently

$$|e_i| L_i \leq B_0 + M_H \sum_{l \neq i} L_l, \quad (\text{G.25})$$

$$\gamma_H \sum_{i \in S} L_i \leq B_0 + M_H \sum_{i \notin S} L_i \quad (\text{G.26})$$

whenever S has one sign and avoids the possible small-drift index. If $R_i = V_{a-1} + \sum_{l \neq i} L_l + (|\lambda_{a-1}| + |\lambda_b|)/M_H$, then

$$|e_i| \leq M_H R_i / L_i, \quad |\sigma - \theta_{n_i}(z)| \leq \frac{H M_H R_i}{(\log 2) L_i}, \quad \theta_n(z) = 1 - \frac{\log(E_n(z)/n)}{\log(n+z)}.$$

If the small-drift index is first or last, the other lengths satisfy $\sum_{l \neq i} L_l \leq (B_0 + |e_i| L_i) / \gamma_H$.

Proof. Vary the common product, differentiating the child weight as well. The natural recursion gives $\partial_\sigma G_i = E_{n_i}(z)/(n_i + z)$ on the interval and its transported multiple before the interval. Stationarity, with the two boundary multipliers retained, is exactly (G.24). The transport relation $Z_i \beta_i = Z_{i+1} \beta_i^{1-s_i}$ gives $I_{\text{in}} \leq Z_a V_{a-1}$; all displayed coefficients are bounded by M_H , proving its absolute bound. Now

$$\frac{e_i}{n_i} = \exp\{(\sigma - \theta_{n_i}(z)) \log(n_i + z)\} - 1.$$

The normalized-secant estimate implies $0 < \theta_n(z) < 1$ and $\theta_{n'}(z) - \theta_n(z) \geq d_H(n' - n)$ for $n' > n$. (The secant here is $L(z, z+n) = E_n/n$.) The integer capacities decrease in physical order, so the asserted sign pattern follows; at most one threshold is within $d_H/3$ of σ . Outside that interval the last exponential has absolute difference from one at least γ_H , giving the uniform separation. Moving the remaining terms in (G.24) proves (G.25) and (G.26). Finally the derivative of e_i in σ is at least $(\log 2)/H$, yielding the threshold-distance bound. At an extreme small-drift index all the others have one sign, giving the final claim. $\square$

On a maximal platform the preceding lemma and the increasing old coefficients show that the full-rank slopes $g_i(r_i) = -e_i$ change from positive to negative at most once. Their integral is zero by (G.15). This proves the root-mean assertion used in Proposition G.3.

G.6 Which weak transitions can remain

All histories in this subsection retain the original names and deadlines. Their pure siblings are resolved when they leave the star spine. Let $e = 1 - \sigma \geq e_0 > 0$. The following argument explains why a small reference budget forces both the input and the output names.

Proposition G.7 (Native weak transitions). *There are $\epsilon_H, c_H > 0$ such that a synchronous native weak edge $A \rightarrow B$, of ranks $0 \leq q < R \leq r_i$, satisfies*

$$|g_i(R) - g_i(q)| \leq \epsilon_H \implies \bar{C}_i^-, \bar{C}_i^+ \geq c_H L_i,$$

for every lawful endpoint prefix and suffix context, unless A is the full current eligible alphabet and B is its true deepest q -name suffix together with the child. There is at most one exceptional q for a fixed full rank. Across native blocks these exceptional outputs are strictly ordered as in Lemma G.5.

Proof. We first keep track of the full compression certificate on the original groups. Reading an endpoint rank profile backwards makes both ranks and capacities nondecreasing. Use the algorithm of Lemma G.2. Old groups have the same largest capacity and their coefficients are nonincreasing in this reading order. An old update $A \rightarrow D$ gives $(E_D - E_A)/(D - A) > 1/\kappa_{\text{current}}$; every processed old coefficient is at least the current one, so its benchmark value also decreases. The threshold is at least the natural largest-capacity threshold and therefore preserves all preceding native comparisons. The resulting identity is

$$\bar{C} = \Lambda_K + \sum_j L_j \Delta_j, \quad \Delta_j = g_j(d_j) - g_j(\min(c_j, K)) \geq 0. \quad (\text{G.27})$$

During this algorithm the benchmark *value* at every already processed group only decreases. Splitting a group at a cut preserves this argument, since repeated equal coefficients cause no difficulty.

Set $\eta = 1/(4H^2)$, $\nu = e_0/H$, and initially take $\epsilon \leq 1/(16H^2)$. Since $g_i'' \leq -H^{-2}$, an integer strictly between q and R has value greater than both weak endpoint values by η , whereas an integer outside $[q, R]$ has value less than both by η . To see the latter, compare its secant with the secant on $[q, R]$, whose absolute slope is at most $\epsilon/(R - q)$; strong concavity separates the slopes by at least $(R - k)/(2H^2)$ on the left (and symmetrically on the right). Thus if $\min(r_i, K) \notin \{q, R\}$, either (G.27) is impossible or $\Delta_i \geq \eta$, which pays ηL_i .

Suppose now that this clipped value is q or R , with $q \geq 1$. Every future rank is at most q . The reverse benchmark must reach q strictly after block i : if its incoming value were $A < q$, its value on block i would be strictly below both weak endpoint values, so block i would not update it; subsequent updates only decrease this value and cannot finish at either weak endpoint. At the first reverse crossing $A \rightarrow q$, occurring at capacity h , every already processed future group of capacity $c \geq q$ whose actual rank is less than q gains at least

$$g_c(A) - g_c(q) > (q - A) \left[\left(\frac{h + z}{c + z} \right)^e - 1 \right] \geq \frac{e_0}{H} = \nu. \quad (\text{G.28})$$

Here $h > c$; a later forward rank cannot rise back to q . The last bound uses $h \geq c + 1$ and $\log(1 + 1/(c + z)) \geq 1/H$. It follows that

$$\bar{C} \geq \min\{\eta L_i, \nu D_B\}, \quad D_B = \sum_{j > i: c_j \geq q, |B \cap E_j| < q} L_j.$$

More strongly, in the accepted benchmark cases the full decomposition retains $\Lambda_K + \nu D_{\text{rank}}$, where D_{rank} sums all future groups of capacity at least q and actual rank below q . This includes both expired names and premature refinements.

Keeping the multiplier now turns this side-length payment into a current-length payment. If $K = q$ and B is wrong, its capacity- q group has rank below q , so $\bar{C} \geq \Lambda_q + \nu T_q$. By Lemma G.4, this is at least

$$c_{\text{curv}} L_i, \quad c_{\text{curv}} = \frac{\min(1, 2\nu/\log H)}{2H^2}.$$

If $K = R < r_i$, the reverse algorithm first reaches R at block i or earlier, of capacity at least $r_i > R$. The same comparison as (G.28) pays ν on the already processed capacity- R group, giving the same conclusion. Old coefficients are no greater than the corresponding natural one, as required for this comparison.

It remains to consider a wrong output with $R = r_i = r$ and $K \geq r$. Set

$$\epsilon^* = \min\{(16H^2)^{-1}, (\log 2)/(192H^2)\}, \quad \gamma = \frac{\log 2}{48H^2}, \quad \mu = \min(\nu, \gamma), \quad M = 2H(1 + \log H).$$

At every future capacity $q < h < r$, rank q costs at least γ against the full benchmark by (G.22); rank below q costs ν by (G.28). The capacity- q group has rank below q because B is wrong. Consequently, with $T' = \sum_{a=q}^{r-1} T_a$,

$$\bar{C} \geq \Lambda_K + \mu T'. \quad (\text{G.29})$$

Set

$$W = \sum_{c_j \geq r} \kappa_j L_j, \quad A_H = \frac{2H^2 \log H}{\log 2}, \quad \epsilon_H = \min\{\epsilon^*, 1/(16HA_H)\}.$$

With $\ell = L(q + z, r + z)$ and $e' = \log \ell / \log(r + z)$,

$$0 < e' < 1, \quad |e' - e| \leq \frac{2\epsilon_H}{(r - q) \log 2}.$$

For an earlier/current capacity $c \geq r$, put $n = \min(c, K) \geq r$. Since $\kappa_j \leq (c + z)^{-e}$, normalized-secant monotonicity yields

$$\begin{aligned} g_j(q) - g_j(n) &\leq \kappa_j(n - q)[(n + z)^{e'} - (c + z)^e] \\ &\leq A_H \epsilon_H \kappa_j. \end{aligned}$$

The last inequality uses $n - q \leq H$ and $\partial_v(n + z)^v \leq H \log H$ on $[0, 1]$; it is one-sided and remains valid if $e' < e$. Smaller capacities between q and r contribute at most MT' , and capacities at most q contribute zero. Thus

$$\Lambda_q \leq \Lambda_K + A_H \epsilon_H W + MT'.$$

The Green estimate (G.21) on $[q - 1, r]$, evaluated at q , gives $\Lambda_q + \log H T' \geq (r - q)W/(2H)$. Combining these inequalities and (G.29) shows

$$\frac{7W}{16H} \leq \Lambda_K + (M + \log H)T' \leq \max\{1, (M + \log H)/\mu\} \bar{C}.$$

Since $W \geq L_i/H$, this pays $c_{\text{wrong}} L_i$, where $c_{\text{wrong}} = 7 \min\{1, \mu/(M + \log H)\}/(16H^2)$. The case of a genuine deadline completion, with full rank r on the preceding blocks, is included (then $K \leq r$); no lower bound on intervening side lengths has been imposed.

For a proper input $q < R < r_i$, the following time exchange pays the current block even when its output is correct. At the right endpoint b_i choose $c \in A \setminus B$ and an eligible $d \notin A$, already cut at $\tau_d \leq l_i$. Both names have deadlines at least b_i . Move only c to τ_d to form one comparison history, and move only d to b_i to form a second. Both are lawful. Between τ_d and b_i their ranks are $k(t) - 1$ and $k(t) + 1$, surrounding the original $k(t)$, and remain between zero and eligible capacity. Proposition G.3 gives nonnegative comparison costs, while exact subtraction gives

$$2\overline{C} - \overline{C}^- - \overline{C}^+ = \int_{\tau_d}^{b_i} \kappa(t)[E_{k(t)+1} - 2E_{k(t)} + E_{k(t)-1}] dt \geq L_i/H^2.$$

The right endpoint costs at least $L_i/(2H^2)$; the left differs by $L_i[g_i(R) - g_i(q)]$, so both cost at least ηL_i . This also covers $q = 0$. Taking the minimum of the three positive constants proves the stated selection for every context and hence for its finite minimum. Strict concavity makes the weak matching from a fixed full rank unique, and Lemma G.5 gives the ordering across blocks. $\square$

Proposition G.8 (Old weak transitions). *Every weak transition in an original old group of a successor platform has reference endpoint costs at least $c_H L_i$. For a full-input transition both costs are at least $c_H V_{a-1}$. These assertions hold for every lawful star-history context, including $z = 1$ at a terminal platform.*

Proof. Let the incoming platform have full exterior rank r , $S = r + z$, and $V = V_{a-1} > 0$. Its averaged incoming coefficient is $\kappa_{\text{av}} = I_{\text{in}}/(SV)$. At the preceding strict node, write $p \geq 1$, $u < 1$, $x = S^u$, $A = E_p(x)/(p+x)$ and $\beta = x/(p+x)$. Stationarity, with $m = -\lambda_{\text{preceding}} \leq 0$, gives

$$I_{\text{in}} = \beta(pV + m)/A \leq \frac{x}{D(p, x)}V, \quad D(p, x) = E_p(x)/p.$$

The entropy integral and concavity give

$$D(p, S^u) \geq S^{u-1}(1 + \log S) + \frac{p}{2H}, \quad D(p, x) \leq 1 + \log H.$$

Indeed integrate $1 + \log(x+t)$ for $0 \leq t \leq p$, using $\log(1+t/x) \geq t/(x+p)$, and then use the tangent bound for S^{1-u} . It follows that

$$\kappa_{\text{av}}(1 + \log S) \leq 1 - \delta_0, \quad \delta_0 = \frac{p}{2H(1 + \log H)}.$$

Put $\beta_H = 1/[4H(1 + \log H)^2]$, and use a weak threshold at most $\min\{\beta_H/4, \kappa_*/(8H)\}$. For a full-input edge $r \rightarrow q$, weakness implies $\kappa_i \geq (1 - \epsilon)/(1 + \log S)$, hence $\kappa_i - \kappa_{\text{av}} \geq \beta_H$. Old coefficients increase, so

$$\sum_{l < i} L_l(\kappa_i - \kappa_l) \geq V(\kappa_i - \kappa_{\text{av}}).$$

Move just the names in $A \setminus B$ to time zero, leaving the future B cuts fixed. The comparison is lawful and nonnegative. If the original transition time is τ , exact subtraction is

$$\overline{C}_\tau - \overline{C}_{\text{early}} = (E_r - E_q) \sum_{l < i} L_l(\kappa_i - \kappa_l) + [g_i(r) - g_i(q)]\tau.$$

Since $E_r - E_q \geq 1$, $\tau \leq V$, both endpoints cost at least $\beta_H V/2$. If instead $q < R < r$, use the same legal two-cut exchange as in Proposition G.7. Now its second difference is at least $\kappa_* L_i/H$, giving both endpoint costs at least $\kappa_* L_i/(4H)$.

For the eventual actual-word application, an admitted endpoint row $C_v \geq c_0 \overline{C}_v - h$, together with the original named regularity estimate $E_i(t) \geq -C_H \vartheta L_i - C_H K(1 + L_i^\zeta)$, converts this to $c'_H L_i - C$. Choose ϑ first small, then use Young's inequality on L_i^ζ ($\zeta < 1$). Nonweak edges use their favored endpoint. Multiple coordinates share the same finite deterministic cost budget. $\square$

G.7 A terminal suffix without proper weak binary edges

Proposition G.9. *In a terminal maximal platform with positive native lengths there is a deterministic native suffix whose total length is at least $c_H T$, and which has no proper full-input weak binary edge. Its earliest largest original block has length at least $c_H T/H$. The suffix and the tie rule depend only on the natural profile.*

Proof. Let its natural alphabet sizes be $H \geq h_1 > \dots > h_m \geq 2$, $r_i = h_i - 1$, $\kappa_i = h_i^{\sigma-1}$, and $e_i = h_i^\sigma \log h_i - (h_i - 1) = -g_i(r_i)$. The exact terminal balance gives

$$D := \sum_i L_i e_i = nV_{\text{old}} - \frac{E_n(1)}{h_1} I_{\text{in}} \geq b_H V_{\text{old}}, \quad b_H = \frac{1}{2H^2(1 + \log H)^2}.$$

For completeness, when $V_{\text{old}} > 0$ insert $I_{\text{in}} = \beta(pV_{\text{old}} + m)/A$, $m \leq 0$, from the preceding proof. The inequality $E_p(x)/p \geq 1 + \log x + p/[2(x+p)]$ gives a surplus of at least $np/[2H^2(1 + \log H)^2]$ times V_{old} . When there is no incoming block, $D = 0$.

If $r \rightarrow q$ is proper ($1 \leq q < r$) and $|g_i(r) - g_i(q)| \leq \epsilon_w \leq 1/(4H^3)$, then $-g_i'' \geq H^{-2}$ and $g_i(0) = 0$ imply

$$g_i(q) - \frac{q}{r} g_i(r) \geq \frac{q(r-q)}{2H^2}.$$

Writing $v = g_i(r) - g_i(q)$ shows $g_i(r) \geq qr/(2H^2) + rv/(r-q) \geq 1/(4H^2) =: c_w$. Thus every such weak block has $e_i \leq -c_w$. The thresholds α_h solving $e_h(\sigma) = 0$ are strictly increasing in h , by Lemma G.6. Choose disjoint fixed closed neighborhoods $|\sigma - \alpha_h| \leq \tau$ in which $|e_h| \leq c_w/2$; this is possible since $|\partial_\sigma e_h| \leq H(\log H)^2$. Outside these neighborhoods the nonzero drifts have a uniform absolute lower bound $g_0 > 0$.

If a present threshold is near σ , begin the suffix there; otherwise begin at the first positive drift, which exists since $D \geq 0$. The head has negative drift at most $-g_0$, while the suffix has at most one small drift followed by positive drifts and therefore contains no proper weak binary block. Its drift sum obeys

$$D_{\text{tail}} \geq b_H V_{\text{old}} + g_0 L_{\text{head}} \geq 0.$$

Since $|e_i| \leq H(1 + \log H)$, its length is a fixed fraction of $V_{\text{old}} + L_{\text{head}} + L_{\text{tail}} = T$. The largest of its at most H original blocks has length at least $c_H T/H$; choosing the earliest one resolves ties deterministically. $\square$

This suffix supplies an unread reservoir for the one uncolored root count. It does not impose a within-block terminal path event.

G.8 Physical prefix guards

Keep the original groups, including old groups, indexed by $g = 1, \dots, m$, $m \leq H$, with capacities r_g and transported coefficients κ_g . Zero-length original groups may retain their indices; the coefficients are those of the same fixed complete-chain optimizer. For $\mu_g = \kappa_g(r_g + z)L_g$ and $\beta_g = z/(r_g + z)$, fix actual total and child counts in the own-group windows

$$|n_g - \mu_g| + |k_g - \beta_g \mu_g| \leq B\sqrt{1 + \mu_g}. \quad (\text{G.30})$$

The child positions and the old word may also be fixed. Define

$$w_g = \log(r_g + z)n_g - \log z k_g - r_g L_g,$$

$$\begin{aligned}\zeta_g &= \log(r_g + z)(n_g - \mu_g) - \log z(k_g - \beta_g \mu_g), \\ Y &= \sum_g w_g = \sum_g \zeta_g, \quad b_0 = a_0 - Y, \quad Z_l = \sum_{g>l} \zeta_g.\end{aligned}\tag{G.31}$$

The equality for Y is reference balance, not conditional expectation. The entrance a_0 is the actual entrance, with all its dependence on the old word retained.

Lemma G.10 (Clipped positive prefixes). *Let m_l be the least reference cost of a lawful endpoint rank profile positive through group l and zero thereafter; set $m_0 = 0$. Then*

$$m_l = \min_{1 \leq R \leq r_a} \sum_{g \leq l} L_g g_g(\min(r_g, R)) \geq c_H \min(V_l, T - V_l), \quad m_m = 0.\tag{G.32}$$

If $a_0, b_0 \geq -h_0$, then, for each fixed $\theta > 0$,

$$b_0 + Z_l + \theta m_l \geq -h_{\text{bin}} \quad (0 \leq l \leq m),\tag{G.33}$$

where h_{bin} depends only on H, B, h_0, θ .

Proof. Apply original-group reverse compression to the positive prefix, starting the benchmark at one instead of zero. Every update still preserves the already processed comparisons: for old groups its threshold is $1/\kappa_{\text{current}}$, with larger coefficients already processed, and for native groups it is the same secant comparison as before. Thus every positive rank profile dominates some clipped profile. Conversely, retain the deepest R eligible names, delete each on expiry, and cut the rest at V_l . This realizes the clipped prefix lawfully and proves the equality in (G.32). Each candidate is nonnegative by the full-history certificate; no stationarity for a truncated system is invoked.

For fixed R put $k_g = \min(r_g, R)$ and

$$f_g = g_g(k_g) = k_g(1 - S_g^{t_g}), \quad t_g = \frac{\log(\kappa_g E_{k_g}(z)/k_g)}{\log S_g}, \quad S_g = r_g + z.$$

Across old groups and the first native transition, $t_{g+1} - t_g \geq d_*/\log H$. At native capacity h write

$$t_g = \sigma - T_h(R), \quad T_h(R) = 1 - \frac{\log(E_{\min(h, R)}(z)/\min(h, R))}{\log(h + z)}.$$

For $h \leq R$, Lemma G.1 gives $\partial_h T_h(R) \geq 1/(24H^2)$; for $h \geq R$,

$$\partial_h T_h(R) = \frac{\log(E_R/R)}{(h + z) \log^2(h + z)} \geq \frac{\log(2 \log 2)}{H \log^2 H}.$$

The function is continuous at R , and native capacities fall by positive integers. Therefore every original transition raises t_g by at least

$$d_0 = \min \left\{ \frac{d_*}{\log H}, \frac{1}{24H^2}, \frac{\log(2 \log 2)}{H \log^2 H} \right\} > 0.$$

Set $c_H = 1 - 2^{-d_0/3}$. If $f_g \leq c_H$, then $t_g \geq -d_0/3$; hence every later $f_v \leq 1 - 2^{2d_0/3} \leq -c_H$. Thus either all prefix coefficients exceed c_H , or all tail coefficients are at most $-c_H$. In the first case its cost is at least $c_H V_l$; in the second the retained multiplier gives

$$\sum_{g \leq l} L_g f_g = \Lambda_R - \sum_{g > l} L_g f_g \geq c_H (T - V_l).$$

Minimizing proves (G.32). At $l = m$, $R = r_a$ gives zero.

Own-group centrality implies for every consecutive group set I

$$\left| \sum_{g \in I} \zeta_g \right| \leq C_B \sqrt{1 + \sum_{g \in I} L_g}.$$

Use both exact expressions for the same remainder,

$$b_0 + Z_l = b_0 + \sum_{g > l} \zeta_g = a_0 - \sum_{g \leq l} \zeta_g.$$

Their lower bounds give $-h_0 - C_B \sqrt{1 + D_l}$, where $D_l = \min(V_l, T - V_l)$. The single inequality $C_B \sqrt{D_l} \leq \theta c_H D_l + C_B^2 / (4\theta c_H)$ proves (G.33), with $h_{\text{bin}} = h_0 + C_B + C_B^2 / (4\theta c_H)$. $\square$

Only original physical endpoints occur in this lemma. Splitting a weak block artificially would destroy the strict increase of t_g . Its actual full-stop row is

$$C_{\text{stop}}(l) = b_0 + \sum_{g > l} w_g = a_0 - \sum_{g \leq l} w_g = C_{\text{stop}}^{\text{nat}}(l) + b_0 + Z_l, \quad C_{\text{stop}}^{\text{nat}}(l) = \sum_{g \leq l} L_g g_g(r_g). \quad (\text{G.34})$$

In particular the two extreme rows require the actual bounds on a_0, b_0 ; fresh exterior names cannot repair them.

It is useful to see why a carry bound alone is insufficient here. At an internal native endpoint j , let n_0 be the incoming count, C_j^{route} the count routed to the next child, and K_j^{past} the permanent-child count in all old and native groups through j . Then

$$\begin{aligned} T_j &= \log S_a n_0 + \sum_{i=a}^j \log S_i n_i - \log S_{j+1} C_j^{\text{route}} - \sum_{l=a}^j V_l, \\ W_j &= W_{\text{in}} + T_j - \sum_{l=a}^j y_l, \\ H_j &= \log S_{j+1} C_j^{\text{route}} - \log z K_j^{\text{past}} - r_{j+1} V_j. \end{aligned}$$

Direct cancellation yields

$$\sum_{g \leq j} w_g = T_j + H_j, \quad C_{\text{stop}}(j) = a_0 + W_{\text{in}} - W_j - \sum_{l=a}^j y_l - H_j. \quad (\text{G.35})$$

Fresh naming changes T_j, H_j oppositely. The old routed-child risk in H_j is not a later native wall. Lemma G.10 absorbs this remainder directly, using centrality of each old physical group; an aggregate old window would not suffice at an old internal endpoint.

G.9 A common count box for every endpoint

Proposition G.11 (Displaced exterior counts). *Fix all original clocks, total and permanent-child counts, child binary words, and old prerefinement data, with (G.30). Assume the natural full balance and nonnegativity of every completed named-star reference history. Conditional on this data, fresh exterior labels in group g are independent uniform labels from its nested eligible set D_g ,*

of size r_g . There is an event of conditional probability at least $c_{\text{box}} > 0$ on which, simultaneously for every completed endpoint history v ,

$$C_v \geq (1 - \theta)\overline{C}_v + b_0 + Z_{l(v)} + T_{\text{good},v} - C_\theta. \quad (\text{G.36})$$

Here $0 < \theta < 1$ is fixed, $l(v)$ is the last positive-rank group, and $T_{\text{good},v}$ is a nonnegative sum of square-root exterior counts defined in the proof. If the guards $b_0 + Z_l + \theta' m_l \geq -h_{\text{bin}}$ hold, the constants can be chosen so that the same event satisfies

$$C_v \geq c_0 \overline{C}_v - h_{\text{bin}} - C \quad \text{for every } v \quad (\text{G.37})$$

for any fixed $c_0 < 1$. For every nonnegative prerefinement-measurable weight F_0 , $\mathbb{E}(F_0 \mathbf{1}_{\text{box}}) \geq c_{\text{box}} \mathbb{E} F_0$.

Proof. Write $N_g = n_g - k_g$ for the exterior count and n_{gc} for its count of name c . Let c_g be the shallowest eligible name. A proper row of rank $0 < q < r_g$ is called good on this group if it omits c_g , and bad if it retains c_g . A bad row has a missing deeper name d with later deadline. Its cut is no later than the left endpoint of the group, whereas c_g is retained through the right endpoint. Move the cut of c_g to that of d , and, in a second comparison, move that of d to the cut of c_g . Both histories are lawful. The exact second difference is at least $\kappa_g L_g / H$, so reference nonnegativity gives $\overline{C}_v \geq \kappa_g L_g / (2H)$ for each bad group. There are at most H such groups, hence

$$\overline{C}_v \geq \frac{\kappa_*}{2H^2} \sum_{g \text{ bad}} L_g. \quad (\text{G.38})$$

For a proper rank- q row retaining A_g , exact expansion of its entropy costs gives

$$C_v = b_0 + \overline{C}_v + Z_{l(v)} + \sum_{g \text{ proper}} e_{vg} - \sum_{g \text{ proper}} \log(q + z) \sum_{c \in A_g} (n_{gc} - N_g / r_g), \quad (\text{G.39})$$

where

$$e_{vg} = \left[\log S_g - \frac{q}{r_g} \log(q + z) \right] (n_g - \mu_g) - \left(1 - \frac{q}{r_g} \right) \log(q + z) (k_g - \beta_g \mu_g).$$

There is no fresh-name error for a full or empty row. By centrality, $|e_{vg}| \leq C_B(1 + \sqrt{N_g})$ and $N_g \leq C_B(1 + L_g)$.

Choose a fixed small $\epsilon > 0$ and then a large fixed K . For sufficiently large N_g , require every nonshallow count to lie in the interval

$$n_{gc} = N_g / r_g - K \sqrt{N_g} + [-\epsilon, \epsilon] \sqrt{N_g} \quad (c \neq c_g), \quad (\text{G.40})$$

with the shallow count determined by the total. A good proper row then gains $qK\sqrt{N_g} + O_H(\epsilon\sqrt{N_g})$, while a bad row loses at most $C_H K \sqrt{N_g}$. Choosing K after B makes the gain pay the good binary errors and leaves $\sqrt{N_g}$ on every sufficiently large good group. Define their sum to be $T_{\text{good},v}$. For large N_g these boxes are interior, contain $\gg_H N_g^{(r_g-1)/2}$ lattice points, and each point has multinomial mass $\asymp_H N_g^{-(r_g-1)/2}$ by Stirling's formula with all coordinates in fixed central windows. Their probability is therefore bounded below. On bounded groups choose a supported balanced type; its minimum positive probability is a fixed constant. A group with $r_g = 1$ has no exterior choice. The finitely many independent group boxes thus have conditional probability at least c_{box} .

The total remaining bad error is at most $C_K \sum_g \text{bad} \sqrt{1 + L_g}$. Apply Young's inequality once and use (G.38) to bound it by $\theta \bar{C}_v + C_\theta$. Substitution in (G.39) proves (G.36) simultaneously for the finite family. Taking $\theta = \theta' = (1 - c_0)/2$ and using $\bar{C}_v \geq m_{l(v)}$ yields (G.37). More economically, the guard can use $\min_{v:l(v)=l}(\theta \bar{C}_v + T_{\text{good},v})$, which is at least θm_l . Finally the probability bound holds for every fixed unread environment; conditioning proves the weighted assertion. It applies to the whole type event before any fractional power. The input likelihood, fixed by the original totals and binary child words, is unchanged. $\square$

A named regularity event whose conditional failure probability is at most $c_{\text{box}}/2$ for every fixed child word may be intersected with this box. The weighted assertion does not permit a weight that reads the fresh exterior labels.

Lemma G.12 (Monotone transport to the box). *Fix the same original clocks and child binary words. Let the source be any subset of exterior count types with $|n_{gc} - N_g/r_g| \leq K_{\text{src}} \sqrt{1 + N_g}$. There is a target distribution of constant conditional mass in a fixed wider central window, satisfying the endpoint box, and a coupling from each source type to each target type that changes only nonshallow names into the shallowest name. For every nonnegative functional increasing under these replacements,*

$$\mathbb{E}[\mathbf{1}_{\text{target}} \Phi] \geq c_{\text{tar}} \mathbb{E}[\mathbf{1}_{\text{source}} \Phi]. \quad (\text{G.41})$$

Proof. Choose (G.40) with $K > \sqrt{2}K_{\text{src}} + \epsilon$, enlarged as required for the box. Every target nonshallow count m_c is then at most its source count n_c . For bounded $N < N_0$ choose the all-shallow target, of probability at least H^{-N_0} ; this lies in a fixed wider central window. The large-count Stirling estimate proves target mass at least $c_{\text{tar}} > 0$. Given a uniform source word of type n , retain a uniformly selected m_c of the n_c occurrences of each nonshallow c , and change the rest to the shallow name 0. The resulting word is uniform of type m . Indeed a fixed target word has $m_0!/[n_0! \prod_{c \neq 0} (n_c - m_c)!]$ source preimages; multiplying by the source word probability $n_0! \prod_{c \neq 0} n_c! / N!$ and the retention probability $\prod_{c \neq 0} \binom{n_c}{m_c}^{-1}$ gives $m_0! \prod_{c \neq 0} m_c! / N!$, its required uniform probability. All original slots, times and child marks stay fixed. Monotonicity therefore gives $\mathbb{E}(\Phi \mid m) \geq \mathbb{E}(\Phi \mid n)$ for every source and target pair. Average over the normalized source acceptance, then over target types and use source mass at most one. This proves (G.41); truncation handles unbounded Φ . The target endpoint support follows from Proposition G.11 and the already fixed two-entrance guards. No within-block path or filtered regularity assertion is part of this transport. $\square$

G.10 The strict-exit packet on original counts

Proposition G.13 (Diagonal strict exits). *Let maximal platforms be indexed in decreasing products $\sigma_1 > \dots > \sigma_M$, with $u_j = \sigma_{j+1}/\sigma_j$ and $d_j = 1/u_j - 1$. Let T_j be the native length of proper platform j , and put $V_j = \sum_{a \leq j} T_a$. Fix $C_y \geq 0$ and target numbers $0 \leq y_j \leq C_y \log(2 + T_j)$ for $j < M$, held fixed throughout the count selection and transport. For grouped platform outputs Z_j , define $\eta_j = \sum_{a \leq j} (\sigma_a/\sigma_j)(Z_a - y_a)$ for $1 \leq j < M$. Suppose $V_j \leq C_H T_j$ and $T_j \geq T_0(H, C_y)$ for the proper platforms in question. Conditional on admitted original uncolored count windows, there is a count-only acceptance, in fixed full-leaf central windows, of mass*

$$\asymp_{H, C_y} \prod_{j < M} \frac{R_j}{\sqrt{T_j}}, \quad R_j = \max\{R_0, \min(\sqrt{T_j}, d_j^{-1})\}, \quad (\text{G.42})$$

on which all strict carries satisfy $aR_j \leq \eta_j \leq bR_j$ for fixed $0 < a < b$. The constants and central windows depend only on H, C_y . This acceptance can be transported through the proper endpoint boxes, with a further factor depending only on H, C_y and without changing the carries.

Proof. Choose in each proper platform an original block B_j of length at least T_j/H . In it choose its current shallow name t_j and the first shallow fine leaf c_j of the immediate child. Set $A_j = \log S_{a_j}$ and $b_j = u_j A_{j+1} = \log z_j$. These are uniformly bounded above and away from zero. The two original probabilities obey $p_{c_j}/p_{t_j} = S_{\text{child}}^{u_j-1}$ and both lie in a fixed interior probability interval. Coarsen just the pair $\{c_j, t_j\}$, fixing its total count m_j , all coarsened words and all other leaf words. Its unread split count K_j , counting t_j , has the original binomial law $\text{Bin}(m_j, p_j)$, independently for different j .

For $2 \leq j < M$, these carries satisfy $\eta_j = Z_j - y_j + \eta_{j-1}/u_{j-1}$. Changing one c_j to t_j raises Z_j by b_j , lowers Z_{j+1} by A_{j+1} and changes no other grouped output. The expansion

$$\eta_l = \sum_{a \leq l} \frac{\sigma_a}{\sigma_l} (Z_a - y_a)$$

therefore cancels the two effects for $l > j$, giving the simultaneous exact diagonalization

$$\eta_j = h_j + b_j K_j. \quad (\text{G.43})$$

There is a fixed central coarsening event of constant probability on which all coarsened categories and all other original-block leaf counts lie in their own-scale central windows. It leaves all K_j unread. On it $m_j \asymp_H T_j$ and $|h_j + b_j m_j p_j| \leq C_{H, C_y} \sqrt{T_j}$. To verify the last estimate, the natural grouped proper output has mean zero at a strict boundary; the sum of finitely many own-group errors through j is $O_H(\sqrt{1 + V_j})$. Since $T_a \leq V_j \leq C_H T_j$ for $a \leq j$, the weighted sum of targets through j is $O_{H, C_y}(\log(2 + T_j))$ and is absorbed by the displayed square-root bound.

Choose R_0 large enough that $(b - a)R_0 > 2 \sup_j b_j$, then $T_0 = T_0(H, C_y) \geq R_0^2$ large enough for all resulting binomial windows to be interior. For each coarsening, the condition $aR_j \leq h_j + b_j K_j \leq bR_j$ is an interval of $\asymp_{H, C_y} R_j$ integers within $C_{H, C_y} \sqrt{m_j}$ of its mean. Each such atom has mass $\asymp_{H, C_y} m_j^{-1/2}$, so independence of the unread split counts proves (G.42). The full fine-leaf counts still lie in fixed enlarged central windows. The event is an explicit polytope of original counts with the full band mass just computed.

Its actual entrances are $a_{0j} = y_j + \eta_j \geq 0$ for $j < M$, with $b_{01} = 0$ when $M > 1$ and $b_{0j} = \eta_{j-1}/u_{j-1} \geq 0$ for $2 \leq j < M$. Apply Lemma G.12 successively to the finitely many proper owners. Each replacement preserves that owner's permanent-child word, all preceding coarse words and the finer child word, hence every Z_j and η_j . The source acceptance remains inside the linear integral at each step. The resulting subprobability measure has mass reduced only by a constant depending on H, C_y ; its bounded density can be normalized to an acceptance in $[0, 1]$. This is an unfiltered count transport. A later path condition is imposed only after this count source has been built. Finally $0 \leq d_j \eta_j \leq C_{H, C_y}$ on the bands, so the exact full-word likelihood ratios in (G.9) are bounded above and below; the native products themselves have not been replaced or perturbed. $\square$

A weight measurable with respect to the coarsened words can be retained in the conditional diagonal calculation. A weight reading the unread K_j or their paths cannot be inserted at that stage.

G.11 Local endpoint representatives and terminal admission

Lemma G.14 (A local representative of a common endpoint pair). *Suppose the actual minima of the two endpoints on original block i satisfy, throughout an admitted whole-count support,*

$$aX_i \leq x_i \leq bX_i, \quad aY_i \leq y_i \leq bY_i, \quad y_i - x_i = \Delta_i(K_i),$$

where $0 < a \leq b$ are fixed, $X_i, Y_i > 0$ are deterministic scores, and K_i is the original local binary count. Then

$$x_i^-(K_i) = \max\{aX_i, aY_i - \Delta_i(K_i)\}, \quad y_i^-(K_i) = x_i^-(K_i) + \Delta_i(K_i) \quad (\text{G.44})$$

satisfy the same scale bounds from below and are no greater than the actual minima coordinatewise. They have exactly the original endpoint difference and read only the local count (and its fixed original total).

Proof. The actual x_i is at least both entries in the first maximum; adding its fixed difference proves the right-end comparison. The second maximum is equivalently $\max\{aX_i + \Delta_i(K_i), aY_i\}$, which proves its lower bound. For a native edge the exact difference is

$$\Delta_i(K_i) = (r_i - q_i)L_i - n_i \log S_i + K_i \log B_i.$$

Every outside context uses this same difference. Its centered empirical bridge is also the same original binary bridge $E_i(t)$. Thus the chord between (G.44) lies below every actual endpoint chord. If $\theta > 0$ is the fixed allocation fraction, the sufficient wall $\theta \ell_i^-(t) + E_i(t) \geq g d_i(t)^\gamma$ (or its fixed intercept version) implies all corresponding allocated walls on the admitted count support. Choose the fixed shift in the scores so that their lower bounds exceed the scalar reserve threshold. The scalar all-reserve theorem then evaluates this local pair on the same scale, using its required interior type and compact original clock. For bounded blocks use the admitted finite support. A deterministic-mark bridge uses the ordinary scalar theorem rather than a zero-variance substitution in the binary theorem. $\square$

This construction avoids choosing two independent corners of an endpoint rectangle. The path event alone does not assert the count support on which the comparisons hold.

Lemma G.15 (Forgetting private terminal names). *Let π be the genuine full-leaf word law after fixing original uncolored totals. Write C for the complete coarse word retaining all proper names and treating the entire last platform as one compound symbol, and V for its private refinement. Then*

$$\pi(dC, dV) = \pi_c(dC) \pi_t(dV \mid C).$$

For a count-only acceptance $0 \leq A(C, V) \leq 1$ of mass W , its conditional average $A_c(C) = \mathbb{E}_{\pi_t}[A(C, V) \mid C]$ has the same mass. If A is invariant under within-block permutations, A_c depends only on coarse counts. Every condition depending only on C that holds on the source still holds where $A_c > 0$.

Proof. Conditional terminal labels are uniform among genuinely eligible names at each original slot, independently between slots. Applying the same within-block permutation to C and V preserves this conditional law, proving the count assertion. The mass identity is conditional expectation. If a coarse condition fails at C , the original acceptance vanishes for almost every refinement there, so its average also vanishes. The new measure is $A_c(C) \pi(dC, dV)$, leaving V at

its genuine conditional law. Thus proper endpoints, strict exits, proper canonical binary types, own-group coarse centrality and proper coarse paths survive. An old test or arbitrary weight of the private V need not survive. After a count transport this averaging is done after its bounded-density normalization, and introduces no further mass loss or fractional power. $\square$

Proposition G.16 (Prefix source before the unread suffix). *Condition on admitted own-scale original uncolored totals before the terminal suffix of Proposition G.9. There is a count-only prefix acceptance $0 \leq A_{\text{pre}} \leq 1$, independent of the uncolored counts and paths in that suffix, of mass*

$$W \asymp_H \prod_{j < M} w_j, \quad w_j = R_j / \sqrt{T_j}, \quad (\text{G.45})$$

with all proper strict-exit bands and endpoint boxes, all admitted proper binary types, and own-group terminal-prefix central and exterior count windows. Its original full-leaf type law is retained. The constants can be fixed uniformly for every suffix count vector in the central lower construction of the root theorem.

Proof. The diagonal packet and each proper-owner count transport read only blocks through their proper owner. In particular they read no terminal native block and can be formed with zero common target before any suffix count. Their mass is (G.45). Apply Lemma G.15 to forget the private terminal names; this preserves the mass and the proper coarse conditions. The absence of suffix *uncolored* count dependence follows from the preceding construction, not merely from this marginalization. Refine the terminal lists already present in the prefix with their genuine conditional uniform law. Require permanent-0 counts in fixed sufficiently wide own-group central windows, and then impose the displaced exterior count boxes on these prefix groups. Binomial centrality and Proposition G.11 give a fixed conditional factor on each of at most H groups. These restrictions do not affect any proper coarse condition, and read no newly sampled clock. The resulting acceptance has the asserted mass. All constants are chosen to cover the uniform central windows of the root construction, so this construction works for each of its later admitted root count vectors. Within every exact full-leaf type the bridge words retain their original uniform permutations. $\square$

Proposition G.17 (Terminal counts after the one root atom). *Use the zero common-target prefix source above and the genuine uncolored Euler root band $-r_* W_0 \leq X_1 \leq 0$. After the one suffix reservoir count has been selected by the root theorem, the terminal endpoint box can be imposed with a conditional probability bounded below uniformly for every admitted coarse prefix and root count vector. It changes neither that root count nor any proper coarse word. With $a_0 = 0$ its terminal entrance is*

$$b_0 = \eta_{\text{prev}}/u_{\text{prev}} - X_1/r_* \geq 0, \quad 1 + b_0 \asymp_H 1 + R_{\text{prev}}/u_{\text{prev}}. \quad (\text{G.46})$$

For an absent proper predecessor set $R_{\text{prev}} = 0$. The admitted rows satisfy

$$C_v \geq b_0 + \zeta_{\text{tail},v} + \frac{3}{4}\overline{C}_v + G_v - C_{\text{box}}, \quad (\text{G.47})$$

where G_v is the good-group square-root gain.

Proof. The exact carry relation for the whole terminal platform is $Z_{\text{term}} = X_1/r_* - \eta_{\text{prev}}/u_{\text{prev}}$. Thus $a_0 = 0$, $b_0 = -Z_{\text{term}}$ give (G.46); the root error is bounded and the proper exit is in its positive band. Condition now on the coarse prefix and every original uncolored count, including

the already chosen root reservoir count. Each old group's actual entering terminal total is in its own-scale coarse window. Refine terminal names by the genuine conditional law of Lemma G.15. Require the permanent-name 0 count in a sufficiently wide central interval. Binomial Chebyshev bounds give a fixed simultaneous probability, using the supported finite convention on bounded groups. With these counts and positions fixed, impose Proposition G.11. Its full-stop guards follow deterministically from Lemma G.10 and the two nonnegative entrances. For terminal $z = 1$, each full-group logarithmic sum is $\log(r_g + 1)n_g$, independent of all refinements. Hence neither Z_{term} , the uncolored root, nor b_0 changes. Proper binary words are coarse and remain fixed. The stronger row (G.36), with $\theta = 1/4$, is exactly (G.47). This is a count admission; terminal within-block paths are imposed subsequently. $\square$

G.12 Actual endpoint scores and their intrinsic normalization

We now distinguish the raw local cost from the cost that enters a native wall. This distinction matters when a platform inherits a large entrance from its predecessor. For a named endpoint star template v , let $A_{v,g}$ be its retained exterior names in group g , $q_g = |A_{v,g}|$, and $l = l(v)$ its last positive-rank group. Its reference cost is

$$\bar{C}_v = \sum_g L_g g_g(q_g).$$

Let $\mathcal{B}_v$ and $\mathcal{G}_v$ be respectively its bad and good proper groups, as in Proposition G.11, and put

$$L_{\text{bad}} = \sum_{g \in \mathcal{B}_v} L_g, \quad L_{\text{zero}} = T - V_l, \quad S_{\text{good}}(v) = \sum_{g \in \mathcal{G}_v} \sqrt{L_g}.$$

All other groups are full and incur no exposed-name discrepancy. The reference estimates and the displaced box give, simultaneously,

$$\begin{aligned} \bar{C}_v &\geq a_H L_{\text{bad}}, & \bar{C}_v &\geq b_H \min(V_l, T - V_l), \\ C_v &\geq b_0 + Z_l + \frac{3}{4} \bar{C}_v + G_v - C_{\text{box}}, & G_v &\geq c_H S_{\text{good}}(v) - C_H. \end{aligned} \quad (\text{G.48})$$

The last comparison follows from own-scale centrality: on a large group $N_g \geq c_H L_g$, whereas bounded N_g forces bounded L_g .

Lemma G.18 (Early and late corner payment). *On this single box, every named endpoint star corner obeys*

$$V_l \leq T/2 : \quad C_v \geq a_0 + c_H V_l - C_H, \quad (\text{G.49})$$

$$V_l \geq T/2 : \quad C_v \geq b_0 + c_H L_{\text{zero}} + c_H L_{\text{bad}} + c_H S_{\text{good}}(v) - C_H. \quad (\text{G.50})$$

The all-zero and complete-deadline corners have costs exactly a_0 and b_0 , respectively.

Proof. In the early case use $b_0 + Z_l = a_0 - \sum_{g \leq l} \zeta_g$. Its error is $O(\sqrt{1 + V_l})$ and is absorbed once by the positive prefix charge in (G.48), retaining a fixed fraction of V_l . In the late case use $b_0 + Z_l = b_0 + \sum_{g > l} \zeta_g$; only the zero-rank tail has a square-root error. Split the $3\bar{C}_v/4$ budget into two equal pieces for L_{bad} and L_{zero} , absorb that error with half the latter piece, and retain the good-group gain. The extreme identities follow directly from (G.34). The same proof, with a fixed constant, includes $T = 0$. $\square$

Here is the exact cancellation behind two-sided scores. With $N_g(A)$ the exterior count in A , write

$$H_g = \log S_g n_g - \log z k_g, \quad H_g(A) = \log(q_g + z)[k_g + N_g(A)] - \log z k_g.$$

For an empty group $H_g(A) = 0$; for a full group it is H_g exactly. The notation $\overline{H}_g$ below means the deterministic natural reference potential, including unconditioned old routing. Set

$$\nu_{vg} = [H_g - H_g(A_{v,g})] - [\overline{H}_g - \overline{H}_g(A_{v,g})].$$

Then $\nu_{vg} = 0$ identically on every full group, and direct expansion of $C_v = a_0 + \sum_{g \leq l} [q_g L_g - H_g(A_{v,g})]$ gives

$$C_v = a_0 - \sum_{g \leq l} \zeta_g + \overline{C}_v + \sum_{g \text{ proper}} \nu_{vg} = b_0 + \sum_{g > l} \zeta_g + \overline{C}_v + \sum_{g \text{ proper}} \nu_{vg}. \quad (\text{G.51})$$

Every repaired named count is within $C_H \sqrt{1 + L_g}$ of its own natural group mean. Therefore

$$|\nu_{vg}| \leq C_H \sqrt{1 + L_g}, \quad \left| \sum_{g \in J} \zeta_g \right| \leq C_H \sqrt{1 + \sum_{g \in J} L_g} \quad (\text{G.52})$$

for consecutive J . In the late formula the discrepancies of large full groups have canceled, rather than contributing a spurious $\sqrt{T}$ error.

Lemma G.19 (Raw finite scores). *Suppose $a_0, b_0 \geq 0$ lie in fixed comparable intervals of deterministic scales $A_{\text{in}}, B_{\text{in}}$, respectively, allowing scale zero for an identically zero entrance. For a fixed sufficiently large h , set*

$$S_v^{\text{raw}} = \begin{cases} h + A_{\text{in}} + \overline{C}_v, & V_l \leq T/2, \\ h + B_{\text{in}} + \overline{C}_v + S_{\text{good}}(v), & V_l > T/2. \end{cases}$$

Then $c_H S_v^{\text{raw}} \leq C_v + h \leq C_H S_v^{\text{raw}}$ simultaneously on every admitted count type.

Proof. For an early corner all proper groups are in the prefix, and (G.51)–(G.52) give $C_v \leq a_0 + \overline{C}_v + C_H \sqrt{1 + V_l}$. Since $V_l \ll_H \overline{C}_v$, this is $a_0 + C'_H \overline{C}_v + C'_H$. For the lower bound use the first expression for $b_0 + Z_l$ in (G.48), discard $G_v \geq 0$, and absorb its square root with at most $\overline{C}_v/4$ to get

$$C_v \geq a_0 + \frac{1}{2} \overline{C}_v - C_H. \quad (\text{G.53})$$

For a late corner, the stronger box row similarly gives

$$C_v \geq b_0 + \frac{1}{2} \overline{C}_v + c_H S_{\text{good}}(v) - C_H. \quad (\text{G.54})$$

The tail formula gives an upper bound by $b_0 + \overline{C}_v$ plus the square roots of the zero tail, bad groups and good groups. The first two are bounded by $C_H(1 + \overline{C}_v)$; the last are $C_H S_{\text{good}} + C_H$. A fixed large h completes both comparisons, including the literal extreme rows. $\square$

These raw scores describe C_v . They are not the native reserve scores, which must account for the normalization in the full cost.

For later use we spell out how the raw payment interacts with exact competitor classes. Let A be a finite nonnegative complete-type weight in the own-group central support, $W = \mathbb{E}_\pi A$, and let $\mathcal{E}_v = \mathcal{B}_v \cup \mathcal{G}_v \cup \{g > l(v)\}$ be its exposed groups. Full groups have one cell containing all

eligible names. The exact count-reveal estimate (H.1) therefore gives, in the original class of the retained anchor,

$$W_v := \mathbb{E}_\pi[A \mid \text{Class}_v] \leq C_H W \prod_{g \in \mathcal{E}_v} (1 + n_g)^{(a_g - 1)/2}. \quad (\text{G.55})$$

There is no full-group count factor and no class-probability factor. If a predeclared selected-bridge cost satisfies $D_g \geq c_D(1 + n_g)^{-A_D}$ (and $D_g = 1$ on unselected groups), then (G.50) pays every polynomial in (G.55) on a late corner, giving

$$e^{-C_v} W_v \leq C_H e^{-b_0} W \prod_{g \in \mathcal{E}_v} D_g. \quad (\text{G.56})$$

Indeed $n_g \ll_H 1 + L_g$, and a positive fraction of L_g on a bad or zero-tail group, or of $\sqrt{L_g}$ on a good group, dominates any fixed multiple of $\log(1 + L_g)$ up to a fixed constant. The same argument works for any fixed positive fraction of $C_v - b_0$. Under the same-class bank hypotheses of (H.9), apply the bank theorem with every $\mathcal{E}_v$ slot exposed and depth zero elsewhere. Its bounds $q_{g,0} \leq C_H D_g$ then give, for an acceptance $0 \leq F \leq A$ with the stated path tests,

$$e^{-C_v} \mathbb{E}_\pi[F \mid \text{Class}_v] \leq C_H e^{-b_0} W \prod_{g \text{ selected}} D_g. \quad (\text{G.57})$$

An early corner instead uses the literal bound $q_v = \mathbb{E}_\pi[F \mid \text{Class}_v] \leq 1$ for $0 \leq F \leq 1$. It yields $e^{-C_v} q_v \leq C_H e^{-a_0} e^{-c_H V_i}$. In the normalization $a_0 = Y, b_0 = 0$, an independently verified same-family target condition

$$e^{-Y} \leq C_{\text{target}} D_{\text{global}}, \quad D_{\text{global}} = W \prod_{g \text{ selected}} D_g, \quad (\text{G.58})$$

pays all early corners. Combining with (G.57), and the dimension-bounded number of endpoint templates, gives for fixed $\delta > 0$

$$\sum_v e^{-\delta C_v} q_v^\delta \leq C_\delta D_{\text{global}}^\delta.$$

Condition (G.58) is an additional target input, not a consequence of the count box. In particular a short prefix is never asked to pay count factors from an unrelated long zero-rank tail. The type envelope A may be unbounded as long as the actual early-corner acceptance F is bounded by one.

Theorem G.20 (Intrinsic endpoint reserve). *Let V_{old} be the total original old length and assume*

$$a_0, b_0 \geq 0, \quad b_0 \leq B\sqrt{1 + V_{\text{old}}}. \quad (\text{G.59})$$

Define the intrinsic cost

$$K_v = Y - I_v = C_v - b_0 \quad \text{when } C_v = a_0 - I_v, \quad Y = a_0 - b_0.$$

For every lawful endpoint template with $t_v = V_{l(v)} \geq V_{\text{old}}$, set

$$S_v^{\text{intr}} = \begin{cases} h + A_{\text{in}} + \bar{C}_v, & t_v \leq T/2, \\ h + \bar{C}_v + S_{\text{good}}(v), & t_v > T/2. \end{cases} \quad (\text{G.60})$$

For a fixed sufficiently large h , uniformly on the same count support,

$$c_H S_v^{\text{intr}} \leq K_v + h \leq C_H S_v^{\text{intr}}, \quad K_v \geq c_0 \bar{C}_v - C_0. \quad (\text{G.61})$$

The assertion also holds for terminal $z = 1$ using Proposition G.17; no proper-child branching gap is needed for it.

Proof. If $V_{\text{old}} \leq t_v \leq T/2$, the prefix reference charge and (G.59) give, by one Young inequality, $b_0 \leq B\sqrt{1+t_v} \leq \bar{C}_v/4 + C_B$. Subtracting b_0 from (G.53), while retaining a_0 with coefficient one, yields

$$K_v \geq a_0 + \frac{1}{4}\bar{C}_v - C_B.$$

Its upper bound follows from the exact prefix formula and $b_0 \geq 0$: $K_v \leq a_0 + \bar{C}_v + C_H\sqrt{1+t_v} \leq a_0 + C'_H\bar{C}_v + C'_H$. The case $t_v = V_{\text{old}} = 0$ is covered by the fixed constant. For $t_v > T/2$, subtract b_0 in the exact tail formula itself:

$$K_v = Z_{l(v)} + \bar{C}_v + \sum_{g \text{ proper}} \nu_{vg}.$$

The proof of (G.54) now gives directly $K_v \geq \bar{C}_v/2 + c_H S_{\text{good}} - C_H$; its upper proof gives $K_v \leq C_H(\bar{C}_v + S_{\text{good}}) + C_H$. Only the zero-tail and bad-group errors are absorbed, and full-group errors cancel exactly. These inequalities prove the theorem after increasing h once. $\square$

On the zero-target packet, $b_0 = \eta_{\text{prev}}/u_{\text{prev}}$ satisfies (G.59): the preceding native length is at most V_{old} , its exit scale is bounded by a fixed multiple of its square root, and u_{prev} is bounded below. Terminal root rounding adds only a fixed constant. The complete-deadline template has $K_v = 0$ and intrinsic score h . The all-zero template has $K_v = a_0 - b_0$ and is excluded when $V_{\text{old}} > 0$; that exclusion is essential. The late score in (G.60) contains no B_{in} or preceding exit reserve.

Corollary G.21 (Legal native pairs). *For a genuine native synchronous edge in block $i = [l_i, b_i]$ having at least one exterior cut, retain its lawful outside endpoint contexts. Let v^-, v^+ denote their left and right completions and set*

$$\begin{aligned} X_i^{\text{intr}} &= \min_v S_{v^-}^{\text{intr}}, & Y_i^{\text{intr}} &= \min_v S_{v^+}^{\text{intr}}, \\ x_i^{\text{intr}} &= \min_v (K_{v^-} + h), & y_i^{\text{intr}} &= \min_v (K_{v^+} + h). \end{aligned}$$

These actual minima are in the corresponding common scale boxes. For a canonical full-to-deepest- q edge they satisfy

$$y_i^{\text{intr}} - x_i^{\text{intr}} = \Delta_i = (r_i - q)L_i - \log S_i n_i + \log(q + z)K_{i,B}. \quad (\text{G.62})$$

A sufficient local pair, reading only this original local type, is

$$x_i^- = \max\{c_H X_i^{\text{intr}}, c_H Y_i^{\text{intr}} - \Delta_i\}, \quad y_i^- = x_i^- + \Delta_i. \quad (\text{G.63})$$

Proof. The actual name cut in $A \setminus B$ ensures every endpoint template has $t_v \geq l_i \geq V_{\text{old}}$, including a legal cut at the old/native boundary. Theorem G.20 applies to every member, so taking minima proves the common boxes. Moving the synchronous cut across block i changes only its original rank and logarithmic count contribution. This is exactly (G.62), independent of outside context. Subtracting the same b_0 from both raw endpoints changes no increment; therefore the minimum right value equals the minimum left value plus Δ_i . Lemma G.14 proves (G.63). All original types, words and clocks are retained. Choose h before any fixed reserve contraction so that the contracted endpoints exceed its scalar threshold. $\square$

Proposition G.22 (Allocation in a native-containing chamber). *In a lawful activation chamber containing a cut assigned to a native physical block, the simultaneous cost allocation is valid with*

$K = C - b_0$. If E_i is the original centered empirical bridge in coordinate i , L_{intr} its synchronized multilinear endpoint chord, and ℓ_i^{intr} its legal outside-context minimum chord shifted once by C_0 , then, for $\theta = 1/(2H)$,

$$K_{\text{sync}} + C_0 \geq \frac{1}{2}(L_{\text{intr}} + C_0) + \sum_i [\theta \ell_i^{\text{intr}} + E_i]. \quad (\text{G.64})$$

For a surviving native canonical edge its sufficient support is

$$\theta \ell_i^{\text{intr}}(t) + E_i(t) \geq g d_i(t)^\gamma - h_W, \quad (\text{G.65})$$

and the local pair (G.63) suffices for it.

Proof. Fix one native cut of the chamber. Every endpoint vertex retains that cut at a legal native boundary, hence has $t_v \geq V_{\text{old}}$. By Theorem G.20, every such vertex obeys $K_v \geq c_0 \bar{C}_v - C_0$ with $\bar{C}_v \geq 0$. The synchronization identity is unchanged on subtracting the constant b_0 throughout this same-word chamber:

$$K_{\text{sync}} = L_{\text{intr}} + \sum_i E_i, \quad L_{\text{intr}} = L_{\text{multi}} - b_0.$$

The endpoint minima, shifted once, satisfy $0 \leq \ell_i^{\text{intr}} \leq L_{\text{intr}} + C_0$. There are at most H coordinates, so $\theta \sum_i \ell_i^{\text{intr}} \leq (L_{\text{intr}} + C_0)/2$; substitution proves (G.64). There is no leftover $-\theta b_0$ term because the cost was normalized before allocation.

Strong edges and reference-paid weak edges use the same named regularity bounds and $K_v \geq c_0 \bar{C}_v - C_0$. For the remaining native canonical edges retain (G.65). An old coordinate uses its genuine old charge (Proposition G.8) and the same regularity estimate. For a proper atomic child, the branching normal form and occupancy estimate then apply with this normalized cost: synchronization differences, spread losses and pure-cell lifetimes are unchanged by subtracting the chamber constant. Divide the available reference, spread and curve budgets by the same fixed fractions as in that allocation; every original cut and its multiplicity remains. At $z = 1$ this argument first concerns star histories, and extension to branching uses the separate floating-spine argument with this repaired endpoint input. Entirely old chambers are outside the stated domain. $\square$

A final two-color computation makes the normalization concrete. On its last native block $[l, T]$, with every other fine name cut at its own deadline, moving only the final boundary to t gives the exact global fine cost

$$K_{\text{fullstop}}(t) = \log 2 N((t, T]) - (T - t). \quad (\text{G.66})$$

Put $D = \log 2 N((l, T]) - (T - l)$. After a fixed shift its forward pair is $(h + D, h)$ and its reverse pair $(h, h + D)$. This context is among the legal outside contexts of the same $q = 0$ coordinate. Its shifted chord therefore dominates the repaired minimum chord at both ends, and hence throughout the block. Since $\ell_{\text{min}}^{\text{intr}} \geq 0$, the sufficient allocated wall implies

$$\theta \ell_{\text{min}}^{\text{intr}} + E \geq g d^\gamma - h_W \implies K_{\text{fullstop}} + h \geq g d^\gamma - h_W.$$

More generally two lawful pairs at one exact count have the same increment and differ only by a constant translation; their lower pair is still lawful. The ordinary all-reserve theorem evaluates the pair in (G.66) only after verifying positivity, its exact root count and unread-clock conditioning. Up to fixed buffers it has the form $(1, 1 + D)$, and contains no incoming b_0 term.

H The joint lower measure and its transport

This appendix supplies the probability and measure constructions used in Section 9. We keep the original physical blocks throughout. Conditional on its uncolored total n_i , a block has n_i uniform points on its interval and all $n_i + 1$ spacings. Its fine labels have their original categorical law. Conditional on a complete named count vector, they are a uniform multiset permutation. Constants in this appendix depend on the fixed dimension and the fixed compact scalar, count, and wall parameters.

H.1 Revealing part of an original word

The following elementary estimate lets us retain an entire coupled count band when an original comparison class reveals some of its labels.

Lemma H.1 (Central counts after partial revelation). *In block i , let N_i labels be independent with alphabet size $a_i \leq H$ and probabilities $p_{i,c} \geq p_* > 0$. The blocks are independent under their original law π . Let $A(K) \geq 0$ be a finite weight on their complete count vectors, supported on*

$$|K_{i,c} - N_i p_{i,c}| \leq B\sqrt{1 + N_i}, \quad W = \mathbb{E}_\pi A.$$

Fix m_i slots in block i . For any positive-probability event Q measurable in just these slots, including coupled or partial signatures,

$$\mathbb{E}_\pi[A \mid Q] \leq CW \prod_{i:m_i > 0} \left(\frac{1 + N_i}{1 + N_i - m_i} \right)^{(a_i-1)/2} \leq CW \prod_{i:m_i > 0} (1 + m_i)^{(a_i-1)/2}. \quad (\text{H.1})$$

If $m_i \leq N_i/2$ in every affected block, the ratio is bounded by a fixed constant. A block with no revealed slots incurs no factor.

Proof. For a fixed a -letter law in this compact simplex, its multinomial mass $P_{n,p}(k)$, with $d = a - 1$, satisfies

$$\sup_k P_{n,p}(k) \leq C(1 + n)^{-d/2}, \quad P_{N,p}(k) \geq c(1 + N)^{-d/2} \quad \text{on the stated central window.} \quad (\text{H.2})$$

For the first bound, Fourier inversion, with the last coordinate set to zero, bounds a coefficient by the integral of $|p_a + \sum_{j \leq d} p_j e^{it_j}|^n$. The square of this modulus is at most

$$1 - 2p_*^2 \sum_{j \leq d} (1 - \cos t_j) \leq 1 - c \sum_{j \leq d} t_j^2, \quad |t_j| \leq \pi.$$

Its integral is $O((1 + n)^{-d/2})$. For the lower bound, Stirling's inequalities give $P_{N,p}(k) \geq cN^{-d/2} \exp\{-ND(k/N \| p)\}$ when N is large. The entropy Hessian is bounded on the segment between p and k/N , so its exponent is $O(B^2)$. Bounded N is covered by compactness and feasibility. The case $a = 1$ is deterministic.

Condition first on every revealed label, with count vector b_i . The remaining labels still have their original independent laws, and

$$\pi(K = k \mid \text{revealed labels}) = \prod_i P_{N_i - m_i, p_i}(k_i - b_i).$$

For $m_i = 0$, its factor cancels exactly against the original type mass. For the other blocks, (H.2) bounds their ratio by the first product in (H.1). Multiply by $A(k)$ and sum the whole coupled type vector. Finally, conditioning on Q is a convex mixture of these revealed assignments. The last inequality follows from $(1+N)/(1+N-m) \leq 1+m$. If $W = 0$, all the relevant feasible type weights vanish, and the conclusion is immediate. $\square$

For a fixed original history and anchor, a literal signature class is an event Q of this kind: it reads only the slots with nontrivial partitions. This applies to that fixed class under the original law. It does not change that law to a law conditioned on an adaptively chosen history. In particular, if a class fully exposes blocks in I and reads m_j additional slots elsewhere, its count cost is at most

$$C \prod_{i \in I} (1 + N_i)^{(a_i-1)/2} \prod_{j \notin I} (1 + m_j)^{(a_j-1)/2}.$$

Only counts in actually revealed blocks or strips appear.

H.2 Two free intervals in the original class

We next bound the path part after the complete type and exposed labels have been fixed. Consider one selected block of length P , total $n \asymp P$, and binary risk

$$R(t) = ct - aN(t) + bN_B(t), \quad c, b, a - b > 0,$$

with these parameters in fixed compact intervals. Every accepted word has an actual endpoint pair satisfying

$$0 \leq x \leq C_{\text{box}}X, \quad 0 \leq y \leq C_{\text{box}}Y, \quad y - x = cP - an + bk, \quad x + R(t) \geq -h, \quad (\text{H.3})$$

where $X, Y \geq h_* > 0$. All additional walls and refinements can remain in the acceptance. Assume the uncolored endpoint bound

$$N([0, t]) + N([P - t, P]) \leq C_{\text{occ}}(1 + t), \quad 0 \leq t \leq P/2. \quad (\text{H.4})$$

The original class may read prescribed slots in endpoint strips of width $C_s(1 + d)$, but is unrestricted between them. This last condition is a statement about its literal signatures.

Put

$$\mathcal{T} = \frac{(1+X)(1+Y)}{n}, \quad p_0 = \frac{a - cP/n}{b}, \quad \rho_0 = \frac{(cP/(n+1))^2}{(cP/(n+1))^2 + b^2 p_0(1 - p_0)}. \quad (\text{H.5})$$

On the rare branch $\mathcal{T} < t_0$, the accepted interior type and (H.3) imply that p_0 belongs to a fixed compact subinterval of $(0, 1)$. Define $\Psi = \mathcal{T}^{2\chi(\eta, \rho_0)}$ on this branch and $\Psi = 1$ otherwise, where η lies in a fixed compact subinterval of $(0, 1)$.

Fix $0 < \gamma < 1/2$ and

$$D_n = [C_{\text{cut}} \log(2 + n)]^{1/\gamma}.$$

Set $q_d = 1$ if the branch is saturated, if $n < n_0$, or if $d > D_n$. Here n_0 is chosen so that, in the remaining cases, the endpoint strips contain at most a fixed small fraction of n slots and the following two intervals fit in the free middle. Choose their starting and ending boundaries s_d, t_d just inside those strips. Write

$$U_d = 1 + X + d, \quad V_d = 1 + Y + d, \quad M_d = \max(U_d, V_d), \quad L_d = \varepsilon_0 \min\{P, (n/M_d)^2\}. \quad (\text{H.6})$$

Take intervals $[s_d, s_d + L_d]$, $[t_d - L_d, t_d]$. The constant ε_0 is small enough that they are disjoint and their combined occupancy is at most half the unread slots, by (H.4). Grid rounding changes only fixed allowances. If the resulting length is below a fixed scalar threshold, again set $q_d = 1$.

In the remaining case set

$$p_{\pm} = p_0 \pm C_1(X + Y + 1 + d)/n.$$

Reduce t_0 and increase n_0 to keep these probabilities uniformly interior. Let q_L be the iid Bernoulli p_+ probability on the actual slots of the left interval that

$$cv - aN([s_d, s_d + v]) + bN_B([s_d, s_d + v]) \geq -C_2U_d, \quad 0 \leq v \leq L_d.$$

Let q_R be the iid Bernoulli p_- probability on the right interval, read backwards, that

$$-cv + aN([t_d - v, t_d]) - bN_B([t_d - v, t_d]) \geq -C_2V_d.$$

Define $q_d = \mathbf{1}_{(H.4)} q_L q_R$. These iid probabilities are upper bounds for the original conditional law, not the law of the accepted word.

To prove this, fix all revealed labels and a complete named type. If m slots and e binary successes have been revealed, the remaining binary word is a uniform $(k - e)$ -subset of $N = n - m$ slots. Its fraction $p' = (k - e)/(n - m)$ satisfies

$$|k - np_0| \leq C(X + Y), \quad |p' - p_0| \leq C(X + Y + 1 + d)/n,$$

so $p_- \leq p' \leq p_+$. Let $M_J \leq N/2$ be the combined number of slots in both free intervals. Relative to iid Bernoulli p' , their joint density under the exact subset law is

$$\frac{\mathbb{P}\{\text{Bin}(N - M_J, p') = Np' - j\}}{\mathbb{P}\{\text{Bin}(N, p') = Np'\}} \leq C. \quad (\text{H.7})$$

The numerator is bounded by a maximum atom and the denominator is a central atom. This is one comparison for both intervals and holds for every assignment on them. The endpoint identity and occupancy bound show that (H.3) requires both displayed tests, with the stated allowances. Under the iid law the intervals are independent. The left event increases and the right event decreases with binary successes, so coordinatewise uniform coupling gives

$$\mathbb{P}\{\text{accepted path} \mid \text{complete type, revealed labels}\} \leq Cq_d(G). \quad (\text{H.8})$$

Additional named restrictions within either binary list do not change this uniform binary-subset marginal.

For several blocks, fix the whole type and all exposed labels before multiplying (H.8). Suppose $0 \leq F \leq A(K)$ times the necessary path indicators, where A is supported on the original central windows and $W = \mathbb{E}_{\pi} A$. Applying Lemma H.1 only after the full type sum gives

$$\mathbb{E}_{\pi}[F \mid Q] \leq CW R_{\text{strip}} \prod_{j \in I} (1 + n_j)^{(a_j - 1)/2} \prod_{\substack{i \notin I \\ i \text{ selected}}} q_{i, d_i}(G_i), \quad (\text{H.9})$$

where I denotes fully exposed blocks and

$$R_{\text{strip}} = \prod_{i \notin I: m_i > n_i/2} (1 + m_i)^{(a_i - 1)/2}.$$

At active depths, $m_i \leq \varepsilon n_i$, so this extra factor only concerns the discarded large-depth regime. Conditioning on a partial-signature event Q last averages the uniform bound over its exact revealed assignments. There is no factor $1/\pi(Q)$, and the type acceptance need not factor across blocks.

H.3 Moments of the free-interval costs

For $d \leq D_n$, the functions just constructed satisfy

$$\mathbb{E}_{G|n} q_d(G)^\eta \leq C(1+d)^C \Psi. \quad (\text{H.10})$$

Here and below $A(K)$ and deterministic parameters must be fixed before these unread clocks for a clock moment assertion. The class bound itself is pointwise after fixing the clocks.

The exact-total uniform clock restricted jointly to the two free intervals has density, relative to a rate n/P Poisson clock,

$$\frac{\mathbb{P}\{\text{Pois}(n(1-\kappa)) = n - N_J\}}{\mathbb{P}\{\text{Pois}(n) = n\}} \leq C, \quad \kappa = 2L_d/P < 1/2. \quad (\text{H.11})$$

Conditional point locations also agree. Only after this single joint comparison are the two reference restrictions independent. Drop the occupancy indicator for the upper bound. The auxiliary risk drifts have magnitude $O((X+Y+1+d)/n)$, whose product with $\sqrt{L_d}$ is bounded. Theorem 6.18, for either sign, therefore gives

$$\mathbb{E} q_d^\eta \leq C \min \left\{ 1, \left(\frac{U_d}{\sqrt{1+L_d}} \right)^{2\chi_+} \right\} \min \left\{ 1, \left(\frac{V_d}{\sqrt{1+L_d}} \right)^{2\chi_-} \right\}. \quad (\text{H.12})$$

The one-end theorem is applied to the original positive mark pair in both orientations, with compact drift changes; all original spacings were retained before (H.11).

If $M_d \leq \sqrt{n}$, then $1+L_d \asymp n$ and the product of the reserve ratios is $\asymp U_d V_d/n$. If $M_d > \sqrt{n}$, then $1+L_d \asymp (n/M_d)^2$; the larger reserve factor is bounded by one and the smaller ratio is $\asymp U_d V_d/n$. The exponent changes satisfy

$$|\chi_+ - \chi(\eta, \rho_0)| + |\chi_- - \chi(\eta, \rho_0)| \leq C\{\mathcal{T} + (1+d)/n\}.$$

Every nontrivial reserve ratio in (H.12) is at least $c \min(1, \mathcal{T})$. Multiplying its negative logarithm by the last bound is uniformly bounded, because $\mathcal{T} |\log \mathcal{T}|$ is bounded and $d \leq D_n$. Replace both exponents by $\chi(\eta, \rho_0)$ and use $U_d V_d \leq (1+d)^2(1+X)(1+Y)$ to prove (H.10). If the free interval was too short for the one-end theorem, then $M_d \gg n$, and $d = o(\sqrt{n})$ forces one of X, Y to have order n . Thus $\mathcal{T}$ has a fixed positive lower bound and the convention $q_d = 1$ is valid. Bounded $n < n_0$ is treated in the same upper estimate; this does not enlarge the support convention of the source lower.

Fix $b_0 > 0$ and an integer B_0 large enough to sum the later pivot multiplicities. Choose D_0 with $D_0 \eta_- > 2 \sup \chi + 1$, and set

$$D^{\text{raw}}(G) = (2+n)^{-D_0} + \sum_{d \geq 0} (1+d)^{B_0} e^{-b_0 d^\gamma} q_d(G), \quad D(G) = \min(1, D^{\text{raw}}(G)). \quad (\text{H.13})$$

Choose C_{cut} after D_0 so that the η -powers of the numerical coefficient tail beyond D_n sum to at most $C(2+n)^{-D_0 \eta}$, uniformly in η . Subadditivity and (H.10) then give

$$(2+n)^{-D_0} \leq D \leq 1, \quad \mathbb{E} D^\eta \leq C \Psi. \quad (\text{H.14})$$

At depth zero (H.8) gives $q \leq CD$ for each supported complete type. Clipping at one preserves this bound because $q \leq 1$.

Large depths are paid using their actual geometric penalty, not a false small moment for $q_d = 1$. If that penalty includes cd^γ , increase C_{cut} so that

$$e^{-cd^\gamma/2}(1+n)^{H/2} \leq C(2+n)^{-D_0}, \quad d > D_n. \quad (\text{H.15})$$

This pays both the selected bridge floor and any large-strip reveal ratio in that block, leaving half the penalty to sum pivots. An unselected block with $m > n/2$ instead has $n \leq 2m \leq C(1+d)$, so its reveal ratio is a polynomial in its actual depth. For active depths use $q_d \leq CD(1+d)^{-B_0}e^{b_0d^\gamma}$, choosing b_0 below the available geometric coefficient. When several blocks share a depth parameter, divide its fixed penalty among the dimension-bounded number of factors before choosing these constants.

A related count extraction will sometimes be useful in a completely free interior interval. Conditional on every label outside it, a band of at most $C(1+R)$ possible counts there has probability at most $\min(1, C(1+R)/\sqrt{1+N_I})$, irrespective of its center. This is the binomial maximum-atom bound in (H.2). The whole band is summed before taking a power. If this interval and the two path intervals are disjoint with total length less than half the original block, the single density comparison (H.11) applies to all three at once. The count factor has moment $O(\min(1, (1+R)/\sqrt{1+P})^\eta)$; atypically small occupancy has exponentially small probability. This explains the ordinary count-window factor without replacing the original class by three independently conditioned experiments.

H.4 Named regularity on each retained word

Lemma H.2 (Uniform named refinement). *Let a fixed parent list of $n \leq C(1+L)$ points in an original block satisfy, at the fixed grid boundaries,*

$$|N((a, b]) - (n/L)(b-a)| \leq \theta_{\text{in}}(b-a) + K_{\text{in}}\{1 + d_L(a)^\zeta + d_L(b)^\zeta\}, \quad (\text{H.16})$$

where $0 < \zeta < 1/2$ and the grid has bounded boundary density. For every feasible full refinement type (k_c) , its original uniform multiset refinement satisfies, simultaneously in the finitely many names,

$$|N_c((a, b]) - (k_c/L)(b-a)| \leq \theta_{\text{out}}(b-a) + K_{\text{out}}\{1 + d_L(a)^\zeta + d_L(b)^\zeta\} \quad (\text{H.17})$$

with conditional probability at least $1 - \varepsilon$. Here θ_{in} is sufficiently small in terms of the prescribed $\theta_{\text{out}} > 0$, and K_{out} is sufficiently large in terms of the fixed data and ε . The assertion holds for every fixed parent word, however it was previously selected.

Proof. If an interval contains m parent positions, its count of name c is hypergeometric with mean αm , where $\alpha = k_c/n$. Sampling without replacement has exponential moments bounded by the corresponding independent Bernoulli moments. One elementary verification applies the elementary-symmetric-mean inequality to the n numbers e^t or 1, and handles the lower tail by complementation. Chernoff's argument gives

$$\mathbb{P}\{|N_c - \alpha m| > u\} \leq 2 \exp\{-cu^2/(m+u)\}.$$

The displacement of the mean from $(k_c/L)(b-a)$ is bounded by (H.16), and $m \leq C(b-a) + K_{\text{in}}(1 + d_L(a)^\zeta + d_L(b)^\zeta)$. After the stated choices, the probability of violating (H.17) is at most

$$C \exp\{-c_1(b-a) - c_2 K_{\text{out}}(1 + d_L(a)^\zeta + d_L(b)^\zeta)\}.$$

The sum over all grid endpoint pairs is uniformly finite. Pairs near an original endpoint are summed by the convergent series $\sum_{j \geq 0} e^{-cj^\zeta}$; pairs in opposite halves also pay $e^{-c_1(b-a)}$. Increasing K_{out} makes the finite union over names and blocks have probability at most ε . Bounded blocks have bounded counts, absorbed by the fixed intercept; empty lists have their unique empty refinement. $\square$

Apply this lemma to both actual lists of a retained binary word, conditioning on that word and its complete type. It supplies named REG with constant conditional success before any rare bridge is averaged. If natural rather than empirical rates are used, a central type changes the rate by $O(L^{-1/2})$, absorbed by a spare linear slope above a fixed length threshold. The lemma concerns this per-word refinement only; any later function of the selected spatial lists remains in its own conditional measure.

H.5 Admission of the macroscopic recoloring intervals

Consider a prescribed original block with $P, n \asymp T$, complete central fine type $|K_c - np_c| \leq B\sqrt{1+n}$, and a fixed interior interval I of length at least ρP , a fixed positive distance from both endpoints. For two specified actual fine names s, t , require

$$m = N(I) \geq c_I n, \quad |K_{I,s} - p_s m| + |K_{I,t} - p_t m| \leq \frac{m}{\log(2+T)}. \quad (\text{H.18})$$

The same requirement may be imposed in finitely many prescribed intervals. Conditional on n , the uncolored occupancy failure has probability $O(e^{-cn})$. Conditional on the complete type and a clock with $m \geq c_I n$, the named occupancy is hypergeometric, with mean mK_c/n . Its distance from $p_c m$ is $O(m/\sqrt{n})$, which is less than $m/(8 \log(2+T))$ above a fixed threshold. The same Chernoff bound as in Lemma H.2 gives, at each full central type,

$$\mathbb{P}_{G,\pi}\{(\text{H.18}) \text{ fails} \mid K\} \leq C \exp\{-cT/\log^2(2+T)\}. \quad (\text{H.19})$$

Let $q(K, G)$ already include the allocated curved path, binary REG, and full named REG, with

$$q \leq CD, \quad \mathbb{E}q^\eta \geq c\Psi, \quad \mathbb{E}D^\eta \leq C\Psi, \quad \Psi \geq c_0(2+n)^{-B_1}. \quad (\text{H.20})$$

The all-reserve and contracted-chord bridge theorems 8.1 and 8.5, followed by Lemma H.2, supply the lower assertion. The free-interval construction supplies the other two assertions. Let q_{good} also impose (H.18), and put $q_{\text{bad}} = q - q_{\text{good}}$. Subadditivity, then Jensen, give

$$\begin{aligned} \mathbb{E}q_{\text{good}}^\eta &\geq \mathbb{E}q^\eta - \mathbb{E}q_{\text{bad}}^\eta \\ &\geq c\Psi - (\mathbb{E}q_{\text{bad}})^\eta \geq c\Psi - C \exp\{-c\eta T/\log^2(2+T)\} \geq c'\Psi. \end{aligned} \quad (\text{H.21})$$

The final inequality follows after one fixed threshold enlargement, uniformly in the compact range of η , from the polynomial lower bound. All walls and REG restrictions were already inside q , and the dominating cost D does not change. The same argument applies to an unselected regular block with $D = \Psi = 1$. This local fractional estimate is taken before any later scale or root probability is integrated. For large T , (H.18) also guarantees at least cn occurrences of each of its two names, enough for every prescribed $O(\log T)$ recoloring.

H.6 One coupled count measure, with the actual owner powers

The count construction in Appendix G precedes the unread positions. More precisely, the diagonal exit bands and their endpoint count transports (G.13, G.11, and G.12) use only the proper owners. Projecting out terminal private names and then filling their prefix lists under the original conditional law gives the whole prefix count measure of G.16. Conditional on every original prefix total in its own central window, this is one positive measure μ on full fine types and count-only state, independent of newly unread prefix positions, with

$$W = \mu(1) \asymp \prod_{j < m} w_j. \quad (\text{H.22})$$

It retains the displaced good-name box. The conditional terminal completion in G.17 has constant mass for each retained prefix type and each suffix root vector in the specified central construction. Independence from suffix uncolored counts follows from the preceding support of the proper-owner construction.

For each fixed complete type, take the legal pair of G.14, namely (9.5). All its outside contexts have the same actual increment, so it is below both actual intrinsic endpoint minima and belongs to their common positive boxes. Write its path as $\ell_i^- + E_i$, with ℓ_i^- the joining chord. Impose on that original clock

$$\theta \ell_i^-(t) + E_i(t) \geq 2g \min(t, L_i - t)^\gamma - h_W, \quad 0 < \zeta < \gamma < 1/2, \quad 0 < \theta < 1, \quad (\text{H.23})$$

together with binary REG. Conditional on this word, refine both lists by Lemma H.2. On unelected blocks require only the appropriate ordinary and named REG. Each local mass $q_i(\omega, G_i)$ then satisfies (H.20) at $\eta_i = \sigma_{\text{owner}(i)}$, with the factor Ψ_i^I of (9.6). The necessary flat path follows from (H.23) because its uncontracted chord is larger and nonnegative, so the depth-zero bank bound applies at these intrinsic scales. Bounded supported experiments retain their fixed positive conditional mass and have $D_i = \Psi_i^I = 1$; a scalar parameter is not assigned to a zero count. Applying (H.21) incorporates all prescribed macroscopic recoloring intervals without changing these estimates.

The following measure identity retains the individual owner powers.

Lemma H.3 (Whole-type mixed moment). *Let μ be any finite positive measure on complete types, independent of the original independent environments G_i , and suppose (H.20) holds uniformly in its support, with possibly different $\eta_i \in [\eta_-, \eta_+] \subset (0, 1)$. Then, for $f(G) = \int \prod_i q_i(\omega, G_i) d\mu(\omega)$,*

$$\mathbb{E} \left[f(G) \prod_i D_i(G_i)^{\eta_i - 1} \right] \asymp \mu(1) \prod_i \Psi_i. \quad (\text{H.24})$$

Proof. For each fixed ω and i , Hölder applied to

$$q_i^{\eta_i} = (q_i D_i^{\eta_i - 1})^{\eta_i} (D_i^{\eta_i})^{1 - \eta_i}$$

gives

$$\mathbb{E}[q_i D_i^{\eta_i - 1}] \geq \frac{(\mathbb{E} q_i^{\eta_i})^{1/\eta_i}}{(\mathbb{E} D_i^{\eta_i})^{(1 - \eta_i)/\eta_i}} \gg \Psi_i.$$

The upper bound follows from $q_i \leq C D_i$. These expectations are finite by that same upper bound. At a fixed complete type the original environments are independent, so their local integrands can be multiplied. Tonelli then integrates the resulting uniform two-sided bounds over the entire μ . $\square$

For the present prefix, $f_{\text{pre}}(G) = \int \prod_i q_i(\omega, G_i) d\mu(\omega)$. The owner costs in (9.10) give the deterministic width multiplier $\prod_{j < m} w_j^{\sigma_j - 1}$. Multiplying (H.22) by this multiplier yields exactly $\prod_{j < m} w_j^{\sigma_j}$, while every local path retains its own owner power.

H.7 The ordinary suffix and its one count atom

We detail why the suffix is compatible with the whole prefix construction and with a later common uncolored target. Suppose first that it contains an ordinary weak block. It is the first suffix block. Let M_0 be its length, P the entire preceding original length, and B the following positive-drift length. The clipped-reference and KKT surplus estimates give

$$A_{\text{mean}} = \sum_{g < i} L_g g_g(r_g) \asymp P, \quad B_{\text{mean}} = - \sum_{g > i} L_g g_g(r_g) \asymp B, \quad A_{\text{mean}} + M_0 g_i(r_i) = B_{\text{mean}}. \quad (\text{H.25})$$

Thus $D = A_{\text{mean}} \asymp P$, $P \leq C(M_0 + B)$, and $B \leq C(P + M_0)$, including zero-side cases. This sums all original old groups and does not assign a common drift sign to each of them.

At the weak block's left endpoint, reference clipping gives cost at least $c \min(P, M_0 + B) \gg P$. The full-prefix context has cost exactly A_{mean} and no proper good-group term. At the right endpoint the same argument gives a lower bound $c \min(P + M_0, B) \gg B$, attained up to constants by the full-prefix context of cost B_{mean} . Therefore the intrinsic endpoint minima are

$$X_0^I \asymp 1 + P, \quad Y_0^I \asymp 1 + B. \quad (\text{H.26})$$

In both comparisons the common terminal target is initially zero; there is no incoming-exit reserve in the small endpoint. The constant-mark theorem 8.7, applied to the legal representative with actual increment $r_i M_0 - n \log(r_i + 1)$, gives the local scale

$$\beta_{\min} \asymp \min\{1, (1 + P)(1 + B)/(1 + M_0)\}. \quad (\text{H.27})$$

This uses the deterministic-mark theorem directly.

If the weak block is a largest suffix block, this is the bracket in (9.11), by (H.25). If a later positive block is largest, then $B \geq M_0$, so (H.27) is bounded below. Choose the weak drift threshold smaller than half the fixed positive tail gap. The mean of that largest positive block then dominates the possible negative mean of the weak block, giving $D \gg M$ and saturating the bracket in (9.11) as well. If there is no weak block, every suffix drift is uniformly positive, and $D \gg U$; both statements are again saturated.

Condition on a complete retained prefix. Its uncolored discrepancy Δ_{pre} satisfies $|\Delta_{\text{pre}}| \leq C\sqrt{1 + V_{\text{pre}}}$, and the Euler identity gives the suffix target

$$E_x = D + (x - \Delta_{\text{pre}})/r_m, \quad |E_x - D| \leq C_A \sqrt{1 + D} \quad (\text{H.28})$$

for every prefix-uncolored $0 \leq x \leq A \log(2 + V_{\text{pre}})$, using $D \gg V_{\text{pre}}$. Fix the nonreservoir suffix counts in their own central windows. In a predetermined largest block use the actual new count

$$n_*^x = \left\lfloor \frac{\sum_{\text{suffix}} c_i L_i + E_x - \sum_{i \neq *} w_i n_i}{w_*} \right\rfloor, \quad w_i = \log(r_i + 1), \quad c_i = r_i. \quad (\text{H.29})$$

The support threshold and central root construction of Theorem 8.9 make it nonnegative and central, with Poisson mass $\asymp (1 + U)^{-1/2}$. All its original $n_*^x + 1$ gaps are then sampled. In the

unconditional Poisson experiment this means retaining the event that the sampled total equals (H.29); it does not alter a previously fixed tuple. The bounded floor error is $-C \leq \epsilon \leq 0$.

The count n_*^0 obtained with $x = 0$ is used only to verify the zero-target prefix boxes through the uniform count completion in G.17. No clock at that virtual count is sampled or reused. Increasing x preserves the prefix local endpoint lower bounds: terminal early bounds gain the common entrance, and terminal late intrinsic bounds have neither entrance nor incoming carry. Thus prefix paths, costs, and local pairs are fixed before x is evaluated.

If the weak block is not the reservoir, its local count is unchanged. If it is, put $k = n_*^x - n_*^0 \geq 0$. Then $k \leq C(1 + x)$, and the max formula for the legal pair gives

$$\Delta_x = \Delta_0 - w_*k, \quad x_0^- \leq x_x^- \leq x_0^- + w_*k, \quad a_*Y_0^I \leq y_x^- \leq y_0^-. \quad (\text{H.30})$$

Equation (H.26) absorbs $w_*k \leq C_A \log(2 + P)$ in the left box, while the right endpoint keeps its positive lower box. Its scale is still (H.27); an incoming exit or x has not been added to its small right endpoint.

Complete the terminal full type at its constant conditional mass, then impose the weak ordinary representative's allocated strong wall and uncolored REG. The other suffix blocks have separated drift. Reading a positive slope from the left, or a negative slope from the right, their REG implies the allocated inward wall: absorb $K(1 + t^\zeta) + gt^\gamma$ into a fixed intercept and a fraction of the linear drift. Long central blocks preserve the drift gap; bounded supported blocks use their fixed finite convention. The same argument pays the strong and reference-paid native comparisons. After counts are fixed, the original suffix clocks are independent; there is at most one weak ordinary probability. Fill named REG on each retained clock by Lemma H.2. This constructs a joint completion of order $\mathcal{B}_I = (1 + U)^{-1/2}\beta_0$ on the same word.

For the matching upper and the support needed later, retain the lawful full-stop context for every suffix $q = 0$ comparison: all eligible names remain until its moving cut, then disappear. Its intrinsic cost is the original reverse suffix risk, with no incoming reserve. At a fixed count its centered fluctuation agrees with that of the minimum representative; the latter's positive chord, including its θ -contraction, lies below the full-stop chord. Thus the imposed weak and paid strong walls imply on their actual joint support

$$h_{\text{base}} + R_{\text{tail}}(v) \geq 0 \quad (0 \leq v \leq U) \quad (\text{H.31})$$

for one fixed reserve, large enough for floor and fixed intercept errors. Drop other filters and apply the all-count upper in Theorem 8.9 at this reserve and target (H.28). It gives $O(\mathcal{B}_I)$, with the reservoir denominator retained before its Poisson supremum. Consequently, for every retained complete prefix word and every permitted x , its actual suffix completion kernel is comparable to $\mathcal{B}_I$. Integrating the private prefix gives the pointwise relation

$$\mathbb{E}_{\text{suffix}} f_0^x(G_{\text{pre}}, G_{\text{suffix}}) \asymp \mathcal{B}_I f_{\text{pre}}(G_{\text{pre}}). \quad (\text{H.32})$$

This expectation includes the one root atom, all suffix positions, and the original named refinements. It may be multiplied by any nonnegative prefix-uncolored cost, but it does not scalarize a future private-list weight.

Apply (H.32) first, then Lemma H.3 at each full type, and finally (H.22). This proves the source mixed moment

$$\mathbb{E} \left[f_0^x \prod_j (D_j^{\text{own}})^{\sigma_j - 1} \right] \asymp \mathcal{B}_I \prod_{j < m} w_j^{\sigma_j} \prod_{i \in \mathcal{I}} \Psi_i^I(\sigma_{\text{owner}(i)}). \quad (\text{H.33})$$

The ordinary factor is linear and occurs once.

H.8 Targets fixed by the costs and a literal fine-word density

We specify the order of constants before defining the adaptive targets. First fix compact scalar ranges, θ , $0 < \zeta < \gamma < 1/2$, and the finite count and displacement support. Choose a polynomial floor exponent D_0 as in (H.14); it depends on the compact exponent range and number of factors, not on later fixed wall constants. Use the literal widths $w_j \leq 1$, with $-\log w_j \leq \frac{1}{2} \log T_j$. The floors imply predetermined bounds

$$y_j := -\log D_j^{\text{own}} \leq C_0 + C_1 \log(2 + T_j), \quad y_j \geq 0. \quad (\text{H.34})$$

Only selected head blocks enter the terminal cost, so its target is bounded in terms of V_{pre} . The hierarchy $T_l \leq C_H T_j$ for $l < j$ and bounded reciprocal child entropies now fix all logarithmic transfer ranges described below. Construct the common upper costs simultaneously for these full ranges, then choose the source with spare strong-curve and REG margins. Finally enlarge the macro-block and root support thresholds. These later constants do not alter the floor exponent or logarithmic target bounds.

Set

$$x = \sum_{j=1}^m r_j y_j, \quad r_j = \sigma_j / \sigma_1. \quad (\text{H.35})$$

This is prefix-uncolored and lies in the preassigned range of (H.28). Choose the actual root by (H.29), obtaining $X_1 = x + \epsilon$, $-C \leq \epsilon \leq 0$. Let $F_0^x \in [0, 1]$ be the joint source acceptance just constructed, with private mass f_0^x . Proper targets are still zero at this stage. The suffix is sampled only once, and (H.33) applies to this x .

Let A_j be the logarithm of the first shallow effective alphabet of proper owner j , and put $b_j = u_j A_{j+1}$. In its prescribed macro block take the immediate child's first shallow fine label c_j , and that block's eligible shallow exterior fine label t_j . Both are actual original labels with probabilities bounded below, and their ratio is

$$p(c_j)/p(t_j) = \exp\{(u_j - 1)A_{j+1}\}.$$

The destination need not be the first name of the entire owner. Define, using only the uncolored prefix,

$$e_0 = 0, \quad e_j = \left\lceil \frac{y_j + A_j e_{j-1}}{b_j} \right\rceil, \quad \delta_j = b_j e_j - A_j e_{j-1} - y_j \in [0, b_j), \quad j < m. \quad (\text{H.36})$$

Equation (H.34) and the hierarchy give the predetermined bound $e_j \leq A'_H \log(2 + T_j)$. Condition (H.18) and the final threshold ensure that all these occurrences exist. Choose a uniform e_j -subset of the c_j -slots in its bank and recolor it t_j . The banks lie in disjoint original blocks. This defines a single probability kernel on the full fine word, preserving every clock and the root.

Here is its exact density calculation. In one bank, write s, t for source and destination names, and k', l' for their output counts. Every preimage is obtained by choosing e of the output t -slots, and has $k' + e$ available source slots. Its original word probability is larger by $(p_s/p_t)^e$. Thus the unrestricted push-forward density is

$$R_e(k', l') = (p_s/p_t)^e \frac{\binom{l'}{e}}{\binom{k'+e}{e}}. \quad (\text{H.37})$$

On an output with a retained preimage, expand the e factors. The two bank counts differ from $p_s m, p_t m$ by at most $m/\log(2+T)$, and the shift is at most e . Since both probabilities are uniformly positive,

$$|\log R_e| \leq C\{e/\log(2+T) + e^2/m\} \leq C_A. \quad (\text{H.38})$$

Deleting preimages can only decrease the density. Multiplying the upper bounds over the finitely many disjoint banks gives C_R . If ν_y is the push-forward of πF_0^x , define the actual accepted target by

$$F_y^I = \frac{1}{C_R} \frac{d\nu_y}{d\pi}, \quad 0 \leq F_y^I \leq 1, \quad \mathbb{E}_\pi F_y^I(G, \cdot) = f_0^x(G)/C_R. \quad (\text{H.39})$$

The mass identity holds for every fixed full uncolored environment, including its root indicator and the adaptive vector $e(G)$. Thus (H.39) preserves the integrated mass.

H.9 The reconstructed support and its common upper costs

We verify the necessary target paths directly on the changed slots. Fix a selected block of length P that is recolored. Its canonical retained binary set contains s and excludes t . Every target full type K has the same inverse source type $K - \sum_j e_j v_j$, where v_j is its source-to-destination count shift. Thus its source pair x_0, y_0 is recovered without knowing which slots were changed. If $L_{\text{flip}}(t)$ counts recolorings before time t , the reconstructed process satisfies the exact identity

$$\begin{aligned} \tilde{Z}(t) &= x_0 + (c + be/P)t - aN(t) + bN_{B,\text{target}}(t) \\ &= Z_{\text{source}}(t) - b\{L_{\text{flip}}(t) - et/P\}, \quad \tilde{Z}(0) = x_0, \quad \tilde{Z}(P) = y_0. \end{aligned} \quad (\text{H.40})$$

This follows from $N_{B,\text{target}} = N_{B,\text{source}} - L_{\text{flip}}$.

For every possible chosen subset in the fixed interior bank,

$$|L_{\text{flip}}(t) - et/P| \leq CeP^{-\gamma} \min(t, P-t)^\gamma. \quad (\text{H.41})$$

Before the bank the left side is et/P , after it it is $e(P-t)/P$, and inside it the depth is $\gg P$ and the left side is at most e . The spare $2g$ curve in (H.23) and $e = O(\log P)$ therefore imply

$$\theta \ell_0(t) + E_{\text{target}}(t) \geq g \min(t, P-t)^\gamma - h_W \quad (\text{H.42})$$

above the final threshold. Its endpoints are exactly the original source pair. On an unflipped native block the word and local type are unchanged, and the inverse complete type recovers the same pair. A previous owner's recoloring preserves every point of the later native clock, so the estimate uses that block's own length.

Named REG is preserved with the predetermined spare constants. Any interval meeting the macro bank either has length $\gg P$ or has an endpoint of depth $\gg P$. Its $O(\log P)$ count change is paid respectively by a spare linear slope or a spare depth-to-the- ζ intercept. The empirical-center shift $O(\log P/P)$ is paid by the same margins. The terminal suffix has no recoloring bank; its root, clocks, and fixed-reserve ordinary wall remain unchanged. The support proved here is the source-reconstructed support (H.42). The separate same-word information and carry comparison in Appendix I transfers full fine histories.

To construct one bank envelope in advance for all the permitted transfers, retain U_d, V_d, L_d of (H.6) and let $E_{\max} = \lfloor A \log(2+T) \rfloor$. Use

$$p_\pm = p_0 \pm C_1(X + Y + 1 + d + E_{\max})/n, \quad c_{\max} = c + bE_{\max}/P. \quad (\text{H.43})$$

The left free interval uses speed c_{max} and probability p_+ ; the reversed right interval uses speed c and probability p_- . These opposite choices both enlarge their necessary events. The allowances remain $C_2 U_d, C_2 V_d$. After fixing the true revealed assignments and full target type, (H.40) gives $|K_B - np_0| \leq C(X + Y) + e$; thus its remaining binary fraction lies between p_- and p_+ . The joint exact-subset comparison (H.7) then applies to both intervals on the original target word. This argument uses the direct support (H.42).

The joint clock comparison (H.11) is unchanged. The new drift term times $\sqrt{L_d}$ is at most $C \log(2 + T)/\sqrt{n}$, uniformly bounded because the recolored block is macroscopic. The additional exponent error is $O(E_{max}/n)$, and its product with the logarithm of any active reserve ratio is $O(\log^2(2 + n)/n)$. The proof of (H.10) therefore gives

$$\mathbb{E}(q_d^*)^\eta \leq C(1 + d)^C \Psi_{\text{source}}, \quad \mathbb{E}(D^*)^\eta \leq C \Psi_{\text{source}}, \quad (\text{H.44})$$

using the same floor and coefficient tail. All costs here are fixed from the entire range E_{max} , before any target is evaluated, with the original source reserves.

For the class upper, use the count-only prefix acceptance A_{pre} , extended by one on suffix names, and drop the source's suffix root and terminal filters. The full F_0^x itself is not a clock-independent count weight. Every preimage of a target type has the same inverse type, so positivity and (H.38) bound F_y^I by the shifted weight

$$A_e(K) = A_{\text{pre}}(K - \sum_j e_j v_j).$$

Its whole original type mass W_e satisfies

$$cW \leq W_e \leq CW, \quad W = \mathbb{E}_\pi A_{\text{pre}}. \quad (\text{H.45})$$

Indeed, in one changed block the adjacent multinomial ratio is

$$\frac{\pi(k + ev)}{\pi(k)} = (p_t/p_s)^e \prod_{l=0}^{e-1} \frac{k_s - l}{k_t + l + 1}, \quad \left| \log \frac{\pi(k + ev)}{\pi(k)} \right| \leq C(e/\sqrt{n} + e^2/n).$$

All shifted types are feasible above the fixed threshold, and unchanged block factors cancel exactly. Shift the entire coupled type sum once to prove (H.45). The bound holds simultaneously in every allowed e ; it therefore remains pointwise for $e(G)$, without treating A_e as independent of G . Apply the original conditional free-interval argument at a fixed whole type, sum this whole shifted acceptance, and replace $W_e(G)$ by CW before averaging clocks. This proves (H.9) for the actual transported density with the all-transfer envelopes. Its count factor contains no root atom; that factor remains in the separate source-mass calculation.

H.10 Exact exits, likelihood, and the first mass

The grouped residuals change under (H.36) by

$$Z_j^y - Z_j^0 = b_j e_j - A_j e_{j-1} = y_j + \delta_j \quad (j < m), \quad Z_m^y - Z_m^0 = -A_m e_{m-1}. \quad (\text{H.46})$$

Their r_j -weighted sum is zero because $r_j b_j = r_{j+1} A_{j+1}$. Define the new exits by the genuine identity $Z_j^y = y_j + \eta_j^y - \eta_{j-1}^y / u_{j-1}$. Subtract the source identity, solve its triangular recursion, and get

$$\eta_j^y - \eta_j^0 = \sum_{l \leq j} (\sigma_l / \sigma_j) \delta_l = O_H(1). \quad (\text{H.47})$$

All these changes are nonnegative. The positive source exit bands remain positive and, since $d_j R_j \ll 1$,

$$0 \leq E_y := \sum_{j < m} d_j \eta_j^y \leq C_H, \quad Z_m^y = y_m + \epsilon/r_m - \eta_{m-1}^y/u_{m-1}. \quad (\text{H.48})$$

The latter follows from the unchanged root $X_1 = \sum_j r_j y_j + \epsilon$. In particular, its y_m comes from the actual root; no terminal private transfer is made.

Here is the exact likelihood calculation, including its deterministic normalization. Put $B_{\text{geom}} = \sum_i (h_i - 1)L_i = \sum_{j=1}^k V_j$. The full recursive fine-word law, including the terminal free labels, satisfies, term by term,

$$-\log \pi(\omega) = B_{\text{geom}} + \sum_{j=1}^m Z_j(\omega), \quad X_1 = \sum_{j=1}^m r_j Z_j. \quad (\text{H.49})$$

At an exterior choice the negative log probability is its local log-alphabet; at a child route it is that log-alphabet minus the child's weighted log-alphabet. The child contributions cancel against the next stage, proving both identities. Equal-product internal margins cancel in grouping, by complementarity in (5.46).

Let $q_j = \sigma_j/\sigma_m - 1$, so $q_m = 0$. Eliminating the terminal output in (H.49) gives

$$1 = e^{B_{\text{geom}} + X_1/r_m} \pi(\omega) e^{-\sum_{j < m} q_j Z_j}.$$

Since $q_j - q_{j+1}/u_j = d_j$, the exit identities give $\sum_{j < m} q_j Z_j = \sum_{j < m} q_j y_j + E_y$. Therefore

$$(1/r_m - \sigma_1)X_1 - \sum_{j < m} q_j Z_j = \sum_{j=1}^m (1 - \sigma_j)y_j - E_y + (1/r_m - \sigma_1)\epsilon. \quad (\text{H.50})$$

This uses $(1/r_m - \sigma_1)r_j - q_j = 1 - \sigma_j$, including $j = m$. No exponential in the logarithmic target is discarded. Also, for two retained words in the same uncolored environment, (H.49) and the sum of their exit identities give

$$\frac{\pi(\omega)}{\pi(\omega')} = \exp\{E_y(\omega) - E_y(\omega')\}. \quad (\text{H.51})$$

It is bounded by (H.48), while its actual class mass still requires the separate class estimates.

For a complete cell occupancy g , put $W_g = \prod_{i,c} |c|^{g_{i,c}}/g_{i,c}!$ and $Z_F(g) = \sum_{\omega} F_y^I(g, \omega)$. Under the natural Poisson law the cell intensity is $Z_i = e^{G_i}$. Expanding that law in $e^{-\sum_i h_i L_i} \sum_g W_g Z_F(g)$ gives deterministic factor $e^{-\sum_i h_i L_i + \sum_i Z_i L_i}$, random factor $e^{-\sum_i G_i n_i}$, and $\pi(\omega)^{-1}$. Euler's identity gives $\sum_i G_i n_i = \sigma_1 X_1 + \sum_j s_j V_j$. Combining with (H.49), the deterministic exponent is exactly $-J(L; s)$. Equation (H.50) proves

$$e^{-\sum_i h_i L_i} \sum_g W_g Z_F(g) = e^{-J(L; s)} \mathbb{E}_{\text{nat}} \mathbb{E}_{\pi} \left[F_y^I e^{-E_y} \prod_j (D_j^{\text{own}})^{\sigma_j - 1} e^{(1/r_m - \sigma_1)\epsilon} \right]. \quad (\text{H.52})$$

This is algebra on the actual count/root support. The factors involving E_y and ϵ are bounded above and below. Using the pointwise mass identity (H.39) inside this same expectation, followed by (H.33), evaluates it as

$$e^{-J(L; s)} \mathcal{B}_I \prod_{j < m} w_j^{\sigma_j} \prod_{i \in \mathcal{I}} \Psi_i^I(\sigma_{\text{owner}(i)}). \quad (\text{H.53})$$

The root indicator occurs once, linearly.

To integrate the remaining original prefix totals, let $\mu_i = Z_i L_i$. On their own central windows, $|n_i - \mu_i| \leq B\sqrt{1 + \mu_i}$. The deterministic intrinsic score minima do not depend on n_i . Replacing n_i by μ_i in (9.6), with products frozen, changes p_i^0, ρ_i^0 and the logarithm of its base by $O(\mu_i^{-1/2})$ on the compact rare branch. The base is at least $c/(1 + \mu_i)$, so local Lipschitz continuity of χ gives

$$|\log \Psi_i^I - \log \bar{\Psi}_i^I| \leq C \log(2 + \mu_i) / \sqrt{\mu_i} \leq C. \quad (\text{H.54})$$

If a branch threshold is crossed, both bases are comparable to that fixed threshold and the same conclusion holds. The dimension-bounded number of original Poisson central windows has constant joint mass; the source vanishes off them. Integrating (H.53) gives (10.10). Fixed positive allowances can then be changed to the profile's normalization because all scores are finite minima of sums of nonnegative terms, and all exponents remain in compact ranges.

H.11 Construction on complete cells

We finish by constructing the same measure directly on the complete cell grid used for arithmetic realization. This point is needed because taking a conditional expectation of an old continuous acceptance would not preserve its fractional moments or its comparison-class estimates.

Before counts, positions, or labels are sampled, fix a partition of each original block into cells with endpoints including its physical endpoints, bounded mesh Δ_* , and bounded boundary density. A cell word records the occupancy $g_{i,c}$ and its exchangeable labeled slots. Every final event is invariant under permutations within a cell. Only the declared bounded experiments are used for short blocks, with bounded total, feasible types, and fixed positive retained mass; these blocks are not recoloring banks. A later numerical cutoff for a bank upper estimate does not expand this bounded support convention.

First consider any lift of a fixed cell word to distinct positions inside its cells. For positive marks and either orientation, $Z(t) = x + \varepsilon(\sum_{s \leq t} W_s - ct)$, monotonicity of jumps gives, on $[l, r]$,

$$Z(t) \geq Z(l) - c\Delta_* \quad (\varepsilon = 1), \quad Z(t) \geq Z(r) - c\Delta_* \quad (\varepsilon = -1). \quad (\text{H.55})$$

Use the same half-open convention for all boundary counts. There is no dependence here on the number of points in the cell. Since $g \min(t, L - t)^\gamma$ oscillates by at most $g\Delta_*^\gamma$, a boundary strong wall implies the continuous strong wall on every lift after increasing its allowance by $c_{max}\Delta_* + g\Delta_*^\gamma$. The reverse inclusion requires no loss. Logarithmic soft walls are handled by their fixed local modulus of continuity.

For an allocated bridge $\theta\ell + E$, its endpoints are $\theta x, \theta y$, and its signed-jump representation has speed

$$c_\theta = \theta c + (1 - \theta)S_{\text{total}}/L, \quad (\text{H.56})$$

where S_{total} is its positive mark sum. On the large central experiments this speed stays in a fixed positive compact interval, independent of the sizes of the smaller intrinsic endpoints. The same is true after the compensating change $c + be/L$. Thus (H.55) also applies to the allocated walls at these smaller pairs.

Boundary REG implies continuous REG on every lift. To bound the count on $(s, t]$, expand it to its neighboring outer boundaries for the upper bound and shrink it to the inner boundaries for the lower. Length and mean change by at most $2\Delta_*$ and $2\lambda_{max}\Delta_*$. Also

$$|d_L(s)^\zeta - d_L(s')^\zeta| \leq \Delta_*^\zeta$$

for a neighboring boundary s' . If the inner interval is empty, its length is at most $2\Delta_*$, and nonnegativity gives the required lower bound. This proves the assertion with fixed additional intercept, even though an interior cell's permitted occupancy can grow with its depth.

Define the cell source using boundary allocated walls, boundary binary REG, and its count box from the outset. The continuous relative strong/REG event at the same exact pair is contained in this boundary event. The reverse inclusion just proved provides the continuous necessary upper wall with a fixed larger allowance. Changing such a fixed allowance changes the all-reserve factors by fixed constants only. For named filling, condition on a complete binary word and refine both actual lists. Lemma H.2 applies to each lift with the enlarged fixed REG constants. A boundary binary word is a mixture of these lifts; integrating their uniform success bound gives the same relative retention for the named boundary event.

Construct fresh costs as cell functions. At each active depth choose all strips and free-interval endpoints at original boundaries and round L_d down by at most two meshes. A length below the fixed threshold uses the already justified $q_d = 1$ convention. Use the all-transfer parameters (H.43) and test its two necessary walls at boundaries, with only the fixed mesh allowance. Denote the resulting product by $q_{i,d}^{\text{cell}}(g_i)$. Both the exact-subset comparison (H.7) and the exact-clock comparison (H.11) push to cells with the same density bound. The either-sign pointwise implication (H.55) then gives

$$\mathbb{E}_{g_i|n_i}(q_{i,d}^{\text{cell}})^\eta \leq C(1+d)^C \Psi_i^I(\eta), \quad d \leq D_{n_i}.$$

Use these functions in (H.13), including its literal floor, numerical tail, and clipping, to define $D_i^{I,\text{cell}}(g_i)$. The proofs of (H.14) and (H.44) give

$$q_i^{I,\text{cell}}(\omega, g_i) \leq C D_i^{I,\text{cell}}(g_i), \quad \mathbb{E}(q_i^{I,\text{cell}})^\eta \geq c \Psi_i^I(\eta), \quad \mathbb{E}(D_i^{I,\text{cell}})^\eta \leq C \Psi_i^I(\eta). \quad (\text{H.57})$$

The middle lower bound follows from the contained continuous event, with its proved finite scalar support in bounded-count cases.

The macroscopic recoloring intervals are unions of complete cells, still a fixed positive distance from both block endpoints. Their occupancy is binomial conditional on n_i , and the two-name counts are hypergeometric conditional on the full type and occupancy. Thus (H.19) and the local subtraction (H.21) apply directly to these cell events. They preserve (H.57) before the coupled type or root is integrated.

The exact uncolored cell law at fixed total is

$$\mathbb{P}(g_i | n_i) = n_i! \prod_c \frac{(|c|/L_i)^{g_{i,c}}}{g_{i,c}!}. \quad (\text{H.58})$$

It is the push-forward of the original uniform-point law, equivalently of its full $(n_i + 1)$ -spacing simplex law. Conditional on the occupancy, positions are uniform inside cells, labels have their original categorical law, and conditional on full type the labeled slots have their uniform multiset law. By construction,

$$q_i^{I,\text{cell}}(\omega, G_i) = q_i^{I,\text{cell}}(\omega, g_i), \quad D_i^{I,\text{cell}}(G_i) = D_i^{I,\text{cell}}(g_i).$$

Thus integration over a cell fiber preserves q_i , q_i^η , and $q_i D_i^{\eta-1}$ exactly. At a fixed complete type the block experiments remain independent; Lemma H.3 therefore applies before the full measure μ is integrated.

The owner costs, y_j , and x are now functions of prefix uncolored cell counts alone. For every retained prefix cell word, boundary-to-continuous REG supplies the hypotheses of the suffix

argument on each lift. The actual new reservoir total is still (H.29). Sample its cell occupancy directly by (H.58). Counts, endpoint identities, and the bounded root error are unchanged. Impose the weak ordinary representative's boundary strong wall and boundary REG, with the same constant-mass named completion. The continuous ordinary lower is contained in it; strong blocks are paid by their separated drifts. Conversely the intrinsic full-stop contexts imply the global ordinary wall at boundaries, and (H.55) extends it with one fixed allowance. The all-count root upper still applies. Consequently

$$\mathbb{E}_{\text{suffix}} f_0^{I, \text{cell}, x}(g_{\text{pre}}, g_{\text{suffix}}) \asymp \mathcal{B}_I f_{\text{pre}}^{I, \text{cell}}(g_{\text{pre}}) \quad (\text{H.59})$$

pointwise on the retained prefix. Its root is a single count atom, and the endpoint change of a weak reservoir is still exactly (H.30).

Apply the recoloring kernel to these cell words. Its uniform subset choice commutes with every within-cell permutation, as do the source count, wall, REG, bank, and root events. Formula (H.37) and its bound remain exact, so

$$F_y^{I, \text{cell}} = C_R^{-1} d\nu_y / d\pi \in [0, 1], \quad \mathbb{E}_\pi F_y^{I, \text{cell}}(g, \cdot) = f_0^{I, \text{cell}, x}(g) / C_R \quad (\text{H.60})$$

is cell-invariant. At every boundary the same slot identity (H.40) holds. Macro interiority and the spare strong wall prove (H.42) at these boundaries, then (H.55) extends it to every lift with a fixed allowance uniform in the batch size and target. Named boundary REG is retained by the same macro-length or macro-depth argument, and the terminal suffix is unchanged.

The output and exit identities (H.46)–(H.48) are count identities and incur no cell-rounding error. Apply (H.59), then the local mixed moments from (H.57), and finally the whole count mass (H.22). Equation (H.60) transfers that mass with its fixed normalization. The exact likelihood calculation (H.52) and the central-count comparison (H.54) consequently give the same e^{-J} times the factors in (10.10).

All constants can be fixed in the order stated before (H.34), with the mesh bound included among the initial data: floor exponent and literal widths first, the logarithmic transfer ranges next, then free-interval envelopes and spare strong/REG/refinement and mesh allowances, and the final macro, total, and root thresholds last. The subsequent arithmetic realization may therefore use this bounded cell-invariant density without reading actual prime values inside its slots.

I Original histories and their geometric costs

All comparisons in this appendix retain one original word and its original physical groups. A history assigns an activation time $0 \leq t_e \leq d_e$ to each boundary of an anchored permutation of the fine names. At a physical position one first splits this line at its activated boundaries and then intersects its cells with the eligible alphabet. A competitor belongs to the class $Q_h(\omega)$ when its label is in the anchor's eligible cell at every original slot. For a split $e = (A; B, C)$ define the logarithmic loss

$$\ell_e(a) = \mathbf{1}_{a \in A} \log \frac{\pi(A \cap E)}{\pi(\text{side}_e(a) \cap E)}.$$

It is zero when only one side is eligible. If $B_e(t)$ is its sum on tail slots, the split cost is $C_e(t) = B_e(t) - (d_e - t)$. Product-law telescoping gives $-\log \pi(Q_h) = \sum_e B_e(t_e)$. Use complete-cell boundary tests, processing the cell on the left before its right boundary activates. Equivalent one-sided conventions give the same assertions when used consistently. All original slot representatives, including their multiplicities within a cell, remain in the history sums.

I.1 Legal interpolation on the original time cells

Fix the original permutation, the weak activation order, and a physical block for every variable cut. In a block $[a, b]$ of length $L > 0$ its assigned cuts satisfy $a \leq t_1 \leq \dots \leq t_m \leq b$ and have deadlines at least b . Put $t_0 = a, t_{m+1} = b$ and

$$w_q = (t_{q+1} - t_q)/L, \quad 0 \leq q \leq m.$$

The vertex v_q puts the first q cuts at a and the remaining cuts at b . The weights are nonnegative, sum to one, and $t_r = \sum_q w_q(v_q)_r$, since $\sum_{q < r} w_q = (t_r - a)/L$. Over the physical blocks take the product weights w_v . There are at most 2^n vertices for $n \leq H - 1$ cuts, and only dimension-many permutations, activation orders and block assignments. Every vertex is deadline-legal, although it need not be an available arithmetic pivot.

Reference costs are affine on each physical block. Thus, writing $\overline{M}_h = \sum_e \overline{C}_e$,

$$\overline{M}_h(t) = \sum_v w_v \overline{M}_h(v), \quad \min_t \overline{M}_h(t) = \min_v \overline{M}_h(v). \quad (\text{I.1})$$

Any common offset is subtracted once on both sides. Ties created at a vertex are computed in the chamber's weak order; the entropy telescope makes the total independent of another consistent resolution of those ties. Reference nonnegativity is a separate geometric input.

For the actual word and a split assigned to $[a, b]$, define its empirical bridge, centered at the actual block total, by

$$\mathcal{E}_e(u) = \sum_{s \in [a, u]} \ell_e(\omega_s) - \frac{u - a}{L} \sum_{s \in [a, b]} \ell_e(\omega_s).$$

Direct subtraction cancels all future groups and the deadline, giving

$$C_e(u) = \frac{b - u}{L} C_e(a) + \frac{u - a}{L} C_e(b) - \mathcal{E}_e(u), \quad \sum_e C_e(t_e) = \sum_v w_v \sum_e C_e(v_e) - \sum_e \mathcal{E}_e(t_e). \quad (\text{I.2})$$

Every term uses the same actual anchor, including its old lists. For simultaneous cuts, the chain rule gives the pointwise identity

$$\sum_{e \text{ simultaneous}} \ell_e(c) = \log \frac{\pi(\Pi_-(c))}{\pi(\Pi_+(c))}.$$

They therefore form one empirical bridge of the bundled partition loss. For a synchronous full-to-singleton split in a uniform alphabet this loss is the constant $\log H$; its bridge is purely uncolored.

Suppose the original named counts on a block satisfy

$$|N_c([s, t]) - \nu_c(t - s)| \leq \vartheta(t - s) + K[1 + d(s)^\zeta + d(t)^\zeta], \quad d(u) = \min(u - a, b - u), \quad 0 < \zeta < 1. \quad (\text{I.3})$$

Using the shorter end interval in the empirical bridge yields

$$|\mathcal{E}_e(u)| \leq C_H \vartheta d(u) + C_H K[1 + d(u)^\zeta]. \quad (\text{I.4})$$

For $u - a \leq L/2$, subtract the reference mean in its prefix and whole-block terms; their linear errors sum to at most $2\vartheta(u - a)$, and their endpoint errors to $K[1 + d(u)^\zeta] + 2K(u - a)/L$. The other half uses the negative suffix. Sum over boundedly many colors and bounded loss coefficients. Consequently (I.2) has the corresponding lower error with $\sum_e d(t_e)$ and $\sum_e d(t_e)^\zeta$. A small vertex cost cannot automatically pay these errors; its bundled bridge must then retain an actual path constraint.

I.2 Original lifetimes, projection, and delay

A current original internal cell A is born at s_A and split at t_A . If $a(A)$ is its first original name, every other name $c \in A$ has its unactivated original boundary inside A , so $t_c \geq t_A$ and $d_c \geq t_c \geq t_A$. Only $a(A)$ can expire during its life. An anchor-containing cell loses no names; a pure n_A -cell has eligible size n_A or $n_A - 1$. Singleton leaves contribute no delay.

Let q_h be the number of nonempty eligible cells minus one, and $q_{\text{geom}} = \#\{c \neq * : t_c \leq t < d_c\}$. Every active original boundary starts exactly one nonanchor cell. The second rank counts those whose first name is eligible, and the first those with any eligible name. Hence

$$D_h := \int (q_h - q_{\text{geom}}) = \sum_{A \text{ properpureinternal}} [t_A - \max(s_A, d_{a(A)})]_+ \geq 0. \quad (\text{I.5})$$

This holds for all original permutations, zeros and ties. For the tail-entropy potential

$$F_A(t) = \sum_{s \geq t, \omega_s \in A} \log \frac{\pi_s(A \cap E(s))}{\pi_s(\omega_s)} - \int_t^T (|A \cap E(u)| - 1)_+ du,$$

put

$$\hat{F}_A(t) = F_A(t) + \int_0^t [(n_A - 1) - (|A \cap E(u)| - 1)_+] du.$$

The extra increment on a pure cell's life is exactly its term in (I.5), and is zero for an anchor-containing cell. The exact split telescope can thus be written

$$\sum_e C_e(t_e) = F_{A_0}(t_{\text{root}}) + \sum_{A \text{ properinternal}} [\hat{F}_A(t_A) - \hat{F}_A(s_A)]. \quad (\text{I.6})$$

The root's normalization is unchanged: all original names are eligible before its first split. One may equivalently keep unmodified potentials and the explicit delay, but must not count both.

For a parent alphabet $O \sqcup C$, merge all child-touching cells of a fine partition Π into one parent star cell, retaining pure-outside cells, and intersect its cells with C for the child partition. For a law ν , write $H_\nu(\Pi) = \sum_a \nu(a) \log(\nu(\Pi(a))/\nu(a))$ for residual entropy. If $P(a)$ is the parent name, $Z(a)$ the fine cell and $\bar{Z}(a)$ its parent cell, the conditional entropy chain rule is

$$H_\pi(\Pi) = H_{\pi_p}(\Pi_p) + \pi(C)H_{\pi(\cdot|C)}(\Pi_c) - M(\Pi), \quad M(\Pi) = I(P; Z | \bar{Z}) \geq 0. \quad (\text{I.7})$$

If the child occupies one cell, or no child-touching cell contains an outside name, the contrast is identically zero. Otherwise choose an outside name c in a mixed cell and a different child-touching cell A' . In their merged parent cell Q , the joint atom (c, A') is zero, while its product marginal is $\pi(c)\pi(A')/\pi(Q)^2$. If every fine probability is at least p_* , the elementary relative-entropy inequality $D(P\|Q) \geq 2\|P - Q\|_{\text{TV}}^2$ gives

$$M(\Pi) \geq 2\pi(Q)[\pi(c)\pi(A')/\pi(Q)^2]^2 \geq 2p_*^4. \quad (\text{I.8})$$

The actual per-label contrast is

$$D(a) = \log \frac{\pi_p(\Pi_p(P(a)))}{\pi_p(P(a))} + \mathbf{1}_{a \in C} \log \frac{\pi(\Pi_c(a))}{\pi(a)} - \log \frac{\pi(\Pi(a))}{\pi(a)}.$$

Its mean is M , and it is bounded. It can be negative on child names: in a mixed cell A it is $\log[\pi(Q)\pi(A \cap C)/(\pi(C)\pi(A))]$ there, and $\log(\pi(Q)/\pi(A))$ on outside names. Original named regularity, applied on the dimension-many constant-partition intervals, gives

$$\sum_{\text{slots}} D(\omega_s) \geq cS_{\text{mix}} - C[1 + \sum_{t \text{ mixed endpoints}} d_g(t)^\zeta]. \quad (\text{I.9})$$

Here S_{mix} counts only times with at least two child-touching cells and an outside name in one of them. The original intensities have a positive lower bound. This is a fixed-word regularity estimate, including when the history is chosen adaptively.

Eligible unresolved ranks add exactly. If there are u pure outside cells and v child-touching cells, then

$$|O| + |C| - u - v = (|O| + 1 - (u + 1)) + (|C| - v).$$

Let d_h be the original unactivated-boundary rank and r_h the eligible unresolved rank. Then $D_{\text{orig}} = \int (d_h - r_h) \geq 0$: the number of empty original cells is at most the number of ineligible names. This is the same expired-first delay as above. If effective costs use eligible ranks and the two actual information terms in (I.7), their exact same-word decomposition is

$$-\tilde{I}_h = C_p^{\text{eff}} + C_c^{\text{eff}} + D_{\text{orig}} + \sum_s D(\omega_s).$$

Any entrance/root offset is assigned once.

The projections have lawful realizations when all outside deadlines are at most F and all child deadlines at least F . A fine split with two child-touching children belongs to the child; one with a pure-outside child belongs to the formal parent; a pure-outside split also belongs to the parent. These assignments partition the original event multiset. The child is the original line with outside names deleted and each new boundary time the minimum of the intervening original times. The intervening segments are disjoint and every child boundary is legal. Augment the formal parent by isolating each outside name at its deadline if necessary. Its eligible partition is unchanged: before F no child has died; afterwards only the parent star is eligible. Every augmented nonsingleton contains only unexpired names. Represent it as a lawful anchored line by putting the star first in star cells, and a maximal-deadline name first in pure cells. At a split put the child containing that anchor first. Every right-child anchor is then unexpired. Simultaneous refinements have only dimension-many resolutions.

For these lawful projections $D_p = 0$, $D_c \geq 0$, and $D_{\text{gap}} = D_{\text{orig}} - D_c \geq 0$. Before F the child delay is zero; after F the full and child eligible ranks agree, whereas each unactivated child boundary needs an entire disjoint unactivated original segment. Thus $d_h \geq d_c$ pointwise. Rank additivity gives

$$\sum t_h = \sum t_p + \sum t_c + D_{\text{gap}}, \quad \tilde{\Phi}_h = \Phi_p \Phi_c e^{-D_{\text{gap}}}, \quad \tilde{I}_h = I_p + I_c - D_{\text{gap}} - \sum_s D(\omega_s). \quad (\text{I.10})$$

The child uses the actual selected list.

Cancel common formal-parent and augmented-parent event times, with multiplicity. If the erased times are $e_1 \leq \dots \leq e_j$ and added deadlines $a_1 \leq \dots \leq a_j$, refinement at every time gives $a_l \leq e_l$ and $D_{\text{gap}} = \sum_l (e_l - a_l)$. Retain original pivot labels on all unchanged projected cuts. Endpoint regularity bounds the number of original representatives in $[a+l, a+l+1)$ by $C(1+l)^K$. Hence for a fixed projected history

$$\sum_{\text{erased fiber}} e^{-\delta D_{\text{gap}}} \leq C_H \left(C \sum_{l \geq 0} (1+l)^K e^{-\delta l} \right)^H < \infty. \quad (\text{I.11})$$

Crossing physical endpoints costs only the fixed number of groups. No retained child pivot is renumbered as a child-only representative.

I.3 Realizing added deadlines on the actual grid

Assume original uncolored occupancy obeys, at complete-cell endpoints in each block,

$$N([s, t)) \geq \lambda_0(t - s) - K[1 + d(s)^\zeta + d(t)^\zeta], \quad \lambda_0 > 0.$$

For a physical deadline d , let $r(d)$ be the right boundary of its last occupied cell, or zero if there is none. The empty suffix from $r(d)$ to d has length at most

$$B = \max\{1, (2K/\lambda_0)^{1/(1-\zeta)}\} + HK/\lambda_0.$$

Indeed every whole empty block has length at most K/λ_0 ; the one possible initial partial block of length u satisfies $\lambda_0 u \leq K(1 + u^\zeta)$, giving the first term. Replacing at most H added deadline cuts by $r(d)$ changes no activation status at any original slot. Thus the literal comparison classes and all their weighted masses are identical for every acceptance F , even if it couples old labels, types or future words. The multipliers change by the exact factor

$$\Phi_h/\Phi_{h_r} = e^{-\sum(d-r(d))} \in [e^{-HB}, 1].$$

Apply this only to added parent deadlines. The projection becomes

$$\Phi_h = \Phi_{p_r} \Phi_c e^{-D_{\text{gap}} - E_{\text{round}}}, \quad 0 \leq E_{\text{round}} \leq HB.$$

Retained original and child pivots are unchanged. The finite assignment choices preserve (I.11). Projected parent rows must allow repeated representatives: different deadlines can share the same last occupied cell. The interpolation, signature and geometric bounds below hold in this enlarged positive row because they use named cuts with ties and count tuples by a product of representative counts. No distinct-pivot theorem is invoked on that enlarged parent row.

I.4 Branching and the terminal floating chain

Consider a proper owner with exterior names of relative weight one and permanent child of weight z , where $1 < z_- \leq z \leq z_+$ and $r + z \leq H$. In each original group the reference intensities are κ_g per exterior name and $z\kappa_g$ for the child, with $0 < \kappa_- \leq \kappa_g \leq \kappa_+$. Assume named regularity on all original groups used, including old ones. On the natural complete chain $z \geq \sqrt{2}$ follows from $s_m > 1/2$ and the terminal weight. These are reference intensities even after an old word has been fixed.

For an original history, let $B_h(t) = \{*\} \cup \{c : t_c > t\}$. All its names are eligible. Ordering the names by decreasing activation time gives a lawful star history h^* with the same named cuts and multiplier; its only nonsingleton cell is $B_h(t)$. On the same word,

$$C_h - C_{h^*} = I_{h^*} - I_h = \sum_s L_h(\omega_s), \quad (\text{I.12})$$

where

$$L_h(a) = \mathbf{1}_{a \in B_h} \log \frac{\pi_g(B_h)}{\pi_g(a)} - \log \frac{\pi_g(A_h(a))}{\pi_g(a)}.$$

Suppose the current star cell has u exterior names and the pure original internal cells have original sizes $n_l \geq 2$ and eligible sizes $q_l = n_l - \delta_l$, $\delta_l \in \{0, 1\}$. Then $d_h = u + \sum_l (n_l - 1)$, and the reference contrast per unit time equals

$$\kappa_g \{E_{d_h}(z) - E_u(z) - \sum_l q_l \log q_l\}.$$

Move the $n_l - 1$ nonfirst names into the star one cell at a time. For $k = n_l - 1$ and current mass $w \geq z$, the entropy gain is $E_k(w) - q_l \log q_l$. If the first name is alive, it is at least

$$E_k(z) - E_k(1) \geq (z_- - 1) \log(1 + 1/z_+) =: \gamma_z > 0.$$

If expired, it is at least $(k + 1) \log(k + 1) - k \log k \geq 2 \log 2 > \gamma_z$. Thus the contrast rate is at least c_0 times the number of pure original internal cells. When that number is zero the partitions agree, so the contrast vanishes pointwise.

Let S_{pure} be the sum of these cell lifetimes, and β_l the births of the positive-length components where they are present. Subdivide at activations and physical endpoints and apply (I.3) to the bounded coefficients. Since $d_E(t)^\zeta \leq d_E(\beta_l)^\zeta + |J_l|^\zeta$ inside a component, choosing the slope small and absorbing $K|J_l|^\zeta$ into its linear cost gives

$$C_h \geq C_{h^*} + cS_{\text{pure}} - C[1 + \sum_l d_E(\beta_l)^\zeta]. \quad (\text{I.13})$$

Expired-first delay has already been included through original rank. The remaining birth-depth error cannot be absorbed into an arbitrarily short lifetime; it needs a retained growing wall. Boundary shifts of one fixed cell add its regularity depth error and a bounded geometric term, preserving the same estimate on original representatives. This is a cost comparison and asserts no order between class posteriors.

For a terminal owner all eligible names have equal reference intensity, and the preceding strict-child gap is unavailable. Select instead the right endpoint τ of the last positive-length interval with a nonsingleton eligible cell. If none exists, its entropy is zero and $I_h = -\sum t_e$. Otherwise choose two names in that last cell that remain eligible up to τ from the left and retain one, c , as a spine name. Its ancestral cell contains both for every $t < \tau$. It therefore has eligible size $q_{\text{sp}}(t) \geq 2$ and $d_c \geq \tau$. Clip activations after τ to τ ; this changes no eligible entropy and pays $\sum_e (t_e - \tau)_+$ in geometric delay.

For $t < \tau$, let $B(t)$ consist of the eligible names in the spine cell and the nonfirst names of every other original internal cell. This set is nested. At an off-spine split it loses the right child's first name; at a spine split keeping the left child it does the same; at one keeping the right child it loses the old first name if alive; a spine first-name expiration removes that name at its deadline. Off-spine first-name expirations change nothing. Remove all but c at τ . Equivalently label the refinement tree by representatives, retaining c on its ancestral branch and the original left child on off-spine subtrees. Assign each removed representative its split time, clipped to its own deadline if it has expired and then to τ . The time multiset before clipping is the original one. If $\delta_{\text{sp}}(t)$ indicates an expired spine first name, then

$$|B(t)| - 1 = d_{\text{orig}}(t) - \delta_{\text{sp}}(t), \quad D^* = \int_0^\tau \delta_{\text{sp}}(t) dt + \int_\tau^T d_{\text{orig}}(t) dt \geq 0$$

is exactly the loss of the sum of activation times.

This chain has a lawful representation anchored at the original name 0. If $c = 0$, use decreasing removal times. Otherwise let s_0 be 0's removal time and use the line $0, c$, followed by all other names in decreasing removal order; give the boundary before c time s_0 . It is legal since $s_0 \leq \tau \leq d_c$. Call this comparator h^{sp} ; after s_0 its nonsingleton cell floats away from 0. The exact multiplier discrepancy remains D^* .

If off-spine internal cells have original sizes n_l and eligible sizes $q_l = n_l$ or $n_l - 1$, their $k_l = n_l - 1$ nonfirst names give $|B| = q_{\text{sp}} + \sum_l k_l$. The mean contrast is

$$\kappa_g \{ |B| \log |B| - q_{\text{sp}} \log q_{\text{sp}} - \sum_l q_l \log q_l \}.$$

Absorbing a cell into a mass $w \geq 2$ gains, when its first name is alive, $E_k(w) - (k+1) \log(k+1) = \int_1^w \log(1 + k/u) du \geq \log(1 + 1/H)$; expiration only increases the gain. Let S_{off} be the off-spine lifetime. The same regularity argument as for (I.13) gives

$$C_h \geq C_{h^{\text{sp}}} + D^* + cS_{\text{off}} - C[1 + \sum_l d_E(\beta_l)^\zeta]. \quad (\text{I.14})$$

A birth erased because its first name expired has depth at most its distance from that physical deadline, which is paid by D^* . Absorb these errors into a fraction of D^* and a constant. Other births remain comparison cut times.

For an erased original time mapped to a physical endpoint, its exponentially weighted fiber is bounded. If mapped instead to an interior retained τ , named regularity gives only

$$\sum_{\text{original } t \geq \tau} e^{-a(t-\tau)} \leq C_a[1 + d_E(\tau)^\zeta]. \quad (\text{I.15})$$

There are at most $H-1$ such factors. A retained $d_E(\tau)^\gamma$ wall with $\gamma > \zeta$ will pay their polynomial. No bounded occupancy at an arbitrary interior point is presumed.

The reference cost itself pays the floating interval. This uses nonnegativity of every complete lawful nonincreasing reference rank history with the same entrance normalization. In a group with $r+1$ unit names its integrand is $g_g(d) = d - \kappa_g(d+1) \log(d+1)$. On $[s_0, \tau]$ the floating rank lies in $1 \leq d \leq r-1$. Increasing or decreasing it by one on that whole interval gives two lawful reference competitors: the rank drop at s_0 admits the increase, the capacity bound holds, and both end at zero at τ . Discrete concavity gives

$$2g_g(d) - g_g(d-1) - g_g(d+1) = \kappa_g \Delta^2[(d+1) \log(d+1)] \geq \kappa_*/H.$$

The second difference is the integral of $1/x$ against the unit triangular kernel on $[d, d+2]$. Integrating and using nonnegative competitor costs yields

$$\overline{C}_{h^{\text{sp}}} \geq a_H(\tau - s_0), \quad a_H = \kappa_*/(2H). \quad (\text{I.16})$$

Replace c by 0 in the retained set on $[s_0, \tau]$, removing c at s_0 instead of 0 and keeping all other times. This gives a lawful permanent-0 star h^0 with the same time multiset and reference cost. On the actual word their difference is

$$I_{h^{\text{sp}}} - I_{h^0} = \sum_{[s,t) \subset [s_0, \tau)} \log |B|[N_c([s, t)) - N_0([s, t))].$$

The finite subdivision and regularity therefore give

$$|C_{h^{\text{sp}}} - C_{h^0}| \leq C_H \vartheta(\tau - s_0) + C_K[1 + d_E(s_0)^\zeta + (\tau - s_0)^\zeta]. \quad (\text{I.17})$$

The global reference and depth budgets below, rather than a separate floating-chain closure assertion, pay this difference.

Mixed time is preserved sufficiently well by these comparisons. The augmented proper parent has no delay, so its same-cut star has exactly the same eligible unresolved rank. The terminal comparator has rank at least that of the original terminal partition: before τ they are respectively $d_{\text{orig}} - \delta_{\text{sp}}$ and $d_{\text{orig}} - \delta_{\text{all}}$, with $\delta_{\text{sp}} \leq \delta_{\text{all}}$; afterwards both are zero. Iterating eligible-rank additivity implies that every comparison child has unresolved rank at least its original child. Thus if it has

at least two cells, so did the original. If a proper parent has no pure internal cell, its original and star partitions agree pointwise. Writing M_j, M_j^* for the original and comparison mixed indicators,

$$M_j^*(t) \leq M_j(t) + \mathbf{1}_{\{A_j(t) > 0\}}, \quad \sum_j S_{\text{mix},j}(h^*) \leq \sum_j S_{\text{mix},j}(h) + \sum_{j < m} S_{\text{pure},j}. \quad (\text{I.18})$$

Indeed mixedness means precisely that the merged parent star has an outside name and the child has at least two cells. Consequently a small fixed fraction of original mixed charge can be reserved for the global comparator while retaining half each proper pure-lifetime charge. This is a deterministic comparison of indicators, not of the signed per-label contrast.

I.5 Synchronizing cuts and sharing one endpoint budget

Here first consider a proper coarse owner, with $g_i(d) = d - \kappa_i E_d(z)$ and $\kappa_i \geq \kappa_* > 0$. Suppose all lawful endpoint stars on the supported actual word satisfy $C_v \geq c_0 \bar{C}_v - h_E$, with $\bar{C}_v \geq 0$ and one common normalization. In a fixed chamber, synchronize its m star cuts in a block at their average $\tau = (t_1 + \dots + t_m)/m$, and let $D_i = t_m - t_1$. The actual before/after named sets have ranks $R, q = R - m$ and remain legal. The rank integral is unchanged because $\int_a^b (d(t) - q)/m dt = \tau - a$. Since $g_i''(d) = -\kappa_i/(d + z) \leq -\kappa_*/H$, the mean excess over this chord is at least

$$\frac{\kappa_*}{2H} \int_a^b (d - q)(R - d) dt \geq \frac{\kappa_*}{2H} D_i.$$

For one cut both sides vanish. The actual contrast is supported in $[t_1, t_m]$; named regularity on its finitely many pieces, including the adjacent cell at a nongrid average, bounds the error by a small linear multiple of D_i and $C[1 + d_i(\tau)^\zeta + D_i^\zeta]$. Absorb the spread power into its linear charge. Combining with (I.13), whose births occur at these cuts, gives

$$C_h \geq C_{\text{sync}} + c \sum_i D_i + c_p S_{\text{pure}} - C[1 + \sum_i d_i(\tau_i)^\zeta]. \quad (\text{I.19})$$

No new arithmetic pivot is assigned to τ_i .

The synchronized endpoint domain is a product of two-point choices, each whole bundle placed at its block's left or right endpoint. Let L_{multi} be the interpolation of their actual costs. On the same word,

$$C_{\text{sync}}(\tau) = L_{\text{multi}}(\tau) + \sum_i \mathcal{R}_i(\tau_i),$$

where $\mathcal{R}_i$ is minus the empirical bridge of its actual bundled partition loss, with zero endpoints. Its difference Δ_i between endpoint costs and its remainder are independent of the outside endpoint choices. Thus L_{multi} is an affine sum of one-coordinate functions. Shift every cost by h_E once and put

$$a_i^* = \min_{\text{outside}} (C_{i,L,\text{outside}} + h_E), \quad b_i^* = a_i^* + \Delta_i, \quad \ell_i(t) = (1 - u_i)a_i^* + u_i b_i^*, \quad u_i = (t - a_i)/L_i.$$

Both endpoints are nonnegative, and the same outside choice minimizes both. Positive interpolation gives $L_{\text{multi}} + h_E \geq \ell_i(\tau_i)$ for every i . For $\theta_{\text{alloc}} = 1/(2H)$, set $W_i = \theta_{\text{alloc}} \ell_i + \mathcal{R}_i$. Since there are at most H coordinates,

$$C_{\text{sync}} + h_E \geq \frac{1}{2}(L_{\text{multi}} + h_E) + \sum_i W_i(\tau_i). \quad (\text{I.20})$$

The single endpoint budget is thereby allocated before adding walls.

If $|g_i(R) - g_i(q)| \geq \epsilon > 0$, every reference endpoint at the more expensive side exceeds ϵL_i , and $\ell_i(t) \geq c_0 \epsilon$ times the distance from the cheaper endpoint. Equation (I.4) gives $W_i(t) \geq c \text{dist}(t, \text{cheaper end}) - C$. If both reference endpoints in every legal outside context exceed ηL_i , it gives $W_i(t) \geq c L_i - C$ instead. All remaining weak coordinates retain the actual allocated wall

$$\theta_{\text{alloc}} \ell_i(t) + \mathcal{R}_i(t) \geq g d_i(t)^\gamma - h_W, \quad \gamma > \zeta. \quad (\text{I.21})$$

On native blocks the reference endpoint classification identifies these as full-current-input to deepest-suffix edges. Old weak blocks require their own established charge or support. At a nongrid synchronization time, adjacent-cell regularity adds $C[1 + d_i(t)^\zeta]$, absorbed by half the strong curve; the affine chord changes only by a bounded amount over one cell since its endpoint difference is $O(1 + L_i)$.

Combining (I.19)–(I.21) gives positive spread and pure-lifetime penalties, linear cheaper-end distance penalties for strong coordinates, full-length penalties for paid weak ones, and $(g/2)d_i(\tau_i)^\gamma$ for unpaid weak ones, minus a fixed constant. Their exponential pivot sum is bounded. To verify multiplicity, bin spreads and endpoint distances in unit intervals. Every cut is within its spread of the average, so original endpoint occupancy bounds the number of tuples by a fixed polynomial in these quantities. Exponential or stretched-exponential penalties sum that polynomial, and only dimension-many chambers occur. Repeated representatives in the enlarged parent space obey the same product upper bound. Throughout, the comparison class remains the original one.

I.6 Global endpoint costs and the uncontracted root

Use the actual common-root source of Section 9. Fix throughout the global source and target arguments $0 < \zeta < \gamma < 1/2$, with the regularity slope and the intrinsic wall contractions chosen before constructing the source. Owners have products σ_j , ratios $u_j = \sigma_{j+1}/\sigma_j$, $d_j = u_j^{-1} - 1$ and $r_j = \sigma_j/\sigma_1$. Let T_j be their native lengths and $V_j = \sum_{a \leq j} T_a$. The hierarchy gives $V_j \ll T_j$. Original groups remain separate, including short or zero ones. The actual tuple has root $X_1 = x + \epsilon$, with

$$x = \sum_j r_j y_j, \quad y_j \geq 0, \quad -C \leq \epsilon \leq 0.$$

On its source word the proper targets are zero, the strict exits obey $0 \leq \eta_j \ll \sqrt{1 + T_j}$ and $\sum_{j < m} d_j \eta_j \ll 1$. Set

$$a_j = \eta_j, \quad b_j = \eta_{j-1}/u_{j-1} \quad (j < m), \quad a_m = x/r_m, \quad b_m = \eta_{m-1}/u_{m-1} - \epsilon/r_m,$$

with initial incoming term zero. Then $Z_j^0 = a_j - b_j$ is the exact signed owner output, and $R_0 = \sum_j Z_j^0$ is the full fine residual. In particular the bounded root error remains in a nonnegative b_m . Write $K_j(v) = C_j(v) - b_j = Z_j^0 - I_j(v)$.

A fine endpoint star isolates each nonanchor fine name at a lawful original physical endpoint, keeping all unresolved names in the permanent anchor's cell. Its atomic projections retain the same named cuts. Let t_j be owner j 's last positive cut and $s_j = \mathbf{1}_{\{t_j \leq V_j/2\}}$. The endpoint count estimates for the displaced source boxes give the coefficient-one inequalities

$$K_j(v_j) \geq s_j Z_j^0 + c H_j - C, \quad (\text{I.22})$$

$$K_j(v_j) \geq s_j Z_j^0 + c J_j - C, \quad (\text{I.23})$$

where

$$H_j = s_j t_j + (1 - s_j)[V_j - t_j + L_{\text{bad},j} + S_{\text{good},j}], \quad J_j = \overline{C}_{v_j} + S_{\text{good},j}.$$

The second retains the positive reference terms of the finite-score proof: at early endpoints use the exact prefix expression and absorb its $O(\sqrt{1+t_j})$ centered error into a fraction of reference cost; at late endpoints use the exact tail and pay only its zero-tail square root. The good-group displacement remains positive. Reference cost controls the early prefix and late bad/zero lengths, so $H_j \ll J_j + 1$. These inequalities keep a_m exactly at an early terminal endpoint and b_m exactly at a late one; there is no error of size $\sqrt{V_j}$ from unrelated full groups.

There are no added cuts in a fine star's projections. The exact entropy chain rule gives

$$C_0(h) := R_0 - I_h(\omega_0) = \sum_j K_j(v_j) + \sum_{j,s} D_j(\omega_s).$$

On each genuinely mixed physical group, full named central counts and (I.8) give $cL_g - C\sqrt{1+L_g}$; absorb half its mean. Nonmixed contrasts vanish identically. With

$$J(h) = \sum_j J_j + \sum_j S_{\text{mix},j}, \quad H(h) = \sum_j H_j + \sum_j S_{\text{mix},j},$$

one therefore obtains $C_0(h) \geq \sum_j s_j Z_j^0 + cJ(h) - C$, and also its version with H .

Every adverse flag change $s_j = 0, s_{j+1} = 1$ has a charge at its own scale. If $t_{j+1} \geq t_j/2$, the child's early prefix contributes at least $V_j/4$. Otherwise, on (t_{j+1}, t_j) an eligible parent exterior remains in the fine anchor cell while an immediate-child exterior has become a separate singleton. All those child names are eligible up to $V_j \geq t_j$. This is a genuinely mixed interval of length greater than $V_j/4$. Thus $H \geq V_j/4$, and $J \gg V_j - C$, for every adverse change. Expanding the outputs exactly,

$$\sum_j s_j Z_j^0 = s_m(x + \epsilon)/r_m + \sum_{j < m} \eta_j(s_j - s_{j+1}/u_j).$$

The equal-one terms sum to $-\sum d_j \eta_j = O(1)$. Keep every positive η_j from a 1-to-0 change. Each 0-to-1 loss is $O(\sqrt{1+V_j})$ and is absorbed by a small fraction of its own linear charge. We conclude

$$C_0(h) \geq s_m a_m + \sum_{j < m: s_j=1, s_{j+1}=0} \eta_j + cJ(h) - C. \quad (\text{I.24})$$

The same proof holds when the actual mixed contrast is replaced by any fixed sufficiently small positive multiple of mixed time.

We record the relevant same-word transport calculation. In owner j 's native interior bank of length $\gg T_j$, recolor e_j copies of the immediate child's first shallow leaf to the local shallow exterior. Put $A_j = \log S_j(\text{first})$, $b_j^{\text{flip}} = u_j A_{j+1}$. The construction gives

$$b_j^{\text{flip}} e_j - A_j e_{j-1} = y_j + \delta_j, \quad e_0 = 0, \quad 0 \leq \delta_j < b_j^{\text{flip}}, \quad e_j \ll \log(2 + T_j), \quad (\text{I.25})$$

and the original fine prior ratio is $\log[p(c_j)/p(d_j)] = b_j^{\text{flip}} - A_{j+1}$. Thus the full residual changes by $\Delta R = \sum_{j < m} e_j(b_j^{\text{flip}} - A_{j+1})$. At a regular endpoint bank the possible signatures and per-flip increments of the fine cost are

owner full	0
owner zero, child full	b_j^{flip}
owner zero, child zero	$b_j^{\text{flip}} - A_{j+1}$.

Here “full” means the entire relevant eligible cell, with no mixed state at those two levels. The missing signature, owner full with a split child, is mixed. The increments follow directly by subtracting ΔI from the stated ΔR ; for instance in the middle case $\Delta I = -A_{j+1}$. This keeps the prior change.

Any discrepancy from the ideal coefficient $b_j^{\text{flip}} s_j - A_{j+1} s_{j+1}$ is paid by $J(h) \gg \sqrt{T_j} - C$. The exhaustive cases are: a proper parent or child state contributes an early/bad length or good-group square root in its genuine native or old group; a mixed state contributes the whole group; an early full owner or child has an early prefix reaching that group; a late zero owner has a zero tail containing it; a late zero child has a tail containing its entire native length, which is $\gg T_j$ by the hierarchy. Every per-flip coefficient is bounded. Since $\log(2 + T_j) \leq \epsilon' \sqrt{T_j} + C_{\epsilon'}$, these exceptional banks consume an arbitrarily small fixed part of J . Regular banks keep their exact increments.

The main coefficients telescope without any monotonicity of the flags:

$$\sum_{j < m} e_j (b_j^{\text{flip}} s_j - A_{j+1} s_{j+1}) = \sum_{j < m} s_j (y_j + \delta_j) - s_m A_m e_{m-1}.$$

Since $r_j b_j^{\text{flip}} = r_{j+1} A_{j+1}$, the actual root gives

$$a_m - A_m e_{m-1} = y_m - \sum_{j < m} (r_j / r_m) \delta_j = y_m + O_H(1). \quad (\text{I.26})$$

Thus every fine endpoint star on the target satisfies

$$C_y(h) \geq \sum_{j: s_j=1} y_j + cH(h) - C. \quad (\text{I.27})$$

The early owners can be an arbitrary subset. Each target, including the terminal one, has coefficient one.

For later use, if a late owner’s endpoint template is not full on an original native block g , its charge is at least $c\sqrt{L_g} - C$. Indeed a zero parent pays its zero tail; a proper parent pays its bad length or good square root; a full parent with a nonfull fine class is genuinely mixed on the entire group. For the terminal owner the atomic and fine alphabets agree. These charges add over the finite set of affected groups after reducing the common coefficient.

I.7 Intrinsic chords within a global fine history

Fix a native canonical full-current-input to deepest-suffix edge in owner j and physical block $i = [l_i, b_i]$. First consider a global endpoint context in which the whole immediate child remains intact on the open block. Its varying bundle is exactly the indicated owner exteriors, with deadlines at least b_i ; all other cuts are fixed lawful endpoints. The global endpoint difference and centered bridge are precisely the atomic ones:

$$C_0(h^+) - C_0(h^-) = K_j(v_j^+) - K_j(v_j^-) = \Delta_i, \quad \Delta_i = (r_i - q)L_i - \log S_i n_i + \log(q + z)K_{i,B}.$$

Let ℓ_i^{intr} be the repaired chord from the two actual all-context intrinsic minima, shifted by their fixed reserve h_{intr} . If the child is early, $t_{j+1} \geq b_i \geq L_i$, so both global endpoints have a charge $cL_i + h_*$ after one fixed shift. If the child is late, the intrinsic score upper is

$$K_j(v_j) + h_{\text{intr}} \leq C(h_{\text{intr}} + s_j a_j + J_j).$$

For an early proper owner followed by a late child, (I.24) retains exactly $a_j = \eta_j$; for a late owner J_j suffices. At the terminal owner it retains $s_m a_m$. Hence in these cases both global endpoints dominate a fixed positive multiple of the intrinsic endpoints. Interpolation gives the same chord domination.

Choose a global budget share $\theta_{\text{glob}} > 0$. The linearly paid case plus (I.4) pays a strong inward wall. In the chord case choose in advance $0 < \theta_{\text{intr}} \leq c_* \theta_{\text{glob}}$. The source's actual contracted wall $\theta_{\text{intr}} \ell_i^{\text{intr}} + \mathcal{R}_i \geq g d_i^\gamma - h_W$ then implies the global allocated wall. This is why the contractions are fixed before the source probability construction.

Terminal coordinates require the root to remain outside this allocation. The source supplies same-root representatives with actual local increment and $x_i^- \leq C_A X_i^0$, $y_i^- \leq C_A Y_i^0$, where the deterministic scores X_i^0, Y_i^0 are computed at zero terminal target. For a terminal prefix these follow by a virtual zero-root count completion of its actual type; for the ordinary reservoir the actual root count stays fixed and the extra left reserve is $O_A(\log(2 + V_{\text{pre}})) \ll X_i^0$, while the right stays in its fixed box. No virtual clock is used. After subtracting $a_m s_m$ from a completed terminal endpoint, (I.24) dominates its zero-target score. Consequently its root-separated endpoint chord dominates a fixed multiple of the actual source representative chord, even if the separated endpoint difference itself is no longer Δ_i .

Now allow an arbitrary fine star. Synchronize by owner within each original physical block: $\alpha = (j, i)$ has mean time τ_α , spread D_α , and endpoint depth d_α . Atomic concavity, including terminal $z = 1$, gives

$$\sum_j K_j(h_j) \geq \sum_j K_j(h_j^{\text{sync}}) + c_s \sum_\alpha D_\alpha - C[1 + \sum_\alpha d_\alpha^\zeta].$$

For a fine star its mixed time has the exact formula $S_{\text{mix},j} = (t_j - \theta_j)_+$, where t_j is the owner's largest exterior cut and θ_j is the smallest cut of any nonanchor fine name in its entire child. All child names are eligible until $V_j \geq t_j$. Maxima, minima and positive parts are Lipschitz, so synchronization changes total mixed time by at most $C_H \sum D_\alpha$. The actual mixed contrast is at least $c_M \sum S_{\text{mix},j}$ minus $C[1 + \sum (d_\alpha^\zeta + D_\alpha^\zeta)]$: every nontrivial error endpoint is an original cut, within its spread of its mean. Choose a small fixed $\epsilon_{\text{mix}} > 0$ relative to c_M, c_s . The fine cost therefore dominates

$$F(\tau) = \sum_j K_j(h_j^{\text{sync}}) + \epsilon_{\text{mix}} \sum_j S_{\text{mix},j}(h^{\text{sync}})$$

plus a positive spread charge and the stated depth error.

In each physical block order all its synchronized owner-group times. On this ordered chamber the mixed term is affine. Use ordered-simplex endpoint weights within that block, and products only across distinct blocks. If F_v is the endpoint value and $a = a_m$, then exactly

$$F(\tau) = a\mathcal{S} + \hat{L} + \sum_\alpha \mathcal{R}_\alpha, \quad \mathcal{S} = \sum_v \omega_v s_m(v), \quad \hat{L} = \sum_v \omega_v (F_v - a s_m(v)). \quad (\text{I.28})$$

The enhanced endpoint argument applies with $\epsilon_{\text{mix}} S_{\text{mix}}$ and yields

$$F_v - a s_m(v) \geq \sum_{j:s_j=1, s_{j+1}=0} \eta_j + cJ(v) - C, \quad \hat{L} + C \geq c \sum_v \omega_v J(v) \geq 0.$$

The coordinate marginal remains $\mathbb{P}(v_\alpha = b_i) = (\tau_\alpha - l_i)/L_i$. The root term $a\mathcal{S}$ is not included in $\hat{L}$.

A proper native weak coordinate can have a partially split child; its $\mathcal{R}_\alpha$ is still the atomic projection bridge. Fix only interpolation choices in other physical blocks. Let t_{out} be the largest

child-owner cut there, including fixed boundary cuts. It is at most l_i or at least b_i . If $t_{\text{out}} \geq b_i$, the child's flag is fixed. An early child pays cL_i at every current vertex; a late child leaves either the positive owner exit for an early owner or its J_j for a late one, so the intrinsic chord comparison applies. If $t_{\text{out}} \leq l_i$ and there are child cuts in this block, a right child cut is entirely old and its reference cost is at least $cb_i \geq cL_i$. If all child cuts are left but the parent bundle is right, the whole block is mixed. Thus every parent-right vertex has charge cL_i , and the coordinate marginal pays $c(\tau_\alpha - l_i)$. With no child cut in this block, its flag is fixed: an early child gives the same mixed payment and a late one allows the intrinsic comparison. Averaging the outside choices gives

$$\hat{L} + C \geq c \min\{\tau_\alpha - l_i, \ell_\alpha^{\text{intr}}(\tau_\alpha)\}. \quad (\text{I.29})$$

The first branch plus regularity pays the inward curve; the second plus the same actual $\mathcal{R}_\alpha$ is supplied by the contracted source wall. The identity $\min(A, B) + E = \min(A + E, B + E)$ completes this case.

Strong coordinates have a fixed reference drift gap and are paid from their cheaper endpoint. Old weak and native proper-input or wrong-name weak coordinates have two-ended reference charge cL_i . These statements apply vertex by vertex using the actual marginals. Remaining terminal coordinates use the zero-target representative comparison above, including the actual ordinary root count. Choose $\theta_{\text{glob}} \leq 1/(4H^2)$ and all source contractions small enough in advance. For every coordinate,

$$\theta_{\text{glob}}(\hat{L} + C) + \mathcal{R}_\alpha \geq g_0 d_\alpha^\gamma - C_0.$$

There are at most H^2 shares. Their sum leaves at least three quarters of $\hat{L} + C$. After absorbing the earlier d^ζ errors,

$$C_0(h) \geq a\mathcal{S} + c \sum_v \omega_v J(v) + c \sum_\alpha (d_\alpha^\gamma + D_\alpha) - C. \quad (\text{I.30})$$

Every empirical bridge has been paid once.

I.8 Selecting an endpoint without losing the root

Choose in each physical block an ordered-simplex vertex of maximum weight, with a fixed tie rule. Their product $v^\#$ is lawful and has weight at least $(H + 1)^{-H}$. A chosen endpoint side has marginal probability at least $1/(H + 1)$. If it is the nearer side, its movement is d_α ; if farther, that marginal is d_α/L_i , so its movement is at most $(H + 1)d_\alpha$. Original unsynchronized cuts add their spread. In particular

$$\sum_v \omega_v J(v) \geq (H + 1)^{-H} J(v^\#), \quad |t_c - v_c^\#| \leq C_H(d_\alpha + D_\alpha). \quad (\text{I.31})$$

A coordinatewise nearest vertex need not have positive weight bounded below; the maximum-weight choice avoids that issue.

Suppose $0 \leq a \leq A \log(2 + T)$, where $T = V_m$. Then for every $\epsilon > 0$,

$$a|\mathcal{S} - s_m(v^\#)| \leq \epsilon \sum_v \omega_v \bar{C}_m(v) + \epsilon \sum_\alpha d_\alpha^\gamma + C_\epsilon. \quad (\text{I.32})$$

Only the unique physical block $[l, b]$ crossing $T/2$ can vary the flag, unless another fixed-late cut forces all flags to zero. The coordinate marginals and controlled movement give $|\mathcal{S} - s_m(v^\#)| \leq C_H \sum d_\alpha/L$ in this block, as follows: for an early representative use the union bound for a right

choice; for a late one use one chosen-right coordinate to bound the all-left event. If $L \geq T/4$, then $A \log(2+T)d/L \leq A \log(2+4L)L^{-\gamma}d^\gamma$, whose coefficient is arbitrarily small at large L ; bounded L costs a constant. If $L < T/4$, both endpoints lie inside $(T/4, 3T/4)$, and every contributing endpoint has last terminal cut there. The clipped-prefix reference lower $\bar{C}_m(v) \geq c_H \min(t_v, T-t_v)$ therefore pays $c_H T$. It absorbs $A \log(2+T)$ with an arbitrarily small share and a constant. This proves (I.32) and keeps the full root coefficient. Applying it to (I.30) gives

$$C_0(h) \geq as_m(v^\#) + cJ(v^\#) + c \sum_{\alpha} (d_{\alpha}^{\gamma} + D_{\alpha}) - C.$$

We need a slightly stronger weighted form. Put

$$F_{\epsilon}^0(h^*) = \sum_j K_j^0(h_j^*) + \epsilon \sum_{j < m} S_{\text{mix},j}(h^*), \quad B(h^*) = \sum_j \bar{C}_j(h_j^*), \quad \Pi(h^*) = \sum_{\alpha} (D_{\alpha} + d_{\alpha}^{\gamma}).$$

For fixed sufficiently small $\epsilon > 0$, the same proof yields

$$F_{\epsilon}^0(h^*) \geq as_m(v^\#) + c[J(v^\#) + B(h^*) + \Pi(h^*)] - C. \quad (\text{I.33})$$

Indeed mixed-time Lipschitz and atomic synchronization apply directly to this deterministic mixed-length term. At endpoints any fixed positive mixed coefficient pays adverse flag changes. Reserve positive shares of reference and spread, in addition to all bridge allocations:

$$0 \leq B(h^*) - B(h^{\text{sync}}) \leq C_H \sum_{\alpha} D_{\alpha}.$$

The lower is concavity; the upper follows because the rank profiles differ only between each group's first and last cut, with uniformly bounded integrands. These shares retain $B(h^*)$, while further shares pay J , curves, spread and the flag interpolation. All allocations are fixed before constructing the source.

I.9 From a general fine history to its source and target costs

Proposition I.1 (Original-history source geometry). *On the source support just constructed, every lawful original fine history h has a lawful comparison star h^* and endpoint representative $v = v^\#$ such that*

$$C_0(h) \geq as_m(v) + c\Lambda(h) - C, \quad \Lambda = J(v) + B(h^*) + \Pi(h^*) + D_{\text{gap}} + D_* + S_{\text{pure}} + S_{\text{off}} \geq 0. \quad (\text{I.34})$$

For every fixed $\delta > 0$, the associated geometric envelope has a uniformly bounded sum over all original histories and original pivot representatives. All constants are independent of length ratios and strict product gaps.

Proof. Recursively apply the lawful projection (I.10), keeping original time labels and selected lists. The exact identity is

$$C_0(h) = \sum_j K_j^0(h_j) + D_{\text{gap}} + M_{\text{orig}},$$

with $M_{\text{orig}} \geq c_M S_{\text{orig}} - C[1 + \sum_{\text{original cuts } t} d_E(t)^{\zeta}]$ from (I.9). Apply (I.13) to proper owners and (I.14), (I.17) to the terminal one. Their signed output normalizations cancel in each same-word difference. They retain positive proper pure lifetimes, terminal off-spine lifetimes and true clipping delay D_* . The remaining terminal loss is

$$C_H \vartheta T_{\text{float}} + C_K [1 + T_{\text{float}}^{\zeta} + \sum_{\text{births}} d_E(\beta)^{\zeta}], \quad B(h^*) \geq a_H T_{\text{float}}.$$

If there is no terminal nonsingleton interval, choose its comparison cuts zero, $D_* =$ the sum of its original times, and $S_{\text{off}} = T_{\text{float}} = 0$; this case is exact. By (I.18), a sufficiently small $\epsilon < \min(c_M/2, c_p/4)$ retains $\epsilon S_{\text{mix}}(h^*)$ while leaving positive shares of original mixed and pure time. The weighted quantity in (I.33) is therefore available.

Every remaining depth error has a specific original charge. An erased projection event is paired to a physical deadline with its distance in D_{gap} . A terminal event is either retained, clipped to its deadline, or clipped to τ . In the last case $d_E(e) \leq d_E(\tau) + (e - \tau)$, with $e - \tau$ in D_* ; τ is a retained terminal cut when it is not a physical endpoint. The same alternatives treat expired-first births. Retained cuts lie within their group spread of its synchronized time. Thus all errors are bounded by

$$C[1 + D_{\text{gap}}^\zeta + D_*^\zeta + \sum_\alpha (d_\alpha^\zeta + D_\alpha^\zeta)]. \quad (\text{I.35})$$

Choose the regularity slope small enough that a reserved fraction of $B(h^*) \geq a_H T_{\text{float}}$ pays the linear floating term. Young's inequality pays its sublinear term with another fraction. Use positive delays and spread for their ζ -powers, and the $\gamma > \zeta$ inward curves for d_α^ζ . Retain a fixed positive share of every budget. This proves (I.34).

The projection fibers sum by (I.11). Terminal fibers at a physical endpoint sum by the same bound; at interior τ they incur the finite polynomial in (I.15), paid by the retained inward and spread budgets. Remaining cuts lie in the endpoint strips (I.31); their polynomial multiplicity is summed by exponential spread and stretched-exponential depth weights. Finite assignment chambers, names, tie resolutions and permutations only change the constant. Generated repeated deadline representatives remain in the enlarged positive row. This proves summability without replacing an original class by a comparison class. $\square$

Proposition I.2 (Original-history target geometry). *For the literal transported bounded density F_y on the same actual root tuple, every word of positive weight and every original fine history satisfy, with the constructions above,*

$$C_y(h) := R_y - I_h(\omega_y) \geq \sum_{j:s_j(v)=1} y_j + c\Lambda(h) - C. \quad (\text{I.36})$$

Its geometric envelope has the same original-pivot summability.

Proof. Fix any retained source preimage of the target word. Such a preimage exists for every positive value of the literal transported density. For a comparison fine star define $F_\epsilon^q = \sum_j K_j^q + \epsilon S_{\text{mix}}$ for $q = 0, y$. Mixed time depends only on names and cuts, so exactly

$$\Delta F_\epsilon(h^*) = \Delta R - \sum_j \Delta I_j(h_j^*).$$

Each atomic information is a sum over original slots, zero when a slot does not route to that owner; the formula remains exact when recoloring changes child membership. Compare this change with the change at the endpoint representative v on the same source/image pair. The residual ΔR cancels. The bounded atomic mark vector changes only if a flipped slot meets a movement strip. Bank j has depth $\gg T_j$ in its original block; such an intersection forces $\Pi(h^*) \geq c_H T_j^\gamma - C_H$. Since it contains $O(\log(2 + T_j))$ flips, for every fixed $\alpha > 0$

$$|\Delta F_\epsilon(h^*) - \Delta F_\epsilon(v)| \leq \alpha \Pi(h^*) + C_\alpha. \quad (\text{I.37})$$

At a regular endpoint bank, the three per-flip increments computed above apply to $\sum_j K_j$ itself: the relevant mixed contrasts vanish identically, ancestor exteriors have expired, and deeper

owners read neither changed shallow name. Exceptional signatures have $J(v) \gg \sqrt{T_j} - C$, by the exhaustive endpoint cases already proved. Their logarithmic number consumes any small fixed fraction of $J(v)$. Hence

$$\Delta F_\epsilon(v) \geq \sum_{j < m} e_j(b_j^{\text{flip}} s_j - A_{j+1} s_{j+1}) - \alpha J(v) - C_\alpha.$$

Apply (I.33), (I.37), and the exact telescope (I.26); nonnegative δ_j only help. For small α ,

$$F_\epsilon^y(h^*) \geq \sum_{j: s_j(v)=1} y_j + c[J(v) + B(h^*) + \Pi(h^*)] - C. \quad (\text{I.38})$$

The terminal root has coefficient one, without a comparison of different root tuples or a monotonicity claim for target local minima.

The target retains the original named regularity with prescribed spare margins. An interval meeting a flipped interior bank either has length $\gg T_j$ or an endpoint of depth $\gg T_j$; otherwise it cannot reach that bank. A spare linear slope in the first case, or a spare ζ -depth coefficient in the second, pays $O(\log(2 + T_j))$ flips after fixing a sufficiently large owner threshold. Routed atomic counts are sums of finitely many fine counts. Thus choose target regularity constants first and construct the source inside them; no clock or reference intensity changes.

Project the original history on this target word. Its exact identity is $C_y(h) = \sum_j K_j^y(h_j) + D_{\text{gap}} + M_y$. Use the proper and terminal same-word comparisons and mixed-time domination exactly as in the preceding proof. The available weighted budget is now (I.38). The floating reference cost pays the linear and sublinear floating errors; (I.35) is paid by the same true delays, spread and inward curves. Positive shares remain, proving (I.36). Fiber counting is unchanged and keeps every original representative. This argument holds simultaneously for all histories on the chosen source/image pair; it does not compare class posteriors. $\square$

I.10 The actual comparison class and the complete row

Set, for the predeclared environmental costs and count widths,

$$\begin{aligned} D_j^{\text{own}} &= w_j \prod_{i: \text{owner}(i)=j} D_i \quad (j < m), & D_m^{\text{own}} &= \prod_{i: \text{owner}(i)=m} D_i, \\ y_j &= -\log D_j^{\text{own}}, & D_{\text{all}} &= \prod_j D_j^{\text{own}}. \end{aligned}$$

These targets are fixed before reading the private word. Fix its actual class $Q_h(\omega)$ and the bounded transported acceptance $F = F_y$. It has the actual root, named count support, strict exits, reconstructed source tests, and bounded full relative likelihood established in the transport construction.

In a physical block $g = [l_g, b_g]$, choose an integer endpoint-strip radius $r_g \geq 1$ dominating, by a dimension factor, the depths of all original cuts and all movement strips from h^* to v ; cap it at $L_g/2$ when necessary. From the exact depth bookkeeping,

$$\sum_g r_g^\gamma \leq C_H(1 + \Lambda). \quad (\text{I.39})$$

If $r_g < L_g/4$, the original partition is constant on the middle interval and h^* there agrees with v . These are partition facts, independent of label probabilities. Proper projection and star

comparison preserve eligible unresolved rank. The only possible increase for h^* is from expired terminal off-spine first names, whose number is at most the original terminal off-spine internal-cell count. Thus, pointwise,

$$0 \leq r(h^*) - r(h) \leq m_{\text{off}}.$$

If v is full but the original middle is not, this rank difference is at least one over an interval of length $L_g/2$, so $S_{\text{off}} \geq L_g/2$. If v is nonfull on a late owner's native block, its endpoint charge is $J(v) \geq c\sqrt{L_g} - C$, as proved above. If the strips occupy a quarter block, their own L_g^γ charge pays. Every late block is therefore geometrically paid or the original class is full outside its endpoint strips. Any missing polynomial floor on a paid block is absorbed by an arbitrarily small fixed fraction of Λ at its own length scale.

For a proper late owner's count bank, either those strips reach its interior or the original class is full on the bank. In the first case Λ has a positive power of T_j and pays the width floor $w_j \gg (2 + T_j)^{-1/2}$. Let J_{free} be any set of remaining banks. Condition on the labels read by the actual class, revealing their full labels if needed, then on pair-coarsened data in these banks and every label outside the relevant two-name pairs. The remaining counts are independent original binomials, on at least cT_j trials on accepting support. Their exact diagonal formulas are $\eta_j = h_j + b_j^{\text{flip}} K_j$. All h_j and target-band centers are measurable in this one fixed sigma-field, independent of every remaining K_j . Uniform binomial maximum atoms at arbitrary centers therefore give

$$\mathbb{P}(\text{all remaining exit restrictions} \mid \text{fixed data}) \leq C \prod_{j \in J_{\text{free}}} w_j. \quad (\text{I.40})$$

Too-small pair totals have empty accepting support. Releasing the fixed data preserves the bound. Early-owner blocks and arbitrary observed signatures need no full-block reveal penalty.

For an unpaid selected path block, condition first on every complete target type and all revealed strip labels. The remaining labels have the original uniform multiset law. The reconstructed-source estimates, uniform over every permitted inverse transfer, give the environmental envelope

$$\mathbb{P}(\text{necessary tests} \mid \text{types, revealed strips}) \leq C q_{i,r_i}(G_i). \quad (\text{I.41})$$

This is the joint two-bank density comparison, not a product of two separately conditioned full bridges. These predeclared envelopes are uniform over the types subsequently summed. Below their logarithmic cutoff, strip occupancy leaves at least half the block for that joint comparison. Above the cutoff take $q = 1$ and pay the predetermined polynomial floor of D_i by the depth charge; supported bounded blocks use $D_i = 1$.

After complete types are fixed, different blocks' remaining multiset permutations are independent. Drop all other restrictions, multiply (I.41), then release the whole coupled type measure and apply (I.40). This proves for the original class

$$\mathbb{E}_\pi[F \mid Q_h(\omega)] \leq C \prod_{j \in J_{\text{free}}} w_j \prod_{i \in I_{\text{free}}} q_{i,r_i}(G_i). \quad (\text{I.42})$$

Every type is summed before any fractional power. Only necessary tests of F have been used; no arbitrary child weight has been multiplied by an averaged path probability.

The same environmental envelopes can be chosen for any sufficiently small fixed $b_0 > 0$ so that $q_{i,r} \leq C D_i e^{b_0 r^\gamma}$. Their stretched-exponential coefficient sum is finite, so clipping and normalization into $[0, 1]$ changes only a fixed constant. Their prescribed polynomial floors and source

moments persist. Choose b_0 small relative to the coefficient in (I.36) and (I.39). Pay all missing late widths and path floors with the local payments above. Then

$$\mathbb{E}_\pi[F \mid Q_h(\omega)] \leq C e^{(c/2)\Lambda} \prod_{j:s_j(v)=0} D_j^{\text{own}}.$$

These choices are noncircular: the floor fixes the logarithmic target range independently of b_0 ; that range fixes the finite geometry; then choose b_0 , and only afterwards construct envelopes and source. Combining with (I.36) gives

$$e^{-C_y(h)} \mathbb{E}_\pi[F \mid Q_h(\omega)] \leq C D_{\text{all}} e^{-c'\Lambda}.$$

For retained compatible competitors the full likelihood ratio is $e^{E_y(\omega) - E_y(\omega')}$, whose amplitude is bounded by the strict-exit weighted likelihood identity. Therefore the true relative posterior

$$q_h^{\text{rel}} = \mathbb{E}_\pi[F(\omega')\pi(\omega)/\pi(\omega') \mid Q_h(\omega)]$$

satisfies the same bound with a fixed extra factor. Raising to a fixed $\delta > 0$ and using original-pivot summability proves

$$\sum_{h \text{ original}} e^{-\delta C_y(h)} (q_h^{\text{rel}})^\delta \leq C_\delta D_{\text{all}}^\delta. \quad (\text{I.43})$$

I.11 Bounded family weights and the arithmetic first mass

Fix an original cell occupancy g , its finite legal label space Ω , and $0 < \delta \leq 1$. An acceptance $F : \Omega \rightarrow [0, 1]$ may couple all slots, but is independent of actual prime values and invariant under within-cell slot permutations. Set

$$Z_F = \sum_Y F(Y), \quad M_h(F; Y) = \sum_{Z \sim_h Y} F(Z), \quad R_F(Y) = \sum_h \Phi_g(h)^\delta M_h(F; Y)^\delta.$$

The finite weighted class identity is exactly

$$\sum_h \Phi_g(h)^\delta \Psi_{h,1+\delta}(F) = \sum_Y F(Y) R_F(Y). \quad (\text{I.44})$$

It is simply the expansion of the class power sum, so a pointwise bound $R_F \leq C$ on $F > 0$ gives a total at most $C Z_F$. For any positive product law π ,

$$M_h(F; Y) = \frac{\pi(Q_h(Y))}{\pi(Y)} q_h^{\text{rel}}(F; Y).$$

This follows by inserting $\pi(Z)/\pi(Z)$ in the finite class sum; bounded F alone does not remove its relative likelihood. A pointwise row for a support indicator also holds for every smaller bounded weight, by positivity; a merely averaged row does not imply this.

Suppose the correlated prime-cell kernel applies and the retained occupancy has squarefree supply $cW_g \leq S_g \leq W_g$, with all tuples inside the required size cutoff. On harmonic tuple measure times Lebesgue space define $f(\mathbf{a}, t) = \sum_Y F(Y) \mathbf{1}_{B_Y(\mathbf{a})}(t)$, where the original boxes have side $b = \log 2$. Then $\int f = b^k S_g Z_F \gg W_g Z_F$. At a point of B_Y , every other containing box overlaps B_Y , so its divisor log vector is within b in each coordinate. The correlated kernel and (I.44) consequently give $\int f^{1+\delta} \ll W_g Z_F$. Hölder on this same unnormalized measure yields

$$\text{meas}\{f > 0\} \geq \frac{(\int f)^{(1+\delta)/\delta}}{(\int f^{1+\delta})^{1/\delta}} \gg W_g Z_F.$$

Its support is in the original divisor union. Occupancies partition the tuple space and can be summed, retaining the linear dependence on the bounded acceptance.

Finally, extra nominal data may be removed in a linear first-mass identity when their support is already cell-measurable. Suppose that, conditional on g , the joint law of (Y, Ξ) and an acceptance $a(Y, \Xi) \in [0, 1]$ are equivariant under within-cell permutations, and accepted realizations have an invariant cell-label support $S(Y)$. Let $\pi_0(Y) = \mathbb{P}_{\text{joint}}(Y \mid g)$ be the actual marginal, and set $F(Y) = \mathbb{E}_{\text{joint}}[a(Y, \Xi) \mid Y, g]$ where $\pi_0(Y) > 0$, with $F(Y) = 0$ otherwise. Then F is invariant and $0 \leq F \leq \mathbf{1}_S$. For every nonnegative Y -measurable weight w ,

$$\sum_Y \pi_0(Y) w(Y) F(Y) = \mathbb{E}_{\text{joint}}[w(Y) a(Y, \Xi) \mid g].$$

For a desired nonnegative counting weight v , take $w = v/\pi_0$ where $\pi_0 > 0$ and $w = 0$ otherwise; the left side is then $\sum_Y v(Y) F(Y)$. Use Tonelli for an outer occupancy integral. The conditional law of Ξ may depend on Y . This preserves linear likelihood factors; it does not commute conditional expectation with a fractional power. Uniform deletion of specified-color slot subsets and count-dependent target laws are permutation equivariant, so the transport has this property whenever its source and boundary support do. The complete-cell construction in the main proof retains its nested fractional quantities before this linear arithmetic projection.

J Prime supply, collisions, and the arithmetic grid

The accepting density has been defined on complete cell words before actual primes are assigned. We first prove the collision estimate for an arbitrary such weight, then show that every supported occupancy has sufficient squarefree harmonic mass. The final comparison keeps the physical optimizer fixed throughout the passage to complete prime cells.

J.1 A fractional collision kernel for correlated label weights

We prove a prime-cell majorant for arbitrary nonnegative label weights, including weights coupling different slots. The triangularization is related to the unweighted argument of [19, Section 7.1]; we give the finite history construction and the complete weighted argument here.

J.1.1 Cells, slots, and the arithmetic moment

Fix $k \geq 1$, put $b = \log 2$, and set

$$\eta_i = \frac{\log(k - i + 2)}{k - i + 1}, \quad \rho_i = e^{\eta_i}, \quad y_0 = 3 \leq y_1 \leq \cdots \leq y_k.$$

Use pairwise disjoint prime cells $D_{i,t} = (\lambda_{i,t-1}, \lambda_{i,t}] \cap \mathbb{P}$ for $1 \leq t \leq v_i$, satisfying

$$\begin{aligned} \lambda_{i,t-1} &\geq 3, & \sum_{p \in D_{i,t}} p^{-1} &\leq \eta_i, \\ \log \lambda_{i,t-1} &\geq c_* \log y_{i-1} \rho_i^t, & \left| \log \frac{\log y_i}{\log y_{i-1}} - \eta_i v_i \right| &\leq A_*. \end{aligned} \tag{J.1}$$

The constants $c_* > 0, A_* < \infty$ are fixed. The elementary greedy cells described below have $c_*^{-1}, A_* \ll_k 1$. Empty blocks with $v_i = 0$ are allowed; the last inequality still applies to them.

Fix integers $g_{i,t} \geq 0$, and define

$$r_i = \sum_t g_{i,t}, \quad R_i = \sum_{h \leq i} r_h, \quad R_0 = 0, \quad G_g = \prod_{i,t} g_{i,t}!, \quad M_g = \prod_i \eta_i^{r_i}.$$

The slots in block i are $\mathcal{B}_i = (R_{i-1}, R_i] \cap \mathbb{Z}$. Assign its first $g_{i,1}$ slots to cell 1, its next $g_{i,2}$ to cell 2, and so on. Write $E_g(q)$ for the assigned cell of a positive slot q , and set $E_g(0) = 0$. No ordering of prime values inside one cell is imposed.

A label is a map Y assigning to a slot $q \in \mathcal{B}_i$ one color in $C_q = \{0, i, i+1, \dots, k\}$. Equivalently,

$$\mathcal{P}_{\mathbf{r}} = \{(Y_1, \dots, Y_k) : Y_j \subseteq \{1, \dots, R_j\}, \quad Y_j \cap Y_h = \emptyset \ (j \neq h)\},$$

where unused slots have color zero. Let $F : \mathcal{P}_{\mathbf{r}} \rightarrow [0, \infty)$ be finite, independent of all prime values, and invariant under every permutation of slots within each assigned cell. No product structure is assumed for F . It may depend on the entire occupancy vector g .

Let $\mathcal{A}(g)$ consist of squarefree tuples $\mathbf{a} = (a_1, \dots, a_k)$ having $g_{i,t}$ prime factors from each $D_{i,t}$, with all prime factors coming from these cells. Choose any named-slot representation $\mathbf{p}$ of such a tuple. For labels Y, Z , put

$$Y \sim_{\mathbf{p}} Z \iff \left| \sum_{q \in Y_j} \log p_q - \sum_{q \in Z_j} \log p_q \right| < b \quad (1 \leq j \leq k).$$

For $1 < P \leq 2$, define the arithmetic moment and its harmonic sum by

$$\begin{aligned} W_F^P(\mathbf{a}) &= \sum_{Y \in \mathcal{P}_{\mathbf{r}}} F(Y) \left(\sum_{Z: Y \sim_{\mathbf{p}} Z} F(Z) \right)^{P-1}, \\ \mathcal{W}_F^P(g) &= \sum_{\mathbf{a} \in \mathcal{A}(g)} \frac{W_F^P(\mathbf{a})}{a_1 \cdots a_k}. \end{aligned} \tag{J.2}$$

Within-cell invariance makes these definitions independent of the chosen representation. Distinct labels correspond to distinct divisor tuples because all the primes are distinct.

Lemma J.1 (Elementary cell supply). *For every ordered cutoff vector above there are cells containing exactly the primes $\max\{k, y_{i-1}\} < p \leq y_i$ and satisfying (J.1) with constants depending only on k .*

Proof. For one block put $u = \max\{k, y_{i-1}\}$ and $\eta = \eta_i$. First greedily partition all primes above u into successive maximal finite cells whose reciprocal mass is at most η . This is possible because every such prime satisfies $p^{-1} < \eta$ and the prime reciprocal sum diverges. Let λ_t be the last prime in complete cell t , with $\lambda_0 = u$, and let q_{t+1} be the first prime in the next cell. The mass H_t of complete cell t satisfies

$$\eta - q_{t+1}^{-1} < H_t \leq \eta.$$

All but at most $Q_k = \lceil 2/\min_i \eta_i \rceil$ cells have $q_{t+1} > Q_k$, and therefore $H_t \geq \eta/2$ for those cells. Mertens's estimate, uniformly for $3 \leq u \leq v$, gives

$$\sum_{u < p \leq v} \frac{1}{p} = \log \frac{\log v}{\log u} + O(1).$$

It follows that $\log \lambda_t \geq c_k \log u e^{\eta t/2}$. Consequently

$$\sum_{t \geq 1} q_{t+1}^{-1} \leq \sum_{t \geq 1} \exp(-c_k \log 3 e^{\eta t/2}) \ll_k 1.$$

Summing the deficits in H_t now gives $\sum_{s \leq t} H_s = t\eta + O_k(1)$, whence

$$\log \lambda_t = \log u e^{t\eta + O_k(1)}.$$

Truncate this infinite sequence at y_i . Retain each nonempty cell intersecting $(u, y_i]$ and replace the endpoint of the last retained cell by y_i . There is at most one incomplete mass, whose additional deficit is at most η . Thus the same cumulative mass argument gives $\log(\log y_i / \log u) = \eta v_i + O_k(1)$ when $y_i > u$, including the case of no retained primes. If $y_i \leq u$, use no cells and note that $\log y_i / \log y_{i-1}$ is bounded above and below in terms of k . Finally $\log u / \log y_{i-1} \asymp_k 1$; the estimates at preceding endpoints, including $t = 1$, give (J.1). $\square$

J.1.2 All histories and their explicit partitions

A history $h = (\sigma, \mathbf{I})$ consists of a permutation $\sigma = (\sigma_1, \dots, \sigma_k)$ of $\{1, \dots, k\}$ and integers satisfying

$$0 \leq I_j \leq R_{\sigma_j}, \quad \text{all positive } I_j \text{ are distinct.} \quad (\text{J.3})$$

Assign a positive pivot I_j to its slot block and every zero pivot to block 1. Define

$$\mathcal{J}_i(h) = \{j : I_j \text{ is assigned to block } i\}, \quad n_i(h) = |\mathcal{J}_i(h)|, \quad J_i(h) = \sum_{s \leq i} n_s(h).$$

Let $\mathfrak{H}_{\mathbf{r}}$ be the histories in (J.3) with $J_i(h) \geq i$ for every i . Necessarily $J_k(h) = k$. No monotonicity of $I_1, \dots, I_k$ is imposed. Zero pivots can precede positive pivots.

Put $A_j(h) = \{\sigma_j, \dots, \sigma_k\}$. For a slot q , partition its color set C_q by the following explicit equivalence:

$$c \equiv_{h,q} c' \iff 1_{c \in A_j(h)} = 1_{c' \in A_j(h)} \text{ for every } j \text{ with } I_j < q. \quad (\text{J.4})$$

Color 0 belongs to none of the $A_j(h)$. Write $\Pi_{h,q}$ for the nonempty equivalence cells. Declare Z compatible with Y , written $Z \equiv_h Y$, if their colors lie in the same cell of $\Pi_{h,q}$ for every slot q . Equivalently,

$$\bigcup_{s=j}^k (Z_{\sigma_s} \cap (I_j, R_k]) = \bigcup_{s=j}^k (Y_{\sigma_s} \cap (I_j, R_k]) \quad (1 \leq j \leq k). \quad (\text{J.5})$$

This equivalence follows simply by checking the membership of each individual slot in the two unions. It includes repeated zero pivots, empty blocks, and arbitrary ties among cell indices.

Define

$$\Phi_g(h) = \prod_{i=1}^k \rho_i^{-(k-J_i(h))v_i - \sum_{j \in \mathcal{J}_i(h)} E_g(I_j)}, \quad (\text{J.6})$$

$$M_F(Y; h) = \sum_{Z: Z \equiv_h Y} F(Z), \quad \Psi_{h,P}(F) = \sum_Y F(Y) M_F(Y; h)^{P-1}. \quad (\text{J.7})$$

All these are explicit finite quantities. The enlarged history set can include histories not generated by any pair; their nonnegative contributions cause no problem in an upper bound.

Theorem J.2 (Correlated fractional collision majorant). *Under (J.1), uniformly for $1 < P \leq 2$ and every weight F specified above,*

$$\boxed{\mathcal{W}_F^P(g) \leq C(k, c_*, A_*) \frac{\prod_i \eta_i^{r_i}}{\prod_{i,t} g_{i,t}!} \sum_{h \in \mathfrak{H}_r} \Phi_g(h)^{P-1} \Psi_{h,P}(F)}. \quad (\text{J.8})$$

For the cells of Lemma J.1 the constant is C_k only. It is independent of $P, \mathbf{y}, \mathbf{r}, g, F$.

Proof. Every arithmetic tuple in $\mathcal{A}(g)$ has exactly G_g named representations by distinct primes in the prescribed cells. Hence, writing $d\mu(\mathbf{p}) = (\prod_q p_q)^{-1}$ for this finite named-prime measure and $\delta = P - 1 \in (0, 1]$, we have the exact identity

$$\mathcal{W}_F^P(g) = \frac{1}{G_g} \sum_Y F(Y) \sum_{\mathbf{p} \text{ distinct}} d\mu(\mathbf{p}) \left(\sum_Z F(Z) 1_{Y \sim_{\mathbf{p}} Z} \right)^\delta. \quad (\text{J.9})$$

In particular the factorial is present before any prime is deleted. The total mass of this measure is at most M_g by (J.1). For any nonnegative function f , finite-measure Hölder (or concavity) gives

$$\sum d\mu f^\delta \leq M_g^{1-\delta} \left(\sum d\mu f \right)^\delta. \quad (\text{J.10})$$

This includes $\delta = 1$; if the measure has zero mass the desired theorem is immediate.

We next prove a linear prime bound for a fixed pair Y, Z . Its canonical history is defined without reference to any prime values. Starting with all coordinate colors remaining, at step j let U_j be the set of slots belonging to exactly one of $Y_c \Delta Z_c$, as c runs through the remaining coordinates. If $U_j \neq \emptyset$, choose $I_j = \max U_j$, let σ_j be the unique remaining coordinate containing that slot in its symmetric difference, and remove σ_j . If $U_j = \emptyset$, choose $I_j = 0$, let σ_j be the largest remaining coordinate, and remove it. Continue for exactly k steps.

At a positive step the pivot lies in $Y_{\sigma_j} \Delta Z_{\sigma_j}$, so $I_j \leq R_{\sigma_j}$. It is absent from every later remaining difference, and therefore cannot be selected again. A zero step need not be followed only by zero steps: deleting its coordinate can create new exclusive slots. Every removed coordinate $c \leq i$ supplies a pivot in the first i blocks, where zeros are counted in block 1. Thus $J_i \geq i$, and the canonical history belongs to $\mathfrak{H}_r$.

For disjoint-coordinate labels the exclusive set just used equals the symmetric difference of the unions of the remaining coordinate sets. Indeed a slot has at most one first-label color and at most one second-label color: if both colors remain and are different it occurs in two differences, and if exactly one remains it occurs in one. Maximality of I_j , including $I_j = 0$, therefore proves (J.5) for this canonical history.

Fix all nonpivot primes. Retain the collision row for each positive pivot step and order these rows and their pivot columns by step. The coefficient of a previously chosen pivot in any later row is zero, and each diagonal coefficient is 1 or -1 . The resulting square matrix is upper triangular, with all entries in $\{0, 1, -1\}$ and dimension at most k . Its inverse has row sums bounded by 2^k : this follows either by back substitution or the finite expansion of the inverse of a unit triangular matrix. Discarding the zero-step rows only enlarges the feasible set. It follows that, for the fixed nonpivot primes, all positive pivot logarithms lie in a product of intervals of length at most $B_k = 2^{k+1}b$. The interval centers depend only on the nonpivot primes and the labels. This is containment in a rectangle, not an independence assertion about collision constraints.

Chebyshev's prime-counting upper bound implies, for a fixed positive B_k , that

$$\sum_{p \in D_{i,t} \cap [X, e^{B_k} X]} \frac{1}{p} \leq \frac{C_k}{\log \max\{3, \lambda_{i,t-1}, X\}} \leq \frac{C_k c_*^{-1} \rho_i^{-t}}{\log y_{i-1}}. \quad (\text{J.11})$$

For the first inequality, if the intersection is nonempty take $L = \max\{\lambda_{i,t-1}, X\} \geq 3$ and contain it in $[L, e^{B_k} L]$; then bound its number of primes by $\pi(e^{B_k} L)$ and each reciprocal by L^{-1} . After integrating the rectangle, discard all distinctness conditions and use the mass bound η_i for every nonpivot slot. At most k factors η_i^{-1} are lost, a constant depending only on k . Thus, for the canonical history $h(Y, Z)$,

$$\sum_{\mathbf{p} \text{ distinct}} d\mu(\mathbf{p}) 1_{Y \sim \mathbf{p} Z} \leq C(k, c_*) M_g \prod_i (\log y_{i-1})^{-n_i(h)} \rho_i^{-\sum_{j \in \mathcal{I}_i(h)} E_g(I_j)}. \quad (\text{J.12})$$

Here a zero pivot deletes no prime. We inserted a factor $(\log y_0)^{-1}$ for each such pivot; their number is at most k , so this only changes the displayed constant.

The physical cutoff factors equal the factors in (J.6) up to the stated constant. More explicitly, summation by parts gives

$$\begin{aligned} \sum_i n_i(h) \log \log y_{i-1} &= k \log \log 3 + \sum_{i < k} (k - J_i(h)) \log \frac{\log y_i}{\log y_{i-1}} \\ &= \sum_i (k - J_i(h)) \eta_i v_i + O(k + k^2 A_*). \end{aligned}$$

Therefore the left side of (J.12) is at most $C(k, c_*, A_*) M_g \Phi_g(h(Y, Z))$.

For fixed Y , partition the competitors Z by their canonical histories, and enlarge each generating class to all $Z \equiv_h Y$. Weights are nonnegative, so

$$\sum_Z F(Z) \sum_{\mathbf{p} \text{ distinct}} d\mu(\mathbf{p}) 1_{Y \sim \mathbf{p} Z} \leq C(k, c_*, A_*) M_g \sum_{h \in \mathfrak{H}_r} \Phi_g(h) M_F(Y; h).$$

Apply (J.10) in (J.9), and use $(\sum_h a_h)^\delta \leq \sum_h a_h^\delta$ for $a_h \geq 0$. The powers of M_g combine exactly to M_g . Taking the preceding constant at least one, its δ power is bounded by the same constant uniformly for $0 < \delta \leq 1$. Finally sum against $F(Y)$ to obtain (J.8). No factor $g_{i,t}$ or falling factorial is introduced when pivot slots are removed: their choices already occur in the history sum, and G_g is the exact original multiplicity. $\square$

J.1.3 The correlated partition term

Let $\mathcal{Q}_h$ be the partition of the whole label space into the equivalence classes of $\equiv_h$. This is the Cartesian partition induced by the slot partitions (J.4), but the weights of its classes need not factor. For every nonnegative label weight, the exact identity is

$$\Psi_{h,P}(F) = \sum_{Q \in \mathcal{Q}_h} \left(\sum_{Y \in Q} F(Y) \right)^P. \quad (\text{J.13})$$

Indeed $M_F(Y; h)$ is constant on Q , with value $\sum_{Z \in Q} F(Z)$, so the anchors in Q supply the remaining first power. If $F(Y) = \prod_q w_{q,Y(q)}$, and only in that additional case for this factorization, the right side becomes

$$\prod_q \sum_{B \in \Pi_{h,q}} \left(\sum_{c \in B} w_{q,c} \right)^P.$$

For a Gibbs weight depending on aggregate counts from several slots, (J.13) must be retained or estimated with its correlations intact. Substituting its one-slot marginals in the last product is not justified.

The theorem is pointwise in g . Thus one may multiply both sides by any nonnegative occupancy test $T(g)$ and sum over g , allowing $F = F_g$ separately at every occupancy. This preserves exactly the nominal harmonic factorial mass M_g/G_g and the same constant. It does not assert a lower bound for the actual prime mass, nor identify the occupancy distribution conditioned on a collision. In particular, a bound for the weighted history and occupancy sum on the right of (J.8), with all required polynomial factors, is still needed to deduce a sharp lower bound for K_k .

J.2 Squarefree mass at every supported occupancy

For a nonempty finite prime cell D , put

$$H_D = \sum_{p \in D} p^{-1}, \quad p_D = \min D, \quad e_g(D) = \sum_{\substack{p_1 < \dots < p_g \\ p_j \in D}} (p_1 \cdots p_g)^{-1}, \quad e_0(D) = 1.$$

Drawing g independent primes with probabilities $1/(pH_D)$ gives the exact identity

$$g!e_g(D) = H_D^g \mathbb{P}\{\text{all draws distinct}\}.$$

Conditional on s distinct previous draws, their total probability mass is at most $s/(H_D p_D)$. If $(g-1)/(H_D p_D) \leq 1/2$, then

$$\mathbb{P}\{\text{all distinct}\} \geq \prod_{s=0}^{g-1} \left(1 - \frac{s}{H_D p_D}\right) \geq \exp\left(-\frac{g(g-1)}{H_D p_D}\right).$$

For a complete greedy cell with target mass η , $0 \leq \eta - H_D < 1/p_{\text{next}} \leq 1/p_D$. If $p_D \geq 2/\eta$ and $(g-1)/(H_D p_D) \leq 1/2$, then $H_D \geq \eta/2$ and

$$1 \geq \frac{g!e_g(D)}{\eta^g} \geq \exp\left(-\frac{2g^2}{\eta p_D}\right). \quad (\text{J.14})$$

This includes $g = 0$.

Use only complete retained cells $D_{i,j}$, omitting the final incomplete cell, and set $p_{i,j} = \min D_{i,j}$, $\eta_* = \min_i \eta_i$, and

$$W_g = \prod_{i,j} \frac{\eta_i^{g_{i,j}}}{g_{i,j}!}, \quad S_g^{\text{all}} = \prod_{i,j} e_{g_{i,j}}(D_{i,j}).$$

The second expression is exactly the squarefree harmonic tuple mass, since the cells are disjoint. Provided the single-cell capacity conditions above hold, multiplication of (J.14) gives

$$1 \geq S_g^{\text{all}}/W_g \geq \exp\left(-\frac{2}{\eta_*} \sum_{i,j} \frac{g_{i,j}^2}{p_{i,j}}\right). \quad (\text{J.15})$$

We verify these conditions uniformly on the accepting support. Let $L_i = \eta_i v_i$, suppose the total count obeys $r_i/L_i \leq D_0$, and each of its h_i fine colors satisfies at cell boundaries

$$\left|N_{i,c}(t) - \frac{t}{L_i} q_{i,c}\right| \leq H_0 + \theta \min(t, L_i - t).$$

An empty block has count zero. Summing over colors gives, for the uncolored cumulative count $G_i(j) = \sum_{s \leq j} g_{i,s}$,

$$\left| G_i(j) - \frac{j}{v_i} r_i \right| \leq h_i H_0 + h_i \theta \eta_i \min(j, v_i - j).$$

Using $g_{i,j} \leq G_i(j)$ and $g_{i,j} \leq r_i - G_i(j-1)$ separately, and putting $A_0 = (k+1)H_0$, $A_1 = (\max_i \eta_i)(D_0 + (k+1)\theta)$, proves

$$g_{i,j} \leq A_0 + A_1 \min(j, v_i - j + 1). \quad (\text{J.16})$$

Only the existence of one accepted labeling is needed. The uncolored occupancy is the same for every labeling, and recoloring changes none of it. The named REG bounds in the construction imply the displayed linear tubes since $t^\zeta \leq 1 + t$ for $0 < \zeta < 1$.

Choose all retained primes greater than a fixed $Q \geq 2/\eta_*$. Each complete cell has mass at least $\eta_i/2$. The previous $j-1$ retained cells and the reciprocal-integer bound imply

$$p_{i,j} \geq \frac{Q}{2} e^{(j-1)\eta_i/2}.$$

Let $q = e^{-\eta_*/2}$ and define

$$B_1 = \frac{A_0}{1-q} + \frac{A_1}{(1-q)^2},$$

$$B_2 = \frac{A_0^2}{1-q} + \frac{2A_0A_1}{(1-q)^2} + \frac{A_1^2(1+q)}{(1-q)^3}.$$

These are the sums of $(A_0 + A_1j)q^{j-1}$ and its squared-polynomial counterpart. Thus

$$\frac{g_{i,j}}{H_{i,j}p_{i,j}} \leq \frac{4B_1}{\eta_*Q}, \quad \sum_{i,j} \frac{g_{i,j}^2}{p_{i,j}} \leq \frac{2kB_2}{Q}.$$

Taking $Q \geq \max(3, 2/\eta_*, 8B_1/\eta_*)$ makes every capacity condition valid, and yields pointwise

$$1 \geq S_g^{\text{all}}/W_g \geq \exp(-4kB_2/(\eta_*Q)) > 0. \quad (\text{J.17})$$

This is a multiplicative bound at every supported occupancy; no absolute probability of repeated primes is subtracted from a rare mass.

The same support has a fixed size bound. Write $P_{i,j} = \max D_{i,j}$. The $v_i - j$ later cells and the uniform Mertens difference estimate give

$$\log P_{i,j} \leq e^{C_M} \log y_i e^{-\eta_i(v_i-j)/2}.$$

The right-hand tube in (J.16) therefore implies

$$\log a_i \leq \sum_j g_{i,j} \log P_{i,j} \leq e^{C_M} B_1 \log y_i.$$

Choose the allowed size exponent at least $A_{\text{tube}} = \max(1, e^{C_M} B_1)$. Then the restricted actual mass equals S_g^{all} on every supported occupancy, proving (11.8). The support constants are fixed before this roughness and grid choice.

J.3 Cancellation and one arithmetic first mass

The relative-class estimate of Appendix I is

$$\sum_h e^{-\delta C_y(h)} (q_h^{\text{rel}})^\delta \leq C \left(\prod_j D_j^{\text{own}} \right)^\delta.$$

The exact outputs give $R_y = \sum_j y_j + \epsilon/r_m - E_y$, with $E_y = \sum_{j < m} d_j \eta_j^y \geq 0$ and $-C \leq \epsilon \leq 0$. Since $\prod_j D_j^{\text{own}} = e^{-\sum_j y_j}$, $e^{R_y} \prod_j D_j^{\text{own}} \leq 1$. Also, with the actual class $Q_h(Y)$,

$$\Phi_g(h) \sum_{Z \in Q_h(Y)} F(Z) = e^{I_h} q_h^{\text{rel}}, \quad C_y(h) = R_y - I_h.$$

Multiplication proves the unnormalized row bound

$$\sum_h \Phi_g(h)^\delta \left(\sum_{Z \in Q_h(Y)} F(Z) \right)^\delta \leq C \quad (F(Y) > 0).$$

Every original pivot is retained. A bounded clock discrepancy changes the constant once per original cut.

On harmonic tuple measure times Lebesgue measure let $f(\mathbf{a}, t) = \sum_Y F(Y) 1_{B_Y(\mathbf{a})}(t)$ and $Z_F = \sum_Y F(Y)$. The weight is bounded, invariant under permutations inside each cell and independent of the actual prime values there. By (J.17), $\int f = (\log 2)^k S_g Z_F \geq c W_g Z_F$. Pointwise, if $t \in B_Y$, every other box containing t has center difference less than $\log 2$ in each coordinate. Consequently

$$f^{1+\delta} \leq \sum_Y F(Y) 1_{B_Y} \left(\sum_{Z: Y \sim_{\mathbf{p}} Z} F(Z) \right)^\delta.$$

Integrate and apply Theorem J.2 and the row bound to obtain $\int f^{1+\delta} \leq C W_g Z_F$. Hölder on its support gives

$$\text{meas}\{f > 0\} \geq \frac{(\int f)^{(1+\delta)/\delta}}{(\int f^{1+\delta})^{1/\delta}} \geq c W_g Z_F.$$

Its support is contained in the original divisor union. Summing over the disjoint occupancies retains exactly the first power of the selected nominal mass.

J.4 Bounded perturbations with the physical optimizer fixed

Fix the geometric, scalar, count, REG and occupancy constants, then choose the roughness and cell grids. The initial roughness deletion has bounded harmonic mass, and omitting the final incomplete cell loses less than one fixed cell mass in each group. The coarse prime sum comparison therefore bounds the total resulting length loss by a constant $R = O_k(1)$. Choose the lift above all source thresholds and above $C_H(1 + R)^2$. At the lifted physical vector L optimize once and keep its products, owner partition, fine probabilities, selected edges and ordinary suffix fixed. The complete-cell lengths L' satisfy $\sum_i |L'_i - L_i| \leq R$. Every original group remains present.

Map each grid block affinely onto its physical block. This preserves the order of activations, deadlines and eligible partitions of every lawful history. All reference integrands are bounded on the compact parameter ranges. Thus, uniformly over the full history space,

$$|\overline{C}_{L',s}(h') - \overline{C}_{L,s}(h)| \leq C_H R, \quad \overline{C}_{L',s}(h') \geq -C_H R,$$

$$|\partial_{s_i} J(L', s) - \partial_{s_i} J(L, s)| \leq C_H R, \quad |J(L', s) - J(L, s)| \leq C_H R. \quad (\text{J.18})$$

The fixed-coefficient reference charges for wrong names, old weak directions, mixed times, pure lifetimes and terminal rank curvature acquire just this bounded additive defect. For example the same two lawful competitors give $\bar{C}_{\text{float}} \geq c_H T_{\text{float}} - C_H R$. The old-mixture node equation has residual $O_H(R)$: divide by the old length to obtain its entropy error and multiply back when making the old-group charge. The final error is additive $O_H(R)$. A nonempty old collection is large after the lift, so the hierarchy $V_{\text{old}} \leq C_H T_{\text{owner}} + C_H R$ retains its fixed-factor form. Compression and interpolation are performed on L' itself; only their KKT lower bounds have this intercept. The exact likelihood, projection and carry identities have no stationarity error.

Two endpoint rows require exact treatment. For one owner let $m_{\text{ref}} = \sum_g (\mathbb{E} H_g - r_g L'_g) = O_H(R)$ and $\zeta_g = H_g - \mathbb{E} H_g$. Its accepted exact output is $Y = a_0 - b_0 = m_{\text{ref}} + \sum_g \zeta_g$, so

$$\begin{aligned} C_v &= a_0 - \sum_{g \leq l} \zeta_g + \bar{C}'_v + \sum_{\text{proper}} \nu_{v,g}, \\ &= b_0 + m_{\text{ref}} + \sum_{g > l} \zeta_g + \bar{C}'_v + \sum_{\text{proper}} \nu_{v,g}. \end{aligned} \quad (\text{J.19})$$

For the all-zero template $C_v = a_0$ exactly. For the complete-deadline template $\nu = 0$ and $\bar{C}'_v = -m_{\text{ref}}$, so $C_v = b_0$ exactly and its intrinsic cost is zero. Neither small endpoint is enlarged by R . Every other endpoint has an own-scale charge. If its last positive time is strictly internal, clipped charging gives at least $c_H \min_g L_g$. If it reaches the end but is not everywhere full, some proper group has either a linear bad-group charge or the displaced good-group square-root charge. Hence

$$\bar{C}_v + S_{\text{good}}(v) \geq c_H \sqrt{\min_g L_g} - C_H. \quad (\text{J.20})$$

The chosen lift absorbs all bounded defects within the already reserved fractions of these charges. The displaced named box is retained; central uncolored counts alone would not imply this step.

The finite context sets at L, L' have identical indexing. Formal perturbed scores may use $\max(0, \bar{C}'_v)$, with the two exceptional templates given their exact scores. Reference errors are $O_H(R)$ and $|\sqrt{L'_g} - \sqrt{L_g}| \leq \sqrt{|L'_g - L_g|}$. Native exit scales are comparable. If an early/late flag changes, $|t_v - T/2| \leq C_H R$. For large T its reference charge is $\geq c_H T - C_H R$, dominating both possible additions; for bounded T , the fixed positive score floor suffices. Thus each paired score, and then its separate minimum, changes by at most a fixed factor. The actual grid pair still uses $\Delta'_i = c_i L'_i - a_i n_i + b_i k_i$ in the max construction.

At natural mean $\mu'_i = Z_i L'_i$, the centered parameter $p_i^0 = (a_i - c_i/Z_i)/b_i$ is unchanged; the gap-clock parameter $L'_i/(\mu'_i + 1)$ changes by $O_H(R)/(1 + L_i)^2$. The local Lipschitz estimate for χ and $|\log T_i| \leq C_H \log(2 + L_i)$ on the rare branch therefore give a bounded exponent-times-log error. Score comparability gives comparability of all binary factors, including threshold crossings. Similarly $w'_j = R'_j/\sqrt{T'_j} \asymp w_j$. The frozen ordinary suffix lengths change by $O_H(R)$; its numerical factor is stable with $1 + \max(D', 0)$ in this comparison. If two largest blocks are comparable, choosing either places the other on a side and gives a fixed positive ballot factor. This numerical observation does not itself admit a negative mean into a probability theorem; the exact root argument below handles that possibility.

J.5 Count bands and the exact root on the final grid

For the same complete integer type, all grouped outputs and all recursive exit coordinates change by deterministic $O_H(R)$ because the count coefficients are fixed. Condition on the common pair-

coarsened field in the strict-exit construction. Each proper macro split remains an independent $\text{Bin}(m_j, p_j)$, with $m_j \asymp T_j$, compact p_j , and the same coefficient in its exit; its effect on every deeper exit cancels exactly. Band centers move by $O_H(R)$ without reading any remaining split.

Translate each split by a deterministic integer of size $O_H(R)$, starting in a fixed interior subband. The fixed width floor exceeds the finitely many lattice steps, so rounding stays inside the original band with a fixed fractional margin. The small change in its width is absorbed by that margin. For a bounded translation the logarithm of the binomial probability ratio on the central range is $O_H(R/\sqrt{m_j} + R^2/m_j)$, directly by factorial ratios. The central windows have spare width since $\sqrt{m_j} \gg R$. The same factorial calculation applies to each finite multinomial thick box: its conditional center and incoming count shift by $O_H(R)$, while its fixed $\sqrt{L_g}$ width is unchanged. This check is made stagewise before any path is sampled.

The natural uncolored means also change by $O_H(R)$. For $|n - \mu| \leq C\sqrt{\mu}$, the logarithm of their Poisson ratio is $n \log(\mu'/\mu) - (\mu' - \mu) = O_H(1)$. Hence integration of the complete count measure retains its true weights and mass $W' \asymp \prod_{j < m} w_j$. Equations (J.19) and (J.20) supply the same strong-box fractions and legal physical-score endpoint boxes. Terminal completion is checked by the same thick-box calculation at every admitted root count, retaining the original root clock.

The actual root count is the floor solving the exact grid equation. At the same other counts it changes by at most $C_H(1+R)$ from the physical-budget floor. Since $V_{\text{pre}} \leq C_H U$ and $U \gg (1+R)^2$, the Euler identity with its bounded residual gives $|n'_* - \mu'_*| \leq C_H \sqrt{U}$. Its Poisson atom is comparable to $U^{-1/2}$, uniformly over retained prefix environments and the allowed logarithmic targets. All $n'_* + 1$ gaps are then sampled at this count.

Let P be the entire original length before the ordinary weak block and B its later positive-drift length. If $P > 0$, the lift makes it large; the physical relation $D \asymp P$ gives $D_{\text{grid}} \geq c_H P - O_H(R) > 0$. The actual ordinary pair remains in the boxes $X \asymp 1 + P$, $Y \asymp 1 + B$, so the ordinary exact-type theorem applies. If $P = 0$, there are no earlier owners or terminal binary head, all common targets vanish, and $x = 0$. The root still sets the total risk to its bounded floor error even if the natural grid mean has either sign $O_H(R)$. If $B > 0$, the later central blocks have actual surplus at least $c_H B - C_H \sqrt{B} - O_H(R) \geq c'_H B$. The weak first-block increment is exactly determined by those counts and the root floor: one legal endpoint remains fixed and the other is comparable to $1 + B$. If $B = 0$, their difference is the bounded root floor and both reserves remain fixed.

Apply the all-reserve ordinary exact-type strong-wall and REG theorem directly to these legal pairs. Its hypotheses are the exact endpoint relation, positive reserves and compact marks, speed and gap parameter; they do not require a nonnegative natural mean. Other suffix blocks retain their separated signed drifts at central counts. Conditional on all counts, the original clocks and their constant-retention named refinements give one joint kernel

$$\asymp U^{-1/2} \min \left\{ 1, \frac{(1+P)(1+B)}{1+M_{\text{weak}}} \right\}$$

when a weak block exists, and the saturated path factor otherwise. The physical positive-tail identities identify this with the frozen ordinary factor. Both sides use the same central root atom and the same full-gap ordinary bridge; a later largest positive block makes the path factor saturated.

All binary blocks now have supported central types, their actual grid increments and positive common boxes. The curved-wall, REG and all-transfer estimates apply at the fixed physical products. Integrate the mixed moments at a fixed full type and then sum the complete type

measure. The exact owner powers and tilt identity give

$$K_k(\log 3 + L) \geq c_H e^{-J(L';s)} \mathcal{P}_{\text{fixed}}(L';s) \geq c_H e^{-J(L;s)} \mathcal{P}_I(L).$$

The first inequality is the first-mass transfer already proved; the second uses (J.18) and the factor comparisons above. Products have remained fixed at the physical optimizer throughout. All REG and occupancy constants remain those used to choose the grid; only the final length threshold and additive geometric constants depend on its loss bound. Finally Theorem 16.1 compares the original vector with its one fixed lift, as used in Theorem 9.1.

K Geometric covers and conditional estimates for the upper bound

This appendix proves the local assertions used in Proposition 14.1 and the ordinary root estimate of Proposition 14.9. All clocks and all eligibility sets in these arguments retain their original block labels.

K.1 Rectangles and deterministic sums at stopping locations

Fix a size-truncated squarefree tuple as in Lemma 2.2 and a genuine conditional fiber of its label law. For every exterior name c , let E_c be its original deadline. In physical logarithmic coordinates, a lawful cut satisfies $0 \leq \tau_c \leq E_c$. The class Q_τ reveals name c at points to the right of τ_c ; all unrevealed eligible names remain in the compound category. Whole batches at an endpoint are treated on the prescribed side. Let π be the original categorical product law on this fiber and set

$$B = \sum_c E_c, \quad \Xi_\tau = \sum_c \tau_c - B - \log \pi(Q_\tau), \quad Z(t) = 1 + \sum_{x \leq t} e^{x-t}.$$

The sum uses all original prefix occurrences, including those that may not enter a selected descendant list. Revealed factors translate the remaining boxes. The unresolved displacement in coordinate c is at most $C_H e^{\tau_c} Z(\tau_c)$. Bounding the remaining child coordinates by their original size rectangles therefore gives

$$\text{Vol}_{\text{norm}}(Q_\tau) \leq C_H \pi(Q_\tau) e^{\Xi_\tau} \prod_c Z(\tau_c). \quad (\text{K.1})$$

In an ancestor atom α , this formula retains its absolute probability $\pi(\alpha)$ and its inherited geometric multiplier. Conditioning divides by $\pi(\alpha)$ only temporarily; reinsertion restores it.

The cuts used for moment estimates are fixed before drawing the occurrence process. To handle first failures, partition a deterministic physical interval $[a, b]$ into cells of length at most Δ , with a bounded number of endpoints per unit interval. If v is the right endpoint of a cell and t belongs to that cell, positivity gives

$$Z(t) \leq e^\Delta Z(v). \quad (\text{K.2})$$

Indeed every term present at t is also present at v , and its kernel changes by e^{v-t} . Suppose first-failure atoms form a lawful antichain and each has capacity at most

$$C\pi(q)e^{-h}\omega(t)Z(t)^d \prod_c Z(\tau_c), \quad d \leq H,$$

where $\omega(t) = (1 + \text{dist}(t, \{a, b\}))^{-A}$, or its one-ended version, and the outside cuts τ_c are deterministic. Within one mesh cell $\omega(t) \leq C_{\Delta, A} \omega(v)$ and the sum of the atom probabilities is at most one. Thus the total failure capacity is bounded by $Ce^{-h}S$, with

$$S = \sum_v \omega(v) Z(v)^d \prod_c Z(\tau_c).$$

The stopped atom probabilities have already been summed in this expression.

For a Poisson process with bounded intensity per unit interval, $Z(v)$ has all fixed moments uniformly in deterministic v . This follows either by its Laplace functional or by splitting into unit intervals and using the exponentially decreasing coefficients. For sufficiently large fixed A , Minkowski's inequality and $\sum_v \omega(v) < \infty$ give $\mathbb{E}S^p \leq C_{H, p}$ for each required fixed p . Finite sums over the endpoint contexts have the same property. A cut within bounded distance of an endpoint is covered by finitely many extra mesh cells. An incoming selected-list sum is bounded pointwise by the original-list sum at the same deterministic threshold.

The probability tests themselves may be made on the complete deterministic mesh, including empty cells. The risk has one-sided jumps and bounded linear speed; survival on that mesh implies the corresponding continuous necessary wall after a fixed reserve increase, using the appropriate side for the jump convention. The rising wall has bounded derivative on the fixed interval, so the same conclusion holds for its first-hit strip. Consequently the fixed-time bank estimates below are only applied at deterministic mesh locations.

Under insertion of q fixed original occurrences, every deterministic $Z(v)$ increases by at most q . Keeping the mesh and contexts fixed therefore gives

$$S(G + \text{insertions}) \leq C_{H, q} S(G). \quad (\text{K.3})$$

The categorical log-likelihood changes at each fixed test by at most $C_H q$, which is paid by the reserve. A Poisson–Mecke expansion of a fixed shot monomial uses finitely many partitions of its occurrences; apply this insertion bound in each term before summing the deterministic locations. This proves the fixed-moment and insertion assertions needed for all the actual stopping covers. In particular it does not estimate a shot directly at an unrestricted data-dependent location.

K.2 Positive prime expansion with the original root

Let $\mathcal{P}_i$ be the original prime ranges, including 2, 3 in $\mathcal{P}_1$, and put

$$H_i = \sum_{p \in \mathcal{P}_i} \frac{1}{p}, \quad B_{\text{geo}} = \sum_i V_i = \sum_i (h_i - 1) L_i.$$

Take fixed positive Z_i, s_i , put $s = s_1 > 0$, and define

$$J = \sum_i L_i + \sum_i s_i V_i - \sum_i Z_i L_i, \quad X(G) = \frac{\sum_i N_i(G) \log Z_i - \sum_i s_i V_i}{s}.$$

Let $\mathbb{P}_Z$ give independent occurrence counts $\text{Pois}(Z_i/p)$ at each $p \in \mathcal{P}_i$. For every nonnegative symmetric functional A on occurrence lists, Tonelli and the Poisson exponential formula give the exact identity

$$e^{-\sum_i L_i} \sum_{\mathbf{n} \geq 0} \prod_i \frac{1}{n_i!} \sum_{p_{i,a} \in \mathcal{P}_i} A(G) \prod_{i,a} \frac{1}{p_{i,a}}$$

$$= e^{-J+\sum_i Z_i(H_i-L_i)} \mathbb{E}_Z[e^{-sX(G)} A(G)]. \quad (\text{K.4})$$

To verify the exponent, the unnormalized list sum equals $e^{\sum_i Z_i H_i} \mathbb{E}_Z[A \prod_i Z_i^{-N_i}]$, and $\prod_i Z_i^{-N_i} = e^{-sX - \sum_i s_i V_i}$.

For the saddle, $\log Z_i = G_i$ and X is exactly the uncolored routing root. At node j , multiplying minus the local log-probability by s_j gives G_j when the occurrence exits and $G_j - G_{j+1}$ when it enters the child. Along each route these contributions telescope to G_i at its original block. Subtracting the physical budgets proves the identity without a stationarity assumption or resampling any child list.

Define $I_0(G) = e^{-B_{\text{geo}} L^{(k+1)}(\mathbf{a}(G))}$ on the squarefree size-truncated support, and $I_0 = 0$ elsewhere. Each squarefree selection occurs $n_i!$ times among ordered lists, so Lemma 2.2 and (K.4) imply

$$K_k(\ell) \leq 2e^{-J+\sum_i Z_i(H_i-L_i)} \mathbb{E}_Z[e^{-sX} I_0(G)]. \quad (\text{K.5})$$

The geometric caps hold first on that actual support. If they give $I_0 \leq CC_l(G)$ there, with each C_l nonnegative on every list, then $I_0 \leq C \min_l C_l$ on all lists, because of the zero extension. Their numerical cap minimum is therefore evaluated on unrestricted repeated lists under the same atomic Poisson law and root.

The uniform prime harmonic estimate gives $|H_i - L_i| \leq C_H$; the discrepancy in (K.5) is bounded since $Z_i \leq H$. Deleting a fixed initial prime set pays its finite Euler product and changes each H_i by only a fixed amount. For later cell completion, if $\mu_c^* \geq \mu_c$ and $\sum_c (\mu_c^* - \mu_c) \leq C_H$, the product Poisson density ratio is at most e^{C_H} . It is applied once to the whole nonnegative integrand, retaining its minimum, powers and root. A change of the physical cell budget is compensated in the external root translate.

K.3 Separation of the rank intervals of rare comparisons

In one maximal owner, write $\varepsilon = 1 - \sigma > 0$ and

$$\kappa_r = (r + z)^{-\varepsilon}, \quad g_r(a) = a - \kappa_r E_a(z).$$

The transported old coefficients are retained individually. They are no larger than the natural coefficient at the first native capacity. Every lawful complete reference history has nonnegative cost by the KKT and reference-compression results of Appendix G.

Choose a fixed sufficiently small $\epsilon_0(H) > 0$. An edge $r \rightarrow q$ is weak when $|g_r(r) - g_r(q)| \leq \epsilon_0$. If m_-, m_+ are the two minimum reference costs, their common increment is $m_+ - m_- = L_i \nu_i$. Both are nonnegative and the endpoint scores dominate $1 + m_-$ and $1 + m_+$. Hence

$$|\nu_i| L_i \leq m_- + m_+ \leq X_i + Y_i.$$

Since the natural mean is comparable to L_i , sufficiently small $(1 + X_i)(1 + Y_i)/(1 + \mu_i)$ implies this fixed weakness condition.

We record the three strict comparisons needed below. The normalized secant estimate of Appendix G is

$$\beta_b(y) = \frac{\log((y \log y - b \log b)/(y - b))}{\log y}, \quad \beta'_b(y) \leq -\frac{1}{24H^2}.$$

For a weak edge $r \rightarrow q$ and an earlier capacity $h > r$, put $T_a = (a + z)^{-\varepsilon}(E_a - E_q)/(a - q)$. Weakness gives $|T_r - 1| \leq \epsilon_0/(r - q)$, whence $\beta_{q+z}(r + z) - \varepsilon \leq 2\epsilon_0/\log 2$. If $\epsilon_0 \leq \log 2/(192H^2)$,

then $\log T_h \leq -\log 2/(48H^2)$. At the other endpoint,

$$g_h(r) - g_h(q) = (r - q)\{1 - (\kappa_h/\kappa_r)T_r\}, \quad \kappa_h/\kappa_r \leq e^{-\varepsilon_{\min}/H}.$$

Taking also $\epsilon_0 \leq (e^{\varepsilon_{\min}/H} - 1)/2$ gives a positive lower bound at both r and h . Concavity then proves

$$g_h(a) - g_h(q) \geq \delta_H > 0 \quad (r \leq a \leq h). \quad (\text{K.6})$$

Reducing the coefficient increases this difference, so it also holds in the relevant old groups. If $q \geq 1$ and $0 \leq a < q$, strict concavity $-g_r'' \geq H^{-2}$ gives

$$g_r(r) - g_r(a) \geq -H\epsilon_0 + \frac{(q-a)(r-a)}{2H^2} \geq \frac{3}{4H^2}$$

after $\epsilon_0 \leq 1/(4H^3)$. Finally the forward secant comparison gives, for $q < h < r$,

$$g_h(q) - g_h(h) \geq \gamma_H > 0. \quad (\text{K.7})$$

All thresholds are chosen before any lengths.

Suppose two weak edges have $r_1 > r_2$ but $q_1 < r_2$. Output ordering gives $q_2 < q_1$, so $q_1 \geq 1$. In any left completion for the second edge, all earlier retained ranks are at least r_2 . Lower every such rank to q_2 , keeping the whole later history. This is a legal complete history. Nonnegativity and (K.6) therefore give $X_2 \geq 1 + \delta_H L_1$.

For a right completion of the first edge, fix the prefix through its block and the suffix after the second block. Between their right endpoints b_1, b_2 , all capacities exceed q_1 , so none of the remaining names expires there. The reference cost, as a function of their ordered cut times, is a constant plus

$$\sum_{a=k_{\text{end}}+1}^{q_1} \int_{b_1}^{\tau_a} \{g_t(a) - g_t(a-1)\} dt.$$

Each summand is concave because $\kappa(t)$ is nondecreasing. Its minimum on the ordered simplex occurs at a vertex, with all cuts at b_1 or b_2 . Thus one constant rank $a \leq q_1$ can be used on that interval without increasing the cost. If $a < q_1$, lower the rank on the first block to a and use the strict concavity comparison above: its original cost was at least $c_H L_1$. If $a = q_1$, replace the intervening ranks by their full eligible capacities and use (K.7): the original cost was at least $c_H L_2$. Both comparisons preserve the complete outside history. Consequently

$$Y_1 \geq 1 + c_H \min(L_1, L_2), \quad \bar{T}_1 \geq c_1 \frac{1 + \min(L_1, L_2)}{1 + L_1}, \quad \bar{T}_2 \geq c_2 \frac{1 + L_1}{1 + L_2}.$$

If $L_2 \geq L_1$ the first ratio is bounded below; otherwise their product is. Hence two sufficiently rare edges must have $q_1 \geq r_2$. Applying this to every pair proves disjointness of the rank intervals $(q_i, r_i]$. The argument also applies to terminal nonzero edges: $z = 1$, but their secant bases are at least two and their marks remain nondegenerate. The terminal zero-output edge remains ordinary.

K.4 The two scales of a proper zero-output comparison

Consider a proper zero-output edge with sufficiently small persistence ratio $\bar{T}$. Let M be the selected native block, A, B the preceding and following native lengths, and V_{old} the complete old length. Put

$$P = V_{\text{old}} + A, \quad T = P + M + B, \quad R_{\text{exit}} = \max\{R_0, \min(\sqrt{1 + A + M + B}, d^{-1})\}.$$

Full input and zero output force the two completion times to be P and $P + M$ in every outside context. The clipped-prefix estimate from Appendix G gives $\bar{C}_v \geq c_H \min(t_v, T - t_v)$. If the block fails to cross $T/2$ strictly, one of its endpoint costs is at least $c_H M$ and the persistence ratio is bounded below. Thus sufficiently small ratio forces $P < T/2 < P + M$.

For upper bounds on the minima, retain every earlier name to its own deadline and remove the current names at the selected endpoint. Both contexts have no good groups. Complete owner balance gives

$$\bar{C}_{\text{full}}^- = \sum_{\text{old}} L_g g_g(r_{\text{top}}) + \sum_{\text{native before}} L_g g_g(r_g), \quad \bar{C}_{\text{full}}^+ = - \sum_{\text{native after}} L_g g_g(r_g).$$

Their costs are nonnegative and at most $C_H P$ and $C_H B$, respectively. Together with the clipped lower bound and the early/late score definition, this proves

$$X \asymp_H 1 + V_{\text{old}} + A + R_{\text{exit}}, \quad Y \asymp_H 1 + B. \quad (\text{K.8})$$

In particular rarity implies $V_{\text{old}} + A + B \leq C_H \tau M$; for a small fixed τ , M dominates the entire old and native length. Uniformly even as $d \downarrow 0$,

$$R_{\text{exit}} \asymp_H \varrho := \max \left\{ \varrho_0, \frac{c_0 \sqrt{1+M}}{1+d\sqrt{1+M}} \right\}.$$

Conversely a sufficiently small $(1+B)(1+V_{\text{old}}+A+\varrho)/(1+M)$ forces that dominance and midpoint crossing, so the scales characterize the same rare region. For the original mark $W = \log(r+z) - \log z \mathbf{1}_{\text{child}}$, complete balance also gives

$$eM = \sum_{\text{old}} L_g g_g(r_{\text{top}}) + \sum_{\text{native} \neq i} L_g g_g(r_g), \quad e = (r+z)^\sigma \mathbb{E}W - r,$$

and therefore $|e|M \leq C_H(V_{\text{old}}+A+B)$. The two reserves in (K.8) remain separate when applying a one-end or two-end probability estimate.

K.5 Necessary one-end tests on original native intervals

Consider a weak edge $r \rightarrow q$ with $q \geq 1$ on $[a, b]$, of length L , in the rare region $\bar{T} < \tau$, where $\tau > 0$ is sufficiently small in terms of H . Write $S = r + z$, $B = q + z$, $\nu = g_r(r) - g_r(q)$, and let $\mathcal{J}$ consist of the later native groups with $q < h < r$. Their total length is $M_{\mathcal{J}}$. We first prove

$$1 + X \geq c_H \sqrt{1+L}, \quad 1 + Y \leq C_H \tau \sqrt{1+L}, \quad M_{\mathcal{J}} \leq C_H Y. \quad (\text{K.9})$$

At a left completion the current block retains a proper deepest subset. If the completion is late, this is a good group and contributes $\sqrt{L}$; if it is early, its last positive time is at least b , and clipped-prefix charging gives a cost at least $c_H L$. This proves the first inequality, and rarity gives the second. For any right completion, compress the cuts between b and the end of $\mathcal{J}$ to the two endpoints, by the same concavity argument as in Section K.3. The compressed rank is some $a' \leq q$. If $a' < q$, lowering the current block costs at least $\delta_H L$; if $a' = q$, filling the intermediate groups costs at least $\gamma_H M_{\mathcal{J}}$. Complete reference nonnegativity thus gives $Y \geq \min(\delta_H L, \gamma_H M_{\mathcal{J}})$. For sufficiently small τ , rarity excludes the first alternative, proving the last claim. These arguments also hold for $z = 1$ and positive q .

The endpoint reference minima differ by νL , so $|\nu|L \leq X + Y$. Choose the deterministic interval length

$$\ell = c_{\text{bank}} \frac{L^2}{(1 + X)^2}. \quad (\text{K.10})$$

By choosing c_{bank} sufficiently small and then τ sufficiently small, (K.9) and rarity imply

$$\ell_{\text{scalar}} \leq \ell \leq L/16, \quad |\nu|\sqrt{\ell} \leq C_H, \quad \frac{1 + Y}{\sqrt{\ell}} \asymp_H \bar{T} := \frac{(1 + X)(1 + Y)}{1 + S^\sigma L}.$$

Indeed rarity gives $1 + X \leq C_H \tau L / (1 + Y)$ and hence $\ell \geq c_H \tau^{-2}$. Neither ℓ nor its location is selected from the sampled word.

Group the deepest q names and the true compound child into the category B , keeping the other names separate. Reveal the complete coarse word in $\mathcal{J}$, and put

$$H_{\mathcal{J}} = \sum_{q < h < r} \{N_h \log(h + z) - K_{B,h} \log B - (h - q)L_h\}.$$

Read the current interval backwards and set

$$W = \log S - \log B \mathbf{1}_B, \quad C(t) = H_{\mathcal{J}} + \sum_{x < t} W_x - (r - q)t.$$

The original rate is S^σ and its mean drift is $-\nu$. Earlier exterior coordinates keep their full original sizes; the intermediate ones keep their shorter deadlines; the B coordinates are projected into their original full rectangle. The unresolved exterior budget is the full budget minus $(r - q)t + \sum_{\mathcal{J}} (h - q)L_h$, while the observed negative log-probability is the mark sum above. Formula (K.1) therefore gives the actual partial capacity $C_H \pi_t e^{C(t)} \text{Shot}_t$. Stop at the first mesh failure of $C(t) \geq -h - A \log(1 + t)$, including $t = 0$. Section K.1 bounds the union of failures by $C S e^{-h}$ with all required moments.

Because $H_{\mathcal{J}} \leq (\log H) N_{\mathcal{J}}$ pointwise, every survivor satisfies the necessary test

$$\sum_{x < t} W_x - (r - q)t \geq -h - C_H N_{\mathcal{J}} - A \log(1 + t), \quad 0 \leq t \leq \ell. \quad (\text{K.11})$$

First make this implication with the entire intermediate word fixed, then release that word with its original mass. The right side now reads only $N_{\mathcal{J}}$; any future private test remains in the joint surviving-word operator.

Let Q_h be the conditional private probability of (K.11). The signed logarithmic-wall one-end estimate, on the original clock with bounded scaled drift, gives

$$\mathbb{E}_{G_i} Q_h^\sigma \leq C_H \min \left\{ 1, \left(\frac{1 + h + C_H N_{\mathcal{J}}}{\sqrt{\ell}} \right)^{2\chi(\sigma, \rho)} \right\}, \quad \rho = \frac{(\mathbb{E}W)^2}{\mathbb{E}W^2}.$$

The intermediate count is independent of this original native clock, with mean at most $C_H M_{\mathcal{J}} \leq C_H Y$. Its fixed moments are at most $C_p(1 + Y)^p$. Integrating over all its counts proves

$$\mathbb{E} Q_h^\sigma \leq C_H (1 + h)^{K \bar{T}^{2\chi(\sigma, \rho)}}. \quad (\text{K.12})$$

To compare with the exact-type parameter, write $a_0 = \log S$, $b_0 = \log B$, $c = r - q$, $\lambda = S^\sigma$, $\mu = \lambda L$, and $p = B/S$. Its centered type parameter and correlation are

$$\bar{p} = (a_0 - c/\lambda)/b_0, \quad \bar{\rho} = \frac{(cL/(\mu + 1))^2}{(cL/(\mu + 1))^2 + b_0^2 \bar{p}(1 - \bar{p})}.$$

The identity $\bar{p} - p = -\nu/(b_0\lambda)$ gives $|\bar{p} - p| \leq C_H(|\nu| + 1/L)$. As $\bar{T} \geq c_H/\sqrt{\ell}$, $|\log \bar{T}| \leq C_H + \frac{1}{2} \log \ell$. The scaled-drift bound and $\ell \leq L/16$ bound the product of this logarithm and the correlation error. Compact local Lipschitz continuity of χ therefore permits either parameter in (K.12) at fixed comparison cost. This includes every count and all $n + 1$ conditional spacings. Fixed insertions change the reserve by at most a constant per point; the Mecke argument of Section K.1 attaches the required shot powers at the same exponent σ .

For a proper $r \rightarrow 0$ edge, assume explicitly $1 + X \geq c_X\sqrt{1 + L}$. The child weight satisfies $z \geq 2^{u_{\min}} > 1$, so the two marks remain nondegenerate. Here $\mathcal{J}$ consists of all later native groups. At a right completion, postpone the current cuts to their individual deadlines. The complete earlier history is unchanged, and the difference of reference costs is $\sum_{0 < h < r} [-g_h(h)]L_h \geq \gamma_H M_{\mathcal{J}}$. Thus $M_{\mathcal{J}} \leq C_H Y$ directly. The preceding proof applies with $B = z$, $c = r$, and the child category in place of the deepest suffix. In particular it supplies precisely the same moment (K.12). The left-score hypothesis is needed: there is no current proper-good group when the output is zero.

The sets consisting of a current block and its intermediate native groups are disjoint for distinct rare edges of an owner, by rank separation, and are disjoint across owners. Their label intervals have length at most $L/16$ and avoid a count middle in the same block. If the count middle lies in an intermediate group, its weight is decreasing in that count and the necessary reserve is increasing. For independent copies N, N' ,

$$2 \operatorname{Cov}(f(N), g(N)) = \mathbb{E}[(f(N) - f(N'))(g(N) - g(N'))] \leq 0$$

when f is decreasing and g increasing. Applying this with all other groups fixed supplies the required separation without an independence assertion for the shared count.

K.6 Leaving a terminal suffix unread

All terminal binary weak blocks lie in the negative ordinary-drift head. The ordinary-suffix geometry gives, before trimming,

$$D_0 \geq b_H V_{\text{old}} + g_0 L_{\text{head}}, \quad U \geq c_H T_{\text{total}}, \quad M_* \geq c_H T_{\text{total}}/H.$$

A terminal necessary test may read intermediate counts within this suffix. If so, start the suffix at the first capacity at most its output q . The removed initial segment has length

$$\delta \leq M_{\mathcal{J}} \leq C_H Y \leq C_H \tau \sqrt{1 + L} \leq C'_H \tau L.$$

For several tests use the latest one; rank separation puts all earlier read sets before it. The old head contains its current length L . Choosing the fixed τ small enough, bounded ordinary drifts give $E_H \delta \leq D_0/4$, $\delta \leq D_0/4$ and $\delta < M_*/2$. The largest original root block is therefore retained and

$$U' = U - \delta \asymp U, \quad M'_* = M_*, \quad A'_* = A_* - \delta, \quad B'_* = B_*, \quad D' = D_0 - \sum_{\text{removed}} e_h L_h \asymp D_0.$$

The new suffix still has $D' \geq c_H V'_{\text{pre}}$ and $U' \geq c_H T_{\text{total}}$. Moreover $1 + D' + A'_*$ is comparable to $1 + D_0 + A_*$, so its ordinary factor is unchanged up to constants. Every clock read by the necessary masks now belongs to the actual prefix of this suffix. When no terminal binary rare edge is present, no trimming is needed.

K.7 Whole-band estimates with two variable reserves

Fix a compact nondegenerate positive binary law W , a compact positive rate ν and speed c , and a power σ in a compact subinterval of $(0, 1)$. Suppose $M, L, R \geq 1$ and the band width r satisfies

$$1 \leq r \leq C\sqrt{M}, \quad r \leq CR, \quad |\nu\mathbb{E}W - c|M \leq C(L + R). \quad (\text{K.13})$$

All interval locations are deterministic. Use a middle interval of length comparable to M , disjoint from the retained end intervals. An additional later-side count N_B is independent, with Poisson mean at most CL . Let the positive left test have reserve at most $C(1 + h + N_B)$ and the reflected negative right test have reserve at most $C(1 + h + R + (1 + |j|)r)$. Both allow a fixed logarithmically lowered wall. After fixing the entire outside private word, assume the whole band permits at most Cr values of the middle binary count.

Put $T = LR/M$ and $\rho_{\text{nat}} = (\mathbb{E}W)^2/\mathbb{E}W^2$. There is a numerical majorant $B_{h,j}(G)$ for the conditional probability of the band and its retained tests such that

$$\mathbb{E}B_{h,j}^\sigma \leq C(1 + h)^K(1 + |j|)^K(r/\sqrt{M})^\sigma \min\{1, T^{2\chi(\sigma, \rho_{\text{nat}})}\}. \quad (\text{K.14})$$

Here the entire permitted count band is summed before the power. To prove this, if $\max(L, R) \leq C_0\sqrt{M}$, retain end intervals of length $M/16$. Otherwise retain only the smaller-reserve end, of length $\ell = \min(M/16, c_*M^2/\max(L, R)^2)$. Conditioning on all outside labels, the middle count is binomial, so its maximum atom gives the pointwise majorant

$$B_{h,j} = \min\{1, Cr/\sqrt{1 + N_{\text{mid}}}\} q_{\text{left}} q_{\text{right}},$$

with an omitted end factor equal to one. The bound is uniform in the outside-dependent center. Release those outside labels only afterwards. The Poisson negative moment supplies $C(r/\sqrt{M})^\sigma$, including small middle counts. On every retained end the scaled drift is bounded by (K.13). Thus the signed one-end estimate gives

$$\mathbb{E}q_H^\sigma \leq C \min\{1, [(1 + H)/\sqrt{1 + \ell}]^{2\chi_c}\}, \quad \chi_c = \chi\left(\sigma, \frac{(c/\nu)^2}{(c/\nu)^2 + \text{Var } W}\right).$$

If both ends remain, independence and the moments of N_B give the path factor $\min(1, (LR/M)^{2\chi_c})$ times the displayed reserve polynomials. The condition $r \leq CR$ pays all translated right bands. If only one end remains, its reserve divided by $\sqrt{\ell}$ is bounded by those polynomials times LR/M , yielding the same result. If ℓ is bounded, $\max(L, R) \geq cM$ and $T \geq c$; discard the path test at fixed cost while retaining the middle atom. On $T < 1$ the drift hypothesis implies $|\chi_c - \chi(\sigma, \rho_{\text{nat}})| \leq C(L + R)/M \leq 2CT$. The replacement costs at most $e^{C|T \log T|}$. This proves (K.14); for $T \geq 1$ both numerical path factors equal one.

For actual prime clocks these assertions follow by a pointwise quantile comparison. On a deterministic physical interval of length ℓ , its harmonic cumulative mass is $t + O_H(1)$ for every initial length t , with the same statement for half-open intervals and their reflections. The harmonic-to-physical quantile map therefore satisfies $|T(s) - s| \leq C_H$. Its image of a homogeneous Poisson process is the required atomic prime process, with repeated occurrences allowed. A positive-jump wall is tested at its pre-jump minimum; a negative wall at its post-jump minimum. During the collapse of any atom the homogeneous partial sums dominate these minima up to the bounded clock error. An actual necessary wall is consequently contained in the homogeneous wall with a fixed reserve enlargement, for every mark word and every atom multiplicity. Logarithmic allowances change by only a fixed amount.

For clarity, the full-count homogeneous estimate used here follows from Theorem 6.11. Let $m = \lfloor \nu\ell/2 \rfloor$. The probability of fewer than m arrivals by ℓ is exponentially small. Otherwise survival implies a rank wall through the first m independent exponential gaps. For the positive sign, survival at the j th arrival implies $cT_j \leq w_+j + h + A \log(1 + T_j)$ and hence $T_j \leq C(j + h + 1)$. For the negative sign, when this latter bound fails the signed risk is already nonnegative; in the complementary case the same logarithmic comparison applies. In either direction the rank-wall reserve is $C(1 + h)$ with a fixed logarithmic allowance. Theorem 6.11 gives $C(1 + h)^{2\chi}(1 + \ell)^{-\chi}$, and the exponentially small exception is absorbed. This proof includes all original counts.

The fixed-width version is especially useful at first hits. Split a deterministic interval $[0, z)$ into its left quarter, middle half, and right quarter. Given both end words, a fixed terminal band admits at most a fixed number of middle low counts. Hence, uniformly over every band center b ,

$$\sup_b q_b(G) \leq C(1 + N_M)^{-1/2} \ell_h(G_L) r_{r_0}(G_R),$$

and independence of these three deterministic restrictions gives

$$\mathbb{E}(\sup_b q_b)^\sigma \leq C(1 + h)^{2\chi} (1 + r_0)^{2\chi} (1 + z)^{-\sigma/2 - 2\chi}. \quad (\text{K.15})$$

There is no extra terminal-height factor. The estimate is uniform in z in a prescribed comparable range, not an estimate for a supremum over z ; stopping covers retain their deterministic summation weights.

Insertion of q fixed occurrences decreases the middle negative count factor, relaxes the positive left reserve by at most qw_+ , and makes the negative right wall stronger. The same middle-binomial argument can leave the original middle marks free after conditioning on the inserted marks. Thus both displayed bounds remain valid with that left reserve shift. To attach shots, for $\beta = 4/\sigma_{\min}$ form

$$C_z(G) = \sum_{h, r_0 \geq 0} (1 + h)^{-\beta} (1 + r_0)^{-\beta} (1 + N_M)^{-1/2} \ell_h(G_L) r_{r_0}(G_R).$$

Fractional subadditivity and (K.15) show $\mathbb{E}C_z^\sigma \leq C(1 + z)^{-\sigma/2 - 2\chi}$, since each reserve exponent is at most -3 . Reindexing after insertion gives $C_z(G + q \text{ points}) \leq C_q C_z(G)$. Expanding an integrably weighted fixed-degree shot polynomial into Poisson occurrence partitions and applying Mecke now preserves this power of $1 + z$. The expansion is valid for atomic intensities as well.

For the proper zero-output application, take the separate scales $L = 1 + B$ and $R = 1 + V_{\text{old}} + A + \varrho$ from (K.8) and band width ϱ . They satisfy all of (K.13), including $\varrho \leq CR$. If $d\sqrt{M} < 1$, then $\varrho \asymp \sqrt{M}$ and $R \geq c\sqrt{M}$; the left-only calculation gives the same bound with a width factor comparable to one, uniformly as $d \downarrow 0$.

K.8 Joint count weights and the conditional routing tower

For each proper owner choose a fixed middle in a largest native block, write n_j for its count, and set $M_j = (1 + d_j \sqrt{1 + n_j})^{-1}$. For each simple edge let $Q_i(h)$ be its necessary-test probability. The only possible overlap between a count middle and these tests is when the middle lies in an intermediate set $\mathcal{J}_i$. With all other clocks fixed, $M_j^{\sigma_j}$ decreases in n_j and $Q_i^{\sigma_{o(i)}}$ increases. The covariance inequality proved above extracts $\mathbb{E}M_j^{\sigma_j}$. Each middle lies in at most one such set by rank separation. The remaining test inputs are disjoint original groups, so

$$\mathbb{E} \left[\prod_{j < m} M_j^{\sigma_j} \prod_i Q_i(h)^{\sigma_{o(i)}} \right] \leq \prod_{j < m} \mathbb{E}M_j^{\sigma_j} \prod_i \mathbb{E}Q_i(h)^{\sigma_{o(i)}}.$$

Splitting a Poisson count below and above half its mean gives

$$\mathbb{E}M_j^{\sigma_j} \leq C_H(1 + d_j\sqrt{1 + \mu_{C_j}})^{-\sigma_j},$$

uniformly even as $d_j \downarrow 0$. Together with (K.12) this gives all proper count and simple native factors at their respective owner powers. Fixed insertions only decrease M_j and shift each reserve by a fixed amount; the same Mecke argument attaches the permitted shots. One unweighted count insertion pays its original mean.

Here is the finite conditional argument that uses these environmental costs. Fix the complete original uncolored word G and let $\mathcal{Q}_j$ reveal its first j coarse owner routes. In a native block of owner j , every earlier owner has already expired; its earlier routes are deterministically child. Thus on every genuine $\mathcal{Q}_{j-1}$ atom its current labels still have probabilities $1/(r + z_j)$ for each exterior name and $z_j/(r + z_j)$ for the child, independently over original slots. Let E_j avoid the count middle and satisfy $\pi_G(E_j \mid \mathcal{Q}_{j-1}) \leq p_j(G)$ uniformly on those atoms; for the terminal owner assume the corresponding conditional bound on E_m . Native-only masks have this property.

For any uncolored-measurable thresholds ξ_j , put $\zeta_j = \eta_j - \xi_j$. Condition on the entire current coarse route outside its middle. Then E_j is fixed and

$$\zeta_j = H_0 - \log z_j K_j, \quad K_j \sim \text{Bin}(n_j, z_j/(r + z_j)),$$

where H_0 may depend arbitrarily on the actual incoming list. The binomial maximum atom and the two exponential half-lattice sums give, uniformly in H_0 and also for $n_j = 0$,

$$\begin{aligned} \mathbb{E}[e^{\zeta_j} 1_{\{\zeta_j \leq 0\}} 1_{E_j} \mid \mathcal{Q}_{j-1}] &\leq CM_j p_j, \\ \mathbb{E}[e^{-d_j \zeta_j} 1_{\{\zeta_j > 0\}} 1_{E_j} \mid \mathcal{Q}_{j-1}] &\leq CM_j p_j. \end{aligned}$$

For the negative side use the geometric sum with ratio $e^{-\log z_j}$; for the positive side use $e^{-d_j \log z_j}$ and cap the bound by one. At $d_j = 0$ the probability bound suffices. The arbitrary center is removed before releasing the outside route. Future child words are not fixed in this calculation.

Backward conditioning now bounds every product of positive half-line factors followed by one negative half-line factor by $C^j \prod_{l \leq j} M_l p_l$, and the fully surviving branch by $C^{m-1} p_m \prod_{l < m} M_l p_l$. For completeness set $a_j = M_j p_j$ for $j < m$, $a_m = p_m$, $r_j = \sigma_j / \sigma_1$, and

$$B_0 = \sum_{j \leq m} r_j \log a_j, \quad \theta_j = r_j^{-1} \sum_{l > j} r_l \log a_l, \quad \xi_j = X/r_j + \theta_j.$$

Stop the actual projected route at the first $\eta_j \leq \xi_j$. The original capacity is $C_H \pi_G(q) e^{P_j}$, and the carry identity gives on this branch

$$e^{P_j} = e^{X+c_j} e^{\zeta_j} \prod_{l < j} e^{-d_l \zeta_l}, \quad c_j = \theta_j - \sum_{l < j} d_l \theta_l = B_0 - \sum_{l \leq j} \log a_l.$$

The terminal branch has $c_m = -\sum_{l < m} d_l \theta_l$ and the same last expression. Summing genuine stopped capacities and using the backward estimates proves for a surviving set $\mathcal{S} \subseteq \bigcap_j E_j$ that

$$I(\mathcal{S}) \leq C_H \min\{1, e^X \prod_j a_j^{r_j}\}, \quad (\prod_j a_j^{r_j})^{\sigma_1} = \prod_{j < m} M_j^{\sigma_j} \prod_j p_j^{\sigma_j}.$$

If a p_j vanishes, the conditional bound makes the finite surviving support empty, and no logarithm is required. For $m = 1$ the direct full-word capacity proves the assertion. This calculation preserves the actual descendant lists and is the simple-owner tower used in the causal proof; hard owners supply their own three conditional kernels in Lemma 14.4.

K.9 The ordinary rectangle and the same-count root integral

We prove Proposition 14.9. Write the unread original suffix as blocks with $h_1 > \dots > h_m \geq 2$, $w_i = \log h_i$, $c_i = h_i - 1$, lengths $L_i \geq 0$, and $U = \sum_i L_i$. Let $0 < t \leq s < 1$, $r = t/s$, in the fixed compact ranges. With the entire prefix fixed, take arbitrary $D > 0$ and $\kappa \in \mathbb{R}$, and set

$$Z = \sum_i (w_i N_i - c_i L_i), \quad X = \kappa + rZ.$$

After the positive enlargement, the N_i are independent Poisson counts with means comparable to L_i . The natural means are $h_i^t L_i$. Conditional on each count there are its original uniform positions, with all $n_i + 1$ gaps. Completed harmonic-cell budgets are used consistently in Z , and their bounded difference from physical budgets is included in κ , keeping X unchanged.

Choose a largest original block $M = L_{i_*}$ and put

$$A = \sum_{i < i_*} L_i, \quad B = \sum_{i > i_*} L_i, \quad B_{\text{act}} = \sum_{i > i_*} (w_i N_i - c_i L_i), \quad A_{\text{act}} = - \sum_{i < i_*} (w_i N_i - c_i L_i).$$

Then $Z = B_{\text{act}} + Z_* - A_{\text{act}}$ and $M \leq U \leq mM$. The two outside sums are independent and retain all their counts.

Three actual rectangles are used before enlargement. A threshold at the beginning of the suffix assigns its primes freely and bounds earlier contributions by their full original rectangles, giving $I \leq Ce^Z$. A threshold within the largest block, with its j smallest prime logarithms retained, gives $I \leq Ce^{B_{\text{act}}} W_{*,N_*}$, where $W_{*,n}$ is the capped ordered-sum rectangle of (13.11). To verify this, the $h_* - 1$ unresolved widths are bounded by the prefix scale times $1 + \sum_{l \leq j} e^{(w_*/c_*) \text{cell}_l}$, whereas the free later labels contribute h_*^{-j} after normalization. Minimize over j , and use the right endpoint rectangle for the cap one. The remaining rectangle is $I \leq C$. Thus, together with a prefix cap $I \leq CDe^X$, the same tuple obeys

$$I \leq C\hat{V}, \quad \hat{V} = \min\{1, e^Z, e^{B_{\text{act}}} W_{*,N_*}, De^X\}. \quad (\text{K.16})$$

For every $n \geq 0$ and $\ell \geq 0$, Lemma 13.6 gives

$$\mathbb{P}(W_{*,n} > e^{-\ell} \mid n) \leq \min \left\{ 1, \frac{C(1+\ell)(C_1 + Z_* + \ell)_+}{1+n} \right\}.$$

For negative ℓ the event is empty.

First fix all outside counts and restrict Z to an interval $[E, E+W]$ of fixed width. There are at most $1 + W/\log 2$ possible N_* . The two full Poisson mode bounds

$$\sup_n p_\mu(n) \leq C(1+\mu)^{-1/2}, \quad \sup_n \frac{p_\mu(n)}{1+n} \leq C(1+\mu)^{-3/2} \quad (\text{K.17})$$

follow from Stirling near the mode and the elementary Poisson tails; they include $n = 0$ and all noncentral counts. The event $e^{B_{\text{act}}} W_{*,N_*} > e^{-\ell}$ has left reserve $1 + \ell + B_{\text{act}}$ and right reserve $C_1 + Z + \ell + A_{\text{act}}$. The endpoint rectangle also requires $Z + \ell \geq 0$. Apply the two bounds in (K.17) separately, then average the independent sides. Since $\mathbb{E}(B_{\text{act}})_+ \leq CB$ and $\mathbb{E}(A_{\text{act}})_+ \leq CA$, this gives

$$\begin{aligned} & \mathbb{P}\{Z \in [E, E+W], e^Z > e^{-\ell}, e^{B_{\text{act}}} W_{*,N_*} > e^{-\ell}\} \\ & \leq C(1+U)^{-1/2} \min \left\{ 1, \frac{(1+\ell+B)(1+(E+\ell)_+ + A)}{1+M} \right\}. \end{aligned}$$

The root atom and ballot denominator have already been obtained from the same N_* .

Put $H_0 = (-\log D - \kappa)/r$ and $w = Z - H_0$. Then $e^{-sX} = D^s e^{-tw}$ and $De^X = e^{rw}$. Positive layer cake in (K.16) gives

$$D^{-s} \mathbb{E}[e^{-sX} \widehat{V}] = \int_0^\infty e^{-\ell} \mathbb{E}[e^{-tw} 1_{\{w > -\ell/r\}} 1_{\{Z+\ell \geq 0\}} 1_{\{W_* > e^{-\ell-B_{\text{act}}}\}}] d\ell.$$

At fixed outside counts, $a = w + \ell/r > 0$ runs along a translated lattice with spacing w_* . Moreover $Z + \ell = H_0 + a - (1/r - 1)\ell$, so its nonnegativity forces $a \geq -H_0$ up to a fixed geometric translation. Uniformly over the lattice translate and $b \geq 0$,

$$\begin{aligned} \sum_{a>0, a \geq -H_0-C} e^{-ta} &\leq C e^{-t(H_0)-}, \\ \sum_{a>0, a \geq -H_0-C} e^{-ta} (C + H_0 + a + b)_+ &\leq C e^{-t(H_0)-} (1 + (H_0)_+ + b). \end{aligned}$$

For $H_0 < 0$, shift by $(H_0)_-$; for $H_0 \geq 0$, start at zero. This explains why no polynomial loss in the negative deficit occurs. Using first the first mode bound and then the second with the ballot tail bounds the conditional integrand before $e^{-\ell}$ by the minimum of

$$C e^{s\ell-t(H_0)-} (1+M)^{-1/2}, \quad C e^{s\ell-t(H_0)-} (1+M)^{-3/2} [1 + \ell + (B_{\text{act}})_+] [1 + (H_0)_+ + (A_{\text{act}})_+].$$

Average the independent sides separately in these two bounds, then take their minimum. Since $1-s$ has a fixed positive lower bound, integration against $e^{-(1-s)\ell}$ yields

$$\mathbb{E}[e^{-sX} \widehat{V}] \leq C D^s e^{-t(H_0)-} (1+U)^{-1/2} \min \left\{ 1, \frac{(1+B)(1+(H_0)_+ + A)}{1+M} \right\}. \quad (\text{K.18})$$

Missing sides contribute exactly one; an empty suffix is treated by the deterministic minimum with the same 1+length conventions.

The outgoing profile needed in the causal theorem is $J_\beta(X + \log D)$ instead of De^X , where $J_\beta(v) = \min\{1, C_\beta e^v (1 + (-v)_+)^{\beta}\}$ and $\beta \geq 0$ is fixed. If $J_\beta(v) > e^{-\ell}$, then

$$v > -L_\beta(\ell), \quad L_\beta(\ell) = \ell + 2\beta \log(2 + \ell) + C'_\beta.$$

Indeed $(1+u)^\beta \leq C e^{u/2}$ first gives $u = (-v)_+ \leq 2\ell + C$, which substituted back gives the stated bound. Now set $q = r(Z - H_0) + L_\beta(\ell) > 0$. The tilt becomes $D^s e^{sL_\beta(\ell)} e^{-sq}$ and $Z + \ell \leq H_0 + q/r$. The endpoint cap still forces $q \geq r(H_0)_-$ up to a fixed translation. The same lattice sums give $e^{-t(H_0)-}$ and the same right reserve. The only change in the layer integration is the weight

$$e^{-\ell+sL_\beta(\ell)} \leq C e^{-(1-s)\ell} (2 + \ell)^{2s\beta},$$

whose integral, also multiplied by $1+\ell$, is uniformly finite. This proves (K.18) with the polynomial profile and hence Proposition 14.9.

For the natural suffix put $D_{\text{mean}} = \sum_i L_i(h_i^t \log h_i - (h_i - 1)) \geq 0$ and

$$\mathcal{B}_I = (1+U)^{-1/2} \min\{1, (1+B)(1+D_{\text{mean}} + A)/(1+M)\}.$$

The preceding bound also gives

$$\mathbb{E}[e^{-sX} \widehat{V}] \leq C D^s \mathcal{B}_I e^{-t(H_0)-} \left[1 + \frac{(H_0 - D_{\text{mean}})_+}{1 + D_{\text{mean}} + A} \right].$$

The Euler root identity is $\kappa = -rD_{\text{mean}} + \Delta_{\text{pre}}$, so $H_0 - D_{\text{mean}} = -(\Delta_{\text{pre}} + \log D)/r$. This is the explicit prefix deviation weight used in the final upper expectation. It remains together with D^s under that expectation; the conditional estimate allows arbitrary correlation among prefix costs, counts and geometric sums.

L The finite geometry with six divisor coordinates

The finite optimization, named-endpoint geometry, and ordinary-suffix construction also apply for $k = 6$. We verify this explicitly rather than importing a dimension-specific formula. Throughout this section $k = 6$, $H = 7$, $h_i = 8 - i$ for $1 \leq i \leq 6$, and initially $L_i > 0$. Original physical groups and original names are retained, including when several consecutive products are equal.

Proposition L.1 (Finite geometric inputs at $k = 6$). *The saddle, original transported coefficients, complete named endpoint minima, and ordinary suffix in Section 3 are well defined for $k = 6$. They satisfy the same finite geometric estimates used in the uniform proof, with constants depending only on 7. Neither an additional optimizer face nor a separate dimension-six branch of the definition of Φ_k is required.*

Proof. We check the ingredients and their boundary cases in turn.

The saddle and its compact range. The perspective recursion for G_i is convex and increasing in its child variable. Hence each $Z_i = e^{G_i}$ is convex, and J is concave on the closed ordered simplex. At a bottom plateau with product at most $1/2$, raising that plateau has derivative bounded below by a positive sum of terms $h - 1 - h^{1/2} \log h$. These are positive for every $h > 1$: with $t = \sqrt{h}$, the function $t^2 - 1 - 2t \log t$ vanishes at 1 and has derivative $2(t - 1 - \log t) > 0$. This also excludes zero products.

For a top plateau write n for an exterior rank and z for its permanent-child weight. Its entropy increment is

$$E_n(z) = (n + z) \log(n + z) - z \log z, \quad n \geq 1, \quad z \geq 1, \quad n + z \leq 7.$$

It satisfies $E_n(z) \geq 2n \log 2$. The derivative, with respect to the exponent, of $(n + z)^s E_n(z)/(n + z)$ is at most $7(\log 7)^2$. Thus lowering a top product at least $1 - \epsilon_7$ improves J , where

$$\epsilon_7 = \frac{2 \log 2 - 1}{14(\log 7)^2} > 0. \quad (\text{L.1})$$

Positivity here is analytic: $\log 2 = \int_1^2 dt/t > 1/2$. Also $\epsilon_7 < 1/2$, for instance using $\log 2 < 1$ and $\log 7 > 1$. We have proved $1/2 < s_6 \leq s_1 < 1 - \epsilon_7$.

Uniqueness does not require six or more intermediate ranks. To see this directly, set $S_6 = 2$ and $S_i = 1 + \exp(G_{i+1}/s_i)$ for $i < 6$. In a direction v put $g_i = DG_i[v]$, $Q_i = D^2 G_i[v, v]$, and write $a_i = \log S_i$, $u_i = s_{i+1}/s_i$, and $\beta_i = S_{i+1}^{u_i}/S_i$. Define $A_i = a_i - \beta_i u_i a_{i+1}$ for $i < 6$. Differentiation gives

$$g_i = A_i v_i + \beta_i g_{i+1}, \quad Q_i = \beta_i Q_{i+1} + \frac{\beta_i(1 - \beta_i)}{s_i} (g_{i+1} - u_i a_{i+1} v_i)^2.$$

At the terminal node $g_6 = (\log 2)v_6$ and $Q_6 = 0$. All coefficients are compactly bounded on the product range just proved. Since $g_i = a_i v_i + \beta_i (g_{i+1} - u_i a_{i+1} v_i)$, backward induction yields $D^2 Z_i[v, v] \geq c_7 \sum_{j \geq i} v_j^2$. Consequently $-D^2 J[v, v] \geq c_7 \sum_j V_j v_j^2 > 0$ for $v \neq 0$. This proves uniqueness, also on every equality subspace.

Thresholds and genuine old flow. For every real $h > 1$, set $\alpha_h = \log((h - 1)/\log h)/\log h$. With $x = (\log h)/2$,

$$\alpha_h = \frac{1}{2} + \frac{\log(\sinh x/x)}{2x}.$$

The second term is positive. Its derivative has the sign of $N(x) = x \coth x - 1 - \log(\sinh x/x)$; here $N(0+) = 0$ and $N'(x) = 1/x - x/\sinh^2 x > 0$. Thus

$$\frac{1}{2} < \alpha_2 < \alpha_3 < \cdots < \alpha_7 < 1. \quad (\text{L.2})$$

The upper inequality follows from $\log h > 1 - 1/h$. Using the exact transported identity

$$Z_g \prod_{l=g}^{j-1} \beta_l = Z_j \prod_{l=g}^{j-1} \beta_l^{1-s_l} \leq Z_j$$

in the bottom-plateau derivative strengthens the lower product bound to $s_6 > \alpha_2$. If that plateau contains a rank greater than two, one native term is strictly positive; if it consists only of rank two, the old transport inequality is strict and the old length is positive. These exhaust the complete $7, 6, \dots, 2$ chain.

Every adjacent product ratio exceeds $1/2$. A proper compound weight therefore satisfies $z > \sqrt{2}$, whereas the terminal weight is exactly 1. Original old coefficients satisfy

$$7^{-8} \leq \kappa_g \leq 1, \quad \kappa_{g+1} - \kappa_g \geq d_* > 0, \quad d_* := \min \left\{ \frac{\epsilon_7}{49}, 7^{-8} [1 - (6/7)^{\epsilon_7}] \right\}.$$

These statements concern natural reference coefficients; they do not resample an already revealed old list. Platform stationarity and the entropy integral give the unchanged hierarchy

$$V_{a-1} \leq C_7 \sum_{i=a}^b L_i, \quad C_7 = 2 \cdot 7^3 (1 + \log 7)^3.$$

Indeed $E_p(x)/p \geq 1 + \log x + p/[2(x+p)]$ gives the incoming surplus at least $np/[98(1 + \log 7)^2]$ at every strict platform boundary. It is positive without a lower bound on the product gap.

KKT reference geometry and all named contexts. On any owner put $g_g(q) = q - \kappa_g E_q(z)$. The exact clipped benchmarks satisfy

$$\Lambda_R = \sum_g L_g g_g(\min(r_g, R)) = \lambda_{b-R} \geq 0, \quad \Lambda_0 = \Lambda_{r_a} = 0.$$

For old groups the increasing coefficients permit their average over the first region, with the non-negative summation-by-parts correction $\int (\bar{\kappa} - \kappa(t)) E_{d(t)}(z) dt$. Reverse rank compression then gives each completed reference history as a nonnegative sum of a clipped benchmark and compression losses. The compression argument uses only $z \geq 1$, integer ranks, and the decreasing normalized secant

$$\frac{d}{dy} \frac{\log[(y \log y - b \log b)/(y - b)]}{\log y} \leq -\frac{1}{24 \cdot 49} \quad (1 \leq b < y \leq 7).$$

For completeness, at $b = 1$ this follows by writing $F(t) = \log \int_0^1 e^{tu} du$: the tilted density is at least $1/7$, so $F''(t) \geq 1/84$ and $(tF'(t) - F(t))/(e^t t^2) \geq 1/(24 \cdot 49)$. Increasing b decreases the displayed derivative, by the normalized-secant lemma proved in Appendix G.

All subsequent rank margins remain strictly positive. In particular $E''(q) = 1/(q+z) \geq 1/7$, native $\kappa_g \geq 1/7$, and

$$\Lambda_q + \frac{\log 7}{2} \sum_{r_g=q} L_g \geq \frac{1}{14} \sum_{r_g>q} \kappa_g L_g \geq \frac{L_i}{98} \quad (r_i > q).$$

This is the discrete second-difference identity with its two nonnegative neighboring KKT margins; it does not assume global concavity of Λ . The legal two-cut exchange pays a proper input, the normalized-secant update and this same identity pay a wrong retained suffix, and the true old

average surplus pays an old weak full input. Their thresholds can be chosen as the same finite minima of positive dimension constants as in the general proof. When a rank interval is empty its comparison is vacuous, rather than requiring a nonexistent intermediate rank.

For the short-side endpoint estimate an explicit positive choice is

$$d_0 = \min \left\{ \frac{d_*}{\log 7}, \frac{1}{1176}, \frac{\log(2 \log 2)}{7(\log 7)^2} \right\}, \quad c_* = 1 - 2^{-d_0/3} > 0.$$

For each clipped rank the normalized exponent increases by at least d_0 between consecutive original groups. Encountering a coefficient $f_g \leq c_*$ forces every later coefficient to be at most $-c_*$. Together with $\Lambda_R \geq 0$ this proves $\min_R B_R(V_l) \geq c_* \min(V_l, V_b - V_l)$. This assertion is only at original physical endpoints; a weak block is not artificially split to obtain it.

The named enumeration in the profile consequently applies literally. There are at most 21 canonical edges and at most 7^6 choices of outside cuts for each edge. Each removed name has a deadline at least the right endpoint of its assigned block. Even the left completion has a cut at that block's left endpoint, so its last positive time is at least V_{a-1} . This includes equality at the old/native boundary and time zero for a top owner. The incoming-free score comparison is therefore on its correct domain. It still subtracts the same incoming constant from both endpoints and has the same exact binary increment; all named contexts, not only deepest-suffix outside contexts, occur.

An owner of rank one causes no exception: a proper rank-one owner has its one edge $1 \rightarrow 0$ with $z > \sqrt{2}$, and a terminal rank-one owner has no genuine binary edge. Its zero-output edge is ordinary. For every retained binary edge, if $a = \log(r + z)$, $b = \log(q + z)$ and $p = (q + z)/(r + z)$, then

$$p, 1 - p \geq \frac{1}{7}, \quad b \geq \frac{\log 2}{2}, \quad \frac{1}{7} \leq a - bp \leq \log 7.$$

Thus its two-mark variance is positive uniformly, as is its mean, and its natural ρ lies in a fixed compact subinterval of $(0, 1)$. No degenerate terminal $q = 0$ law is inserted in the binary product.

The original branching comparisons also retain positive margins. For a proper owner one can take $\gamma = (\sqrt{2} - 1) \log(8/7) > 0$ in the pure-cell entropy gain, because its compound mass is at least $\sqrt{2}$ and at most 7. If the first name of that cell has expired, the gain is instead at least $2 \log 2$. For the terminal owner, where the proper-child gap cannot be used, a floating nonsingleton chain has $1 \leq d(t) \leq r(t) - 1$ throughout its floating interval. Its two legal reference rank changes $d \pm 1$ give the lower bound $7^{-8} T_{\text{float}}/14$ for its reference cost. At $r = 1$ such an interval is empty. Mixed-time domination uses only eligible-rank addition under the lawful parent/child projections. Thus the same positive mixed, spread, and reference shares pay the original birth errors and erased-pivot fibers with at most six original boundaries; no extra small-rank case is omitted. These are deterministic comparisons under the same named regularity and allocated-wall hypotheses as in the general history theorem.

The ordinary suffix. Set $c_w = 1/196$ and use exactly

$$\tau_7 = \min \left\{ \frac{1}{4}, \frac{1}{2744(\log 7)^2}, \frac{1}{4} \min_{2 \leq h < 7} (\alpha_{h+1} - \alpha_h) \right\} > 0.$$

The strict positivity of every gap was proved in (L.2); no numerical gap test is used. For $e_h = h^\sigma \log h - (h-1)$, $|e'_h| \leq 7(\log 7)^2$, so $|e_h| \leq c_w/2$ in its selected threshold interval. The intervals are disjoint. Moreover a genuine terminal binary weak transition $r \rightarrow q \geq 1$ at threshold $\epsilon_w \leq 1/(4 \cdot 7^3)$ obeys $g(r) \geq 1/196$. Indeed strong concavity gives

$$g(q) - \frac{q}{r} g(r) \geq \frac{q(r-q)}{98}, \quad g(r) \geq \frac{qr}{98} - 7\epsilon_w \geq \frac{1}{196}.$$

Its ordinary drift is therefore at most $-c_w$, so it cannot occur in the selected ordinary suffix.

The actual terminal platform balance gives total ordinary mean $D_{\text{all}} \geq b_7 V_{\text{old}}$, where $b_7 = 1/[98(1 + \log 7)^2]$; for empty old input it gives zero. All groups before the selected suffix have drift at most a fixed $-g_0 < 0$, because their finitely many thresholds are separated by τ_7 . Hence

$$D_{\text{suffix}} \geq b_7 V_{\text{old}} + g_0 L_{\text{head}} \geq 0, \quad U \geq c_7 V_6.$$

A positive-drift block exists whenever no present block is in a threshold interval, since otherwise the total mean would be negative. There are at most six suffix blocks, so its largest has length at least $U/6$. The empty-prefix, single-block, and zero total mean cases all remain included. This is precisely the geometric input to the single ordinary root, with the original counts and gaps unchanged. $\square$

Proposition L.2 (Explicit lift envelope at dimension six). *The finite search and actual-parameter-stability comparison in the definition of Φ_k have positive constants at $k = 6$. Their values are*

$$A_6 = 686, \quad B_6 = 174, \quad N = C = 21, \quad E = 25, \quad c = 42, \quad a = A_6 - C = 665.$$

In particular a valid search bound is

$$R = 1 + \max \left\{ 1, \max_i L_i, \frac{105000}{442225}, \frac{2(687 + 5 \log 2)}{665} \right\}.$$

The same formula for Φ_k therefore includes $k = 6$ on all $\ell_i \geq \log 3$, including zero original block lengths.

Proof. Substitution in the displayed dimension formulas gives these exact values. In particular $B_6 < A_6$ and $a > 0$. There are at most 21 binary factors, each bounded below by $(1 + 7L_i)^{-1}$; there are at most five proper exit factors, with $0 < d_j < 1$; and the sole ordinary factor is at least $(1 + V_6)^{-3/2}$. For $\lambda \geq \max_i L_i$ the lifted objective is thus at least $2^{-5} e^{665\lambda} (1 + 42\lambda)^{-25}$. The inequality $\log(1 + x) \leq \sqrt{x}$ gives the stated finite range exactly as in the profile proof. The actual arithmetic stability exponent is $174\lambda + O_6(1)$, strictly less than the chosen envelope penalty. Positive lifted lengths suffice for the geometry above; the infimum and actual stability recover zero blocks. $\square$

Remark L.3 (Scope of the specialization). This verification is of the finite geometric and defining inputs. The scalar bridge, full-type conditioning, original-class row, and arithmetic transfer steps keep their stated hypotheses and their separate proofs at $H = 7$. With these dimension-compact inputs, the same upper, lower, and lift-removal proof gives the dimension-six case using the same Φ_k .

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